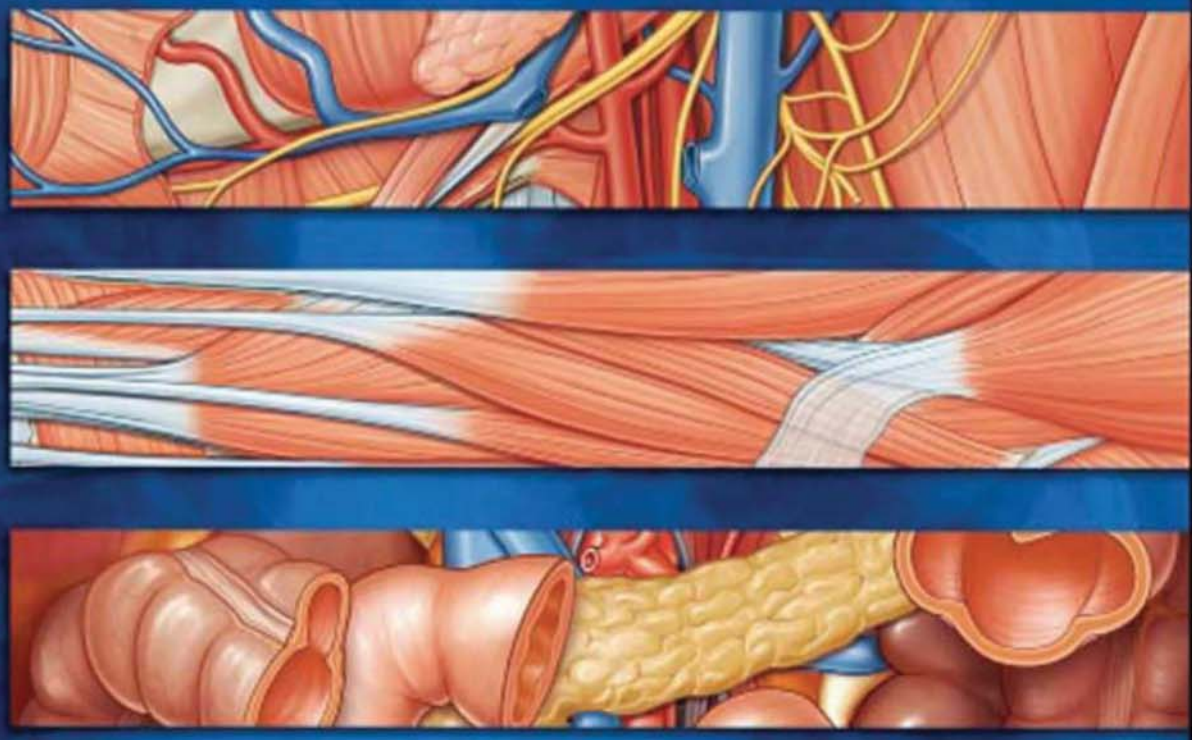
Study smart with Student Consult

GRAY'S ANATOMY

FOR STUDENTS

THIRD EDITION



Richard L. Drake A. Wayne Vogl Adam W. M. Mitchell

CHURCHILL LIVINGSTONE
ELSEVIER



Elsevier | StudentConsult.com

Transform the way you learn.

Compatible with PC, Mac®, most mobile devices, and eReaders, Student Consult allows you to browse, search, and interact with this title – online *and* offline. Redeem your PIN at studentconsult.com today!

SUBASH KC/NMC-15TH/2014

Start using these innovative features today:

- Seamless, real-time integration between devices
- Straightforward navigation and search
- Notes and highlights sharing with other users through social media
- Enhanced images with annotations, labels, and hot spots for zooming on specific details *
- Live streaming video and animations *
- Self-assessment tools such as questions embedded within the text and multiple-format quizzes *

* *some features vary by title*

PIN REDEMPTION INSTRUCTIONS

1. Login or Sign Up at StudentConsult.com
2. Scratch off your PIN code below
3. Enter PIN into the "Redeem a Book Code" box
4. Click "Redeem"
5. Go to "My Library"

Use of the current edition of the electronic version of this book (eBook) is subject to the terms of the nontransferable, limited license granted on StudentConsult.com. Access to the eBook is limited to the first individual who redeems the PIN, located on the inside cover of this book, at StudentConsult.com and may not be transferred to another party by resale, lending, or other means.

For technical assistance: Email: online.help@elsevier.com;
Call: within the US and Canada: 800-401-9962;
outside the US and Canada: +1-314-447-8200

ELSEVIER

MASTER ANATOMY your way.

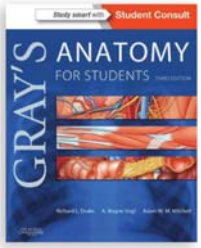
These *Gray's Anatomy* resources work together to promote a **vivid, visual understanding** that will carry you through your dissection labs, course exams, and beyond!

Get the edge you need for an optimal mastery of anatomy.

ORDER TODAY!

800-545-2522

www.elsevierhealth.com

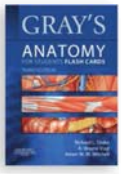


Gray's Anatomy for Students, 3rd Edition

with **STUDENT CONSULT** Online Access

Drake, Vogl & Mitchell

2014. ISBN: 978-0-7020-5131-9

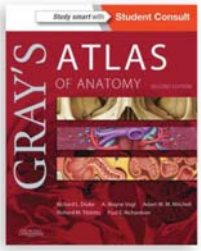


Gray's Anatomy for Students Flash Cards, 3rd Edition

with **STUDENT CONSULT** Online Access

Drake, Vogl & Mitchell

2014. ISBN: 978-1-4557-5898-2



Gray's Atlas of Anatomy, 2nd Edition

with **STUDENT CONSULT** Online Access

Drake et al.

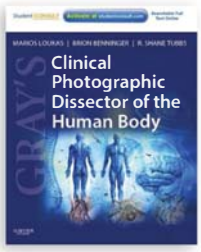
2014. ISBN: 978-1-4557-4802-0



Gray's Anatomy Review

Loukas et al.

2009. ISBN: 978-0-443-06938-3

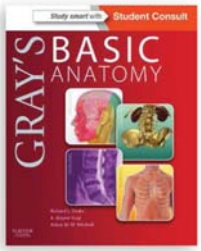


Gray's Clinical Photographic Dissector of the Human Body

with **STUDENT CONSULT** Online Access

Loukas, Benninger & Tubbs

2013. ISBN: 978-1-4377-2417-2

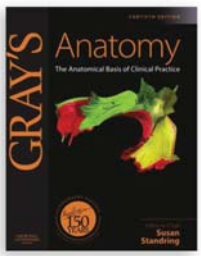


Gray's Basic Anatomy

with **STUDENT CONSULT** Online Access

Drake, Vogl & Mitchell

2013. ISBN: 978-1-4557-1078-2



Gray's Anatomy, 40th Edition

Expert Consult — Online and Print

Standring

2009. ISBN: 978-0-443-06684-9

This page intentionally left blank

SUBASH KC/NMC-15TH/2014

GRAY'S

Anatomy
for students

SUBASH KC/NMC-15TH/2014

SUBASH KC/NMC-15TH/2014

Richard L. Drake, PhD, FAAA
Director of Anatomy
Professor of Surgery
Cleveland Clinic Lerner College of Medicine
Case Western Reserve University
Cleveland, Ohio

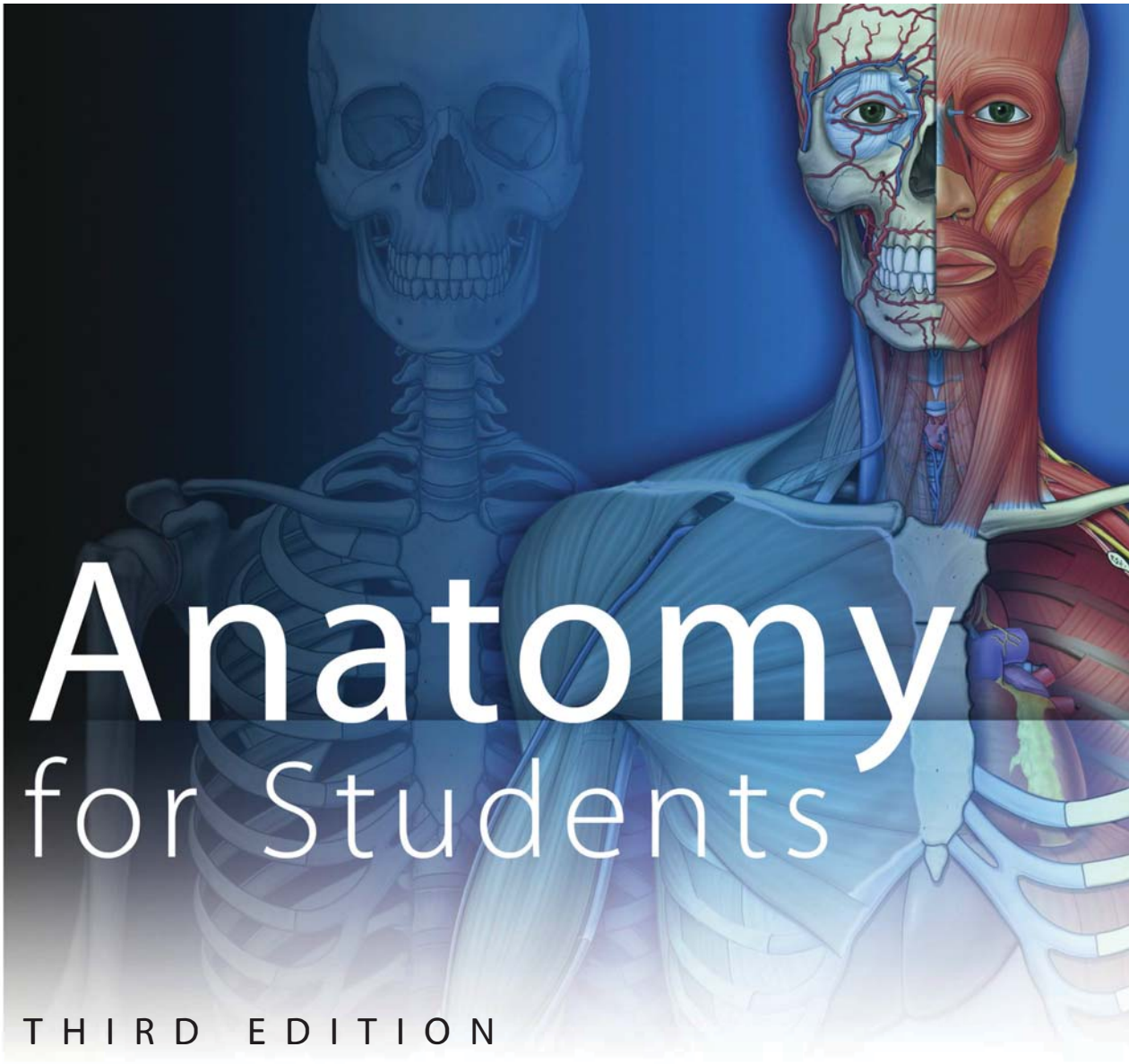
A. Wayne Vogl, PhD, FAAA
Professor of Anatomy and Cell Biology
Department of Cellular and Physiological Sciences
Faculty of Medicine
University of British Columbia
Vancouver, British Columbia, Canada

Adam W. M. Mitchell, MB BS, FRCS, FRCR
Consultant Radiologist
Chelsea and Westminster Hospital
Honorary Senior Lecturer Imperial College
London, United Kingdom

Illustrations by
Richard Tibbitts and **Paul Richardson**

Photographs by
Ansell Horn

GRAY'S



Anatomy for Students

THIRD EDITION

SUBASH KC/NMC-15TH/2014

CHURCHILL LIVINGSTONE
ELSEVIER

CHURCHILL
LIVINGSTONE
ELSEVIER

1600 John F. Kennedy Blvd.
Ste. 1800
Philadelphia, PA 19103-2899

GRAY'S ANATOMY FOR STUDENTS, THIRD EDITION
International Edition

ISBN: 978-0-7020-5131-9
ISBN: 978-0-7020-5132-6

Copyright © 2015, 2010, 2005 by Churchill Livingstone, an imprint of Elsevier Inc.

No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or any information storage and retrieval system, without permission in writing from the publisher. Details on how to seek permission, further information about the Publisher's permissions policies and our arrangements with organizations such as the Copyright Clearance Center and the Copyright Licensing Agency, can be found at our website: www.elsevier.com/permissions.
This book and the individual contributions contained in it are protected under copyright by the Publisher (other than as may be noted herein).

Notices
Knowledge and best practice in this field are constantly changing. As new research and experience broaden our understanding, changes in research methods, professional practices, or medical treatment may become necessary. Practitioners and researchers must always rely on their own experience and knowledge in evaluating and using any information, methods, compounds, or experiments described herein. In using such information or methods they should be mindful of their own safety and the safety of others, including parties for whom they have a professional responsibility.
With respect to any drug or pharmaceutical products identified, readers are advised to check the most current information provided (i) on procedures featured or (ii) by the manufacturer of each product to be administered, to verify the recommended dose or formula, the method and duration of administration, and contraindications. It is the responsibility of practitioners, relying on their own experience and knowledge of their patients, to make diagnoses, to determine dosages and the best treatment for each individual patient, and to take all appropriate safety precautions.
To the fullest extent of the law, neither the Publisher nor the authors, contributors, or editors, assume any liability for any injury and/or damage to persons or property as a matter of products liability, negligence or otherwise, or from any use or operation of any methods, products, instructions, or ideas contained in the material herein.

ISBN: 978-0-7020-5131-9

VP Global Medical Education Content: Madelene Hyde
Senior Manager, Content Development: Rebecca Gruliow
Publishing Services Manager: Patricia Tannian
Senior Project Manager: John Casey
Design Direction: Lou Forgione

Printed in Canada

Last digit is the print number: 9 8 7 6 5 4 3 2 1

SUBASH KC/NMC-15TH/2014

Acknowledgments

First, we would like to collectively thank those who agreed to examine earlier drafts of the book—anatomists, educators, and student members of the editorial review board from around the world. Your input was invaluable.

We'd also like to thank Richard Tibbitts and Paul Richardson for their skill in turning our visual ideas into a reality that is not only a foundation for the acquisition of anatomical knowledge, but also is beautiful.

Thanks must also go to Madelene Hyde, Bill Schmitt, Rebecca Gruliow, John Casey, and all the team at Elsevier for guiding us through the preparation of this book.

We'd also like to thank Professor Richard A. Buckingham of the Abraham Lincoln School of Medicine, University of Illinois for the provision of Fig. 8.114B. Finally, because we worked separately, distanced by, in some cases, thousands of miles, there are various people who gave local support, whom we would like to make mention of individually. We've gratefully listed them here:

Dr. Leonard Epp, Dr. Carl Morgan, Dr. Robert Shellhamer, and Dr. Robert Cardell who profoundly influenced my career as a scientist and an educator.

Richard L. Drake

Dr. Sydney Friedman, Dr. Elio Raviola, and Dr. Charles Slonecker, for their inspiration and support and for instilling in me a passion for the discipline of Anatomy.

Dr. Murray Morrison, Dr. Joanne Matsubara, Dr. Brian Westerberg, Laura Hall, and Jing Cui, for contributing images for the chapter on the head and neck.

Dr. Bruce Crawford and Logan Lee, for help with images for the surface anatomy of the upper limb.

Professor Elizabeth Akesson and Dr. Donna Ford, for their enthusiastic support and valuable critiques.

Dr. Sam Wiseman, for contributing surgical and other images in the abdomen and head and neck chapters.

A. Wayne Vogl

Dr. Sahar Nasseri (senior specialist registrar) for help with the images and text and adding critical analysis on modern imaging.

Drs. J. Healy, J. Lee, G. Rajeswarren, R. Pearce and B. Roberton for their support and constructive criticism.

The radiology staff at Chelsea and Westminster Hospital and The Fortius Clinic.

In particular, Mr. Andrew Williams, FRCS, whose relentless and unflagging support has been invaluable (and he mended my leg!).

Adam W. M. Mitchell

SUBASH KC/NMC-15TH/2014

Dedications

To my wife, Cheryl, who has supported me; and my parents, who have guided me.

—Richard L. Drake

To my family, to my professional colleagues and role models, and to my students—this book is for you.

—A. Wayne Vogl

To Cathy, Max, and Elsa

—Adam W. M. Mitchell

This page intentionally left blank

SUBASH KC/NMC-15TH/2014

Preface

The 3rd edition of Gray's Anatomy for Students builds on the past and looks toward the future.

It maintains the goals and objectives of the 1st and 2nd editions while at the same time continuing to incorporate input from our readers and adjusting the content to align with the evolving educational environment.

One of the major focuses of our attention as we prepared the 3rd edition was clinical content. The reason for this is that learning in context has become an important theme not only in medical education but in anatomical sciences education generally. We dealt with the clinical content in two ways. First, we reevaluated and updated the "In the clinic" boxes, clinical material in the body of the text, clinical cases at the end of the chapters and in the online resources, and in the surface anatomy section. Second, we added new clinical material so that the reader would have up-to-date examples relating anatomical information to clinical situations.

In addition to updating and revising the clinical content, the section on cranial nerves has been significantly expanded. Understanding these important components of the nervous system is critical for students at every level. To facilitate student learning a new figure summarizing the location, function, and distribution of each cranial nerve has been added, as has a new figure summarizing how major structures, including cranial nerves, enter and leave the cranial cavity. We also have included a new figure illustrating the clinical importance of vascular structures in and around the cranial cavity.

Another new feature in this edition relates to review materials. While these useful study aids have always been available on Student Consult as an online resource, in this edition the appropriate review materials for each chapter are listed at the beginning of that chapter. This information includes an online anatomy and embryology self-study course, medical clinical cases, physical therapy clinical cases, self-assessment questions, and more.

We believe that with these changes the 3rd edition of *Gray's Anatomy for Students* is a much improved version of the 2nd edition and hope that the book will continue to be a valuable learning resource for students.

Richard L. Drake
A. Wayne Vogl
Adam W. M. Mitchell
December 2013

SUBASH KC/NMC-15TH/2014

This page intentionally left blank

SUBASH KC/NMC-15TH/2014

About the Book

The idea

In the past 20 years or so, there have been many changes that have shaped how students learn human anatomy in medical and dental schools and in allied health programs, with curricula becoming either more integrated or more systems based. In addition, instructional methods focus on the use of small group activities with the goals of increasing the amount of self-directed learning, and acquiring the skills for the life-long acquisition of knowledge. An explosion of information in every discipline has also been a force in driving curricular change as it increases the amount to be learned without necessarily increasing the time available. With these changes, we felt it was time for a new text to be written that would allow students to learn anatomy within the context of many different curricular designs, and within ever-increasing time constraints.

We began in the fall of 2001 by considering the various approaches and formats that we might adopt, eventually deciding upon a regional approach to anatomy with each chapter having four sections. From the beginning, we wanted the book to be designed with multiple entry points, to be targeted at introductory level students in a broad spectrum of fields, and to be a student-oriented companion text for *Gray's Anatomy*, which is aimed at a more professional audience. We wrote the text first and subsequently constructed all the artwork and illustrations to complement and augment the words. Preliminary drafts of chapters, when complete, were distributed to an international editorial board of anatomists, educators, and anatomy students for review. Their comments were then considered carefully in the preparation of the final book.

The text is not meant to be exhaustive in coverage, but to present enough anatomy to provide students with a structural and functional context in which to add further detail as they progress through their careers. *Gray's Anatomy* was used as the major reference, both for the text and for the illustrations, during the preparation of this book, and it is the recommended source for acquiring additional detail.

The book

Gray's Anatomy for Students is a clinically oriented, student-friendly textbook of human anatomy. It has been prepared primarily for students in a variety of professional programs (e.g., medical, dental, chiropractic, and physical therapy programs). It can be used by students in traditional, systemic, combined traditional/systemic, and problem-based curricula and will be particularly useful to students when lectures and laboratories in gross anatomy are minimal.

ORGANIZATION

Using a regional approach, *Gray's Anatomy for Students* progresses through the body in a logical fashion, building on the body's complexities as the reader becomes more comfortable with the subject matter. Each chapter can be used as an independent learning module, and varying the sequence will not affect the quality of the educational experience. The sequence we have chosen to follow is back, thorax, abdomen, pelvis and perineum, lower limb, upper limb, and head and neck.

We begin with "The body," which contains an overview of the discipline of gross anatomy and an introduction to imaging modalities and general body systems. We follow this with the back because it is often the first area dissected by students. The thorax is next because of its central location and its contents (i.e., the heart, the great vessels, and the lungs). This also begins a progression through the body's cavities. The abdomen and pelvis and perineum follow logically in sequence from the thorax. Continuing downward toward the feet, the lower limb is next, followed by the upper limb. The last region discussed is the head and neck. This region contains some of the most difficult anatomy in the body. Covering all other regions first gives the student the opportunity to build a strong foundation from which to understand this complex region.

CONTENT

Each regional anatomy chapter consists of four consecutive sections: conceptual overview, regional anatomy, surface anatomy, and clinical cases.

The conceptual overview provides the basis on which information in the later sections is built. This section can be read independently of the rest of the text by students who require only a basic level of understanding and can also be read as a summary of important concepts after the regional anatomy has been mastered.

The regional anatomy section provides more detailed anatomy along with a substantial amount of relevant clinical correlations. It is not an exhaustive discussion but instead provides information to a level that we feel is necessary for understanding the organization of the region.

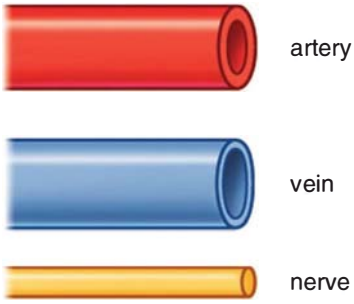
SUBASH KC/NMC-15TH/2014

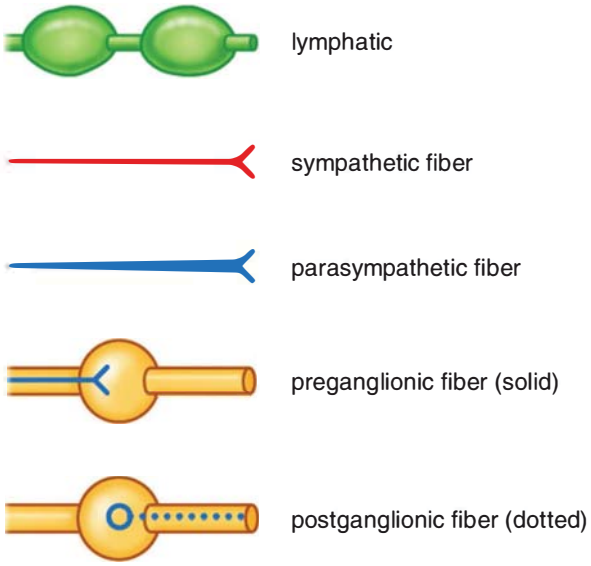
Throughout this section, two levels of clinical material are provided. Clinical hooks are fully integrated with the main anatomical text and function to relate ("hook") the anatomy discussed directly to a clinical application without taking students out of their train of thought and without disrupting the flow of the text. Although fully integrated with the anatomical text, these passages are differentiated from it by the use of green highlighting. "In the clinic" summaries provide students with useful and relevant clinical information demonstrating how applying anatomical knowledge facilitates the solving of clinical problems. These are spread throughout the text close to the most relevant anatomical discussion.

Surface anatomy assists students in visualizing the relationship between anatomical structures and surface landmarks. This section also provides students with practical applications of the anatomical information, combining visual inspection with functional assessment, as occurs during any type of patient examination.

The final section of each chapter consists of clinical cases. These cases represent the third level of clinical material in the book. In these cases the clinical problem is described, and a step-by-step process of questions and answers leads the reader to the resolution of the case. The inclusion of these cases in each chapter provides students with the opportunity to apply an understanding of anatomy to the resolution of a clinical problem.

Illustrations are an integral part of any anatomy text. They must present the reader with a visual image that brings the text to life and presents views that will assist in the understanding and comprehension of the anatomy. The artwork in this text accomplishes all of these goals. The illustrations are original and vibrant, and many views are unique. They have been designed to integrate with the text, present the anatomy in new ways, deal with the issues that students find particularly difficult, and provide a conceptual framework for building further understanding. To ensure that the illustrations of the book work together and to enable students to cross-refer from one illustration to another, we have used standard colors throughout the book, except where indicated otherwise.





The position and size of the artwork was one of the parameters considered in the overall design of each page of the book.

Clinical images are also an important tool in understanding anatomy and are abundant throughout the text. Examples of state-of-the-art medical imaging, including MRIs, CTs, PETs, and ultrasound, as well as high-quality radiographs, provide students with additional tools to increase their ability to visualize anatomy in vivo and, thus, increase their understanding.

What the book does not contain

Gray's Anatomy for Students focuses on gross anatomy. While many curricula around the world are being presented in a more integrated format combining anatomy, physiology, histology, and embryology, we have focused this textbook on understanding only the anatomy and its application to clinical problems. Except for some brief references to embryology where necessary for a better understanding of the anatomy, material from other disciplines is not included. We felt that there are many outstanding textbooks covering these subject areas, and that trying to cover everything in a single book would produce a text of questionable quality and usefulness, not to mention enormous size!

Terminology

In any anatomical text or atlas, terminology is always an interesting issue. In 1989, the Federative Committee on Anatomical Terminology (FCAT) was formed and

SUBASH KC/NMC-15TH/2014

was charged with developing the official terminology of the anatomical sciences. The *Terminologia Anatomica* (2nd edition, Thieme, Stuttgart/New York, 2011) was a joint publication by this group and the 56 member associations of the International Federation of Associations of Anatomists (IFAA). We have chosen to use the terminology presented in this publication in the interest of uniformity. Other terminology is not incorrect; we just felt that using terminology from this single, internationally recognized source would be the most logical and straightforward approach.

Although we use anatomical terms for orientation as much as possible, we also use terms such as "behind" or "in front of" occasionally to make the text more readable. In these cases, the context clarifies the meaning.

Anatomical use of adverbs

During the writing of this book, we had many long discussions about how we were going to describe anatomical relationships as clearly as possible, but maintain the readability of the text. One issue that arose continually in our discussions was the correct use of the "-ly" adverb form of anatomical orientation terms, such as anterior, posterior, superior, inferior, lateral, and medial. We reached the following consensus:

-ly adverbs *e.g., anteriorly, posteriorly*, have been used to modify (describe) verbs in passages where an action or direction is mentioned. For example, "The trachea passes inferiorly through the thorax."

circumstantial adverbs, *e.g., anterior, posterior*, have been used to indicate the fixed location of an anatomical feature. For example, "The trachea is anterior to the esophagus."

Furthermore, both usages may occur in the same passage. For example, "The trachea passes inferiorly through the thorax, anterior to the esophagus."

We have very much enjoyed the process of putting this book together. We hope that you enjoy using it to the same degree.

Richard L. Drake
A. Wayne Vogl
Adam W. M. Mitchell

SUBASH KC/NMC-15TH/2014

This page intentionally left blank

SUBASH KC/NMC-15TH/2014

Index of Clinical Content

1 The body

In the Clinic

Determination of skeletal age 14
Bone marrow transplants 15
Bone fractures 16
Avascular necrosis 16
Osteoporosis 17
Epiphyseal fractures 18
Degenerative joint disease 22
Joint replacement 24
The importance of fascias 25
Muscle paralysis 26
Muscle atrophy 26
Muscle injuries and strains 26
Atherosclerosis 27
Varicose veins 28
Anastomoses and collateral circulation 28
Lymph nodes 31
Dermatomes and myotomes 37
Referred pain 48

Clinical Case

Appendicitis 50

2 Back

In the Clinic

Spina bifida 74
Vertebroplasty 74
Scoliosis 75
Kyphosis 76
Variation in vertebral numbers 76
The vertebrae and cancer 77
Osteoporosis 77
Back pain 79
Herniation of intervertebral discs 79
Joint diseases 80
Ligamenta flava 82
Vertebral fractures 82
Surgical procedures on the back 84
Nerve injuries affecting superficial back muscles 99
Lumbar cerebrospinal fluid tap 106
Herpes zoster 109
Back pain—alternative explanations 110

Clinical Cases

Sciatica versus lumbago 118
Cervical spinal cord injury 118
Psoas abscess 119
Dissecting thoracic aneurysm 119
Sacral tumor 120

3 Thorax

In the Clinic

Axillary tail of breast 141
Breast cancer 141
Cervical ribs 150
Collection of sternal bone marrow 152
Rib fractures 152
Surgical access to the chest 160
Thoracostomy (chest) tube insertion 160
Intercostal nerve block 160
Pleural effusion 167
Pneumothorax 167
Imaging the lungs 178
High-resolution lung CT 178
Bronchoscopy 178
Lung cancer 179
Pericarditis 184
Pericardial effusion 184
Constrictive pericarditis 184
Valve disease 197
Clinical terminology for coronary arteries 201
Heart attack 202
Classic symptoms of heart attack 203
Are heart attack symptoms the same in men and women? 203
Common congenital heart defects 203
Cardiac auscultation 204
Cardiac conduction system 206
Ectopic parathyroid glands in the thymus 212
Venous access for central and dialysis lines 215
Using the superior vena cava to access the inferior vena cava 215
Coarctation of the aorta 217

SUBASH KC/NMC-15TH/2014

Thoracic aorta 217
Aortic arch and its anomalies 217
Abnormal origin of great vessels 217
The vagus nerves, recurrent laryngeal nerves, and hoarseness 221
Esophageal cancer 224
Esophageal rupture 225

Clinical Cases

Cervical rib 241
Lung cancer 242
Chest wound 242
Myocardial infarction 243
Broken pacemaker 246
Coarctation of the aorta 247
Aortic dissection 247
Pneumonia 249
Esophageal cancer 250
Venous access 251

4 Abdomen

In the Clinic

Surgical incisions 278
Laparoscopic surgery 279
Cremasteric reflex 299
Masses around the groin 301
Peritoneum 305
The greater omentum 308
Epithelial transition between the abdominal esophagus and stomach 315
Duodenal ulceration 315
Examination of the upper gastrointestinal tract 316
Examination of the bowel lumen 316
Examination of the bowel wall and extrinsic masses 316
Meckel's diverticulum 318
Computed tomography (CT) scanning and magnetic resonance imaging (MRI) 318
Carcinoma of the stomach 318
Appendicitis 322
Congenital disorders of the gastrointestinal tract 325
Bowel obstruction 326
Diverticular disease 327
Ostomies 327
Annular pancreas 336
Pancreatic cancer 336
Segmental anatomy of the liver 339
Gallstones 341
Jaundice 341
Spleen disorders 342
Vascular supply to the gastrointestinal system 351
Hepatic cirrhosis 356
Surgery for obesity 365
Psoas muscle abscess 371
Diaphragmatic hernias 372
Hiatus hernia 373
Urinary tract stones 380
Urinary tract cancer 381
Nephrostomy 382
Kidney transplant 383
Investigation of the urinary tract 385
Abdominal aortic stent graft 389
Inferior vena cava filter 391
Retroperitoneal lymph node surgery 393

Clinical Cases

Traumatic rupture of the diaphragm 410
Chronic thrombosis of the inferior vena cava 410
Liver biopsy in patients with suspected liver cirrhosis 411
Hodgkin's lymphoma 412
Inguinal hernia 413
Ureteric stone 413
Intraabdominal abscess 414
Complications of an abdominoperineal resection 415
Carcinoma of the head of the pancreas 417
Caval obstruction 418
Diverticular disease 418
Endoleak after endovascular repair of abdominal aortic aneurysm 419
Metastatic lesions in the liver 420

5 Pelvis and perineum

In the Clinic

Bone marrow biopsy 444
Pelvic fracture 446
Common problems with the sacro-iliac joints 448
Pelvic measurements in obstetrics 454
Defecation 456
Episiotomy 460

SUBASH KC/NMC-15TH/2014

Digital rectal examination 462
Carcinoma of the colon and rectum 462
Bladder stones 465
Suprapubic catheterization 466
Bladder cancer 466
Bladder infection 469
Urethral catheterization 469
Testicular tumors 470
Vasectomy 472
Prostate problems 474
Ovarian cancer 477
Imaging the ovary 477
Hysterectomy 478
Tubal ligation 479
Carcinoma of the cervix and uterus 480
The recto-uterine pouch 481
Pudendal block 491
Prostatectomy and impotence 495
Abscesses in the ischio-anal fossae 504
Hemorrhoids 504
Urethral rupture 512

Clinical Cases

Varicocele 527
Sciatic nerve compression 528
Pelvic kidney 528
Left common iliac artery obstruction 529
Iatrogenic ureteric injury 530
Ectopic pregnancy 530
Uterine tumor 531
Uterine fibroids 532

6 Lower limb

In the Clinic

Pelvic fractures 553
Femoral neck fractures 557
Intertrochanteric fractures 558
Femoral shaft fractures 558
Varicose veins 569
Deep vein thrombosis 569
Vascular access to the lower limb 573
Trendelenburg's sign 577
Intramuscular injections 581
Compartment syndrome 590
Muscle injuries to the lower limb 599
Peripheral vascular disease 603
Soft tissue injuries to the knee 613
Degenerative joint disease/osteoarthritis 614
Examination of the knee joint 614
Anterolateral ligament of the knee 615
Achilles tendon rupture 623
Neurological examination of the legs 625
Footdrop 633
Fracture of the talus 638
Ankle fractures 641
Bunions 645
Morton's neuroma 661

Clinical Cases

Varicose veins 672
Knee joint injury 673
Fracture of neck of femur 676
Deep vein thrombosis 677
Ruptured calcaneal tendon 678
Popliteal artery aneurysm 679
Anterior talofibular ligament tear 680

7 Upper limb

In the Clinic

Fracture of the proximal humerus 705
Fractures of the clavicle and dislocations of the acromioclavicular and sternoclavicular joints 711
Dislocations of the glenohumeral joint 712
Rotator cuff disorders 712
Inflammation of the subacromial (subdeltoid) bursa 713
Quadrangular space syndrome 720
"Winging" of the scapula 727
Imaging the blood supply to the upper limb 737
Trauma to the arteries of the upper limb 737
Subclavian/axillary venous access 737
Injuries to the brachial plexus 747
Breast cancer 749
Rupture of biceps tendon 755
Blood pressure measurement 756
Radial nerve injury in the arm 763
Median nerve injury in the arm 763
Supracondylar fracture of the humerus 766
Pulled elbow 766
Developmental changes in the elbow joint 767
Fracture of the head of radius 768

SUBASH KC/NMC-15TH/2014

"Tennis" and "golfer's" elbow (epicondylitis) 768
Elbow arthritis 768
Ulnar nerve injury at the elbow 768
Construction of a dialysis fistula 770
Fractures of the radius and ulna 774
Transection of the radial or ulnar artery 783
Fracture of the scaphoid and avascular necrosis of the proximal scaphoid 797
Carpal tunnel syndrome 798
Snuffbox 801
De Quervain's syndrome 802
Tenosynovitis 802
Trigger finger 802
Allen's test 814
Venipuncture 814
Ulnar nerve injury 816
Radial nerve injury 818

Clinical Cases

Shoulder problem after falling on an outstretched hand 829
Winged scapula 829
Brachial plexus nerve block 830
Complication of a fractured first rib 830
Median nerve compression 831
Immobilizing the extensor digitorum muscle 831
Torn supraspinatus tendon 832
How to examine the hand 833
Shoulder joint problem 834

8 Head and neck

In the Clinic

Medical imaging of the head 871
Fractures of the skull vault 872
Hydrocephalus 877
Cerebrospinal fluid leak 878
Meningitis 878
Brain tumors 878
Stroke 883
Endarterectomy 885
Intracerebral aneurysms 885
Scalp and meninges 890
Head injury 891
Types of intracranial hemorrhage 891
Emissary veins 893
Concussion 893
Clinical assessment of patients with head injury 893
Treatment of head injury 893
Cranial nerve lesions 901
Overview of cranial nerves 901
Parotid gland 913
Facial nerve [VII] palsy (Bell's palsy) 921
Trigeminal neuralgia 921
Scalp laceration 926
Orbital fracture 928
Horner's syndrome 931
Examination of the eye 940
Glaucoma 948
Cataracts 948
Ophthalmoscopy 949
High-definition optical coherence tomography 951
Examination of the ear 957
Swimmer's ear 957
Surfer's ear 958
Tympanic membrane perforation 958
Mastoiditis 961
Lingual nerve injury 987
Dental anesthesia 989
Fascial planes of the head and neck 1004
Central venous access 1005
Jugular venous pulse 1013
Thyroid gland 1020
Thyroidectomy 1021
Thyroid gland pathology 1021
Ectopic parathyroid glands 1022
Recurrent laryngeal nerve palsy 1034
Clinical lymphatic drainage of the head and neck 1040
Tracheostomy 1065
Laryngoscopy 1065
Deviated nasal septum 1077

Clinical Cases

Multinodular goiter 1129
Parotid duct calculus 1130
Extradural hematoma 1131
Stenosis of the internal carotid artery 1132
Posterior communicating artery aneurysm 1133
Recurrent epistaxis 1133
Complication of orbital fracture 1134
Pituitary macroadenoma 1135

SUBASH KC/NMC-15TH/2014

Contents

1 The body

What is anatomy? 2
How can gross anatomy be studied? 2
Important anatomical terms 2

Imaging 5
Diagnostic imaging techniques 5
Nuclear medicine imaging 8

Image interpretation 10
Plain radiography 10
Computed tomography 10
Magnetic resonance imaging 11
Nuclear medicine imaging 11

Safety in imaging 11

Body systems 12
Skeletal system 12
Cartilage 12
Bone 13
Joints 18
Skin and fascias 24
Skin 24
Fascia 24
Muscular system 25
Cardiovascular system 27
Lymphatic system 29
Lymphatic vessels 29
Lymph nodes 30
Lymphatic trunks and ducts 30
Nervous system 31
Central nervous system 31
Functional subdivisions of the CNS 32
Somatic part of the nervous system 33
Visceral part of the nervous system 38
Other systems 48

Clinical case 50

2 Back

Conceptual overview 53
General description 53
Functions 54
Support 54
Movement 54
Protection of the nervous system 55
Component parts 56
Bones 56
Muscles 57
Vertebral canal 59
Spinal nerves 60
Relationship to other regions 61
Head 61
Thorax, abdomen, and pelvis 62
Limbs 62
Key features 62
Long vertebral column and short spinal cord 62
Intervertebral foramina and spinal nerves 63
Innervation of the back 63

Regional anatomy 64
Skeletal framework 64
Vertebrae 64
Intervertebral foramina 72
Posterior spaces between vertebral arches 73
Joints 77
Joints between vertebrae in the back 77
Ligaments 80
Anterior and posterior longitudinal ligaments 80
Ligamenta flava 80
Supraspinous ligament and ligamentum nuchae 81
Interspinous ligaments 82
Back musculature 84
Superficial group of back muscles 84
Intermediate group of back muscles 90
Deep group of back muscles 92
Suboccipital muscles 97
Spinal cord 99
Vasculature 100
Meninges 103
Arrangement of structures in the vertebral canal 104
Spinal nerves 106

Surface anatomy 111
Back surface anatomy 111
Absence of lateral curvatures 111
Primary and secondary curvatures in the sagittal plane 112
Useful nonvertebral skeletal landmarks 112
How to identify specific vertebral spinous processes 114
Visualizing the inferior ends of the spinal cord and subarachnoid space 115
Identifying major muscles 116

Clinical cases 118

SUBASH KC/NMC-15TH/2014

3 Thorax

Conceptual overview 123

General description 123

Functions 124
- Breathing 124
- Protection of vital organs 124
- Conduit 124

Component parts 124
- Thoracic wall 124
- Superior thoracic aperture 126
- Inferior thoracic aperture 126
- Diaphragm 127
- Mediastinum 128
- Pleural cavities 128

Relationship to other regions 130
- Neck 130
- Upper limb 130
- Abdomen 130
- Breast 131

Key features 132
- Vertebral level TIV/V 132
- Venous shunts from left to right 132
- Segmental neurovascular supply of thoracic wall 134
- Sympathetic system 136
- Flexible wall and inferior thoracic aperture 136
- Innervation of the diaphragm 138

Regional anatomy 139

Pectoral region 139
- Breast 139
- Muscles of the pectoral region 142

Thoracic wall 143
- Skeletal framework 143
- Intercostal spaces 150

Diaphragm 161
- Arterial supply 162
- Venous drainage 162
- Innervation 162

Movements of the thoracic wall and diaphragm during breathing 162

Pleural cavities 162
- Pleura 163
- Lungs 167

Mediastinum 180
- Middle mediastinum 180
- Superior mediastinum 210
- Posterior mediastinum 222
- Anterior mediastinum 230

Surface anatomy 231

Thorax surface anatomy 231

How to count ribs 231

Surface anatomy of the breast in women 232

Visualizing structures at the TIV/V vertebral level 232

Visualizing structures in the superior mediastinum 234

Visualizing the margins of the heart 235

Where to listen for heart sounds 236

Visualizing the pleural cavities and lungs, pleural recesses, and lung lobes and fissures 236

Where to listen for lung sounds 238

Clinical cases 241

4 Abdomen

Conceptual overview 255

General description 255

Functions 256
- Houses and protects major viscera 256
- Breathing 258
- Changes in intraabdominal pressure 258

Component parts 259
- Wall 259
- Abdominal cavity 260
- Inferior thoracic aperture 262
- Diaphragm 262
- Pelvic inlet 263

Relationship to other regions 263
- Thorax 263
- Pelvis 263
- Lower limb 264

Key features 265
- Arrangement of abdominal viscera in the adult 265
- Skin and muscles of the anterior and lateral abdominal wall and thoracic intercostal nerves 268
- The groin is a weak area in the anterior abdominal wall 269
- Vertebral level LI 271
- The gastrointestinal system and its derivatives are supplied by three major arteries 271
- Venous shunts from left to right 273
- All venous drainage from the gastrointestinal system passes through the liver 274
- Abdominal viscera are supplied by a large prevertebral plexus 276

Regional anatomy 277

Surface topography 277
- Four-quadrant pattern 277
- Nine-region pattern 278

SUBASH KC/NMC-15TH/2014

Abdominal wall 280
Superficial fascia 280
Anterolateral muscles 282
Extraperitoneal fascia 288
Peritoneum 288
Innervation 289
Arterial supply and venous drainage 291
Lymphatic drainage 292
Groin 292
Inguinal canal 294
Inguinal hernias 299
Abdominal viscera 303
Peritoneum 303
Peritoneal cavity 304
Organs 310
Arterial supply 343
Venous drainage 354
Lymphatics 358
Innervation 358
Posterior abdominal region 366
Posterior abdominal wall 367
Viscera 373
Vasculature 387
Lymphatic system 392
Nervous system in the posterior abdominal region 394
Sympathetic trunks and splanchnic nerves 394

Surface anatomy 402
Abdomen surface anatomy 402
Defining the surface projection of the abdomen 402
How to find the superficial inguinal ring 403
How to determine lumbar vertebral levels 404
Visualizing structures at the LI vertebral level 405
Visualizing the position of major blood vessels 406
Using abdominal quadrants to locate major viscera 407
Defining surface regions to which pain from the gut is referred 408
Where to find the kidneys 409
Where to find the spleen 409

Clinical cases 410

5 Pelvis and perineum

Conceptual overview 423
General description 423
Functions 423
Contain and support bladder, rectum, anal canal, and reproductive tracts 423
Anchors the roots of the external genitalia 425
Component parts 426
Pelvic inlet 426
Pelvic walls 426
Pelvic outlet 428
Pelvic floor 429
Pelvic cavity 429
Perineum 430
Relationship to other regions 432
Abdomen 432
Lower limb 433
Key features 434
The pelvic cavity projects posteriorly 434
Important structures cross the ureters in the pelvic cavity 435
The prostate in men and the uterus in women are anterior to the rectum 436
The perineum is innervated by sacral spinal cord segments 436
Nerves are related to bone 437
Parasympathetic innervation from spinal cord levels S2 to S4 controls erection 438
Muscles and fascia of the pelvic floor and perineum intersect at the perineal body 439
The course of the urethra is different in men and women 440

Regional anatomy 441
Pelvis 441
Bones 441
Joints 446
Orientation 448
Differences between men and women 448
True pelvis 449
Viscera 460
Fascia 481
Peritoneum 481
Nerves 486
Blood vessels 495
Lymphatics 501
Perineum 502
Borders and ceiling 502
Ischio-anal fossae and their anterior recesses 504
Anal triangle 504
Urogenital triangle 506
Somatic nerves 513
Visceral nerves 515
Blood vessels 516
Veins 516
Lymphatics 519

Surface anatomy 520
Surface anatomy of the pelvis and perineum 520
Orientation of the pelvis and perineum in the anatomical position 520

SUBASH KC/NMC-15TH/2014

How to define the margins of the perineum 520
Identification of structures in the anal triangle 522
Identification of structures in the urogenital triangle of women 523
Identification of structures in the urogenital triangle of men 524

Clinical cases 527

6 Lower limb

Conceptual overview 535
General introduction 535
Function 537
Support the body weight 537
Locomotion 537
Component parts 539
Bones and joints 539
Muscles 543
Relationship to other regions 545
Abdomen 545
Pelvis 545
Perineum 545
Key points 545
Innervation is by lumbar and sacral spinal nerves 545
Nerves related to bone 550
Superficial veins 550

Regional anatomy 551
Bony pelvis 551
Proximal femur 554
Hip joint 558
Gateways to the lower limb 562
Nerves 563
Arteries 566
Veins 568
Lymphatics 570
Deep fascia and the saphenous opening 571
Femoral triangle 572
Gluteal region 574
Muscles 574
Nerves 579
Arteries 582
Veins 583
Lymphatics 583
Thigh 583
Bones 584
Muscles 589
Arteries 600
Veins 603
Nerves 603
Knee joint 606
Tibiofibular joint 616
Popliteal fossa 616
Leg 618
Bones 618
Joints 620
Posterior compartment of leg 621
Lateral compartment of leg 628
Anterior compartment of leg 630
Foot 633
Bones 634
Joints 638
Tarsal tunnel, retinacula, and arrangement of major structures at the ankle 646
Arches of the foot 648
Plantar aponeurosis 649
Fibrous sheaths of toes 649
Extensor hoods 650
Intrinsic muscles 650
Arteries 657
Veins 659
Nerves 659

Surface anatomy 663
Lower limb surface anatomy 663
Avoiding the sciatic nerve 663
Finding the femoral artery in the femoral triangle 664
Identifying structures around the knee 664
Visualizing the contents of the popliteal fossa 666
Finding the tarsal tunnel—the gateway to the foot 667
Identifying tendons around the ankle and in the foot 668
Finding the dorsalis pedis artery 669
Approximating the position of the plantar arterial arch 669
Major superficial veins 670
Pulse points 671

Clinical cases 672

7 Upper limb

Conceptual overview 685
General description 685
Functions 686
Positioning the hand 686
The hand as a mechanical tool 689
The hand as a sensory tool 689

SUBASH KC/NMC-15TH/2014

Component parts 690
Bones and joints 690
Muscles 692
Relationship to other regions 693
Neck 693
Back and thoracic wall 694
Key points 695
Innervation by cervical and upper thoracic nerves 695
Nerves related to bone 699
Superficial veins 700
Orientation of the thumb 701

Regional anatomy 702
Shoulder 702
Bones 702
Joints 705
Muscles 713
Posterior scapular region 716
Muscles 717
Gateways to the posterior scapular region 717
Nerves 719
Arteries and veins 719
Axilla 721
Axillary inlet 723
Anterior wall 723
Medial wall 726
Lateral wall 727
Posterior wall 728
Gateways in the posterior wall 730
Floor 731
Contents of the axilla 731
Arm 750
Bones 751
Muscles 754
Arteries and veins 756
Nerves 760
Elbow joint 764
Cubital fossa 768
Forearm 771
Bones 773
Joints 774
Anterior compartment of the forearm 776
Muscles 776
Arteries and veins 782
Nerves 784
Posterior compartment of the forearm 785
Muscles 785
Arteries and veins 791
Nerves 792
Hand 792
Bones 793
Joints 795
Carpal tunnel and structures at the wrist 798
Palmar aponeurosis 800
Palmaris brevis 800
Anatomical snuffbox 800
Fibrous digital sheaths 801
Extensor hoods 802
Muscles 804
Arteries and veins 810
Nerves 814

Surface anatomy 819
Upper limb surface anatomy 819
Bony landmarks and muscles of the posterior scapular region 819
Visualizing the axilla and locating contents and related structures 820
Locating the brachial artery in the arm 821
The triceps brachii tendon and position of the radial nerve 822
Cubital fossa (anterior view) 822
Identifying tendons and locating major vessels and nerves in the distal forearm 824
Normal appearance of the hand 825
Position of the flexor retinaculum and the recurrent branch of the median nerve 826
Motor function of the median and ulnar nerves in the hand 826
Visualizing the positions of the superficial and deep palmar arches 827
Pulse points 827

Clinical cases 829

8 Head and neck

Conceptual overview 837
General description 837
Head 837
Neck 839
Functions 841
Protection 841
Contains upper parts of respiratory and digestive tracts 841
Communication 841
Positioning the head 841
Connects the upper and lower respiratory and digestive tracts 841
Component parts 842
Skull 842
Cervical vertebrae 844

SUBASH KC/NMC-15TH/2014

Hyoid bone 845
Soft palate 846
Muscles 846
Relationship to other regions 847
Thorax 847
Upper limbs 847
Key features 848
Vertebral levels CIII/IV and CV/VI 848
Airway in the neck 849
Cranial nerves 850
Cervical nerves 851
Functional separation of the digestive and respiratory passages 851
Triangles of the neck 854
Regional anatomy 855
Skull 855
Anterior view 855
Lateral view 857
Posterior view 859
Superior view 860
Inferior view 860
Cranial cavity 864
Roof 864
Floor 865
Meninges 873
Cranial dura mater 873
Arachnoid mater 876
Pia mater 877
Arrangement of meninges and spaces 877
Brain and its blood supply 879
Brain 879
Blood supply 879
Venous drainage 886
Cranial nerves 894
Olfactory nerve [I] 896
Optic nerve [II] 896
Oculomotor nerve [III] 897
Trochlear nerve [IV] 897
Trigeminal nerve [V] 898
Ophthalmic nerve [V_1] 898
Maxillary nerve [V_2] 898
Mandibular nerve [V_3] 898
Abducent nerve [VI] 898
Facial nerve [VII] 898
Vestibulocochlear nerve [VIII] 899
Glossopharyngeal nerve [IX] 899
Vagus nerve [X] 900
Accessory nerve [XI] 900
Hypoglossal nerve [XII] 900
Face 904
Muscles 904
Parotid gland 911
Innervation 914
Vessels 916
Scalp 922
Layers 922
Innervation 924
Vessels 925
Lymphatic drainage 926
Orbit 927
Bony orbit 927
Eyelids 928
Lacrimal apparatus 932
Sensory innervation 933
Fissures and foramina 934
Fascial specializations 935
Muscles 936
Vessels 941
Innervation 942
Eyeball 947
Ear 953
External ear 954
Middle ear 958
Internal ear 965
Temporal and infratemporal fossae 972
Bony framework 973
Temporomandibular joints 975
Masseter muscle 977
Temporal fossa 978
Infratemporal fossa 981
Pterygopalatine fossa 992
Skeletal framework 993
Gateways 994
Contents 994
Neck 1000
Fascia 1000
Superficial venous drainage 1003
Anterior triangle of the neck 1006
Posterior triangle of the neck 1023
Root of the neck 1030
Pharynx 1040
Skeletal framework 1041
Pharyngeal wall 1042
Fascia 1045
Gaps in the pharyngeal wall and structures passing through them 1046
Nasopharynx 1046
Oropharynx 1048
Laryngopharynx 1048
Tonsils 1048
Vessels 1049
Nerves 1051
Larynx 1052
Laryngeal cartilages 1053
Extrinsic ligaments 1056
Intrinsic ligaments 1057

SUBASH KC/NMC-15TH/2014

Laryngeal joints 1058
Cavity of the larynx 1059
Intrinsic muscles 1061
Function of the larynx 1064
Vessels 1066
Nerves 1068

Nasal cavities 1069
Lateral wall 1070
Regions 1071
Innervation and blood supply 1072
Skeletal framework 1072
External nose 1074
Paranasal sinuses 1074
Walls, floor, and roof 1076
Nares 1080
Choanae 1081
Gateways 1082
Vessels 1082
Innervation 1085

Oral cavity 1087
Multiple nerves innervate the oral cavity 1088
Skeletal framework 1088
Walls: the cheeks 1091
Floor 1092
Tongue 1095
Salivary glands 1102
Roof—palate 1105
Oral fissure and lips 1113
Oropharyngeal isthmus 1114
Teeth and gingivae 1114

Surface anatomy 1120

Head and neck surface anatomy 1120
Anatomical position of the head and major landmarks 1120
Visualizing structures at the CIII/CIV and CVI vertebral levels 1121
How to outline the anterior and posterior triangles of the neck 1122
How to locate the cricothyroid ligament 1123
How to find the thyroid gland 1124
Estimating the position of the middle meningeal artery 1124
Major features of the face 1125
The eye and lacrimal apparatus 1126
External ear 1127
Pulse points 1128

Clinical cases 1129

SUBASH KC/NMC-15TH/2014

This page intentionally left blank

SUBASH KC/NMC-15TH/2014

1 The Body

ADDITIONAL LEARNING RESOURCES for Chapter 1, The Body, on STUDENT CONSULT (www.studentconsult.com)

- Image Library—illustrations for Chapter 1
- Short Questions—these are questions requiring short responses, Chapter 1

What is anatomy? 2
- How can gross anatomy be studied? 2
- Important anatomical terms 2

Imaging 5
- Diagnostic imaging techniques 5
- Nuclear medicine imaging 8

Image interpretation 10
- Plain radiography 10
- Computed tomography 10
- Magnetic resonance imaging 11
- Nuclear medicine imaging 11

Safety in imaging 11

Body systems 12

Skeletal system 12
- Cartilage 12
- Bone 13
- Joints 18

Skin and fascias 24
- Skin 24
- Fascia 24

Muscular system 25

Cardiovascular system 27

Lymphatic system 29
- Lymphatic vessels 29
- Lymph nodes 30
- Lymphatic trunks and ducts 30

Nervous system 31
- Central nervous system 31
- Functional subdivisions of the CNS 32
- Somatic part of the nervous system 33
- Visceral part of the nervous system 38

Other systems 48

Clinical cases 50

SUBASH KC/NMC-15TH/2014



What is anatomy?

Anatomy includes those structures that can be seen grossly (without the aid of magnification) and microscopically (with the aid of magnification). Typically, when used by itself, the term *anatomy* tends to mean gross or macroscopic anatomy—that is, the study of structures that can be seen without using a microscopic. Microscopic anatomy, also called histology, is the study of cells and tissues using a microscope.

Anatomy forms the basis for the practice of medicine. Anatomy leads the physician toward an understanding of a patient's disease, whether he or she is carrying out a physical examination or using the most advanced imaging techniques. Anatomy is also important for dentists, chiropractors, physical therapists, and all others involved in any aspect of patient treatment that begins with an analysis of clinical signs. The ability to interpret a clinical observation correctly is therefore the endpoint of a sound anatomical understanding.

Observation and visualization are the primary techniques a student should use to learn anatomy. Anatomy is much more than just memorization of lists of names. Although the language of anatomy is important, the network of information needed to visualize the position of physical structures in a patient goes far beyond simple memorization. Knowing the names of the various branches of the external carotid artery is not the same as being able to visualize the course of the lingual artery from its origin in the neck to its termination in the tongue. Similarly, understanding the organization of the soft palate, how it is related to the oral and nasal cavities, and how it moves during swallowing is very different from being able to recite the names of its individual muscles and nerves. An understanding of anatomy requires an understanding of the context in which the terminology can be remembered.

How can gross anatomy be studied?

The term *anatomy* is derived from the Greek word *temnein*, meaning "to cut." Clearly, therefore, the study of anatomy is linked, at its root, to dissection, although dissection of cadavers by students is now augmented, or even in some cases replaced, by viewing prosected (previously dissected) material and plastic models, or using computer teaching modules and other learning aids.

Anatomy can be studied following either a regional or a systemic approach.

- With a **regional approach**, each *region* of the body is studied separately and all aspects of that region are studied at the same time. For example, if the thorax is to be studied, all of its structures are examined. This includes the vasculature, the nerves, the bones, the muscles, and all other structures and organs located in the region of the body defined as the thorax. After studying this region, the other regions of the body (i.e., the abdomen, pelvis, lower limb, upper limb, back, head, and neck) are studied in a similar fashion.
- In contrast, in a **systemic approach**, each *system* of the body is studied and followed throughout the entire body. For example, a study of the cardiovascular system looks at the heart and all of the blood vessels in the body. When this is completed, the nervous system (brain, spinal cord, and all the nerves) might be examined in detail. This approach continues for the whole body until every system, including the nervous, skeletal, muscular, gastrointestinal, respiratory, lymphatic, and reproductive systems, has been studied.

Each of these approaches has benefits and deficiencies. The regional approach works very well if the anatomy course involves cadaver dissection but falls short when it comes to understanding the continuity of an entire system throughout the body. Similarly, the systemic approach fosters an understanding of an entire system throughout the body, but it is very difficult to coordinate this directly with a cadaver dissection or to acquire sufficient detail.

Important anatomical terms

The anatomical position

The anatomical position is the standard reference position of the body used to describe the location of structures (Fig. 1.1). The body is in the anatomical position when standing upright with feet together, hands by the side and face looking forward. The mouth is closed and the facial expression is neutral. The rim of bone under the eyes is in the same horizontal plane as the top of the opening to the ear, and the eyes are open and focused on something in the distance. The palms of the hands face forward with the fingers straight and together and with the pad of the thumb turned 90° to the pads of the fingers. The toes point forward.

Anatomical planes

Three major groups of planes pass through the body in the anatomical position (Fig. 1.1).

SUBASH KC/NMC-15TH/2014

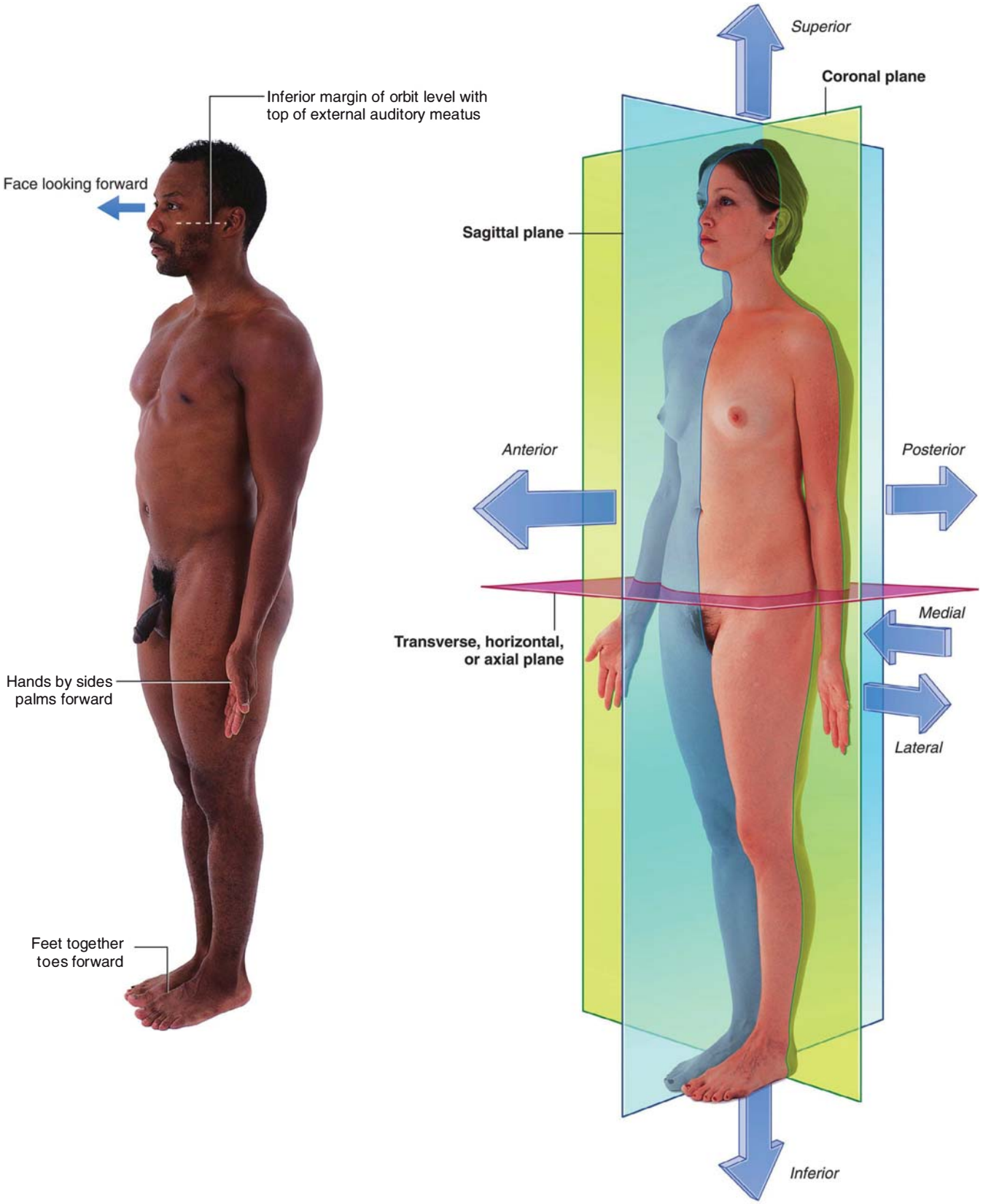


Fig. 1.1 The anatomical position, planes, and terms of location and orientation.

SUBASH KC/NMC-15TH/2014

- **Coronal planes** are oriented vertically and divide the body into anterior and posterior parts.
- **Sagittal planes** also are oriented vertically but are at right angles to the coronal planes and divide the body into right and left parts. The plane that passes through the center of the body dividing it into equal right and left halves is termed the **median sagittal plane**.
- **Transverse, horizontal**, or **axial planes** divide the body into superior and inferior parts.

Terms to describe location

Anterior (ventral) and posterior (dorsal), medial and lateral, superior and inferior

Three major pairs of terms are used to describe the location of structures relative to the body as a whole or to other structures (Fig. 1.1).

- **Anterior** (or **ventral**) and **posterior** (or **dorsal**) describe the position of structures relative to the "front" and "back" of the body. For example, the nose is an anterior (ventral) structure, whereas the vertebral column is a posterior (dorsal) structure. Also, the nose is anterior to the ears and the vertebral column is posterior to the sternum.
- **Medial** and **lateral** describe the position of structures relative to the median sagittal plane and the sides of the body. For example, the thumb is lateral to the little finger. The nose is in the median sagittal plane and is medial to the eyes, which are in turn medial to the external ears.
- **Superior** and **inferior** describe structures in reference to the vertical axis of the body. For example, the head is superior to the shoulders and the knee joint is inferior to the hip joint.

Proximal and distal, cranial and caudal, and rostral

Other terms used to describe positions include proximal and distal, cranial and caudal, and rostral.

- **Proximal** and **distal** are used with reference to being closer to or farther from a structure's origin, particularly in the limbs. For example, the hand is distal to the elbow joint. The glenohumeral joint is proximal to the elbow joint. These terms are also used to describe the relative positions of branches along the course of linear structures, such as airways, vessels, and nerves. For example, distal branches occur farther away toward the ends of the system, whereas proximal branches occur closer to and toward the origin of the system.
- **Cranial** (toward the head) and **caudal** (toward the tail) are sometimes used instead of superior and inferior, respectively.
- **Rostral** is used, particularly in the head, to describe the position of a structure with reference to the nose. For example, the forebrain is rostral to the hindbrain.

Superficial and deep

Two other terms used to describe the position of structures in the body are **superficial** and **deep**. These terms are used to describe the relative positions of two structures with respect to the surface of the body. For example, the sternum is superficial to the heart, and the stomach is deep to the abdominal wall.

Superficial and deep can also be used in a more absolute fashion to define two major regions of the body. The superficial region of the body is external to the outer layer of deep fascia. Deep structures are enclosed by this layer. Structures in the superficial region of the body include the skin, superficial fascia, and mammary glands. Deep structures include most skeletal muscles and viscera. Superficial wounds are external to the outer layer of deep fascia, whereas deep wounds penetrate through it.

SUBASH KC/NMC-15TH/2014

Imaging

Diagnostic imaging techniques

In 1895 Wilhelm Roentgen used the X-rays from a cathode ray tube to expose a photographic plate and produce the first radiographic exposure of his wife's hand. Over the past 30 years there has been a revolution in body imaging, which has been paralleled by developments in computer technology.

Plain radiography

The basic physics of X-ray generation has not changed.

X-rays are photons (a type of electromagnetic radiation) and are generated from a complex X-ray tube, which is a type of cathode ray tube (Fig. 1.2). The X-rays are then collimated (i.e., directed through lead-lined shutters to stop them from fanning out) to the appropriate area, as determined by the radiographic technician. As the X-rays pass through the body they are attenuated (reduced in energy) by the tissues. Those X-rays that pass through the tissues interact with the photographic film.

In the body:

- air attenuates X-rays a little;
- fat attenuates X-rays more than air but less than water; and
- bone attenuates X-rays the most.

These differences in attenuation result in differences in the level of exposure of the film. When the photographic film is developed, bone appears white on the film because this region of the film has been exposed to the least amount of X-rays. Air appears dark on the film because these regions were exposed to the greatest number of X-rays.

Modifications to this X-ray technique allow a continuous stream of X-rays to be produced from the X-ray tube and collected on an input screen to allow real-time visualization of moving anatomical structures, barium studies, angiography, and fluoroscopy (Fig. 1.3).

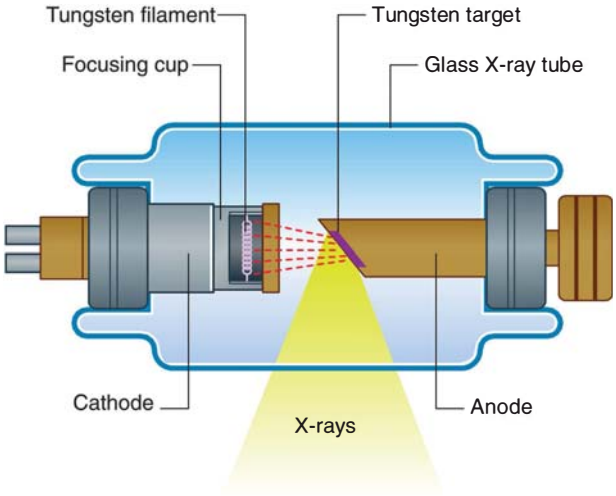


Fig. 1.2 Cathode ray tube for the production of X-rays.



Fig. 1.3 Fluoroscopy unit.

SUBASH KC/NMC-15TH/2014

Contrast agents

To demonstrate specific structures, such as bowel loops or arteries, it may be necessary to fill these structures with a substance that attenuates X-rays more than bowel loops or arteries do normally. It is, however, extremely important that these substances are nontoxic. Barium sulfate, an insoluble salt, is a nontoxic, relatively high-density agent that is extremely useful in the examination of the gastrointestinal tract. When **barium sulfate suspension** is ingested it attenuates X-rays and can therefore be used to demonstrate the bowel lumen (Fig. 1.4). It is common to add air to the barium sulfate suspension, by either ingesting "fizzy" granules or directly instilling air into the body cavity, as in a barium enema. This is known as a double-contrast (air/barium) study.

For some patients it is necessary to inject contrast agents directly into arteries or veins. In this case, iodine-based molecules are suitable contrast agents. **Iodine** is chosen because it has a relatively high atomic mass and so markedly attenuates X-rays, but also, importantly, it is naturally excreted via the urinary system. Intra-arterial and intravenous contrast agents are extremely safe and are well tolerated by most patients. Rarely, some patients have an anaphylactic reaction to intra-arterial or intravenous injections, so the necessary precautions must be taken. Intra-arterial and intravenous contrast agents not only help in visualizing the arteries and veins but because they are excreted by the urinary system, can also be used to visualize the kidneys, ureter, and bladder in a process known as **intravenous urography.**

Subtraction angiography

During angiography it is often difficult to appreciate the contrast agent in the vessels through the overlying bony structures. To circumvent this, the technique of subtraction angiography has been developed. Simply, one or two images are obtained before the injection of contrast media. These images are inverted (such that a negative is created from the positive image). After injection of the contrast media into the vessels, a further series of images are obtained, demonstrating the passage of the contrast through the arteries into the veins and around the circulation. By adding the "negative precontrast image" to the positive postcontrast images, the bones and soft tissues are subtracted to produce a solitary image of contrast only. Before the advent of digital imaging this was a challenge, but now the use of computers has made this technique relatively straightforward and instantaneous (Fig. 1.5).



Fig. 1.4 Barium sulfate follow-through.

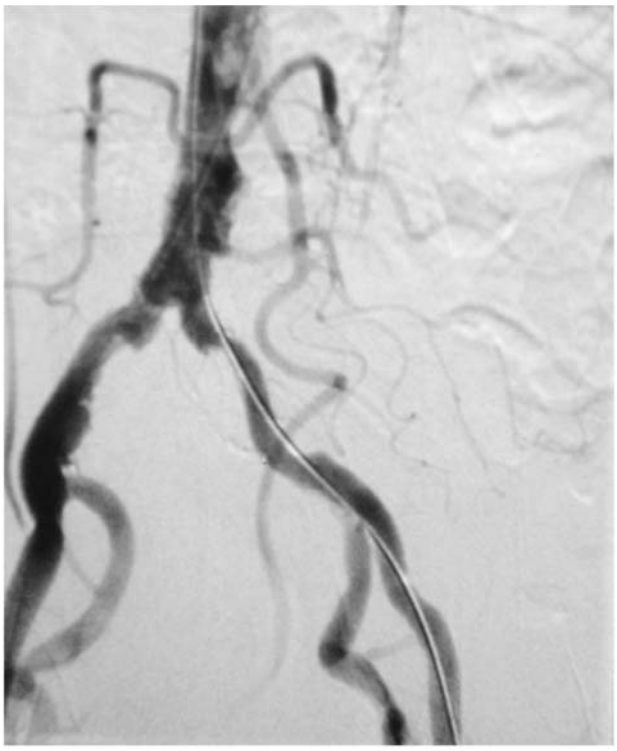


Fig. 1.5 Digital subtraction angiogram.

SUBASH KC/NMC-15TH/2014

Ultrasound

Ultrasonography of the body is widely used for all aspects of medicine.

Ultrasound is a very high frequency sound wave (not electromagnetic radiation) generated by piezoelectric materials, such that a series of sound waves is produced. Importantly, the piezoelectric material can also receive the sound waves that bounce back from the internal organs. The sound waves are then interpreted by a powerful computer, and a real-time image is produced on the display panel.

Doppler ultrasound

Developments in ultrasound technology, including the size of the probes and the frequency range, mean that a broad range of areas can now be scanned.

Traditionally ultrasound is used for assessing the abdomen (Fig. 1.6) and the fetus in pregnant women. Ultrasound is also widely used to assess the eyes, neck, soft tissues, and peripheral musculoskeletal system. Probes have been placed on endoscopes, and endoluminal ultrasound of the esophagus, stomach, and duodenum is now routine. Endocavity ultrasound is carried out most commonly to assess the genital tract in women using a transvaginal or transrectal route. In men, transrectal ultrasound is the imaging method of choice to assess the prostate in those with suspected prostate hypertrophy or malignancy.

Doppler ultrasound enables determination of flow, its direction, and its velocity within a vessel using simple ultrasound techniques. Sound waves bounce off moving structures and are returned. The degree of frequency shift determines whether the object is moving away from or toward the probe and the speed at which it is traveling. Precise measurements of blood flow and blood velocity can therefore be obtained, which in turn can indicate sites of blockage in blood vessels.

Computed tomography

Computed tomography (CT) was invented in the 1970s by Sir Godfrey Hounsfield, who was awarded the Nobel Prize in Medicine in 1979. Since this inspired invention there have been many generations of CT scanners. A CT scanner obtains a series of images of the body (slices) in the axial plane.

The patient lies on a bed, an X-ray tube passes around the body (Fig. 1.7), and a series of images are obtained. A computer carries out a complex mathematical



Fig. 1.6 Ultrasound examination of the abdomen.



Fig. 1.7 Computed tomography scanner.

SUBASH KC/NMC-15TH/2014

transformation on the multitude of images to produce the final image (Fig. 1.8).

Magnetic resonance imaging

Nuclear magnetic resonance imaging was first described in 1946 and used to determine the structure of complex molecules. The complexity of the physics necessary to obtain an image is beyond the scope of this textbook, but the reader should be aware of how the image is produced and the types of images typically seen in routine medical practice.

The process of magnetic resonance imaging (MRI) is dependent on the free protons in the hydrogen nuclei in molecules of water (H_2O). Because water is present in almost all biological tissues, the hydrogen proton is ideal. The protons within a patient's hydrogen nuclei can be regarded as small bar magnets, which are randomly oriented in space. The patient is placed in a strong magnetic field, which aligns the bar magnets. When a pulse of radio waves is passed through the patient the magnets are deflected, and as they return to their aligned position they emit small radio pulses. The strength and frequency of the emitted pulses and the time it takes for the protons to return to their pre-excited state produce a signal. These signals are analyzed by a powerful computer, and an image is created (Fig. 1.9).

By altering the sequence of pulses to which the protons are subjected, different properties of the protons can be assessed. These properties are referred to as the "weighting" of the scan. By altering the pulse sequence and the scanning parameters, T1-weighted images (Fig. 1.10A) and T2-weighted images (Fig. 1.10B) can be obtained. These two types of imaging sequences provide differences in image contrast, which accentuate and optimize different tissue characteristics.

From the clinical point of view:

- Most T1-weighted images show dark fluid and bright fat—for example, within the brain the cerebrospinal fluid (CSF) is dark;
- T2-weighted images demonstrate a bright signal from fluid and an intermediate signal from fat—for example, in the brain the CSF appears white.

MRI can also be used to assess flow within vessels and to produce complex angiograms of the peripheral and cerebral circulation.

Nuclear medicine imaging

Nuclear medicine involves imaging using gamma rays, which are another type of electromagnetic radiation.



Fig. 1.8 Computed tomography scan of the abdomen at vertebral level L2.

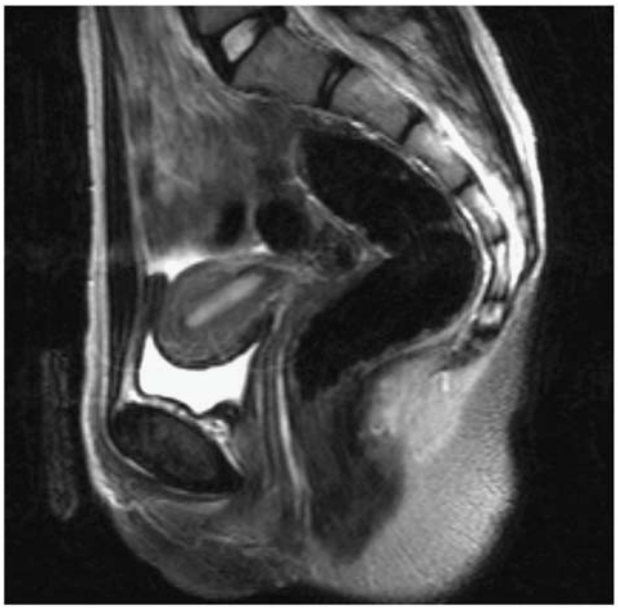


Fig. 1.9 A T2-weighted image in the sagittal plane of the pelvic viscera in a woman.

The important difference between gamma rays and X-rays is that gamma rays are produced from within the nucleus of an atom when an unstable nucleus decays, whereas X-rays are produced by bombarding an atom with electrons.

For an area to be visualized, the patient must receive a gamma ray emitter, which must have a number of properties to be useful, including:

- a reasonable half-life (e.g., 6 to 24 hours),
- an easily measurable gamma ray, and

SUBASH KC/NMC-15TH/2014

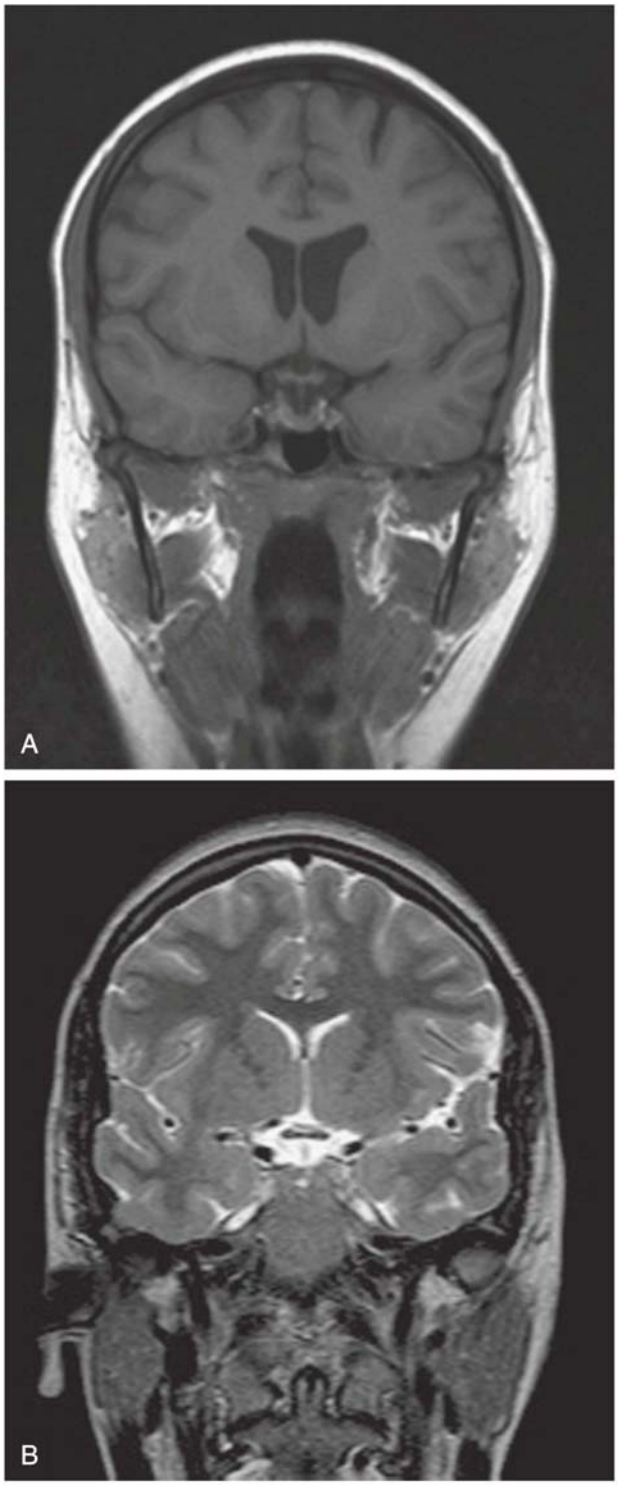


Fig. 1.10 T1-weighted (**A**) and T2-weighted (**B**) magnetic resonance images of the brain in the coronal plane.

- energy deposition in as low a dose as possible in the patient's tissues.

The most commonly used radionuclide (radioisotope) is technetium-99m. This may be injected as a technetium salt or combined with other complex molecules. For example, by combining technetium-99m with methylene diphosphonate (MDP), a radiopharmaceutical is produced. When injected into the body this radiopharmaceutical specifically binds to bone, allowing assessment of the skeleton. Similarly, combining technetium-99m with other compounds permits assessment of other parts of the body, for example the urinary tract and cerebral blood flow.

Depending on how the radiopharmaceutical is absorbed, distributed, metabolized, and excreted by the body after injection, images are obtained using a gamma camera (Fig. 1.11).

Positron emission tomography

Positron emission tomography (PET) is an imaging modality for detecting positron-emitting radionuclides. A positron is an anti-electron, which is a positively charged particle of antimatter. Positrons are emitted from the decay of proton-rich radionuclides. Most of these radionuclides are made in a cyclotron and have extremely short half-lives.

The most commonly used PET radionuclide is fluorodeoxyglucose (FDG) labeled with fluorine-18 (a positron



Fig. 1.11 A gamma camera.

SUBASH KC/NMC-15TH/2014

emitter). Tissues that are actively metabolizing glucose take up this compound, and the resulting localized high concentration of this molecule compared to background emission is detected as a "hot spot."

PET has become an important imaging modality in the detection of cancer and the assessment of its treatment and recurrence.

IMAGE INTERPRETATION

Imaging is necessary in most clinical specialties to diagnose pathological changes to tissues. It is paramount to appreciate what is normal and what is abnormal. An appreciation of how the image is obtained, what the normal variations are, and technical considerations is necessary to obtain a radiological diagnosis. Without understanding the anatomy of the region imaged, it is impossible to comment on the abnormal.

Plain radiography

Plain radiographs are undoubtedly the most common form of image obtained in a hospital or local practice. Before interpretation, it is important to know about the imaging technique and the views obtained as standard.

In most instances (apart from chest radiography) the X-ray tube is 1 m away from the X-ray film. The object in question, for example a hand or a foot, is placed upon the film. When describing subject placement for radiography, the part closest to the X-ray tube is referred to as "anterior" and that closest to the film is referred to as "posterior."

When X-rays are viewed on a viewing box, the right side of the patient is placed to the observer's left; therefore, the observer views the radiograph as though looking at a patient in the anatomical position.

Chest radiograph

The chest radiograph is one of the most commonly requested plain radiographs. An image is taken with the patient erect and placed posteroanteriorly (PA chest radiograph).

Occasionally, when patients are too unwell to stand erect, films are obtained on the bed in an anteroposterior (AP) position. These films are less standardized than PA films, and caution should always be taken when interpreting AP radiographs.

The plain chest radiograph should always be checked for quality. Film markers should be placed on the appropriate side. (Occasionally patients have dextrocardia, which may be misinterpreted if the film marker is placed inappropriately.) A good-quality chest radiograph will demonstrate the lungs, cardiomediastinal contour, diaphragm, ribs, and peripheral soft tissues.

Abdominal radiograph

Plain abdominal radiographs are obtained in the AP supine position. From time to time an erect plain abdominal radiograph is obtained when small bowel obstruction is suspected.

Gastrointestinal contrast examinations

High-density contrast medium is ingested to opacify the esophagus, stomach, small bowel, and large bowel. As described previously (p. 6), the bowel is insufflated with air (or carbon dioxide) to provide a double-contrast study. In many countries, endoscopy has superseded upper gastrointestinal imaging, but the mainstay of imaging the large bowel is the double-contrast barium enema. Typically the patient needs to undergo bowel preparation, in which powerful cathartics are used to empty the bowel. At the time of the examination a small tube is placed into the rectum and a barium suspension is run into the large bowel. The patient undergoes a series of twists and turns so that the contrast passes through the entire large bowel. The contrast is emptied and air is passed through the same tube to insufflate the large bowel. A thin layer of barium coats the normal mucosa, allowing mucosal detail to be visualized (see Fig. 1.4).

Urological contrast studies

Intravenous urography is the standard investigation for assessing the urinary tract. Intravenous contrast medium is injected, and images are obtained as the medium is excreted through the kidneys. A series of films are obtained during this period from immediately after the injection up to approximately 20 minutes later, when the bladder is full of contrast medium.

This series of radiographs demonstrates the kidneys, ureters, and bladder and enables assessment of the retroperitoneum and other structures that may press on the urinary tract.

Computed tomography

Computed tomography is the preferred terminology rather than computerized tomography, though both terms are used interchangeably by physicians.

The general principles of computed tomography are described on p. 7. It is important for the student to understand the presentation of images. Most images are acquired in the axial plane and viewed such that the observer looks

SUBASH KC/NMC-15TH/2014

from below and upward toward the head (from the foot of the bed). By implication:

- the right side of the patient is on the left side of the image, and
- the uppermost border of the image is anterior.

Many patients are given oral and intravenous contrast media to differentiate bowel loops from other abdominal organs and to assess the vascularity of normal anatomical structures. When intravenous contrast is given, the earlier the images are obtained, the greater the likelihood of arterial enhancement. As the time is delayed between injection and image acquisition, a venous phase and an equilibrium phase are also obtained.

The great advantage of CT scanning is the ability to extend and compress the gray scale to visualize the bones, soft tissues, and visceral organs. Altering the window settings and window centering provides the physician with specific information about these structures.

Magnetic resonance imaging

There is no doubt that MRI has revolutionized the understanding and interpretation of the brain and its coverings. Furthermore, it has significantly altered the practice of musculoskeletal medicine and surgery. Images can be obtained in any plane and in most sequences. Typically the images are viewed using the same principles as CT. Intravenous contrast agents are also used to further enhance tissue contrast. Typically, MRI contrast agents contain paramagnetic substances (e.g., gadolinium and manganese).

Nuclear medicine imaging

Most nuclear medicine images are functional studies. Images are usually interpreted directly from a computer, and a series of representative films are obtained for clinical use.

SAFETY IN IMAGING

Whenever a patient undergoes an X-ray or nuclear medicine investigation, a dose of radiation is given (Table 1.1). As a general principle it is expected that the dose given is as low as reasonably possible for a diagnostic image to be obtained. Numerous laws govern the amount of radiation exposure that a patient can undergo for a variety of procedures, and these are monitored to prevent any excess or additional dosage. Whenever a radiograph is booked, the clinician ordering the procedure must appreciate its necessity and understand the dose given to the patient to ensure that the benefits significantly outweigh the risks.

Imaging modalities such as ultrasound and MRI are ideal because they do not impart significant risk to the patient. Moreover, ultrasound imaging is the modality of choice for assessing the fetus.

Any imaging device is expensive, and consequently the more complex the imaging technique (e.g., MRI) the more expensive the investigation. Investigations must be carried out judiciously, based on a sound clinical history and examination, for which an understanding of anatomy is vital.

Table 1.1 The approximate dosage of radiation exposure as an order of magnitude

Examination	Typical effective dose (mSv)	Equivalent duration of background exposure
Chest radiograph	0.02	3 days
Abdomen	1.00	6 months
Intravenous urography	2.50	14 months
CT scan of head	2.30	1 year
CT scan of abdomen and pelvis	10.00	4.5 years

SUBASH KC/NMC-15TH/2014

Body systems

SKELETAL SYSTEM

The skeleton can be divided into two subgroups, the axial skeleton and the appendicular skeleton. The axial skeleton consists of the bones of the skull (cranium), vertebral column, ribs, and sternum, whereas the appendicular skeleton consists of the bones of the upper and lower limbs (Fig. 1.12).

The skeletal system consists of cartilage and bone.

Cartilage

Cartilage is an avascular form of connective tissue consisting of extracellular fibers embedded in a matrix that contains cells localized in small cavities. The amount and kind of extracellular fibers in the matrix varies depending on the type of cartilage. In heavy weightbearing areas or areas prone to pulling forces, the amount of collagen is greatly increased and the cartilage is almost inextensible. In contrast, in areas where weightbearing demands and stress are less, cartilage containing elastic fibers and fewer collagen fibers is common. The functions of cartilage are to:

- support soft tissues,
- provide a smooth, gliding surface for bone articulations at joints, and
- enable the development and growth of long bones.

There are three types of cartilage:

- hyaline—most common; matrix contains a moderate amount of collagen fibers (e.g., articular surfaces of bones);
- elastic—matrix contains collagen fibers along with a large number of elastic fibers (e.g., external ear);
- fibrocartilage—matrix contains a limited number of cells and ground substance amidst a substantial amount of collagen fibers (e.g., intervertebral discs).

Cartilage is nourished by diffusion and has no blood vessels, lymphatics, or nerves.



Fig. 1.12 The axial skeleton and the appendicular skeleton.

SUBASH KC/NMC-15TH/2014

Bone

Bone is a calcified, living, connective tissue that forms the majority of the skeleton. It consists of an intercellular calcified matrix, which also contains collagen fibers, and several types of cells within the matrix. Bones function as:

- supportive structures for the body,
- protectors of vital organs,
- reservoirs of calcium and phosphorus,
- levers on which muscles act to produce movement, and
- containers for blood-producing cells.

There are two types of bone, compact and spongy (trabecular or cancellous). Compact bone is dense bone that forms the outer shell of all bones and surrounds spongy bone. Spongy bone consists of spicules of bone enclosing cavities containing blood-forming cells (marrow). Classification of bones is by shape.

- Long bones are tubular (e.g., humerus in upper limb; femur in lower limb).
- Short bones are cuboidal (e.g., bones of the wrist and ankle).
- Flat bones consist of two compact bone plates separated by spongy bone (e.g., skull).
- Irregular bones are bones with various shapes (e.g., bones of the face).
- Sesamoid bones are round or oval bones that develop in tendons.

Bones are vascular and are innervated. Generally, an adjacent artery gives off a nutrient artery, usually one per bone, that directly enters the internal cavity of the bone and supplies the marrow, spongy bone, and inner layers of compact bone. In addition, all bones are covered externally, except in the area of a joint where articular cartilage is present, by a fibrous connective tissue membrane called the periosteum, which has the unique capability of forming new bone. This membrane receives blood vessels whose branches supply the outer layers of compact bone. A bone stripped of its periosteum will not survive. Nerves accompany the vessels that supply the bone and the periosteum. Most of the nerves passing into the internal cavity with the nutrient artery are vasomotor fibers that regulate blood flow. Bone itself has few sensory nerve fibers. On the other hand, the periosteum is supplied with numerous sensory nerve fibers and is very sensitive to any type of injury.

Developmentally, all bones come from mesenchyme by either intramembranous ossification, in which mesenchymal models of bones undergo ossification, or endochondral ossification, in which cartilaginous models of bones form from mesenchyme and undergo ossification.

SUBASH KC/NMC-15TH/2014

In the clinic

Determination of skeletal age

Throughout life the bones develop in a predictable way to form the skeletally mature adult at the end of puberty. In western countries skeletal maturity tends to occur between the ages of 20 and 25 years. However, this may well vary according to geography and socioeconomic conditions. Skeletal maturity will also be determined by genetic factors and disease states.

Up until the age of skeletal maturity, bony growth and development follows a typically predictable ordered state, which can be measured through either ultrasound, plain radiographs, or MRI scanning. Typically, the nondominant (left) hand is radiographed, and the radiograph is compared to a series of standard radiographs. From these images the bone age can be determined (Fig. 1.13).

In certain disease states, such as malnutrition and hypothyroidism, bony maturity may be slow. If the skeletal bone age is significantly reduced from the patient's true age, treatment may be required.

In the healthy individual the bone age accurately represents the true age of the patient. This is important in determining the true age of the subject. This may also have medicolegal importance.

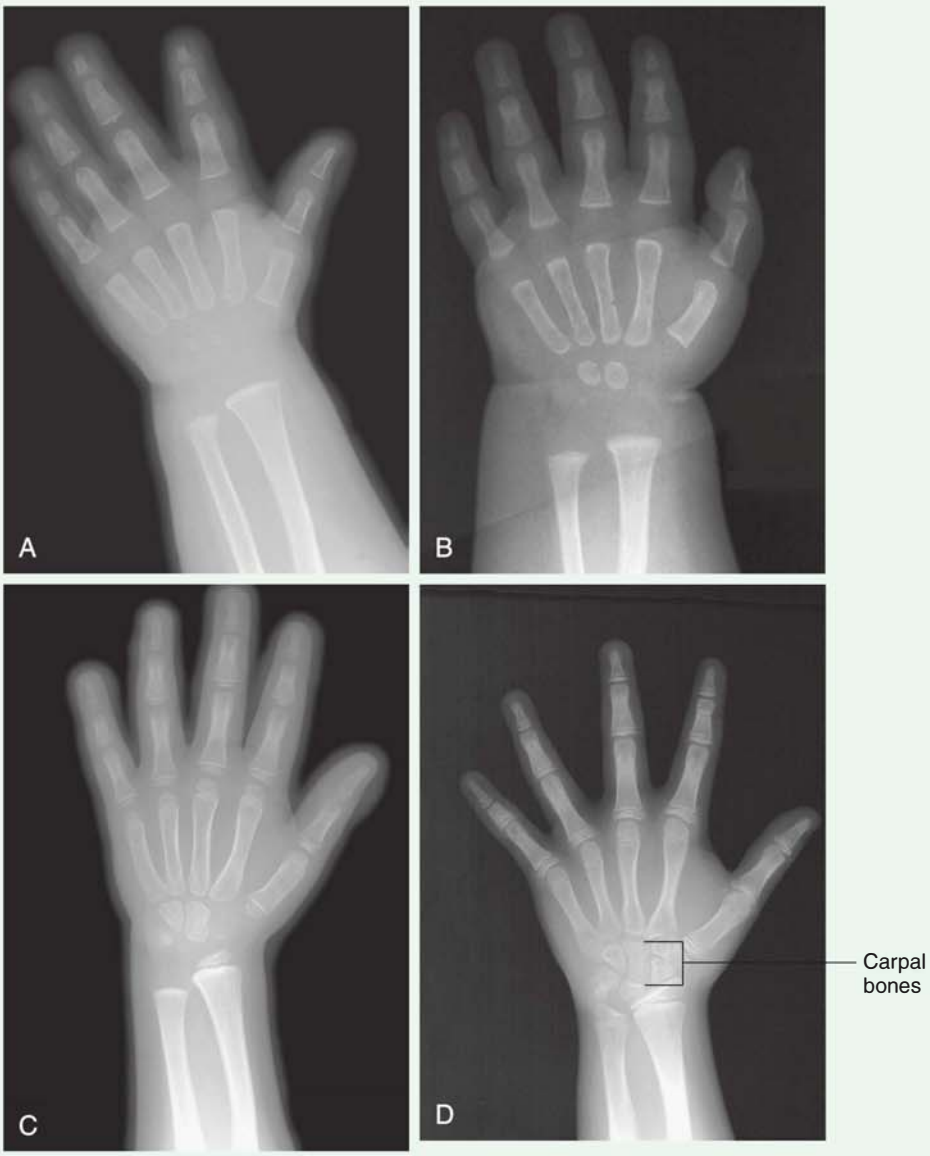


Fig. 1.13 A developmental series of radiographs showing the progressive ossification of carpal (wrist) bones from 3 (**A**) to 10 (**D**) years of age.

SUBASH KC/NMC-15TH/2014

In the clinic

Bone marrow transplants

The bone marrow serves an important function. There are two types of bone marrow, the red marrow (otherwise known as myeloid tissue) and the yellow marrow. Red blood cells, platelets, and most white blood cells arise from within the red marrow. In the yellow marrow a few white cells are made; however this marrow is dominated by large fat globules (producing its yellow appearance) (Fig. 1.14).

From birth most of the body's marrow is red; however, as the subject ages, more red marrow is converted into yellow marrow within the medulla of the long and flat bones.

Bone marrow contains two types of stem cells. Hemopoietic stem cells give rise to the white blood cells, red blood cells, and platelets. Mesenchymal stem cells differentiate into structures that form bone, cartilage, and muscle.

There are a number of diseases that may involve the bone marrow, including infection and malignancy. In patients who develop a bone marrow malignancy (e.g., leukemia) it may be possible to harvest nonmalignant cells from the patient's bone marrow or cells from another person's bone marrow. The patient's own marrow can be destroyed with chemotherapy or radiation and the new cells infused. This treatment is bone marrow transplantation.



Fig. 1.14 T1-weighted image in the coronal plane, demonstrating the relatively high signal intensity returned from the femoral heads and proximal femoral necks, consistent with yellow marrow. In this young patient, the vertebral bodies return an intermediate darker signal that represents red marrow. There is relatively little fat in these vertebrae; hence the lower signal return.

SUBASH KC/NMC-15TH/2014

In the clinic

Bone fractures

Fractures occur in normal bone because of abnormal load or stress, in which the bone gives way. Fractures may also occur in bone that is of poor quality (osteoporosis); in such cases a normal stress is placed upon a bone that is not of sufficient quality to withstand this force and subsequently fractures.

In children whose bones are still developing, fractures may occur across the growth plate or across the shaft. These shaft fractures typically involve partial cortical disruption, similar to breaking a branch of a young tree; hence they are termed "greenstick" fractures (Fig. 1.15).

After a fracture has occurred, the natural response is to heal the fracture. Between the fracture margins a blood clot is formed into which new vessels grow. A jelly-like matrix is formed, and further migration of collagen-producing cells occurs. On this soft tissue framework, calcium hydroxyapatite is produced by osteoblasts and forms insoluble crystals, and then bone matrix is laid down. As more bone is produced, a callus can be demonstrated forming across the fracture site.

Treatment of fractures requires a fracture line reduction. If this cannot be maintained in a plaster of Paris cast, it may require internal or external fixation with screws and metal rods.



Fig. 1.15 Radiograph, lateral view, showing greenstick fractures of the distal radius and distal ulna.

In the clinic

Avascular necrosis

Avascular necrosis is cellular death of bone resulting from a temporary or permanent loss of blood supply to that bone. Avascular necrosis may occur in a variety of medical conditions, some of which have an etiology that is less than clear. A typical site for avascular necrosis is a fracture across the femoral neck in an elderly patient. In these patients there is loss of continuity of the cortical medullary blood flow with loss of blood flow deep to the retinacular fibers. This essentially renders the femoral head bloodless; it subsequently undergoes necrosis and collapses. In these patients it is necessary to replace the femoral head with a prosthesis (Fig. 1.16).

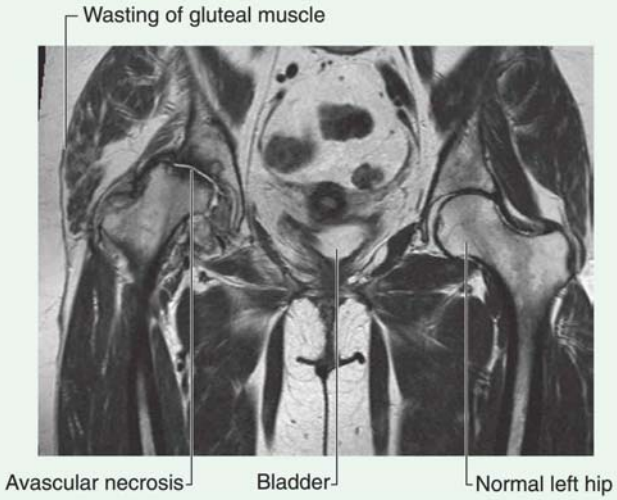


Fig. 1.16 Image of the hip joints demonstrating loss of height of the right femoral head with juxta-articular bony sclerosis and subchondral cyst formation secondary to avascular necrosis. There is also significant wasting of the muscles supporting the hip, which is secondary to disuse and pain.

SUBASH KC/NMC-15TH/2014

In the clinic

Osteoporosis

Osteoporosis is a disease in which the bone mineral density is significantly reduced. This renders the bone significantly more at risk of fracture. Typically, osteoporotic fractures occur in the femoral necks, the vertebrae, and the wrist. Although osteoporosis may occur in men, especially elderly men, the typical patients are postmenopausal women. There are a number of risk factors that predispose bones to develop osteoporosis. These factors include poor diet, steroid usage, smoking, and premature ovarian failure. Treatment involves removing underlying potentiating factors, such as improving diet and preventing further bone loss with drug treatment (e.g., vitamin D and calcium supplements; newer treatments include drugs that increase bone mineral density) (Figs. 1.17 and 1.18).

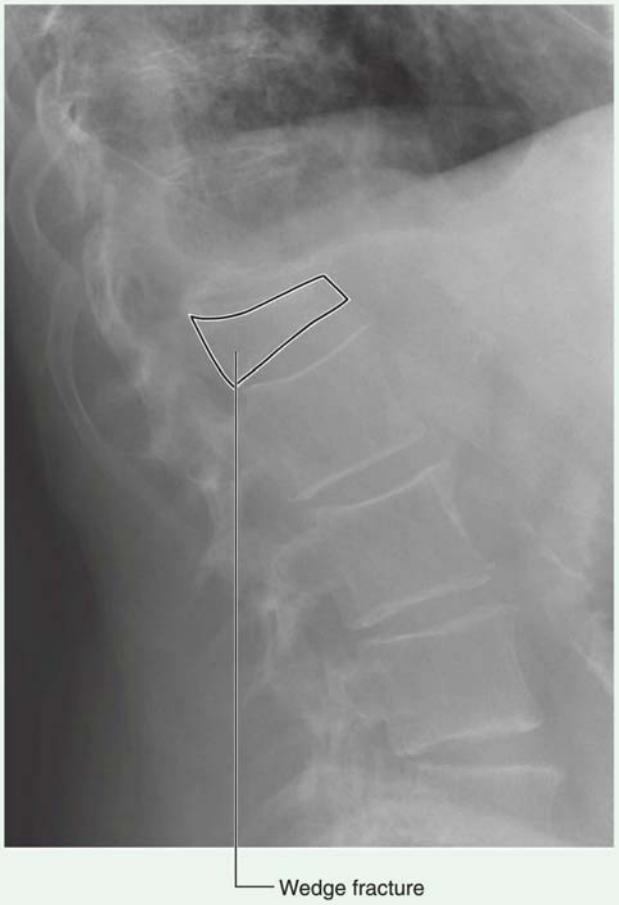


Fig. 1.17 Radiograph of the lumbar region of the vertebral column demonstrating a wedge fracture of the L1 vertebra. This condition is typically seen in patients with osteoporosis.

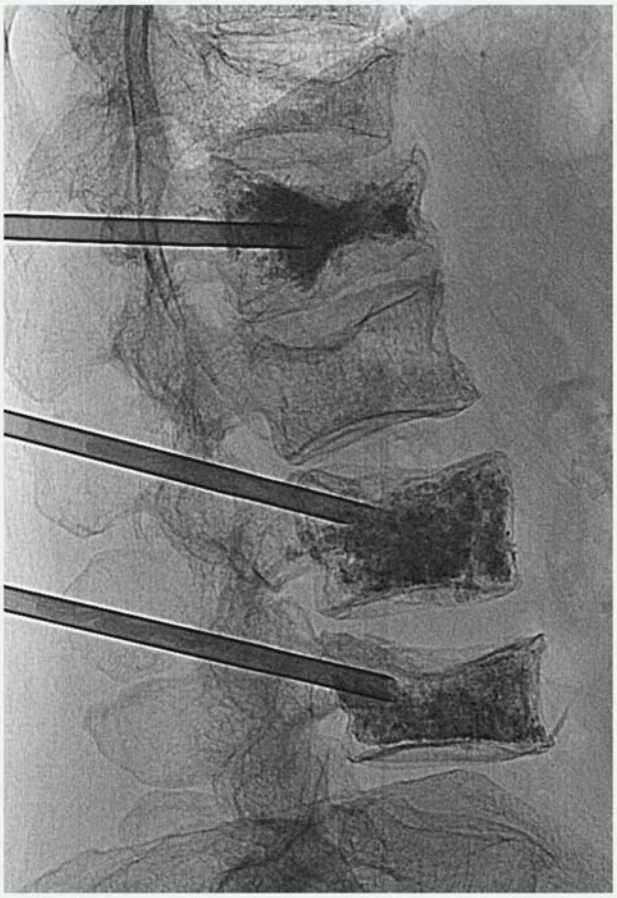


Fig. 1.18 Radiograph of the lumbar region of the vertebral column demonstrating three intrapedicular needles, all of which have been placed into the middle of the vertebral bodies. The high-density material is radiopaque bone cement, which has been injected as a liquid that will harden.

SUBASH KC/NMC-15TH/2014

In the clinic

Epiphyseal fractures

As the skeleton develops, there are stages of intense growth typically around the ages of 7 to 10 years and later in puberty. These growth spurts are associated with increased cellular activity around the growth plate and the metaphyseal region. This increase in activity renders the growth plates and metaphyseal regions more vulnerable to injuries, which may occur from dislocation across a growth plate or fracture through a growth plate. Occasionally an injury may result in growth plate compression, destroying that region of the growth plate, which may result in asymmetrical growth across that joint region. All fractures across the growth plate must be treated with care and expediency, requiring fracture reduction.

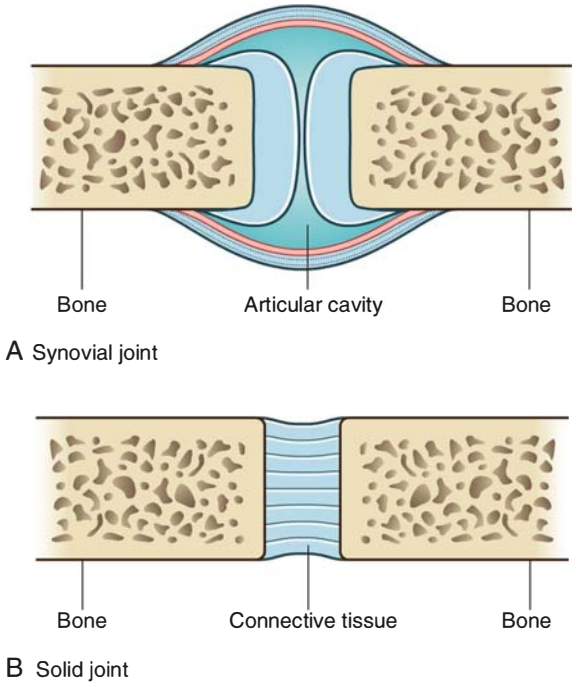


Fig. 1.19 Joints. **A.** Synovial joint. **B.** Solid joint.

Joints

The sites where two skeletal elements come together are termed joints. The two general categories of joints (Fig. 1.19) are those in which:

- the skeletal elements are separated by a cavity (i.e., **synovial joints**), and
- there is no cavity and the components are held together by connective tissue (i.e., **solid joints**).

Blood vessels that cross a joint and nerves that innervate muscles acting on a joint usually contribute articular branches to that joint.

Synovial joints

Synovial joints are connections between skeletal components where the elements involved are separated by a narrow articular cavity (Fig. 1.20). In addition to containing an articular cavity, these joints have a number of characteristic features.

First, a layer of cartilage, usually **hyaline cartilage**, covers the articulating surfaces of the skeletal elements. In other words, bony surfaces do not normally contact one another directly. As a consequence, when these joints are viewed in normal radiographs, a wide gap seems to separate the adjacent bones because the cartilage that covers the articulating surfaces is more transparent to X-rays than bone.

A second characteristic feature of synovial joints is the presence of a **joint capsule** consisting of an inner **synovial membrane** and an outer **fibrous membrane**.

- The synovial membrane attaches to the margins of the joint surfaces at the interface between the cartilage and bone and encloses the articular cavity. The synovial membrane is highly vascular and produces synovial fluid, which percolates into the articular cavity and lubricates the articulating surfaces. Closed sacs of synovial membrane also occur outside joints, where they form synovial bursae or tendon sheaths. Bursae often intervene between structures, such as tendons and bone, tendons and joints, or skin and bone, and reduce the friction of one structure moving over the other. Tendon sheaths surround tendons and also reduce friction.
- The **fibrous membrane** is formed by dense connective tissue and surrounds and stabilizes the joint. Parts of the fibrous membrane may thicken to form ligaments, which further stabilize the joint. Ligaments outside the capsule usually provide additional reinforcement.

Another common but not universal feature of synovial joints is the presence of additional structures within the area enclosed by the capsule or synovial membrane, such as **articular discs** (usually composed of fibrocartilage), **fat pads**, and **tendons**. Articular discs absorb compression forces, adjust to changes in the contours of joint surfaces during movements, and increase the range of movements that can occur at joints. Fat pads usually occur between the synovial membrane and the capsule and move

SUBASH KC/NMC-15TH/2014

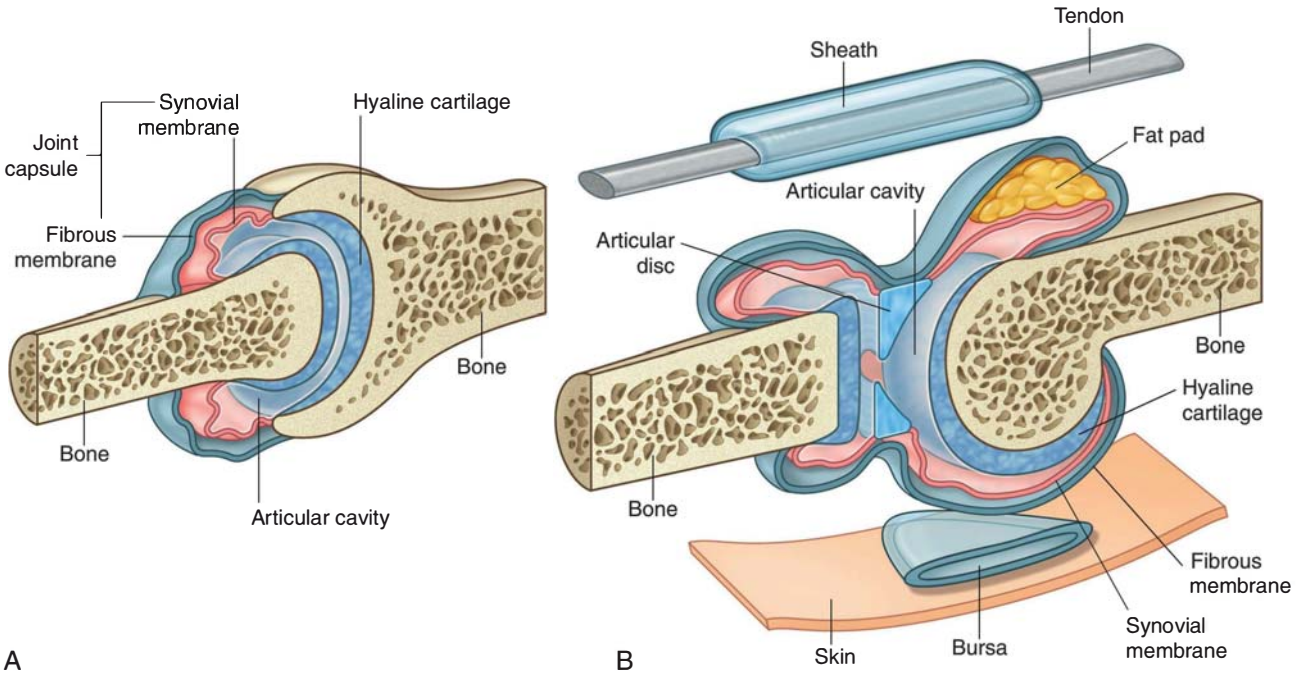


Fig. 1.20 Synovial joints. **A.** Major features of a synovial joint. **B.** Accessory structures associated with synovial joints.

SUBASH KC/NMC-15TH/2014

into and out of regions as joint contours change during movement. Redundant regions of the synovial membrane and fibrous membrane allow for large movements at joints.

Descriptions of synovial joints based on shape and movement

Synovial joints are described based on shape and movement:

- based on the shape of their articular surfaces, synovial joints are described as plane (flat), hinge, pivot, bicondylar (two sets of contact points), condylar (ellipsoid), saddle, and ball and socket;
- based on movement, synovial joints are described as uniaxial (movement in one plane), biaxial (movement in two planes), and multiaxial (movement in three planes).

Hinge joints are uniaxial, whereas ball and socket joints are multiaxial.

Specific types of synovial joints (Fig. 1.21)

- Plane joints—allow sliding or gliding movements when one bone moves across the surface of another (e.g., acromioclavicular joint)
- Hinge joints—allow movement around one axis that passes transversely through the joint; permit flexion and extension (e.g., elbow [humero-ulnar] joint)
- Pivot joints—allow movement around one axis that passes longitudinally along the shaft of the bone; permit rotation (e.g., atlanto-axial joint)
- Bicondylar joints—allow movement mostly in one axis with limited rotation around a second axis; formed by two convex condyles that articulate with concave or flat surfaces (e.g., knee joint)
- Condylar (ellipsoid) joints—allow movement around two axes that are at right angles to each other; permit flexion, extension, abduction, adduction, and circumduction (limited) (e.g., wrist joint)
- Saddle joints—allow movement around two axes that are at right angles to each other; the articular surfaces are saddle shaped; permit flexion, extension, abduction, adduction, and circumduction (e.g., carpometacarpal joint of the thumb)
- Ball and socket joints—allow movement around multiple axes; permit flexion, extension, abduction, adduction, circumduction, and rotation (e.g., hip joint)

Solid joints

Solid joints are connections between skeletal elements where the adjacent surfaces are linked together either by fibrous connective tissue or by cartilage, usually fibrocartilage (Fig. 1.22). Movements at these joints are more restricted than at synovial joints.

Fibrous joints include sutures, gomphoses, and syndesmoses.

- **Sutures** occur only in the skull where adjacent bones are linked by a thin layer of connective tissue termed a *sutural ligament*.
- **Gomphoses** occur only between the teeth and adjacent bone. In these joints, short collagen tissue fibers in the periodontal ligament run between the root of the tooth and the bony socket.
- **Syndesmoses** are joints in which two adjacent bones are linked by a ligament. Examples are the ligamentum flavum, which connects adjacent vertebral laminae, and an interosseous membrane, which links, for example, the radius and ulna in the forearm.

Cartilaginous joints include synchondroses and symphyses.

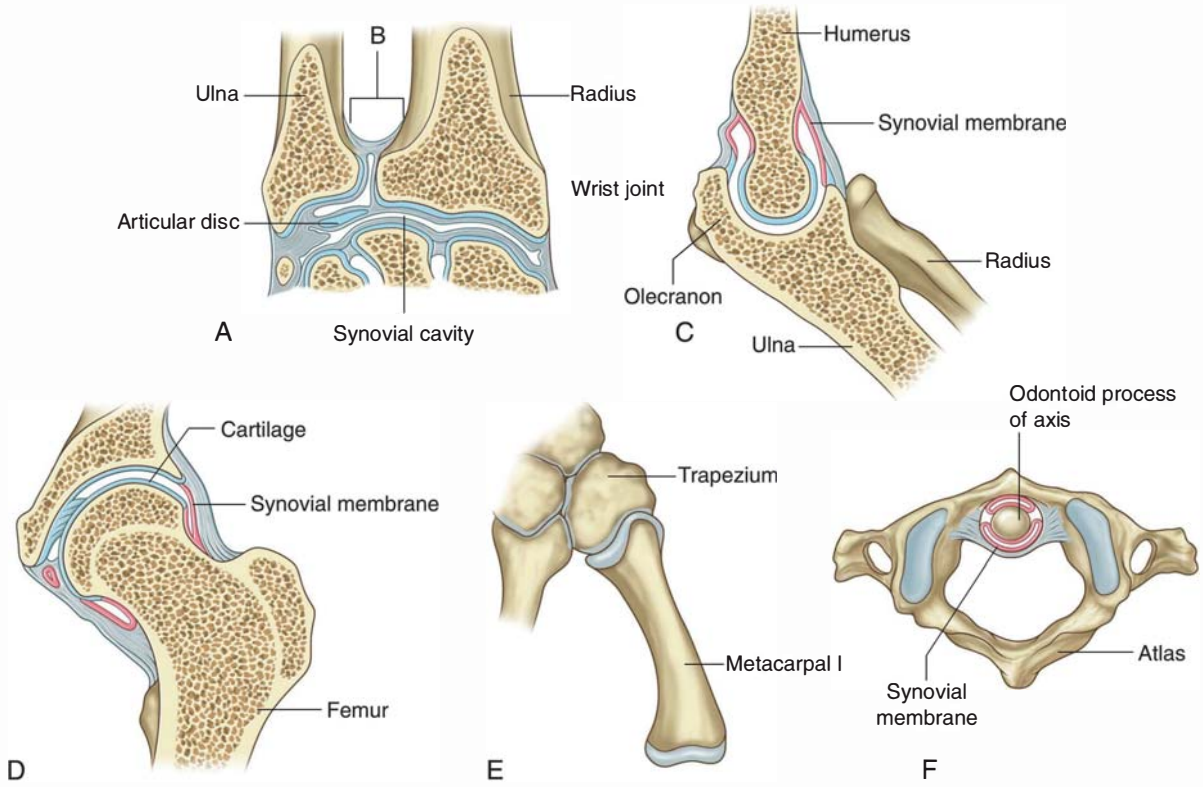


Fig. 1.21 Various types of synovial joints. **A.** Condylar (wrist). **B.** Gliding (radio-ulnar). **C.** Hinge or ginglymus (elbow). **D.** Ball and socket (hip). **E.** Saddle (carpometacarpal of thumb). **F.** Pivot (atlanto-axial).

SUBASH KC/NMC-15TH/2014

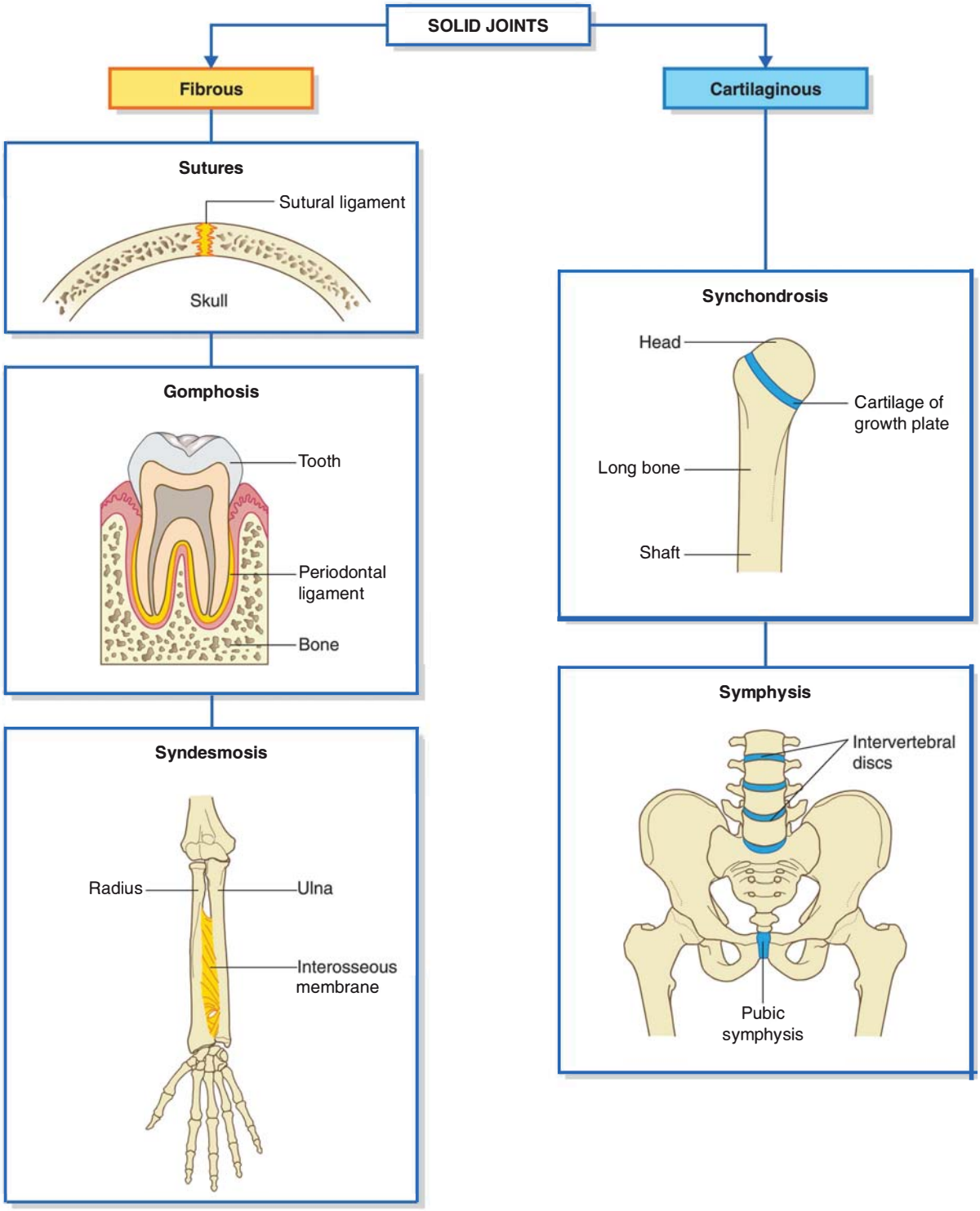


Fig. 1.22 Solid joints.

- **Synchondroses** occur where two ossification centers in a developing bone remain separated by a layer of cartilage, for example, the growth plate that occurs between the head and shaft of developing long bones. These joints allow bone growth and eventually become completely ossified.
- **Symphyses** occur where two separate bones are interconnected by cartilage. Most of these types of joints occur in the midline and include the pubic symphysis between the two pelvic bones, and intervertebral discs between adjacent vertebrae.

SUBASH KC/NMC-15TH/2014

In the clinic

Degenerative joint disease

Degenerative joint disease is commonly known as osteoarthritis or osteoarthrosis. The disorder is related to aging but not caused by aging. Typically there are decreases in water and proteoglycan content within the cartilage. The cartilage becomes more fragile and more susceptible to mechanical disruption (Fig. 1.23). As the cartilage wears, the underlying bone becomes fissured and also thickens. Synovial fluid may be forced into small cracks that appear in the bone's surface, which produces large cysts. Furthermore, reactive juxta-articular bony nodules are formed (osteophytes) (Fig. 1.24). As these processes occur, there is slight deformation, which alters the biomechanical forces through the joint. This in turn creates abnormal stresses, which further disrupt the joint.

In the United States, osteoarthritis accounts for up to one-quarter of primary health care visits and is regarded as a significant problem.

The etiology of osteoarthritis is not clear; however, osteoarthritis can occur secondary to other joint diseases, such as rheumatoid arthritis and infection. Overuse of joints and abnormal strains, such as those experienced by people who play sports, often cause one to be more susceptible to chronic joint osteoarthritis.

Various treatments are available, including weight reduction, proper exercise, anti-inflammatory drug treatment, and joint replacement (Fig. 1.25).

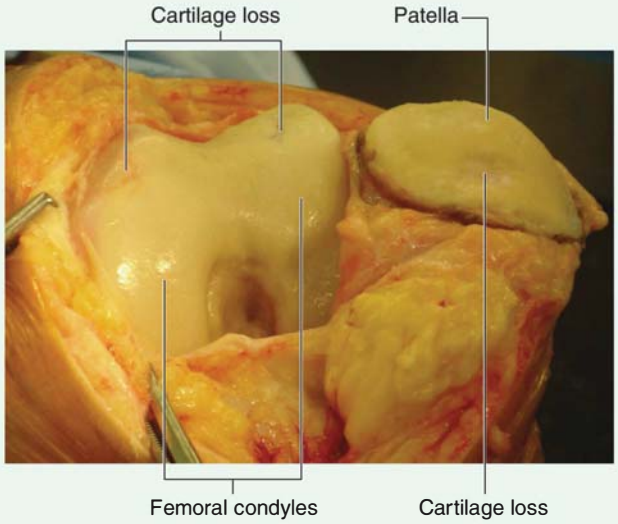


Fig. 1.23 This operative photograph demonstrates the focal areas of cartilage loss in the patella and femoral condyles throughout the knee joint.



Fig. 1.24 This radiograph demonstrates the loss of joint space in the medial compartment and presence of small spiky osteophytic regions at the medial lateral aspect of the joint.

SUBASH KC/NMC-15TH/2014

In the clinic—cont'd



Fig. 1.25 After knee replacement. This radiograph shows the position of the prosthesis.

Arthroscopy

Arthroscopy is a technique of visualizing the inside of a joint using a small telescope placed through a tiny incision in the skin. Arthroscopy can be performed in most joints. However, it is most commonly performed in the knee, shoulder, ankle, and hip joints. The elbow joint and wrist joint can also be viewed through the arthroscope.

Arthroscopy allows the surgeon to view the inside of the joint and its contents. Notably, in the knee, the menisci and the ligaments are easily seen, and it is possible using separate puncture sites and specific instruments to remove the menisci and replace the cruciate ligaments. The advantages of arthroscopy are that it is performed through small incisions, it enables patients to quickly recover and return to normal activity, and it only requires either a light anesthetic or regional anesthesia during the procedure.

SUBASH KC/NMC-15TH/2014

In the clinic

Joint replacement

Joint replacement is undertaken for a variety of reasons. These predominantly include degenerative joint disease and joint destruction. Joints that have severely degenerated or lack their normal function are painful, which can be life limiting, and in otherwise fit and healthy individuals can restrict activities of daily living. In some patients the pain may be so severe that it prevents them from leaving the house and undertaking even the smallest of activities without discomfort.

Large joints are commonly affected, including the hip, knee, and shoulder. However, with ongoing developments in joint replacement materials and surgical techniques, even small joints of the fingers can be replaced.

Typically, both sides of the joint are replaced; in the hip joint the acetabulum will be reamed, and a plastic or metal cup will be introduced. The femoral component will be fitted precisely to the femur and cemented in place (Fig. 1.26).

Most patients derive significant benefit from joint replacement and continue to lead an active life afterward.



Fig. 1.26 This is a radiograph, anteroposterior view, of the pelvis after a right total hip replacement. There are additional significant degenerative changes in the left hip joint, which will also need to be replaced.

SUBASH KC/NMC-15TH/2014

SKIN AND FASCIAS

Skin

The skin is the largest organ of the body. It consists of the epidermis and the dermis. The epidermis is the outer cellular layer of stratified squamous epithelium, which is avascular and varies in thickness. The dermis is a dense bed of vascular connective tissue.

The skin functions as a mechanical and permeability barrier, and as a sensory and thermoregulatory organ. It also can initiate primary immune responses.

Fascia

Fascia is connective tissue containing varying amounts of fat that separate, support, and interconnect organs and structures, enable movement of one structure relative to another, and allow the transit of vessels and nerves from one area to another. There are two general categories of fascia: superficial and deep.

- Superficial (subcutaneous) fascia lies just deep to and is attached to the dermis of the skin. It is made up of loose connective tissue usually containing a large amount of fat. The thickness of the superficial fascia (subcutaneous tissue) varies considerably, both from one area of the body to another and from one individual to another. The superficial fascia allows movement of the skin over deeper areas of the body, acts as a conduit for vessels and nerves coursing to and from the skin, and serves as an energy (fat) reservoir.
- Deep fascia usually consists of dense, organized connective tissue. The outer layer of deep fascia is attached to the deep surface of the superficial fascia and forms a thin fibrous covering over most of the deeper region of the body. Inward extensions of this fascial layer form intermuscular septa that compartmentalize groups of muscles with similar functions and innervations. Other extensions surround individual muscles and groups of vessels and nerves, forming an investing fascia. Near some joints the deep fascia thickens, forming retinacula.

These fascial retinacula hold tendons in place and prevent them from bowing during movements at the joints. Finally, there is a layer of deep fascia separating the membrane lining the abdominal cavity (the parietal peritoneum) from the fascia covering the deep surface of the muscles of the abdominal wall (the transversalis fascia). This layer is referred to as **extraperitoneal fascia**. A similar layer of fascia in the thorax is termed the **endothoracic fascia**.

In the clinic

The importance of fascias

A fascia is a thin band of tissue that surrounds muscles, bones, organs, nerves, and blood vessels and often remains uninterrupted as a three-dimensional structure between tissues. It provides important support for tissues and can provide a boundary between structures.

Fascias have specific dynamic properties and may be relatively elastic where necessary. They contain small blood vessels and tissue receptors and can respond to injury like any other tissue.

Clinically, fascias are extremely important because they often limit the spread of infection and malignant disease. When infections or malignant diseases cross a fascial plain, a primary surgical clearance may require a far more extensive dissection to render the area free of tumor or infection.

A typical example of a fascial layer would be that covering the psoas muscle. Infection within an intervertebral body secondary to tuberculosis can pass laterally into the psoas muscle. Pus fills the psoas muscle but is limited from further spread by the psoas fascia, which surrounds the muscle and extends inferiorly into the groin pointing below the inguinal ligament.

MUSCULAR SYSTEM

The muscular system is generally regarded as consisting of one type of muscle found in the body—skeletal muscle. However, there are two other types of muscle tissue found in the body, smooth muscle and cardiac muscle, that are important components of other systems. These three types of muscle can be characterized by whether they are controlled voluntarily or involuntarily, whether they appear striated (striped) or smooth, and whether they are associated with the body wall (somatic) or with organs and blood vessels (visceral).

- Skeletal muscle forms the majority of the muscle tissue in the body. It consists of parallel bundles of long, multinucleated fibers with transverse stripes, is capable of powerful contractions, and is innervated by somatic and branchial motor nerves. This muscle is used to move bones and other structures, and provides support and gives form to the body. Individual skeletal muscles are often named on the basis of shape (e.g., rhomboid major muscle), attachments (e.g., sternohyoid muscle), function (e.g., flexor pollicis longus muscle), position (e.g., palmar interosseous muscle), or fiber orientation (e.g., external oblique muscle).
- Cardiac muscle is striated muscle found only in the walls of the heart (myocardium) and in some of the large vessels close to where they join the heart. It consists of a branching network of individual cells linked electrically and mechanically to work as a unit. Its contractions are less powerful than those of skeletal muscle and it is resistant to fatigue. Cardiac muscle is innervated by visceral motor nerves.
- Smooth muscle (absence of stripes) consists of elongated or spindle-shaped fibers capable of slow and sustained contractions. It is found in the walls of blood vessels (tunica media), associated with hair follicles in the skin, located in the eyeball, and found in the walls of various structures associated with the gastrointestinal, respiratory, genitourinary, and urogenital systems. Smooth muscle is innervated by visceral motor nerves.

SUBASH KC/NMC-15TH/2014

In the clinic

Muscle paralysis

Muscle paralysis is the inability to move a specific muscle or muscle group and may be associated with other neurological abnormalities, including loss of sensation. Paralysis may be due to abnormalities in the brain, the spinal cord, and the nerves supplying the muscles. Major causes include stroke, trauma, poliomyelitis, and iatrogenic factors.

In the long term, muscle paralysis will produce secondary muscle wasting and overall atrophy of the region due to disuse.

In the clinic

Muscle atrophy

Muscle atrophy is a wasting disorder of muscle. It can be produced by a variety of causes, which include nerve damage to the muscle and disuse.

Muscle atrophy is an important problem in patients who have undergone long-term rest or disuse, requiring extensive rehabilitation and muscle building exercises to maintain normal activities of daily living.

In the clinic

Muscle injuries and strains

Muscle injuries and strains tend to occur in specific muscle groups and usually are related to a sudden exertion and muscle disruption. They typically occur in athletes.

Muscle tears may involve a small interstitial injury up to a complete muscle disruption (Fig. 1.27). It is important to identify which muscle groups are affected and the extent of the tear to facilitate treatment and obtain a prognosis, which will determine the length of rehabilitation necessary to return to normal activity.

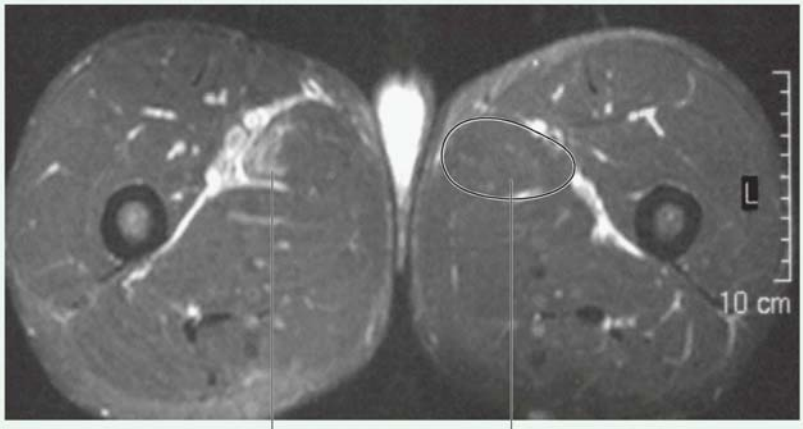


Fig. 1.27 Axial inversion recovery series, which suppresses fat and soft tissue and leaves high signal intensity where fluid is seen. A muscle tear in the right adductor longus with edema in and around the muscle is shown.

SUBASH KC/NMC-15TH/2014

CARDIOVASCULAR SYSTEM

The cardiovascular system consists of the heart, which pumps blood throughout the body, and the blood vessels, which are a closed network of tubes that transport the blood. There are three types of blood vessels:

- arteries, which transport blood away from the heart;
- veins, which transport blood toward the heart;
- capillaries, which connect the arteries and veins, are the smallest of the blood vessels and are where oxygen, nutrients, and wastes are exchanged within the tissues.

The walls of the blood vessels of the cardiovascular system usually consist of three layers or tunics:

- tunica externa (adventitia)—the outer connective tissue layer,
- tunica media—the middle smooth muscle layer (may also contain varying amounts of elastic fibers in medium and large arteries), and
- tunica intima—the inner endothelial lining of the blood vessels.

Arteries are usually further subdivided into three classes, according to the variable amounts of smooth muscle and elastic fibers contributing to the thickness of the tunica media, the overall size of the vessel, and its function.

- Large elastic arteries contain substantial amounts of elastic fibers in the tunica media, allowing expansion and recoil during the normal cardiac cycle. This helps maintain a constant flow of blood during diastole. Examples of large elastic arteries are the aorta, the brachiocephalic trunk, the left common carotid artery, the left subclavian artery, and the pulmonary trunk.
- Medium muscular arteries are composed of a tunica media that contains mostly smooth muscle fibers. This characteristic allows these vessels to regulate their diameter and control the flow of blood to different parts of the body. Examples of medium muscular arteries are most of the named arteries, including the femoral, axillary, and radial arteries.
- Small arteries and arterioles control the filling of the capillaries and directly contribute to the arterial pressure in the vascular system.

Veins also are subdivided into three classes.

- Large veins contain some smooth muscle in the tunica media, but the thickest layer is the tunica externa. Examples of large veins are the superior vena cava, the inferior vena cava, and the portal vein.
- Small and medium veins contain small amounts of smooth muscle, and the thickest layer is the tunica externa. Examples of small and medium veins are superficial veins in the upper and lower limbs and deeper veins of the leg and forearm.
- Venules are the smallest veins and drain the capillaries.

Although veins are similar in general structure to arteries, they have a number of distinguishing features.

- The walls of veins, specifically the tunica media, are thin.
- The luminal diameters of veins are large.
- There often are multiple veins (venae comitantes) closely associated with arteries in peripheral regions.
- Valves often are present in veins, particularly in peripheral vessels inferior to the level of the heart. These are usually paired cusps that facilitate blood flow toward the heart.

More specific information about the cardiovascular system and how it relates to the circulation of blood throughout the body will be discussed, where appropriate, in each of the succeeding chapters of the text.

In the clinic

Atherosclerosis

Atherosclerosis is a disease that affects arteries. There is a chronic inflammatory reaction in the walls of the arteries, with deposition of cholesterol and fatty proteins. This may in turn lead to secondary calcification, with reduction in the diameter of the vessels impeding distal flow. The plaque itself may be a site for attraction of platelets that may "fall off" (embolize) distally. Plaque fissuring may occur, which allows fresh clots to form and occlude the vessel.

The importance of atherosclerosis and its effects depend upon which vessel is affected. If atherosclerosis occurs in the carotid artery, small emboli may form and produce a stroke. In the heart, plaque fissuring may produce an acute vessel thrombosis, producing a myocardial infarction (heart attack). In the legs, chronic narrowing of vessels may limit the ability of the patient to walk and ultimately cause distal ischemia and gangrene of the toes.

SUBASH KC/NMC-15TH/2014

In the clinic

Varicose veins

Varicose veins are tortuous dilated veins that typically occur in the legs, although they may occur in the superficial veins of the arm and in other organs.

In normal individuals the movement of adjacent leg muscles pumps the blood in the veins to the heart. Blood is also pumped from the superficial veins through the investing layer of fascia of the leg into the deep veins. Valves in these perforating veins may become damaged, allowing blood to pass in the opposite direction. This increased volume and pressure produces dilatation and tortuosity of the superficial veins (Fig. 1.28). Apart from the unsightliness of larger veins, the skin may become pigmented and atrophic with a poor response to tissue damage. In some patients even small trauma may produce skin ulceration, which requires elevation of the limb and application of pressure bandages to heal.

Treatment of varicose veins depends on their location, size, and severity. Typically the superficial varicose veins can be excised and stripped, allowing blood only to drain into the deep system.



Fig. 1.28 Photograph demonstrating varicose veins.

In the clinic

Anastomoses and collateral circulation

All organs require a blood supply from the arteries and drainage by veins. Within most organs there are multiple ways of perfusing the tissue such that if the main vessel feeding the organ or vein draining the organ is blocked, a series of smaller vessels (collateral vessels) continue to supply and drain the organ.

In certain circumstances, organs have more than one vessel perfusing them, such as the hand, which is supplied by the radial and ulnar arteries. Loss of either the radial or the ulnar artery may not produce any symptoms of reduced perfusion to the hand.

There are circumstances in which loss of a vein produces significant venous collateralization. Some of these venous collaterals become susceptible to bleeding. This is a considerable problem in patients who have undergone portal vein thrombosis or occlusion, where venous drainage from the gut bypasses the liver through collateral veins to return to the systemic circulation.

Normal vascular anastomoses associated with an organ are important. Some organs, such as the duodenum, have a dual blood supply arising from the branches of the celiac trunk and also from the branches of the superior mesenteric artery. Should either of these vessels be damaged, blood supply will be maintained to the organ. The brain has multiple vessels supplying it, dominated by the carotid arteries and the vertebral arteries. Vessels within the brain are end arteries and have a poor collateral circulation; hence any occlusion will produce long-term cerebral damage.

SUBASH KC/NMC-15TH/2014

LYMPHATIC SYSTEM

Lymphatic vessels

Lymphatic vessels form an extensive and complex interconnected network of channels, which begin as "porous" blind-ended lymphatic capillaries in tissues of the body and converge to form a number of larger vessels, which ultimately connect with large veins in the root of the neck.

Lymphatic vessels mainly collect fluid lost from vascular capillary beds during nutrient exchange processes and deliver it back to the venous side of the vascular system (Fig. 1.29). Also included in this interstitial fluid that drains into the lymphatic capillaries are pathogens, cells of the lymphocytic system, cell products (such as hormones), and cell debris.

In the small intestine, certain fats absorbed and processed by the intestinal epithelium are packaged into protein-coated lipid droplets (**chylomicrons**), which are released from the epithelial cells and enter the interstitial compartment. Together with other components of the interstitial fluid, the chylomicrons drain into lymphatic capillaries (known as **lacteals** in the small intestine) and are ultimately delivered to the venous system in the neck. The lymphatic system is therefore also a major route of transport for fat absorbed by the gut.

The fluid in most lymphatic vessels is clear and colorless and is known as **lymph**. That carried by lymphatic vessels from the small intestine is opaque and milky because of the presence of chylomicrons and is termed **chyle**.

There are lymphatic vessels in most areas of the body except the brain, bone marrow, and avascular tissues such as epithelia and cartilage.

The movement of lymph through the lymphatic vessels is generated mainly by the indirect action of adjacent structures, particularly by contraction of skeletal muscles and pulses in arteries. Unidirectional flow is maintained by the presence of valves in the vessels.

SUBASH KC/NMC-15TH/2014

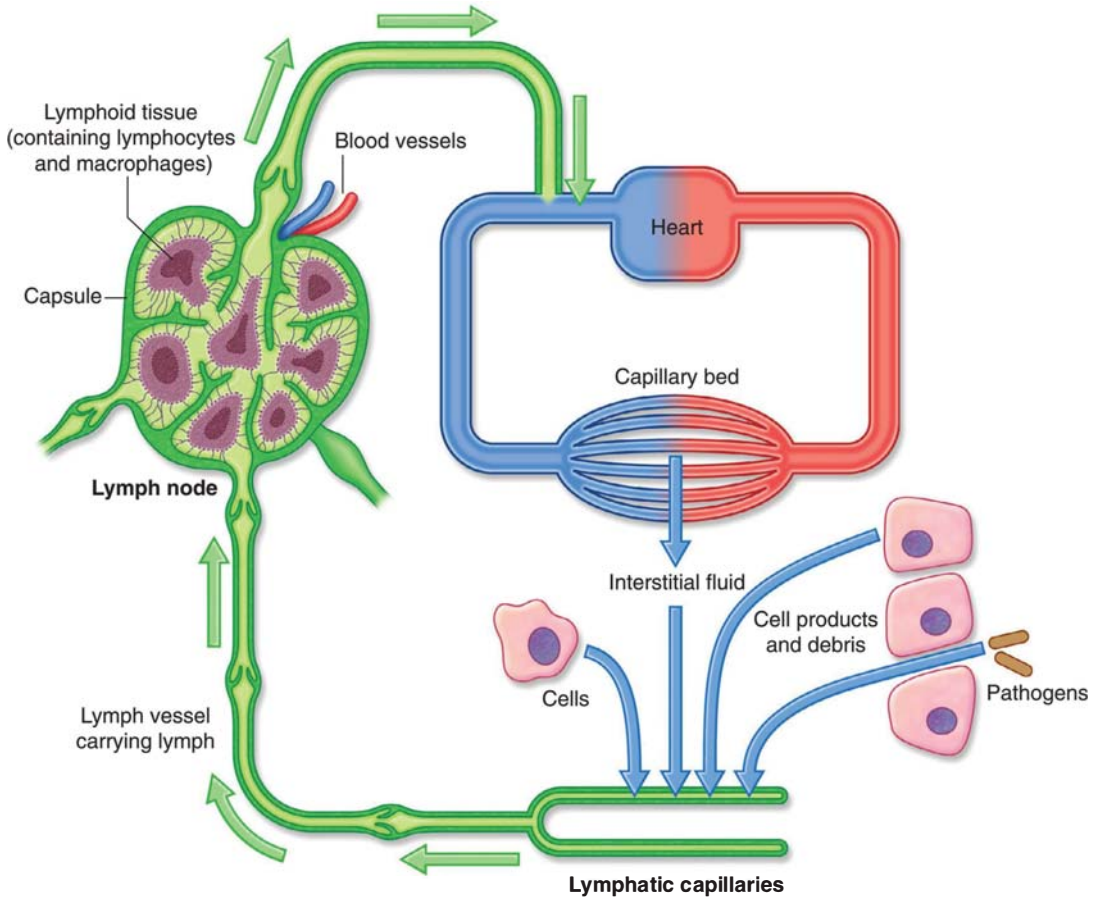


Fig. 1.29 Lymphatic vessels mainly collect fluid lost from vascular capillary beds during nutrient exchange processes and deliver it back to the venous side of the vascular system.

Lymph nodes

Lymph nodes are small (0.1–2.5 cm long) encapsulated structures that interrupt the course of lymphatic vessels and contain elements of the body's defense system, such as clusters of lymphocytes and macrophages. They act as elaborate filters that trap and phagocytose particulate matter in the lymph that percolates through them. In addition, they detect and defend against foreign antigens that are also carried in the lymph (Fig. 1.29).

Because lymph nodes are efficient filters and flow through them is slow, cells that metastasize from (migrate away from) primary tumors and enter lymphatic vessels often lodge and grow as secondary tumors in lymph nodes. Lymph nodes that drain regions that are infected or contain other forms of disease can enlarge or undergo certain physical changes, such as becoming "hard" or "tender." These changes can be used by clinicians to detect pathologic changes or to track spread of disease.

A number of regions in the body are associated with clusters or a particular abundance of lymph nodes (Fig. 1.30). Not surprisingly, nodes in many of these regions drain the body's surface, the digestive system, or the respiratory system. All three of these areas are high-risk sites for the entry of foreign pathogens.

Lymph nodes are abundant and accessible to palpation in the axilla, the groin and femoral region, and the neck. Deep sites that are not palpable include those associated with the trachea and bronchi in the thorax, and with the aorta and its branches in the abdomen.

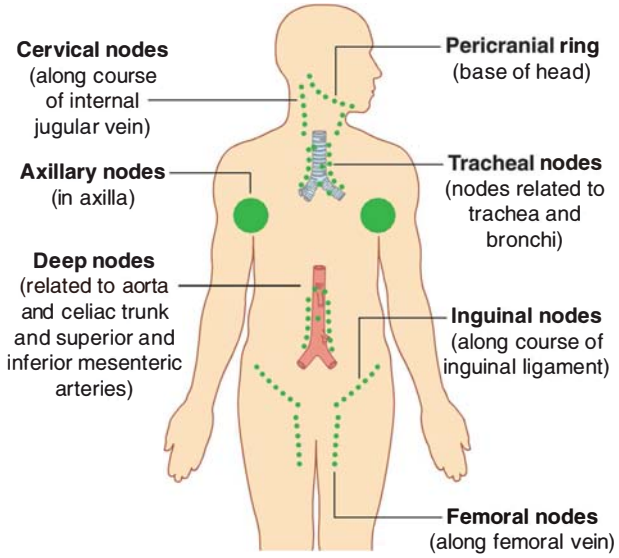


Fig. 1.30 Regions associated with clusters or a particular abundance of lymph nodes.

Lymphatic trunks and ducts

All lymphatic vessels coalesce to form larger trunks or ducts, which drain into the venous system at sites in the neck where the internal jugular veins join the subclavian veins to form the brachiocephalic veins (Fig. 1.31):

- Lymph from the right side of the head and neck, the right upper limb, right side of the thorax, and right side of the upper and more superficial region of the abdominal wall is carried by lymphatic vessels that connect with veins on the right side of the neck.
- Lymph from all other regions of the body is carried by lymphatic vessels that drain into veins on the left side of the neck.

Specific information about the organization of the lymphatic system in each region of the body is discussed in the appropriate chapter.

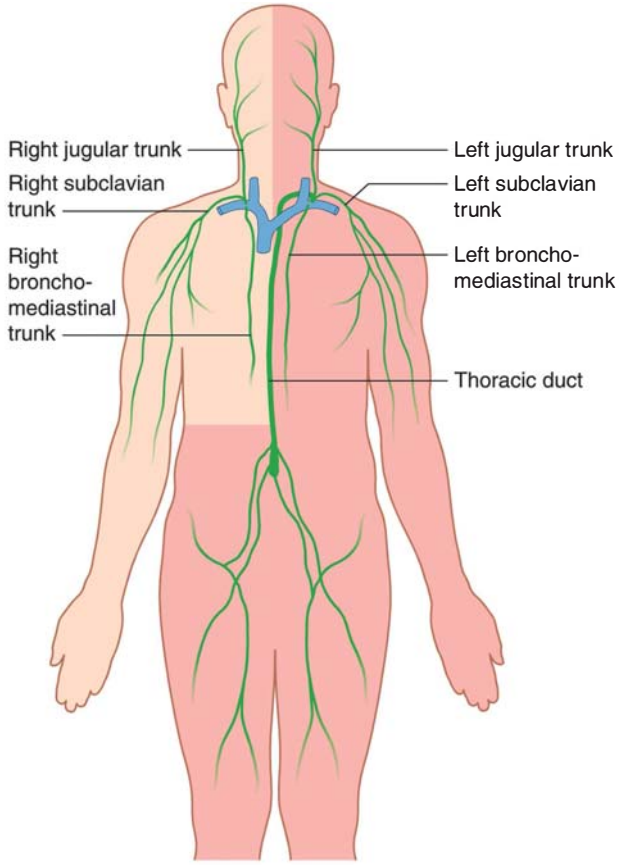


Fig. 1.31 Major lymphatic vessels that drain into large veins in the neck.

SUBASH KC/NMC-15TH/2014

In the clinic

Lymph nodes

Lymph nodes are efficient filters and have an internal honeycomb of reticular connective tissue filled with lymphocytes. These lymphocytes act on bacteria, viruses, and other bodily cells to destroy them. Lymph nodes tend to drain specific areas, and if infection occurs within a drainage area, the lymph node will become active. The rapid cell turnover and production of local inflammatory mediators may cause the node to enlarge and become tender. Similarly, in patients with malignancy the lymphatics may drain metastasizing cells to the lymph nodes. These can become enlarged and inflamed and will need to be removed if clinically symptomatic.

Lymph nodes may become diffusely enlarged in certain systemic illnesses (e.g., viral infection), or local groups may become enlarged with primary lymph node malignancies, such as lymphoma (Fig. 1.32).

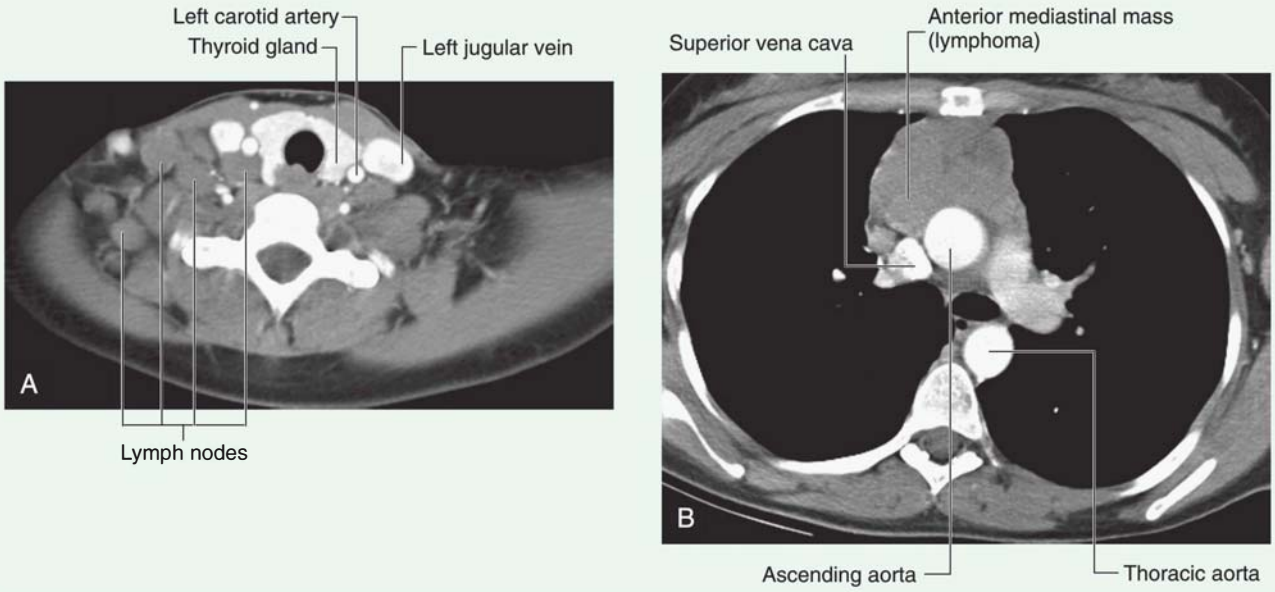


Fig. 1.32 A. This computed tomogram with contrast, in the axial plane, demonstrates the normal common carotid arteries and internal jugular veins with numerous other nonenhancing nodules that represent lymph nodes in a patient with lymphoma. **B.** This computed tomogram with contrast, in the axial plane, demonstrates a large anterior soft tissue mediastinal mass that represents a lymphoma.

SUBASH KC/NMC-15TH/2014

NERVOUS SYSTEM

The nervous system can be separated into parts based on structure and on function:

- structurally, it can be divided into the central nervous system (CNS) and the peripheral nervous system (PNS) (Fig. 1.33);
- functionally, it can be divided into somatic and visceral parts.

The CNS is composed of the brain and spinal cord, both of which develop from the neural tube in the embryo.

The PNS is composed of all nervous structures outside the CNS that connect the CNS to the body. Elements of this system develop from neural crest cells and as outgrowths of the CNS. The PNS consists of the spinal and cranial nerves, visceral nerves and plexuses, and the enteric system. The detailed anatomy of a typical spinal nerve is described in Chapter 2, as is the way spinal nerves are numbered. Cranial nerves are described in Chapter 8. The details of nerve plexuses are described in chapters dealing with the specific regions in which the plexuses are located.

Central nervous system

Brain

The parts of the brain are the cerebral hemispheres, the cerebellum, and the brainstem. The cerebral hemispheres

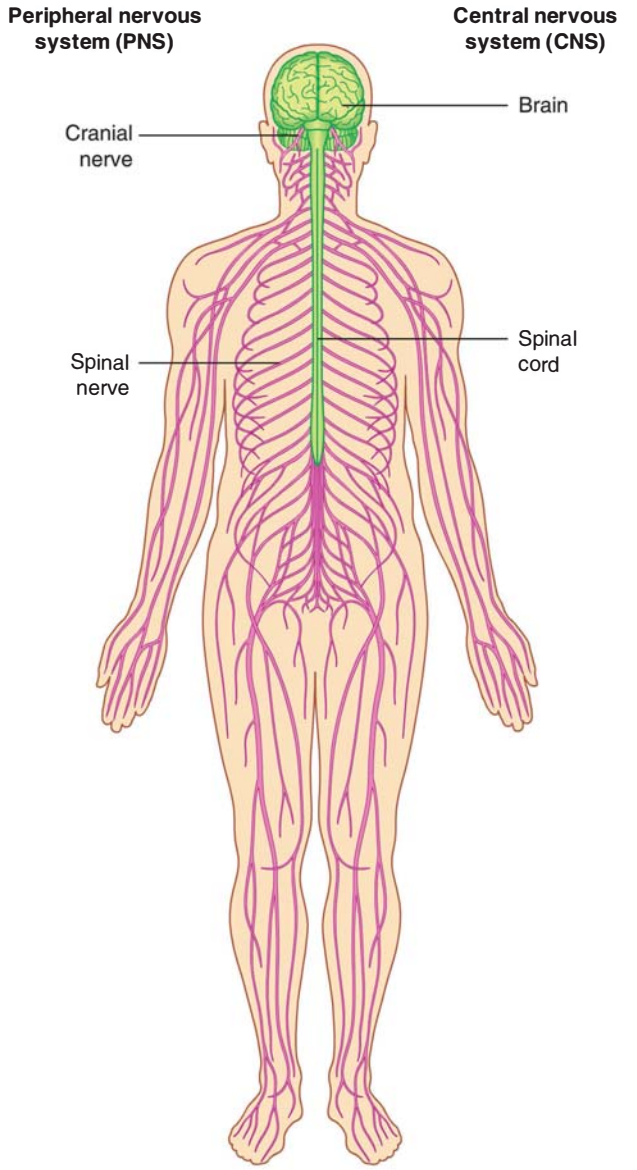


Fig. 1.33 CNS and PNS.

consist of an outer portion, or the **gray matter**, containing cell bodies; an inner portion, or the **white matter**, made up of axons forming tracts or pathways; and the **ventricles**, which are spaces filled with cerebrospinal fluid (CSF).

The cerebellum has two lateral lobes and a midline portion. The components of the brainstem are classically defined as the diencephalon, midbrain, pons, and medulla. However, in common usage today, the term "brainstem" usually refers to the midbrain, pons, and medulla.

A further discussion of the brain can be found in Chapter 8.

Spinal cord

The spinal cord is the part of the CNS in the superior two thirds of the vertebral canal. It is roughly cylindrical in shape, and is circular to oval in cross section with a central canal. A further discussion of the spinal cord can be found in Chapter 2.

Meninges

The meninges (Fig. 1.34) are three connective tissue coverings that surround, protect, and suspend the brain and spinal cord within the cranial cavity and vertebral canal, respectively:

- The dura mater is the thickest and most external of the coverings.
- The arachnoid mater is against the internal surface of the dura mater.
- The pia mater is adherent to the brain and spinal cord.

Between the arachnoid and pia mater is the subarachnoid space, which contains CSF.

A further discussion of the cranial meninges can be found in Chapter 8 and of the spinal meninges in Chapter 2.

Functional subdivisions of the CNS

Functionally, the nervous system can be divided into somatic and visceral parts.

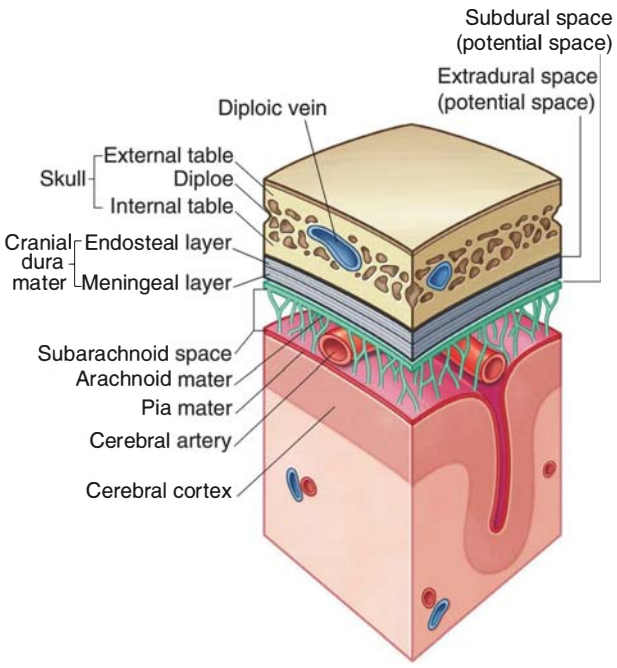


Fig. 1.34 Arrangement of meninges in the cranial cavity.

SUBASH KC/NMC-15TH/2014

- The **somatic part** (*soma*, from the Greek for "body") innervates structures (skin and most skeletal muscle) derived from somites in the embryo, and is mainly involved with receiving and responding to information from the external environment.
- The **visceral part** (*viscera*, from the Greek for "guts") innervates organ systems in the body and other visceral elements, such as smooth muscle and glands, in peripheral regions of the body. It is concerned mainly with detecting and responding to information from the internal environment.

Somatic part of the nervous system

The somatic part of the nervous system consists of:

- nerves that carry conscious sensations from peripheral regions back to the CNS, and
- nerves that innervate voluntary muscles.

Somatic nerves arise segmentally along the developing CNS in association with **somites**, which are themselves arranged segmentally along each side of the neural tube (Fig. 1.35). Part of each somite (the **dermatomyotome**) gives rise to skeletal muscle and the dermis of the skin. As cells of the dermatomyotome differentiate, they migrate into posterior (dorsal) and anterior (ventral) areas of the developing body:

- Cells that migrate anteriorly give rise to muscles of the limbs and trunk (**hypaxial muscles**) and to the associated dermis.
- Cells that migrate posteriorly give rise to the intrinsic muscles of the back (**epaxial muscles**) and the associated dermis.

Developing nerve cells within anterior regions of the neural tube extend processes peripherally into posterior

SUBASH KC/NMC-15TH/2014

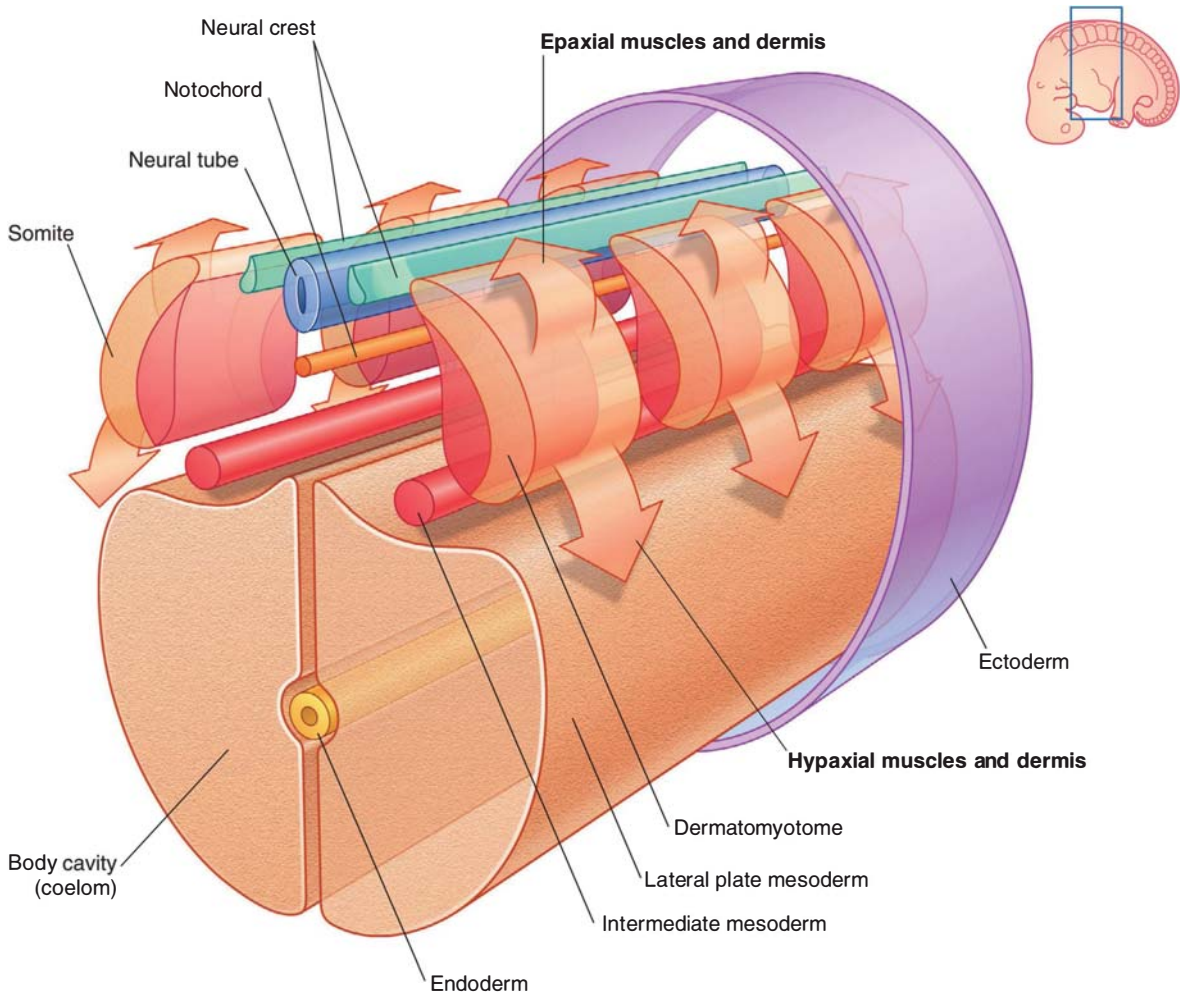


Fig. 1.35 Differentiation of somites in a "tubular" embryo.

and anterior regions of the differentiating dermatomyotome of each somite.

Simultaneously, derivatives of neural crest cells (cells derived from neural folds during formation of the neural tube) differentiate into neurons on each side of the neural tube and extend processes both medially and laterally (Fig. 1.36):

- Medial processes pass into the posterior aspect of the neural tube.
- Lateral processes pass into the differentiating regions of the adjacent dermatomyotome.

Neurons that develop from cells within the spinal cord are **motor neurons** and those that develop from neural crest cells are **sensory neurons**.

Somatic sensory and somatic motor fibers that are organized segmentally along the neural tube become parts of all spinal nerves and some cranial nerves.

The clusters of sensory nerve cell bodies derived from neural crest cells and located outside the CNS form sensory ganglia.

Generally, all sensory information passes into the posterior aspect of the spinal cord, and all motor fibers leave anteriorly.

Somatic sensory neurons carry information from the periphery into the CNS and are also called **somatic sensory afferents** or **general somatic afferents (GSAs)**. The modalities carried by these nerves include temperature, pain, touch, and proprioception. Proprioception is the sense of determining the position and movement of the musculoskeletal system detected by special receptors in muscles and tendons.

Somatic motor fibers carry information away from the CNS to skeletal muscles and are also called **somatic motor efferents** or **general somatic efferents** (**GSEs**). Like somatic sensory fibers that come from the periphery, somatic motor fibers can be very long. They extend from cell bodies in the spinal cord to the muscle cells they innervate.

Dermatomes

Because cells from a specific somite develop into the dermis of the skin in a precise location, somatic sensory fibers originally associated with that somite enter the posterior

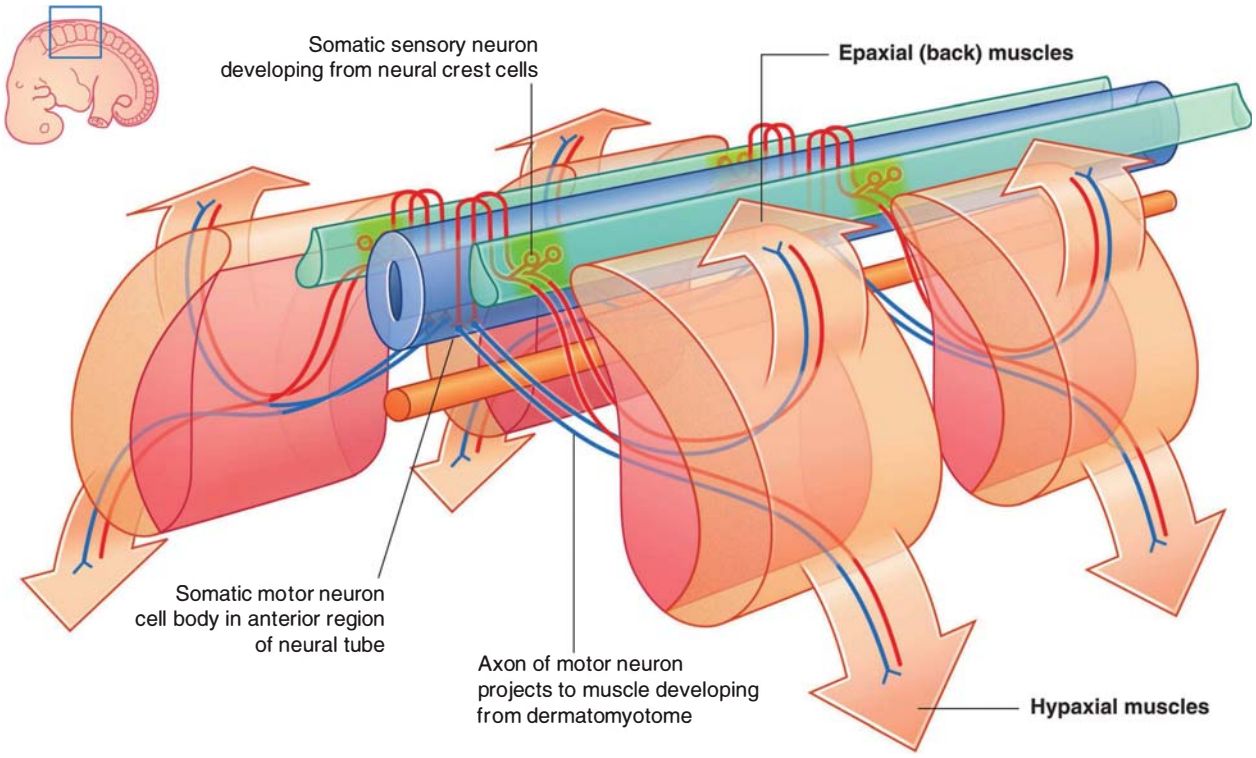


Fig. 1.36 Somatic sensory and motor neurons. Blue lines indicate motor nerves and red lines indicate sensory nerves.

SUBASH KC/NMC-15TH/2014

region of the spinal cord at a specific level and become part of one specific spinal nerve (Fig. 1.37). Each spinal nerve therefore carries somatic sensory information from a specific area of skin on the surface of the body. A **dermatome** is that area of skin supplied by a single spinal cord level, or on one side, by a single spinal nerve.

There is overlap in the distribution of dermatomes, but usually a specific region within each dermatome can be identified as an area supplied by a single spinal cord level. Testing touch in these autonomous zones in a conscious patient can be used to localize lesions to a specific spinal nerve or to a specific level in the spinal cord.

Myotomes

Somatic motor nerves that were originally associated with a specific somite emerge from the anterior region of the spinal cord and, together with sensory nerves from the same level, become part of one spinal nerve. Therefore each spinal nerve carries somatic motor fibers to muscles that originally developed from the related somite. A **myotome** is that portion of a skeletal muscle innervated by a single spinal cord level or, on one side, by a single spinal nerve.

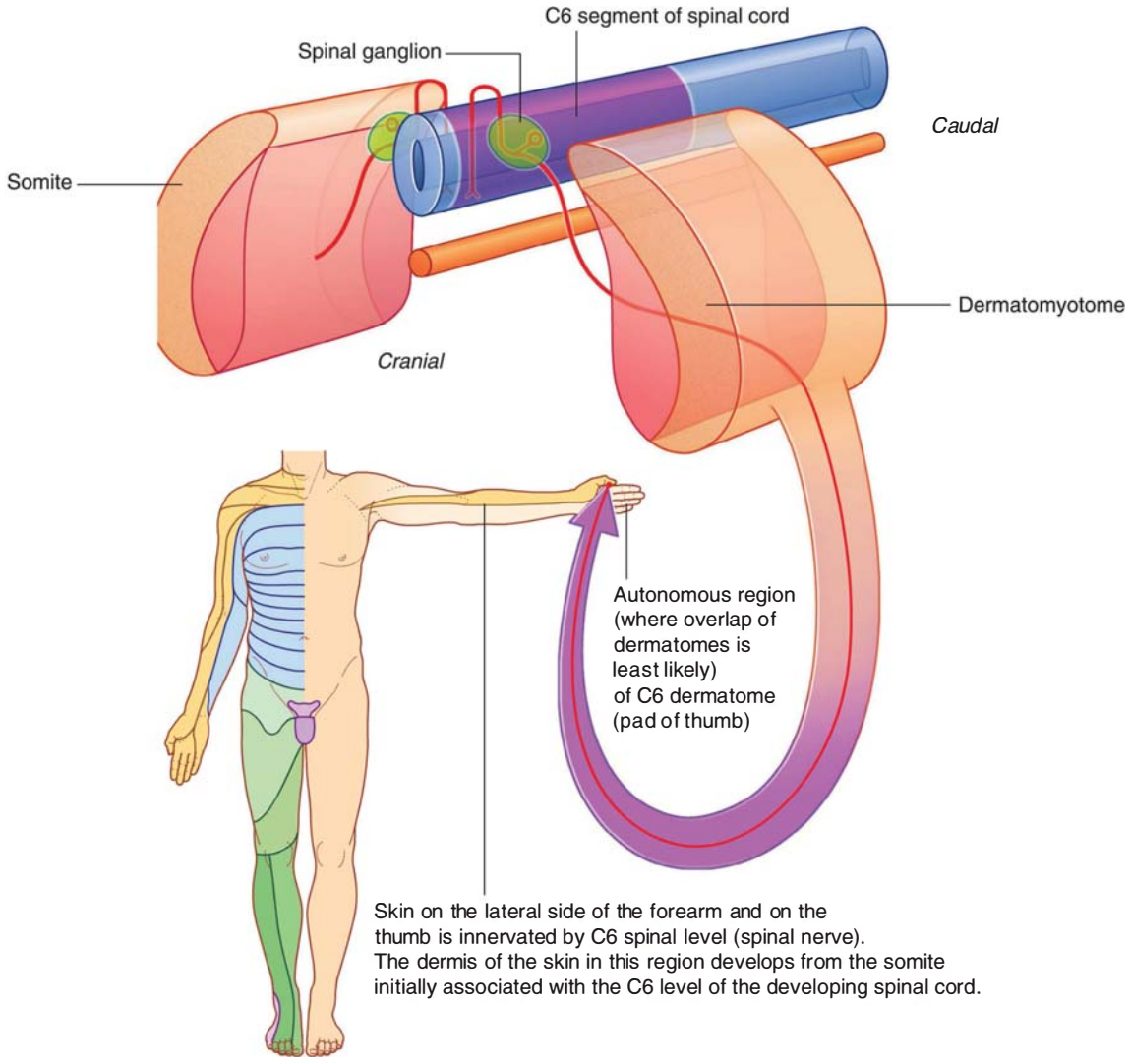


Fig. 1.37 Dermatomes.

SUBASH KC/NMC-15TH/2014

Myotomes are generally more difficult to test than dermatomes because each skeletal muscle in the body is usually innervated by nerves derived from more than one spinal cord level (Fig. 1.38).

Testing movements at successive joints can help in localizing lesions to specific nerves or to a specific spinal cord level. For example:

- Muscles that move the shoulder joint are innervated mainly by spinal nerves from spinal cord levels C5 and C6.
- Muscles that move the elbow are innervated mainly by spinal nerves from spinal cord levels C6 and C7.
- Muscles in the hand are innervated mainly by spinal nerves from spinal cord levels C8 and T1.

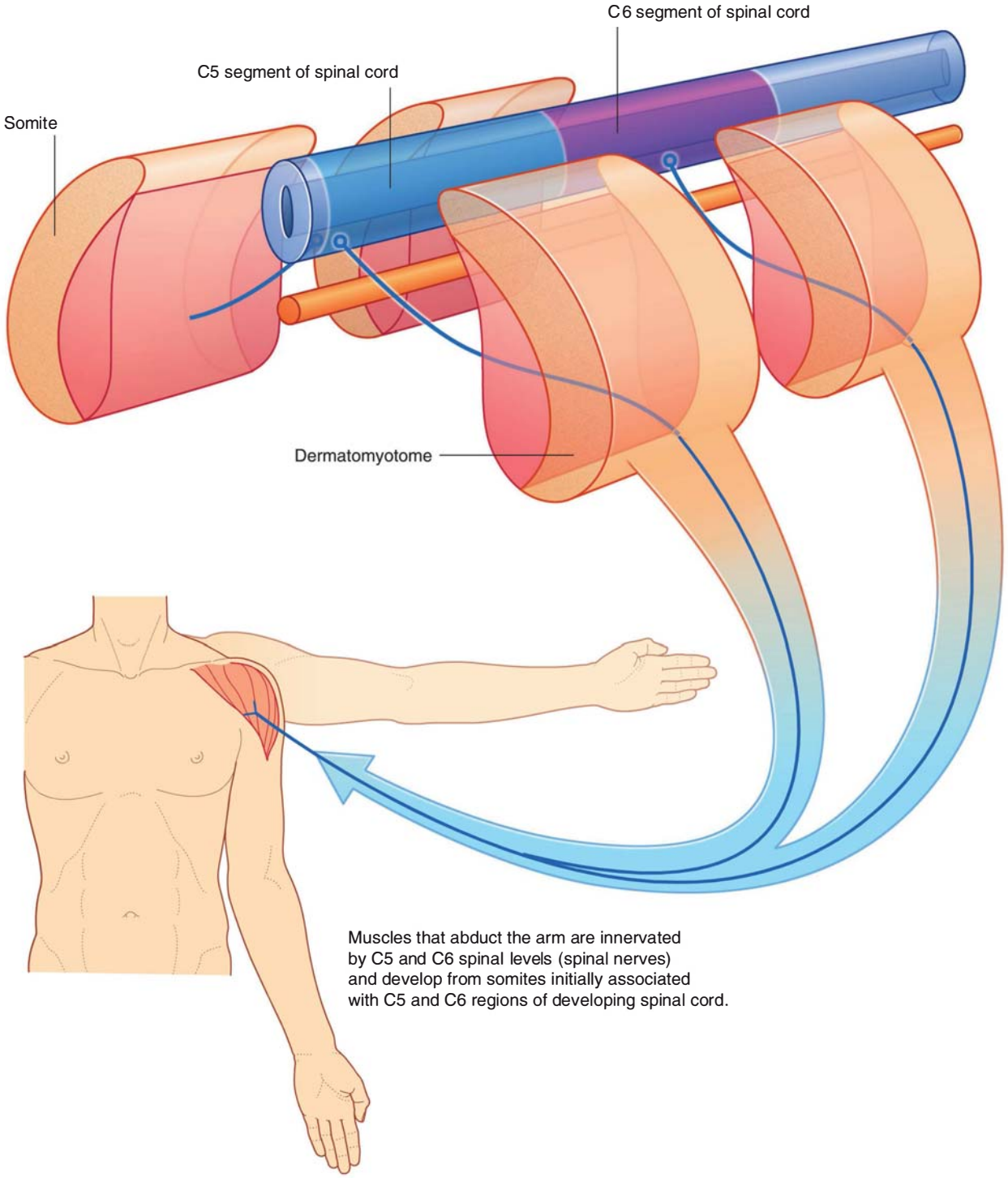


Fig. 1.38 Myotomes.

SUBASH KC/NMC-15TH/2014

In the clinic

Dermatomes and myotomes

A knowledge of dermatomes and myotomes is absolutely fundamental to carrying out a neurological examination. A typical dermatome map is shown in Fig. 1.39.

Clinically, a dermatome is that area of skin supplied by a single nerve or spinal cord level. A myotome is that region of skeletal muscle innervated by a single nerve or spinal cord level. Most individual muscles of the body are innervated by more than one spinal cord level, so the evaluation of myotomes is usually accomplished by testing movements of joints or muscle groups.

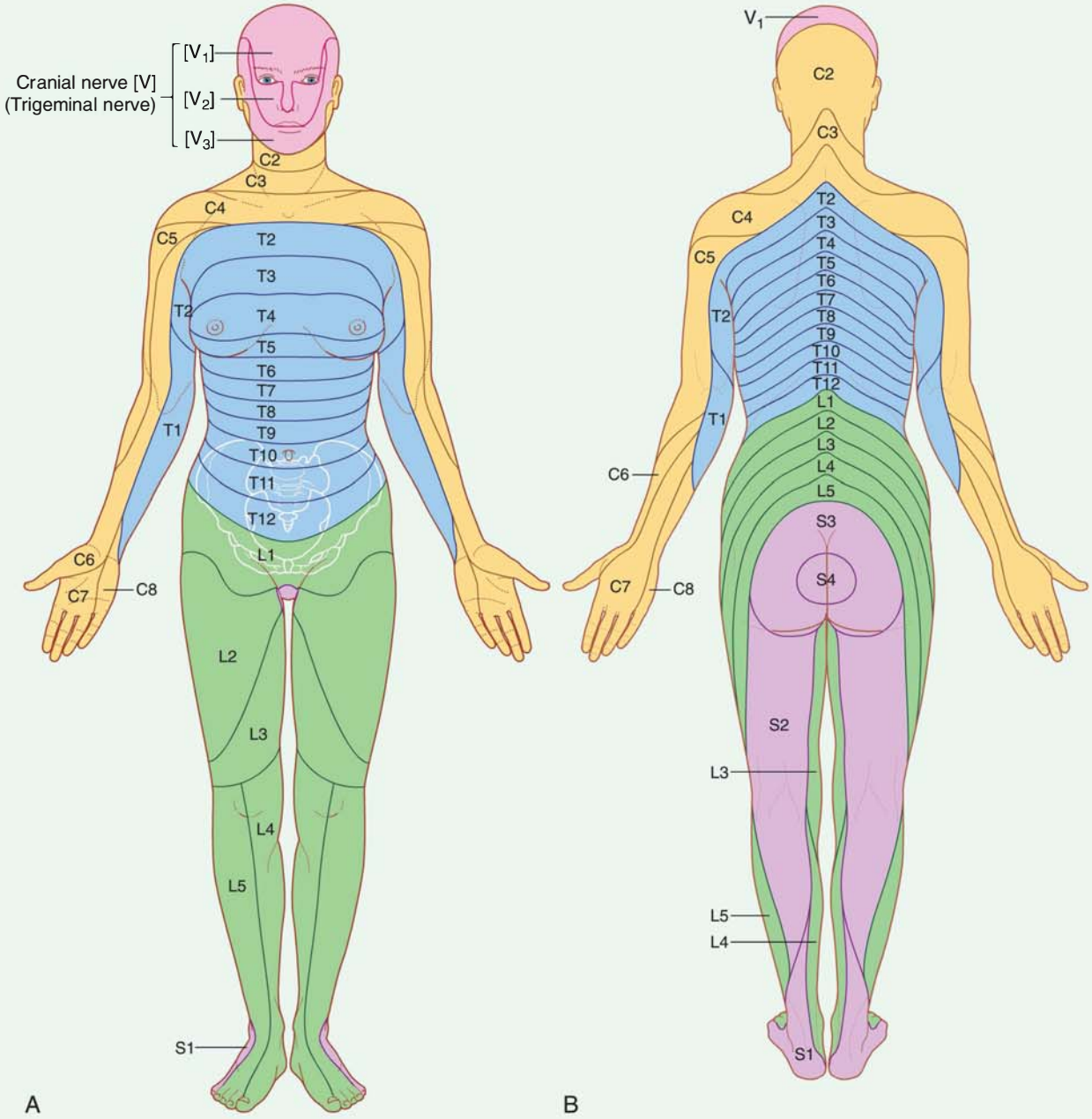


Fig. 1.39 Dermatomes. **A.** Anterior view. **B.** Posterior view.

SUBASH KC/NMC-15TH/2014

Visceral part of the nervous system

The visceral part of the nervous system, as in the somatic part, consists of motor and sensory components:

- Sensory nerves monitor changes in the viscera.
- Motor nerves mainly innervate smooth muscle, cardiac muscle, and glands.

The visceral motor component is commonly referred to as the **autonomic division of the PNS** and is subdivided into **sympathetic** and **parasympathetic** parts.

Like the somatic part of the nervous system, the visceral part is segmentally arranged and develops in a parallel fashion (Fig. 1.40).

Visceral sensory neurons that arise from neural crest cells send processes medially into the adjacent neural tube and laterally into regions associated with the developing body. These sensory neurons and their processes, referred to as **general visceral afferent fibers** (**GVAs**), are associated primarily with chemoreception, mechanoreception, and stretch reception.

Visceral motor neurons that arise from cells in lateral regions of the neural tube send processes out of the

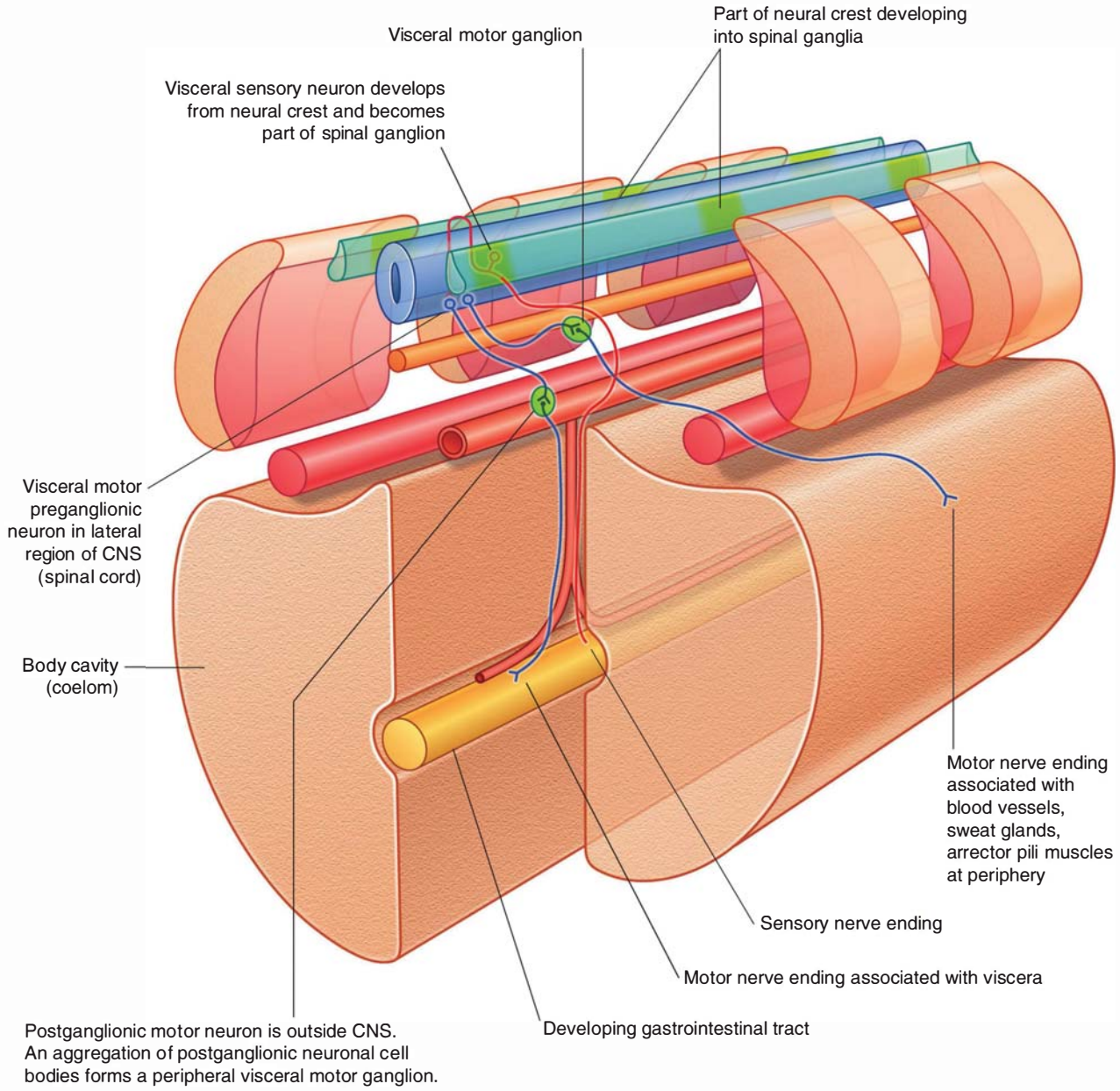


Fig. 1.40 Development of the visceral part of the nervous system.

SUBASH KC/NMC-15TH/2014

anterior aspect of the tube. Unlike in the somatic part, these processes, containing **general visceral efferent fibers (GVEs)**, synapse with other cells, usually other visceral motor neurons, that develop outside the CNS from neural crest cells that migrate away from their original positions close to the developing neural tube.

The visceral motor neurons located in the spinal cord are referred to as preganglionic motor neurons and their axons are called **preganglionic fibers**; the visceral motor neurons located outside the CNS are referred to as postganglionic motor neurons and their axons are called **postganglionic fibers**.

The cell bodies of the visceral motor neurons outside the CNS often associate with each other in a discrete mass called a **ganglion**.

Visceral sensory and motor fibers enter and leave the CNS with their somatic equivalents (Fig. 1.41). Visceral sensory fibers enter the spinal cord together with somatic sensory fibers through posterior roots of spinal nerves. Preganglionic fibers of visceral motor neurons exit the spinal cord in the anterior roots of spinal nerves, along with fibers from somatic motor neurons.

Postganglionic fibers traveling to visceral elements in the periphery are found in the posterior and anterior rami (branches) of spinal nerves.

Visceral motor and sensory fibers that travel to and from viscera form named visceral branches that are separate from the somatic branches. These nerves generally form plexuses from which arise branches to the viscera.

Visceral motor and sensory fibers do not enter and leave the CNS at all levels (Fig. 1.42):

- In the cranial region, visceral components are associated with four of the twelve cranial nerves (CN III, VII, IX, and X).
- In the spinal cord, visceral components are associated mainly with spinal cord levels T1 to L2 and S2 to S4.

Visceral motor components associated with spinal levels T1 to L2 are termed **sympathetic**. Those visceral motor components in cranial and sacral regions, on either side of the sympathetic region, are termed **parasympathetic**:

- The sympathetic system innervates structures in peripheral regions of the body and viscera.
- The parasympathetic system is more restricted to innervation of the viscera only.

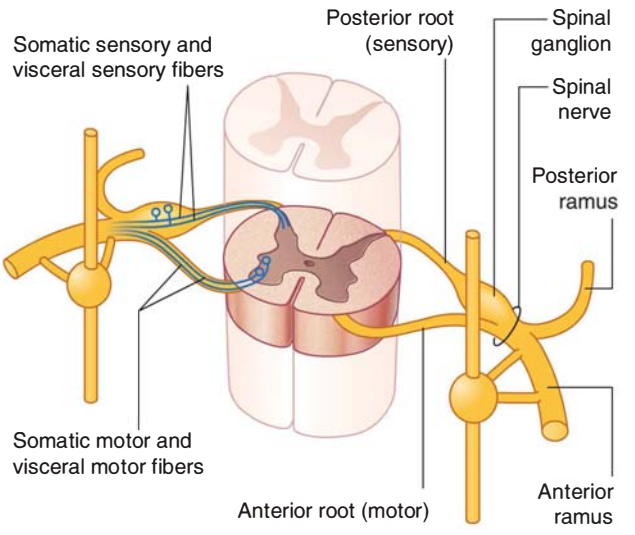


Fig. 1.41 Basic anatomy of a thoracic spinal nerve.

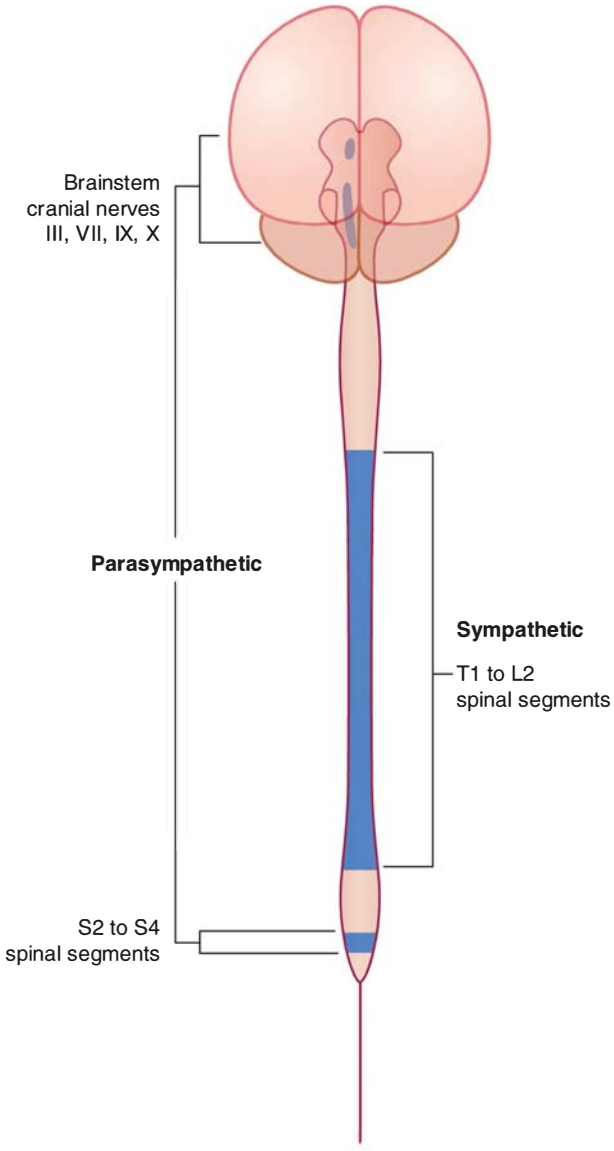


Fig. 1.42 Parts of the CNS associated with visceral motor components.

SUBASH KC/NMC-15TH/2014

Sympathetic system

The sympathetic part of the autonomic division of the PNS leaves thoracolumbar regions of the spinal cord with the somatic components of spinal nerves T1 to L2 (Fig. 1.43). On each side, a paravertebral sympathetic trunk extends from the base of the skull to the inferior end of the vertebral column where the two trunks converge anteriorly to the coccyx at the ganglion impar. Each trunk is attached to the anterior rami of spinal nerves and becomes the route by which sympathetics are distributed to the periphery and all viscera.

Visceral motor preganglionic fibers leave the T1 to L2 part of the spinal cord in anterior roots. The fibers

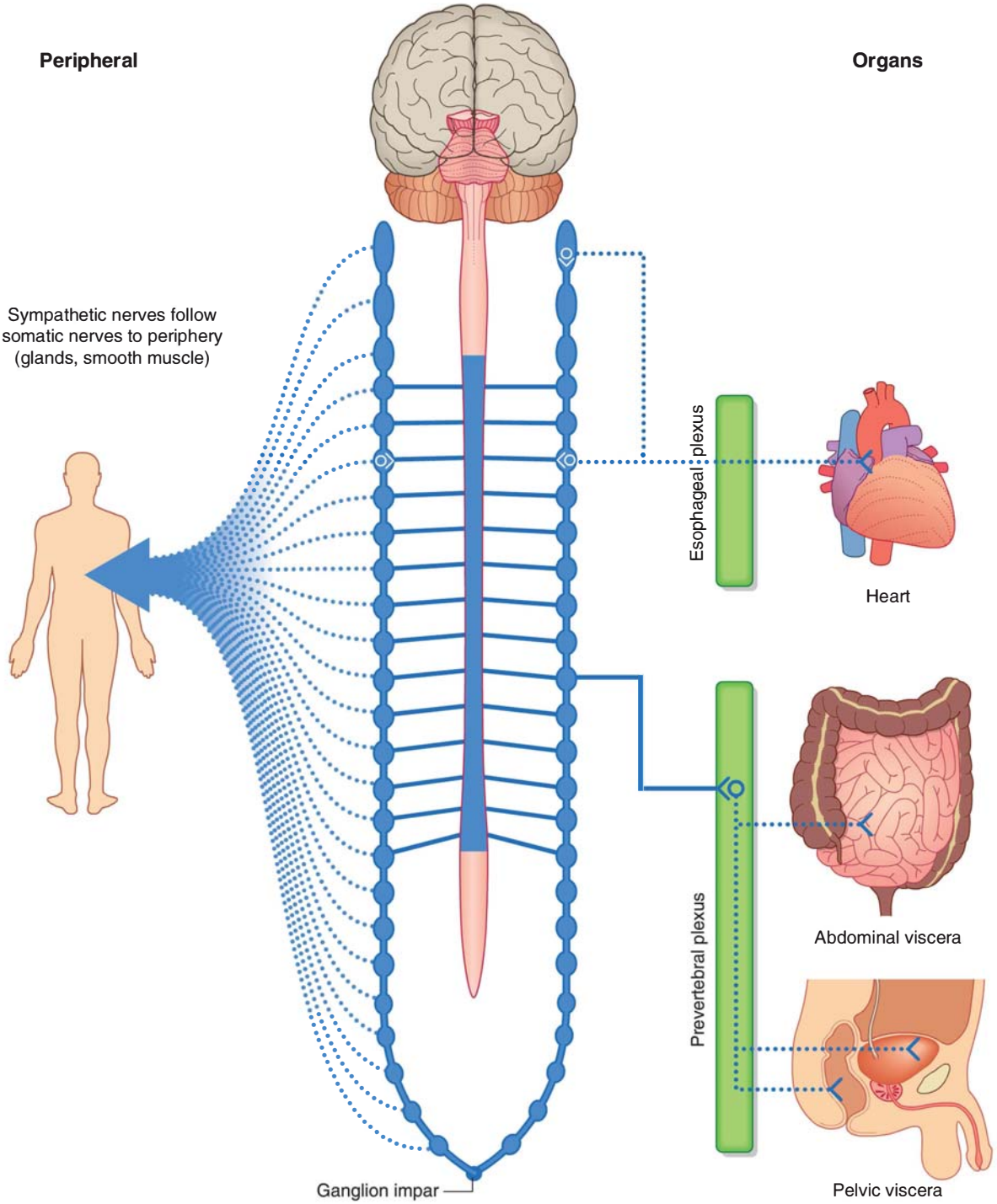


Fig. 1.43 Sympathetic part of the autonomic division of the PNS.

SUBASH KC/NMC-15TH/2014

then enter the spinal nerves, pass through the anterior rami and into the sympathetic trunks. One trunk is located on each side of the vertebral column (paravertebral) and positioned anterior to the anterior rami. Along the trunk is a series of segmentally arranged ganglia formed from collections of postganglionic neuronal cell bodies where the preganglionic neurons synapse with postganglionic neurons. Anterior rami of T1 to L2 are connected to the sympathetic trunk or to a ganglion by a **white ramus communicans,** which carries preganglionic sympathetic fibers and appears white because the fibers it contains are myelinated.

Preganglionic sympathetic fibers that enter a paravertebral ganglion or the sympathetic trunk through a white ramus communicans may take the following four pathways to target tissues:

1. Peripheral sympathetic innervation at the level of origin of the preganglionic fiber

Preganglionic sympathetic fibers may synapse with postganglionic motor neurons in ganglia associated with the sympathetic trunk, after which postganglionic fibers enter the same anterior ramus and are distributed with peripheral branches of the posterior and anterior rami of that spinal nerve (Fig. 1.44). The fibers innervate structures at the periphery of the body in regions supplied by the spinal nerve. The **gray ramus communicans** connects the sympathetic trunk or a ganglion to the anterior ramus and contains the postganglionic sympathetic fibers. It appears gray because postganglionic fibers are nonmyelinated. The gray ramus communicans is positioned medial to the white ramus communicans.

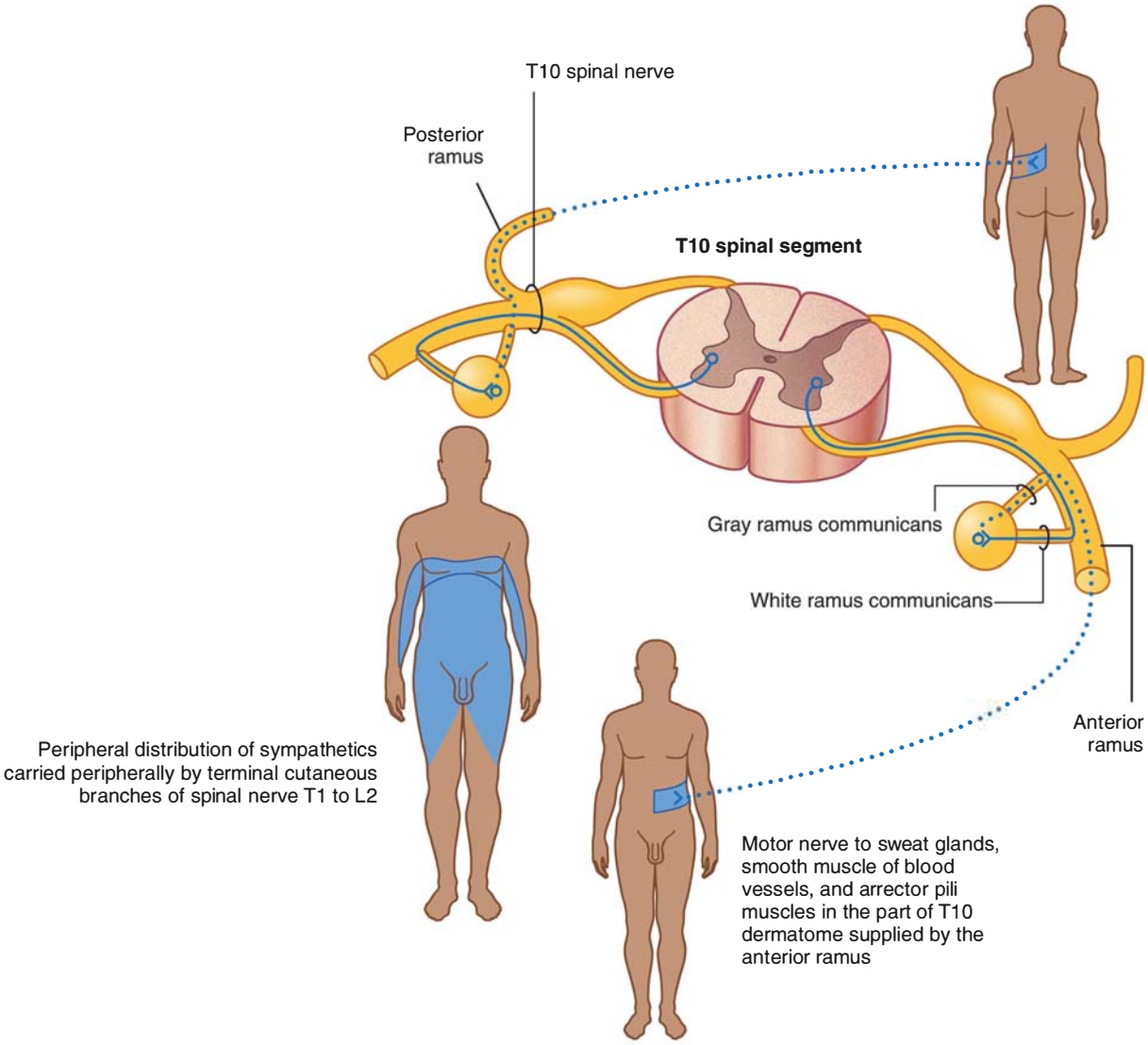


Fig. 1.44 Course of sympathetic fibers that travel to the periphery in the same spinal nerves in which they travel out of the spinal cord.

SUBASH KC/NMC-15TH/2014

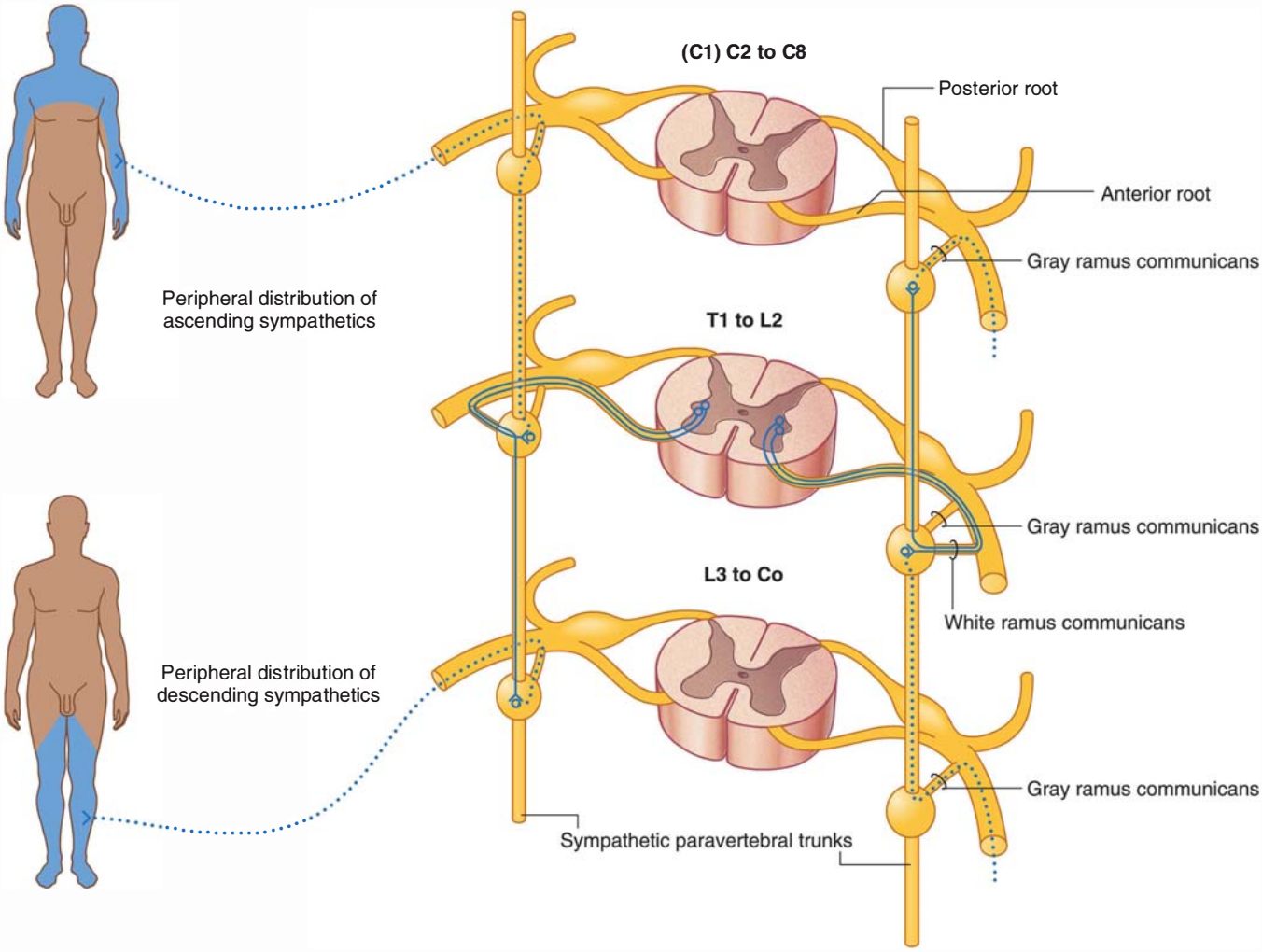


Fig. 1.45 Course of sympathetic nerves that travel to the periphery in spinal nerves that are not the ones through which they left the spinal cord.

2. Peripheral sympathetic innervation above or below the level of origin of the preganglionic fiber

Preganglionic sympathetic fibers may ascend or descend to other vertebral levels where they synapse in ganglia associated with spinal nerves that may or may not have visceral motor input directly from the spinal cord (i.e., those nerves other than T1 to L2) (Fig. 1.45).

The postganglionic fibers leave the distant ganglia via gray rami communicantes and are distributed along the posterior and anterior rami of the spinal nerves.

The ascending and descending fibers, together with all the ganglia, form the **paravertebral sympathetic trunk**, which extends the entire length of the vertebral column. The formation of this trunk, on each side, enables visceral motor fibers of the sympathetic part of the autonomic division of the PNS, which ultimately emerge from only a small region of the spinal cord (T1 to L2), to be distributed to peripheral regions innervated by all spinal nerves.

White rami communicantes only occur in association with spinal nerves T1 to L2, whereas gray rami communicantes are associated with all spinal nerves.

Fibers from spinal cord levels T1 to T5 pass predominantly superiorly, whereas fibers from T5 to L2 pass inferiorly. All sympathetics passing into the head have preganglionic fibers that emerge from spinal cord level T1 and ascend in the sympathetic trunks to the highest ganglion in the neck (the **superior cervical ganglion**), where they synapse. Postganglionic fibers then travel along blood vessels to target tissues in the head, including blood vessels, sweat glands, small smooth muscles associated with the upper eyelids, and the dilator of the pupil.

SUBASH KC/NMC-15TH/2014

3. Sympathetic innervation of thoracic and cervical viscera

Preganglionic sympathetic fibers may synapse with postganglionic motor neurons in ganglia and then leave the ganglia medially to innervate thoracic or cervical viscera (Fig. 1.46). They may ascend in the trunk before synapsing, and after synapsing the postganglionic fibers may combine with those from other levels to form named visceral nerves, such as cardiac nerves. Often, these nerves join branches from the parasympathetic system to form plexuses on or near the surface of the target organ, for example, the cardiac and pulmonary plexuses. Branches of the plexus innervate the organ. Spinal cord levels T1 to T5 mainly innervate cranial, cervical, and thoracic viscera.

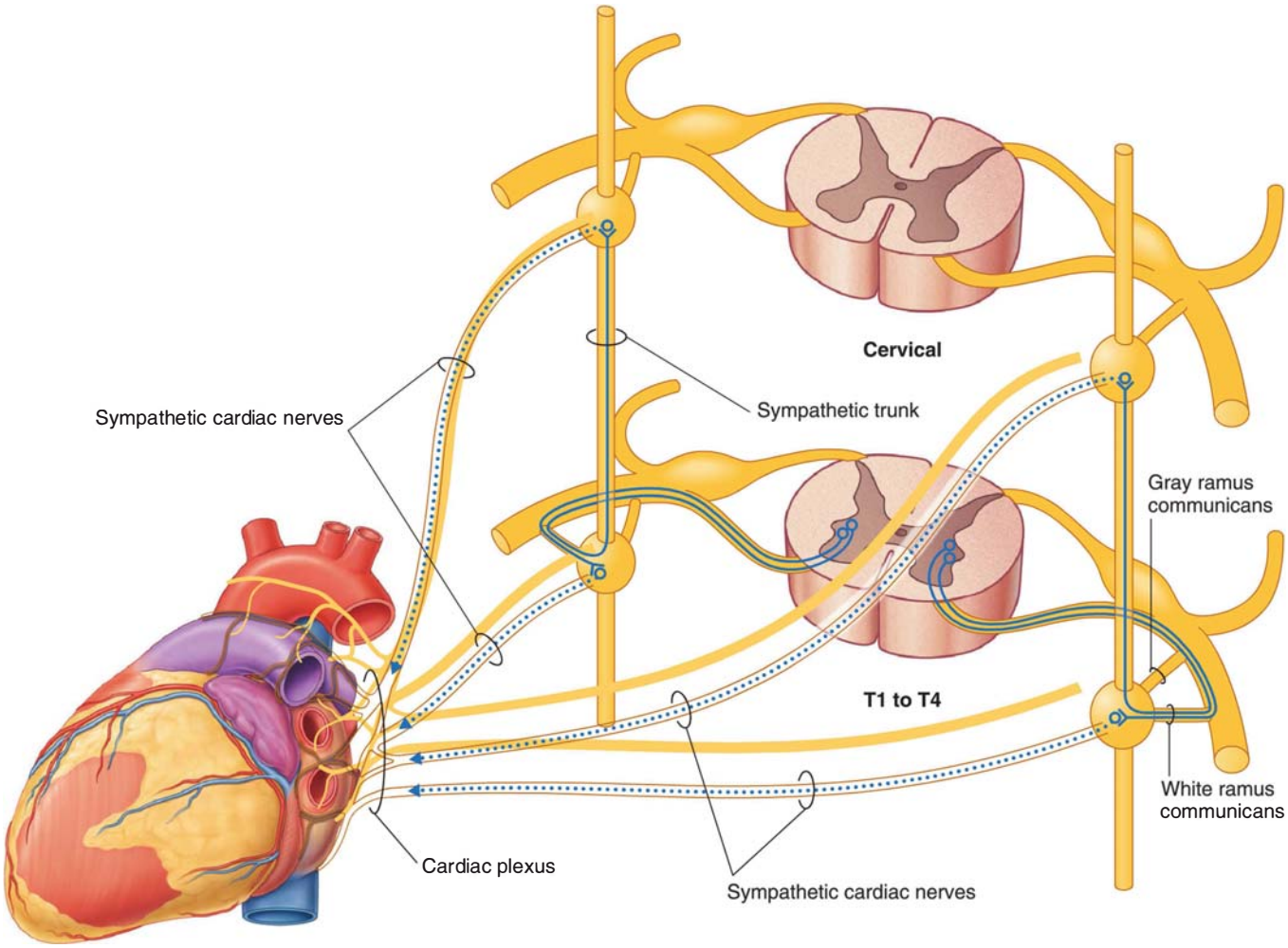


Fig. 1.46 Course of sympathetic nerves traveling to the heart.

SUBASH KC/NMC-15TH/2014

4. Sympathetic innervation of the abdomen and pelvic regions and the adrenals

Preganglionic sympathetic fibers may pass through the sympathetic trunk and paravertebral ganglia without synapsing and, together with similar fibers from other levels, form **splanchnic nerves** (**greater, lesser, least, lumbar**, and **sacral**), which pass into the abdomen and pelvic regions (Fig. 1.47). The preganglionic fibers in these nerves are derived from spinal cord levels T5 to L2.

The splanchnic nerves generally connect with sympathetic ganglia around the roots of major arteries that branch from the abdominal aorta. These ganglia are part of a large prevertebral plexus that also has input from the

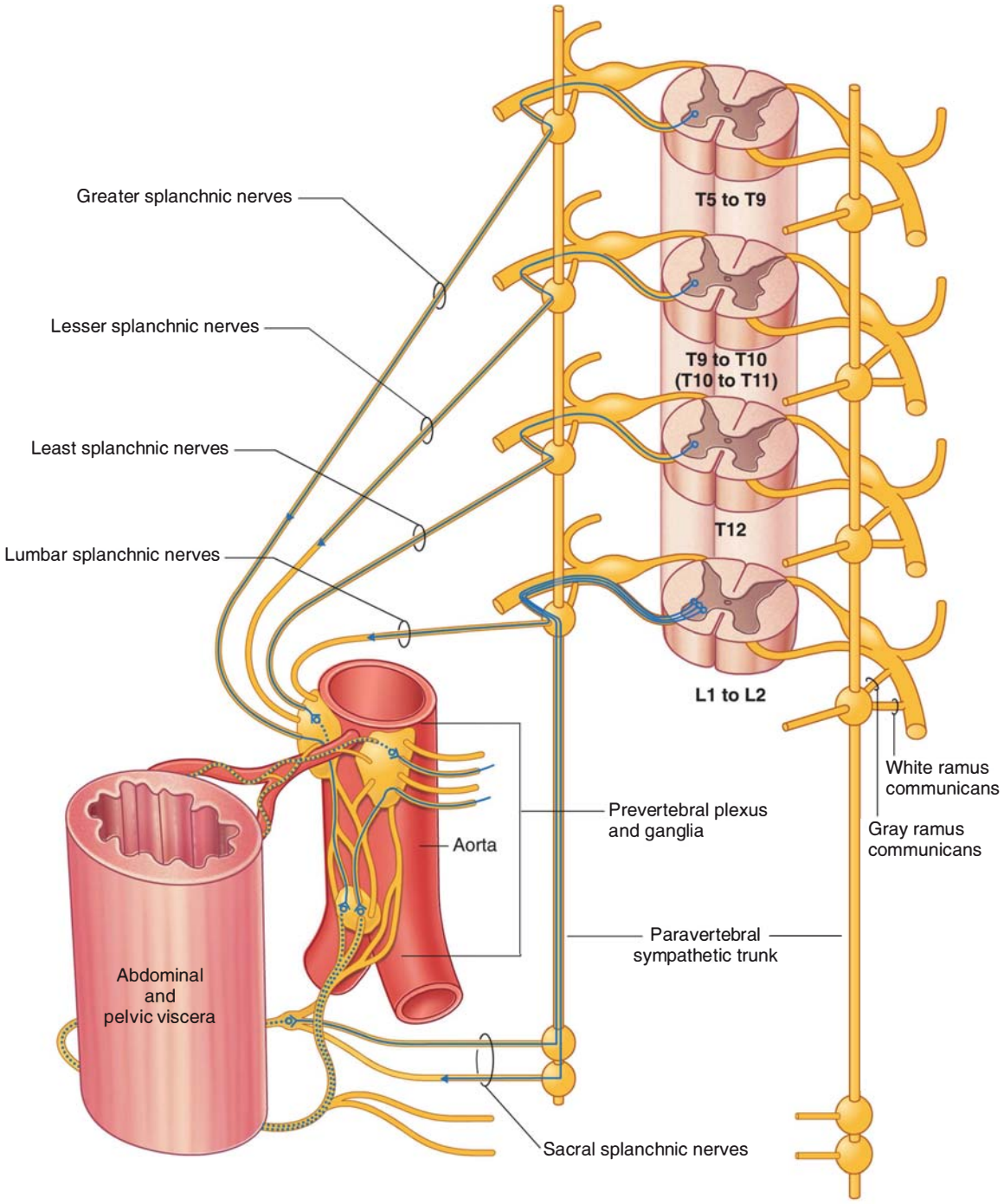


Fig. 1.47 Course of sympathetic nerves traveling to abdominal and pelvic viscera.

SUBASH KC/NMC-15TH/2014

parasympathetic part of the autonomic division of the PNS. Postganglionic sympathetic fibers are distributed in extensions of this plexus, predominantly along arteries, to viscera in the abdomen and pelvis.

Some of the preganglionic fibers in the prevertebral plexus do not synapse in the sympathetic ganglia of the plexus but pass through the system to the adrenal gland, where they synapse directly with cells of the adrenal medulla. These cells are homologues of sympathetic postganglionic neurons and secrete adrenaline and noradrenaline into the vascular system.

Parasympathetic system

The parasympathetic part of the autonomic division of the PNS (Fig. 1.48) leaves cranial and sacral regions of the CNS in association with:

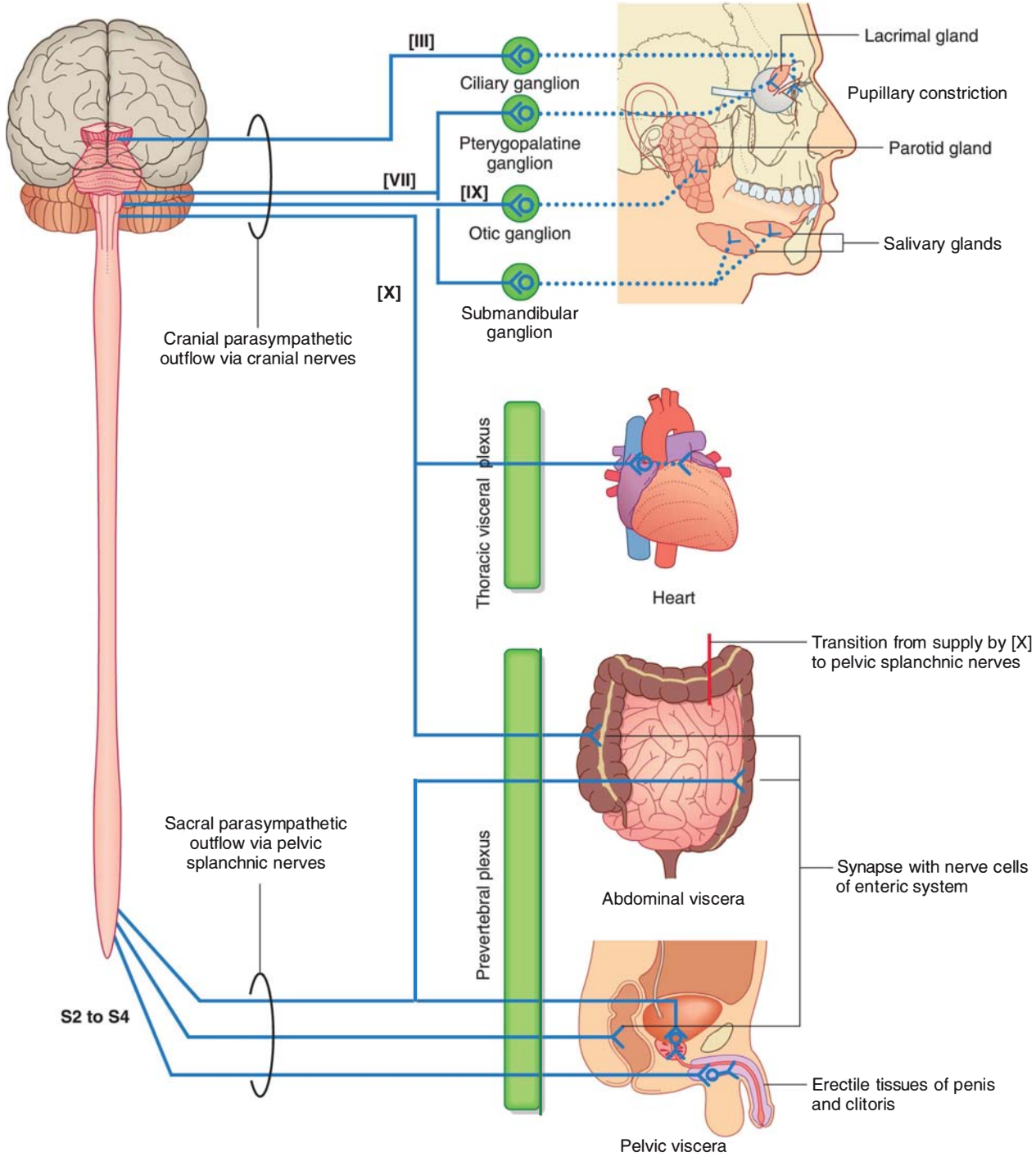


Fig. 1.48 Parasympathetic part of the autonomic division of the PNS.

SUBASH KC/NMC-15TH/2014

- cranial nerves III, VII, IX, and X: III, VII, and IX carry parasympathetic fibers to structures within the head and neck only, whereas X (the vagus nerve) also innervates thoracic and most abdominal viscera; and
- spinal nerves S2 to S4: sacral parasympathetic fibers innervate inferior abdominal viscera, pelvic viscera, and the arteries associated with erectile tissues of the perineum.

Like the visceral motor nerves of the sympathetic part, the visceral motor nerves of the parasympathetic part generally have two neurons in the pathway. The preganglionic neurons are in the CNS, and fibers leave in the cranial nerves.

Sacral preganglionic parasympathetic fibers

In the sacral region, the preganglionic parasympathetic fibers form special visceral nerves (the **pelvic splanchnic nerves**), which originate from the anterior rami of S2 to S4 and enter pelvic extensions of the large prevertebral plexus formed around the abdominal aorta. These fibers are distributed to pelvic and abdominal viscera mainly along blood vessels. The postganglionic motor neurons are in the walls of the viscera. In organs of the gastrointestinal system, preganglionic fibers do not have a postganglionic parasympathetic motor neuron in the pathway; instead, preganglionic fibers synapse directly on neurons in the ganglia of the enteric system.

Cranial nerve preganglionic parasympathetic fibers

The preganglionic parasympathetic motor fibers in CN III, VII, and IX separate from the nerves and connect with one of four distinct ganglia, which house postganglionic motor neurons. These four ganglia are near major branches of CN V. Postganglionic fibers leave the ganglia, join the branches of CN V, and are carried to target tissues (salivary, mucous, and lacrimal glands; constrictor muscle of the pupil; and ciliary muscle in the eye) with these branches.

The vagus nerve [X] gives rise to visceral branches along its course. These branches contribute to plexuses associated with thoracic viscera or to the large prevertebral plexus in the abdomen and pelvis. Many of these plexuses also contain sympathetic fibers.

When present, postganglionic parasympathetic neurons are in the walls of the target viscera.

Visceral sensory innervation (visceral afferents)

Visceral sensory fibers generally accompany visceral motor fibers.

Visceral sensory fibers accompany sympathetic fibers

Visceral sensory fibers follow the course of sympathetic fibers entering the spinal cord at similar spinal cord levels. However, visceral sensory fibers may also enter the spinal cord at levels other than those associated with motor output. For example, visceral sensory fibers from the heart may enter at levels higher than spinal cord level T1. Visceral sensory fibers that accompany sympathetic fibers are mainly concerned with detecting pain.

Visceral sensory fibers accompany parasympathetic fibers

Visceral sensory fibers accompanying parasympathetic fibers are carried mainly in IX and X and in spinal nerves S2 to S4.

Visceral sensory fibers in IX carry information from chemoreceptors and baroreceptors associated with the walls of major arteries in the neck, and from receptors in the pharynx.

Visceral sensory fibers in X include those from cervical viscera, and major vessels and viscera in the thorax and abdomen.

Visceral sensory fibers from pelvic viscera and the distal parts of the colon are carried in S2 to S4.

Visceral sensory fibers associated with parasympathetic fibers primarily relay information to the CNS about the status of normal physiological processes and reflex activities.

SUBASH KC/NMC-15TH/2014

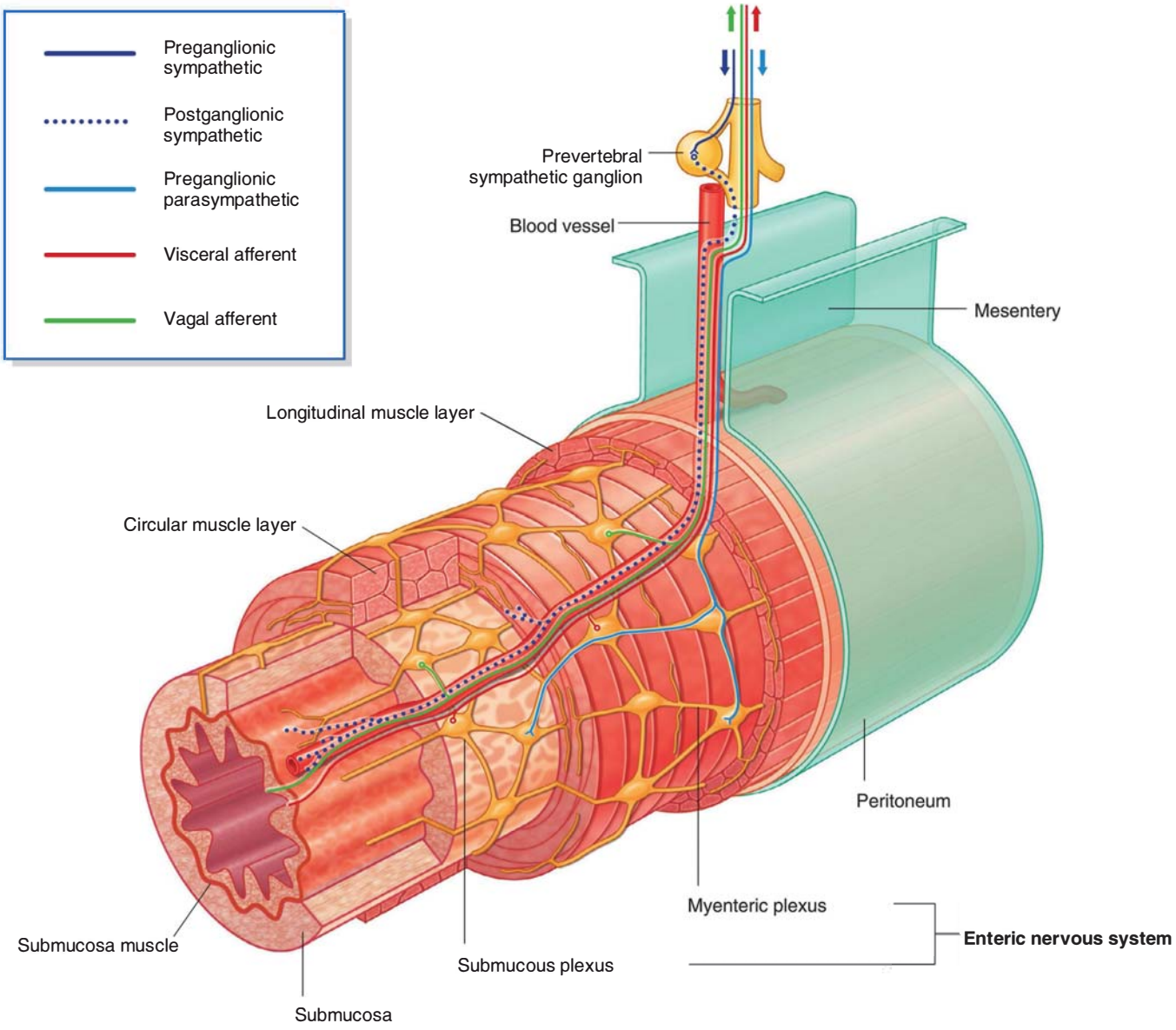


Fig. 1.49 Enteric part of the nervous system.

The enteric system

The enteric nervous system consists of motor and sensory neurons and their support cells, which form two interconnected plexuses, the **myenteric** and **submucous nerve plexuses**, within the walls of the gastrointestinal tract (Fig. 1.49). Each of these plexuses is formed by:

- ganglia, which house the nerve cell bodies and associated cells, and
- bundles of nerve fibers, which pass between ganglia and from the ganglia into surrounding tissues.

Neurons in the enteric system are derived from neural crest cells originally associated with occipitocervical and sacral regions. Interestingly, more neurons are reported to be in the enteric system than in the spinal cord itself.

Sensory and motor neurons within the enteric system control reflex activity within and between parts of the

SUBASH KC/NMC-15TH/2014

gastrointestinal system. These reflexes regulate peristalsis, secretomotor activity, and vascular tone. These activities can occur independently of the brain and spinal cord, but can also be modified by input from preganglionic parasympathetic and postganglionic sympathetic fibers.

Sensory information from the enteric system is carried back to the CNS by visceral sensory fibers.

Nerve plexuses

Nerve plexuses are either somatic or visceral and combine fibers from different sources or levels to form new nerves with specific targets or destinations (Fig. 1.50). Plexuses of the enteric system also generate reflex activity independent of the CNS.

Somatic plexuses

Major somatic plexuses formed from the anterior rami of spinal nerves are the cervical (C1 to C4), brachial (C5 to T1), lumbar (L1 to L4), sacral (L4 to S4), and coccygeal (S5 to Co) plexuses. Except for spinal nerve T1, the anterior rami of thoracic spinal nerves remain independent and do not participate in plexuses.

Visceral plexuses

Visceral nerve plexuses are formed in association with viscera and generally contain efferent (sympathetic and parasympathetic) and afferent components (Fig. 1.50). These plexuses include cardiac and pulmonary plexuses in the thorax and a large prevertebral plexus in the abdomen anterior to the aorta, which extends inferiorly onto the lateral walls of the pelvis. The massive prevertebral plexus supplies input to and receives output from all abdominal and pelvic viscera.

In the clinic

Referred pain

Referred pain occurs when sensory information comes to the spinal cord from one location but is interpreted by the CNS as coming from another location innervated by the same spinal cord level. Usually, this happens when the pain information comes from a region, such as the gut, which has a low amount of sensory output. These afferents converge on neurons at the same spinal cord level that receive information from the skin, which is an area with a high amount of sensory output. As a result, pain from the normally low output region is interpreted as coming from the normally high output region.

Pain is most often referred from a region innervated by the visceral part of the nervous system to a region innervated, at the same spinal cord level, by the somatic side of the nervous system.

Pain can also be referred from one somatic region to another. For example, irritation of the peritoneum on the inferior surface of the diaphragm, which is innervated by the phrenic nerve, can be referred to the skin on the top of the shoulder, which is innervated by other somatic nerves arising at the same spinal cord level.

OTHER SYSTEMS

Specific information about the organization and components of the respiratory, gastrointestinal, and urogenital systems will be discussed in each of the succeeding chapters of this text.

SUBASH KC/NMC-15TH/2014

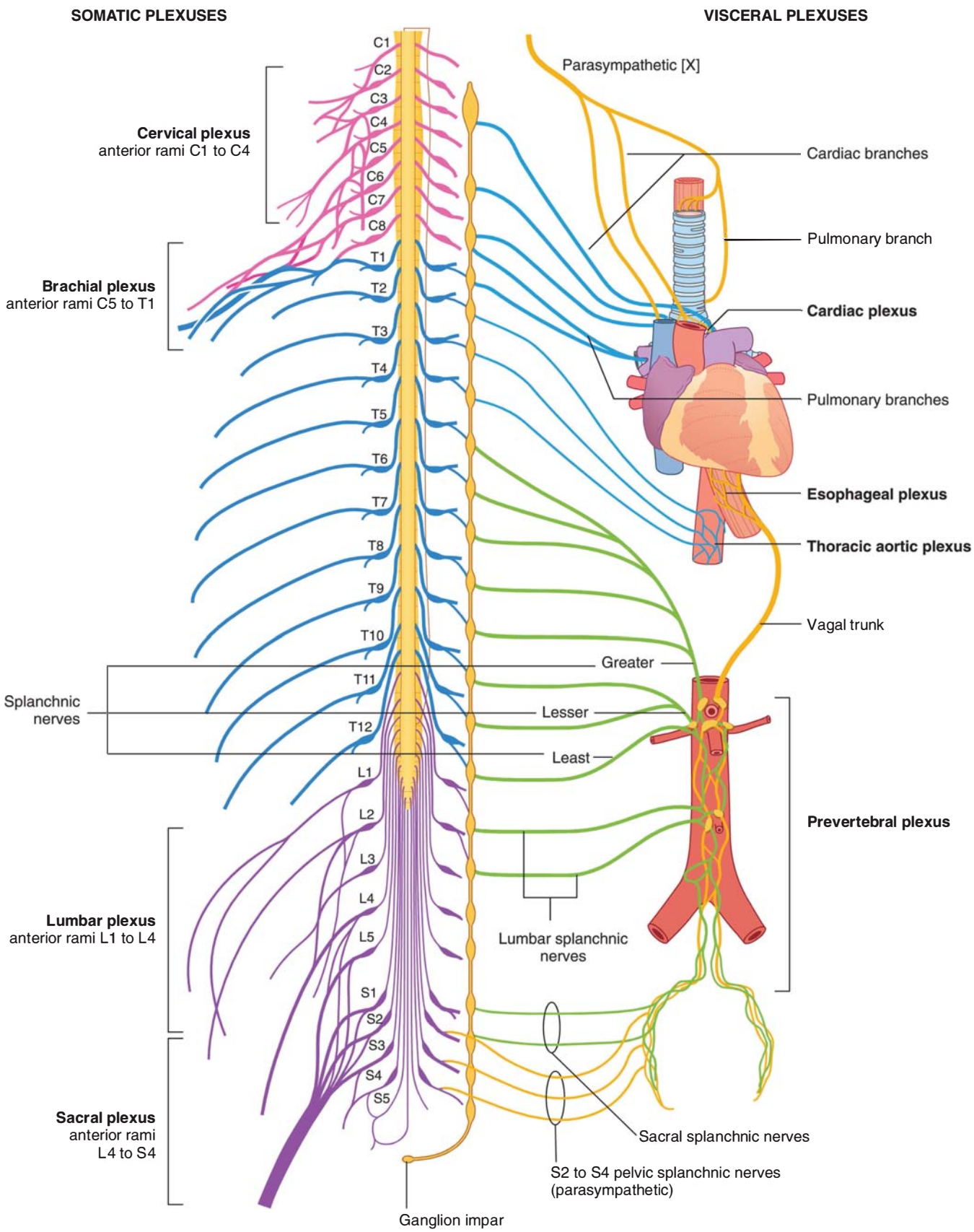


Fig. 1.50 Nerve plexuses.

SUBASH KC/NMC-15TH/2014

Clinical cases

Case 1

APPENDICITIS

A young man sought medical care because of central abdominal pain that was diffuse and colicky. After some hours, the pain began to localize in the right iliac fossa and became constant. He was referred to an abdominal surgeon, who removed a grossly inflamed appendix. The patient made an uneventful recovery.

When the appendix becomes inflamed, the visceral sensory fibers are stimulated. These fibers enter the spinal cord with the sympathetic fibers at spinal cord level T10. The pain is referred to the dermatome of T10, which is in the umbilical region (Fig. 1.51). The pain is diffuse, not focal; every time a peristaltic wave passes through the ileocecal region, the pain recurs. This intermittent type of pain is referred to as colic.

In the later stages of the disease, the appendix contacts and irritates the parietal peritoneum in the right iliac fossa, which is innervated by somatic sensory nerves. This produces a constant focal pain, which predominates over the colicky pain that the patient felt some hours previously. The patient no longer interprets the referred pain from the T10 dermatome.

Although this is a typical history for appendicitis, it should always be borne in mind that the patient's symptoms and signs may vary. The appendix is situated in a retrocecal position in approximately 70% of patients; therefore it may never contact the parietal peritoneum anteriorly in the right iliac fossa. It is also possible that the appendix is long and may directly contact other structures. As a consequence, the patient may have other symptoms (e.g., the appendix may contact the ureter, and the patient may then develop urological symptoms).

Although appendicitis is common, other disorders, for example of the bowel and pelvis, may produce similar symptoms.

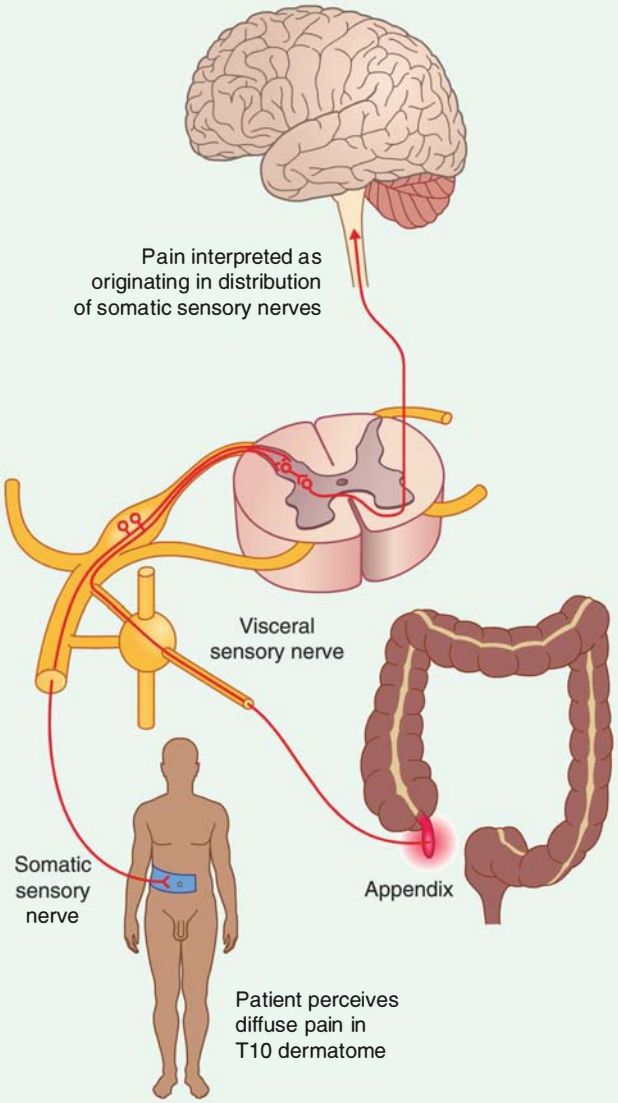


Fig. 1.51 Mechanism for referred pain from an inflamed appendix to the T10 dermatome.

SUBASH KC/NMC-15TH/2014

2 Back

ADDITIONAL LEARNING RESOURCES for Chapter 2, Back, on STUDENT CONSULT (www.studentconsult.com):

- Image Library—illustrations of back anatomy, Chapter 2
- Self-Assessment—National Board style multiple-choice questions, Chapter 2
- Short Questions—these are questions requiring short responses, Chapter 2
- Interactive Surface Anatomy—interactive surface animations, Chapter 2
- PT Case Studies, Chapter 2
  - Low back instability
  - Stenosis
  - Herniated nucleus pulposus
  - Atlanto-occipital dysfunction
  - Atlanto-axial dysfunction
  - Mid-cervical dysfunction
  - Cauda equina syndrome
  - Cervical degenerative joint disease
  - Cervical radiculopathy
- Medical Clinical Case Studies, Chapter 2
  - Ankylosing spondylitis
  - Atlas fracture
  - Cervical facet syndrome
  - Lumbar puncture
  - Spinal cord infarction

FREE Online Anatomy and Embryology Self-Study Course

- Anatomy modules 23 through 25

Conceptual overview 53

General description 53
Functions 54
- Support 54
- Movement 54
- Protection of the nervous system 55

Component parts 56
- Bones 56
- Muscles 57
- Vertebral canal 59
- Spinal nerves 60

Relationship to other regions 61
- Head 61
- Thorax, abdomen, and pelvis 62
- Limbs 62

Key features 62
- Long vertebral column and short spinal cord 62
- Intervertebral foramina and spinal nerves 63
- Innervation of the back 63

Regional anatomy 64

Skeletal framework 64
- Vertebrae 64
- Intervertebral foramina 72
- Posterior spaces between vertebral arches 73

Joints 77
- Joints between vertebrae in the back 77

Ligaments 80
- Anterior and posterior longitudinal ligaments 80
- Ligamenta flava 80
- Supraspinous ligament and ligamentum nuchae 81
- Interspinous ligaments 82

Back musculature 84
- Superficial group of back muscles 84
- Intermediate group of back muscles 90
- Deep group of back muscles 92
- Suboccipital muscles 97

SUBASH KC/NMC-15TH/2014

Spinal cord 99
- Vasculature 100
- Meninges 103
- Arrangement of structures in the vertebral canal 104
- Spinal nerves 106

Surface anatomy ***111***
- Back surface anatomy 111
- Absence of lateral curvatures 111
- Primary and secondary curvatures in the sagittal plane 112
- Useful nonvertebral skeletal landmarks 112
- How to identify specific vertebral spinous processes 114
- Visualizing the inferior ends of the spinal cord and subarachnoid space 115
- Identifying major muscles 116

Clinical cases ***118***

SUBASH KC/NMC-15TH/2014

Conceptual overview

GENERAL DESCRIPTION

The back consists of the posterior aspect of the body and provides the musculoskeletal axis of support for the trunk. Bony elements consist mainly of the vertebrae, although proximal elements of the ribs, superior aspects of the pelvic bones, and posterior basal regions of the skull contribute to the back's skeletal framework (Fig. 2.1).

Associated muscles interconnect the vertebrae and ribs with each other and with the pelvis and skull. The back contains the spinal cord and proximal parts of the spinal nerves, which send and receive information to and from most of the body.

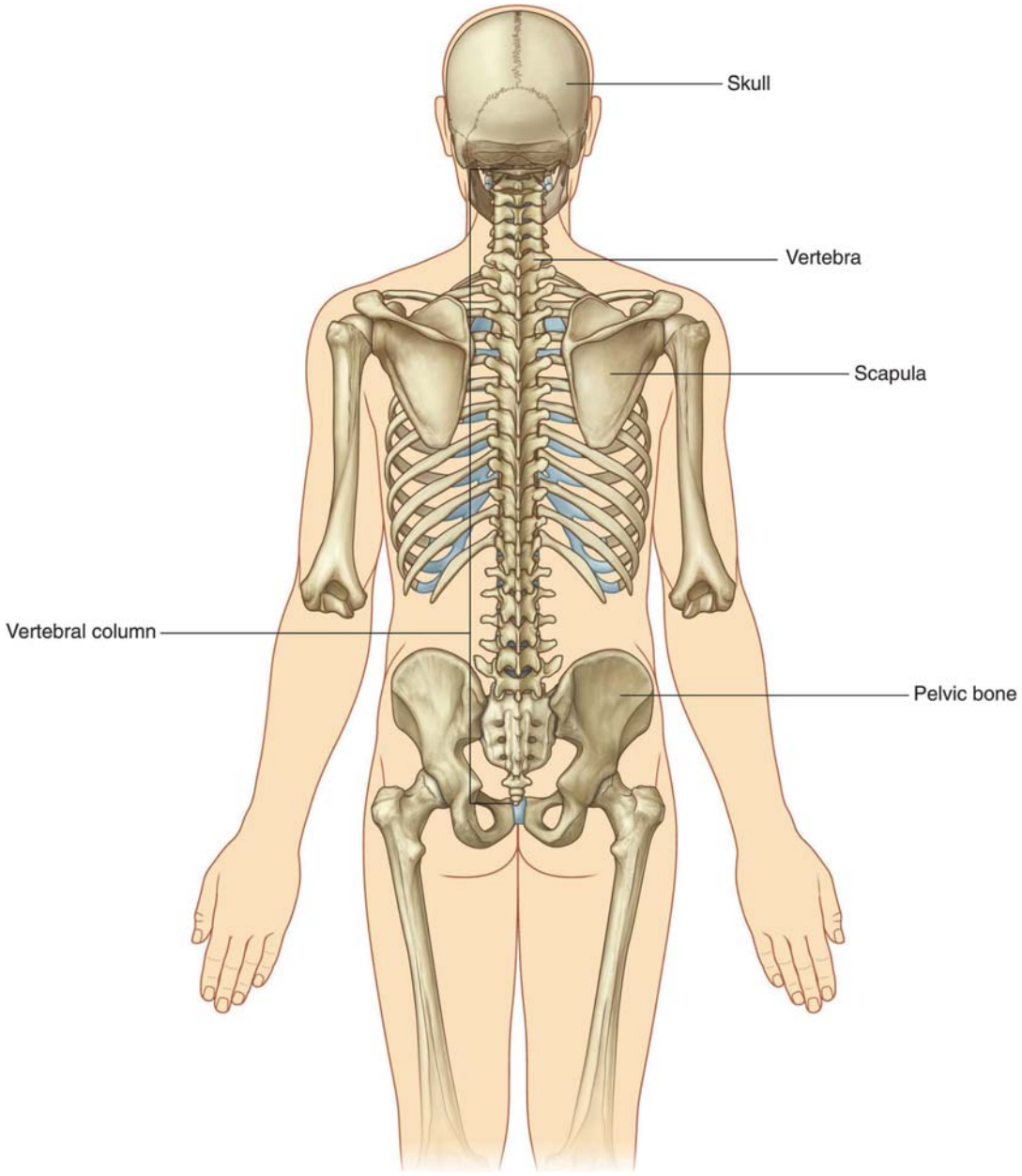


Fig. 2.1 Skeletal framework of the back.

SUBASH KC/NMC-15TH/2014

FUNCTIONS

Support

The skeletal and muscular elements of the back support the body's weight, transmit forces through the pelvis to the lower limbs, carry and position the head, and brace and help maneuver the upper limbs. The vertebral column is positioned posteriorly in the body at the midline. When viewed laterally, it has a number of curvatures (Fig. 2.2):

- The primary curvature of the vertebral column is concave anteriorly, reflecting the original shape of the embryo, and is retained in the thoracic and sacral regions in adults.
- Secondary curvatures, which are concave posteriorly, form in the cervical and lumbar regions and bring the center of gravity into a vertical line, which allows the body's weight to be balanced on the vertebral column in a way that expends the least amount of muscular energy to maintain an upright bipedal stance.

As stresses on the back increase from the cervical to lumbar regions, lower back problems are common.

Movement

Muscles of the back consist of extrinsic and intrinsic groups:

- The extrinsic muscles of the back move the upper limbs and the ribs.
- The intrinsic muscles of the back maintain posture and move the vertebral column; these movements include flexion (anterior bending), extension, lateral flexion, and rotation (Fig. 2.3).

Although the amount of movement between any two vertebrae is limited, the effects between vertebrae are additive along the length of the vertebral column. Also, freedom of movement and extension are limited in the thoracic region relative to the lumbar part of the vertebral column. Muscles in more anterior regions flex the vertebral column.

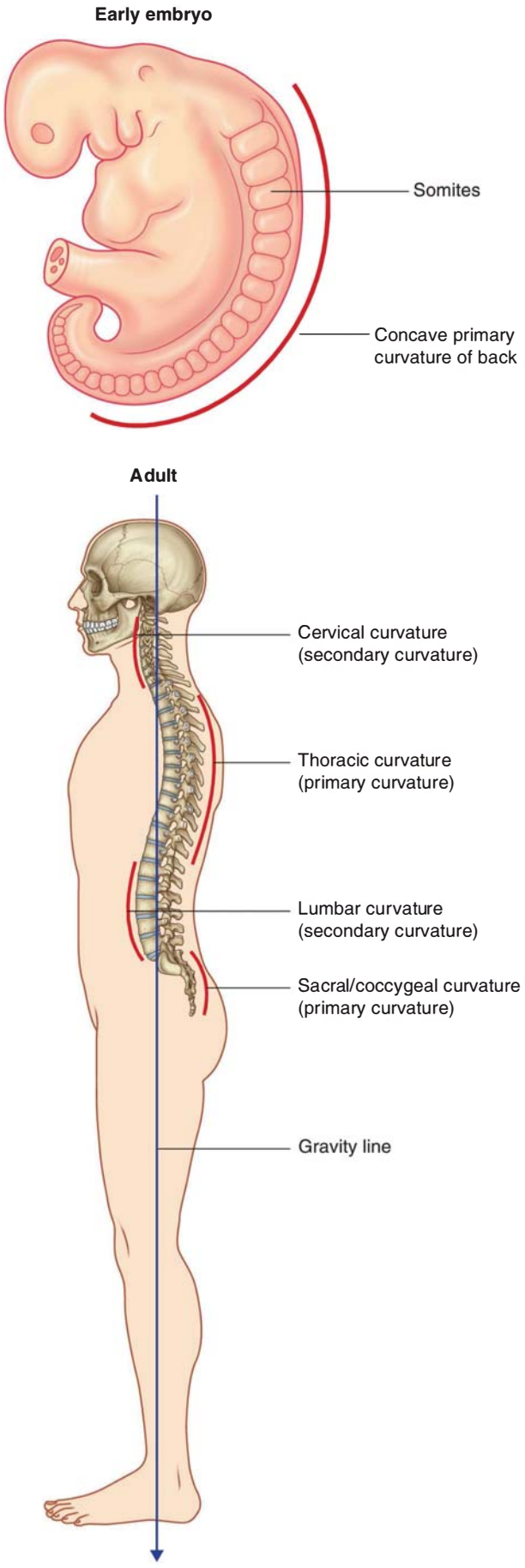


Fig. 2.2 Curvatures of the vertebral column.

SUBASH KC/NMC-15TH/2014

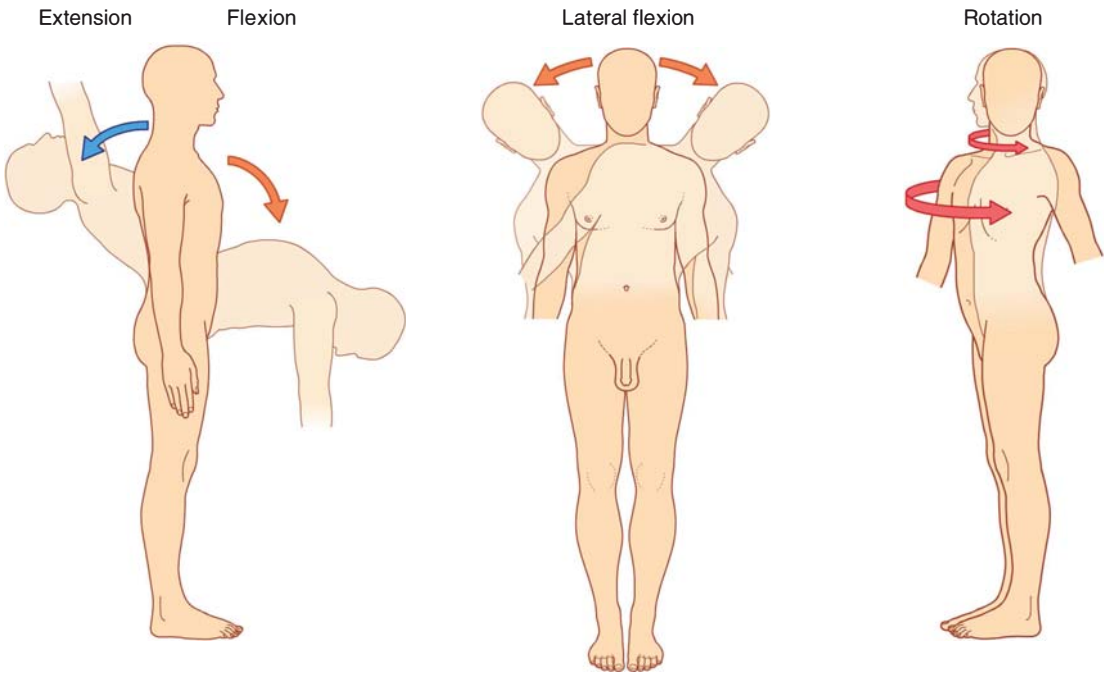


Fig. 2.3 Back movements.

In the cervical region, the first two vertebrae and associated muscles are specifically modified to support and position the head. The head flexes and extends, in the nodding motion, on vertebra CI, and rotation of the head occurs as vertebra CI moves on vertebra CII (Fig. 2.3).

Protection of the nervous system

The vertebral column and associated soft tissues of the back contain the spinal cord and proximal parts of the spinal nerves (Fig. 2.4). The more distal parts of the spinal nerves pass into all other regions of the body, including certain regions of the head.

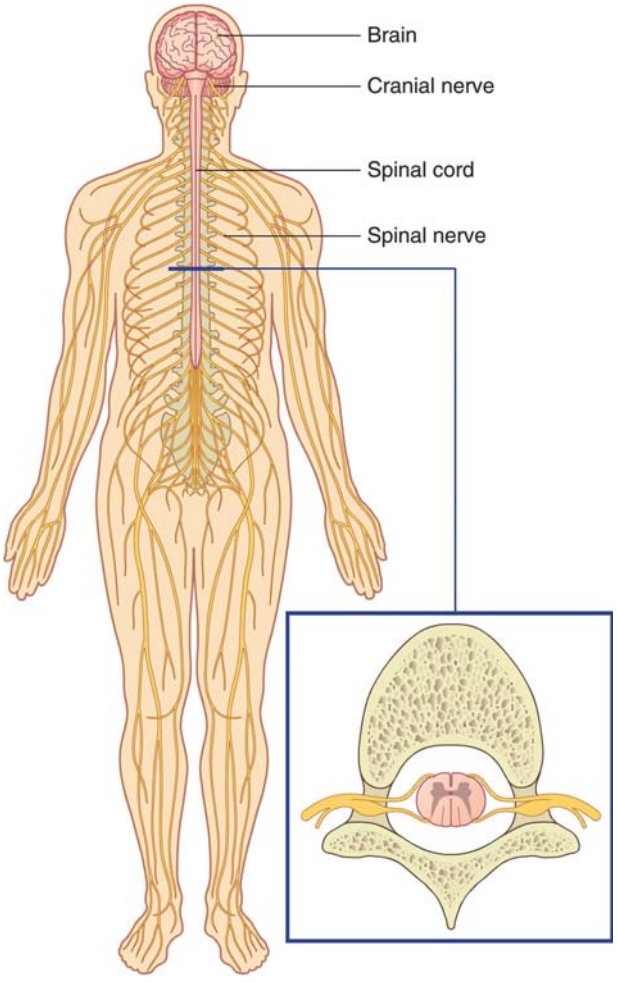


Fig. 2.4 Nervous system.

SUBASH KC/NMC-15TH/2014

COMPONENT PARTS

Bones

The major bones of the back are the 33 vertebrae (Fig. 2.5). The number and specific characteristics of the vertebrae vary depending on the body region with which they are associated. There are seven cervical, twelve thoracic, five lumbar, five sacral, and three to four coccygeal vertebrae. The sacral vertebrae fuse into a single bony element, the sacrum. The coccygeal vertebrae are rudimentary in structure, vary in number from three to four, and often fuse into a single coccyx.

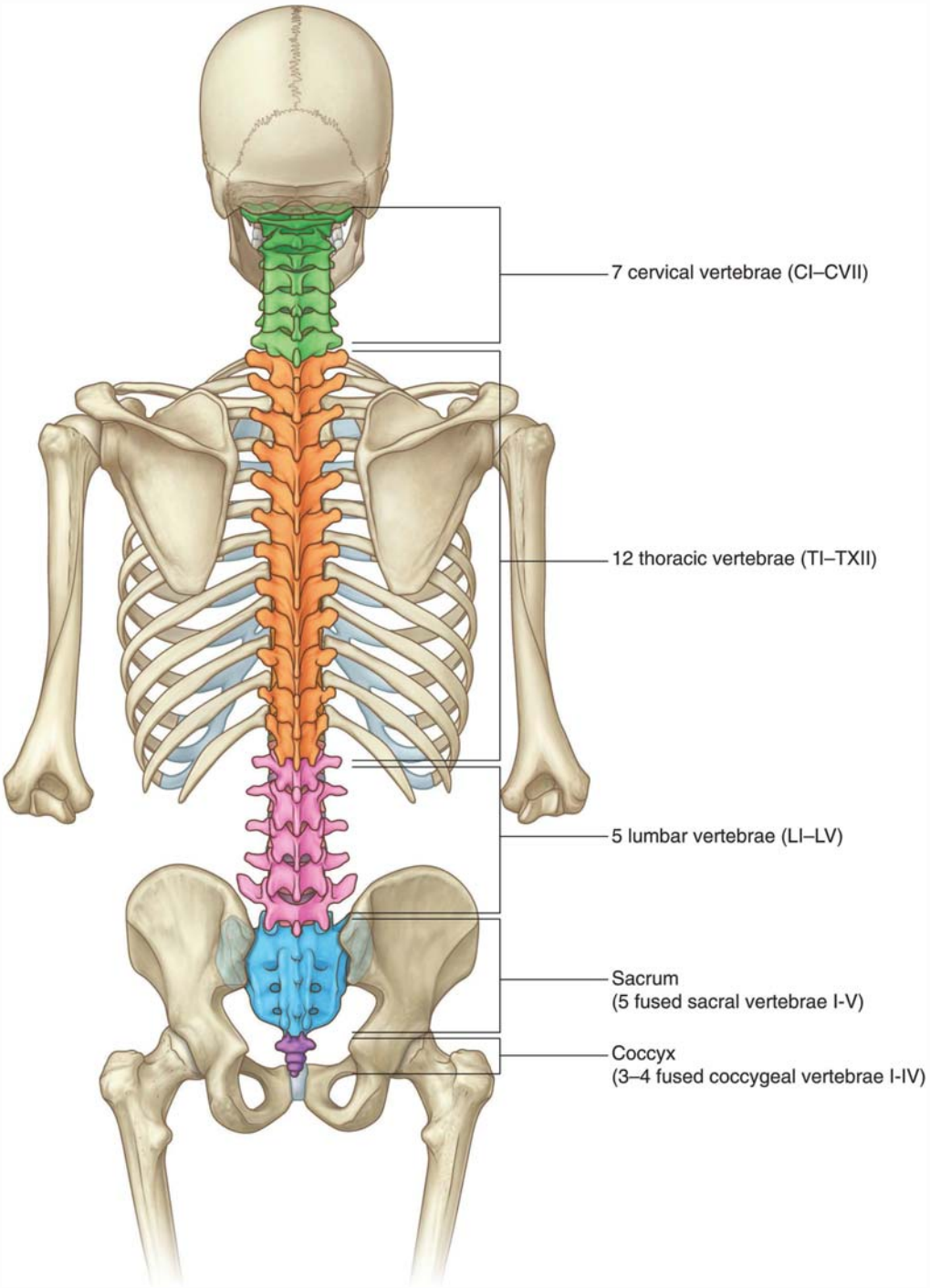


Fig. 2.5 Vertebrae.

SUBASH KC/NMC-15TH/2014

Typical vertebra

A typical vertebra consists of a vertebral body and a vertebral arch (Fig. 2.6).

The vertebral body is anterior and is the major weight-bearing component of the bone. It increases in size from vertebra CII to vertebra LV. Fibrocartilaginous intervertebral discs separate the vertebral bodies of adjacent vertebrae.

The vertebral arch is firmly anchored to the posterior surface of the vertebral body by two pedicles, which form the lateral pillars of the vertebral arch. The roof of the vertebral arch is formed by right and left laminae, which fuse at the midline.

The vertebral arches of the vertebrae are aligned to form the lateral and posterior walls of the vertebral canal, which extends from the first cervical vertebra (CI) to the last sacral vertebra (vertebra SV). This bony canal contains the spinal cord and its protective membranes, together with blood vessels, connective tissue, fat, and proximal parts of spinal nerves.

The vertebral arch of a typical vertebra has a number of characteristic projections, which serve as:

- attachments for muscles and ligaments,
- levers for the action of muscles, and
- sites of articulation with adjacent vertebrae.

A spinous process projects posteriorly and generally inferiorly from the roof of the vertebral arch.

On each side of the vertebral arch, a transverse process extends laterally from the region where a lamina meets a pedicle. From the same region, a superior articular process and an inferior articular process articulate with similar processes on adjacent vertebrae.

Each vertebra also contains rib elements. In the thorax, these costal elements are large and form ribs, which articulate with the vertebral bodies and transverse processes. In all other regions, these rib elements are small and are incorporated into the transverse processes. Occasionally, they develop into ribs in regions other than the thorax, usually in the lower cervical and upper lumbar regions.

Muscles

Muscles in the back can be classified as extrinsic or intrinsic based on their embryological origin and type of innervation (Fig. 2.7).

The extrinsic muscles are involved with movements of the upper limbs and thoracic wall and, in general, are innervated by anterior rami of spinal nerves. The superficial group of these muscles is related to the upper limbs, while the intermediate layer of muscles is associated with the thoracic wall.

All of the intrinsic muscles of the back are deep in position and are innervated by the posterior rami of spinal nerves. They support and move the vertebral column and participate in moving the head. One group of intrinsic muscles also moves the ribs relative to the vertebral column.

SUBASH KC/NMC-15TH/2014

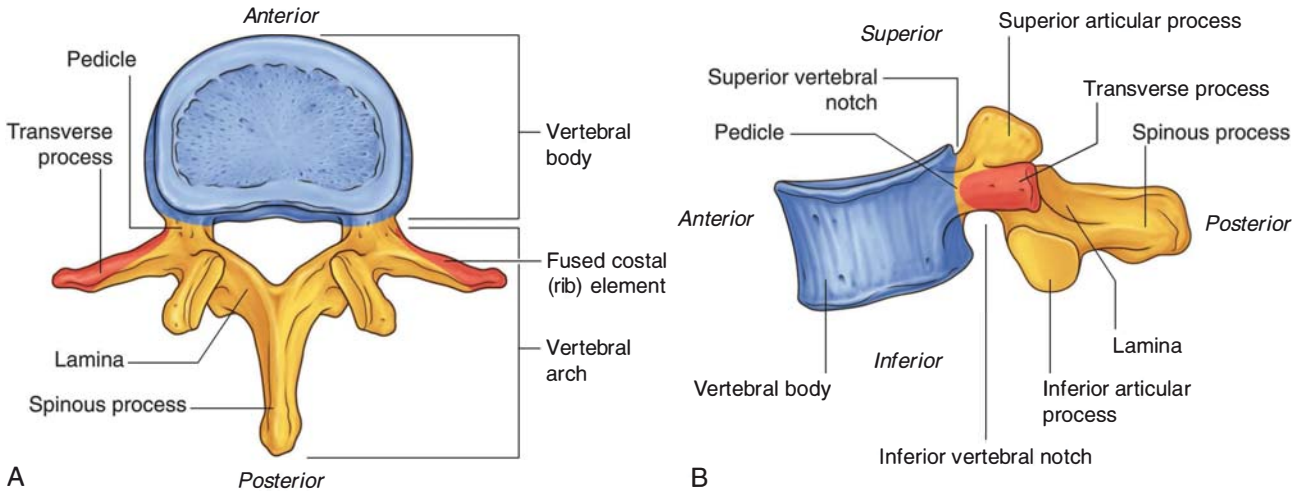


Fig. 2.6 A typical vertebra. **A.** Superior view. **B.** Lateral view.

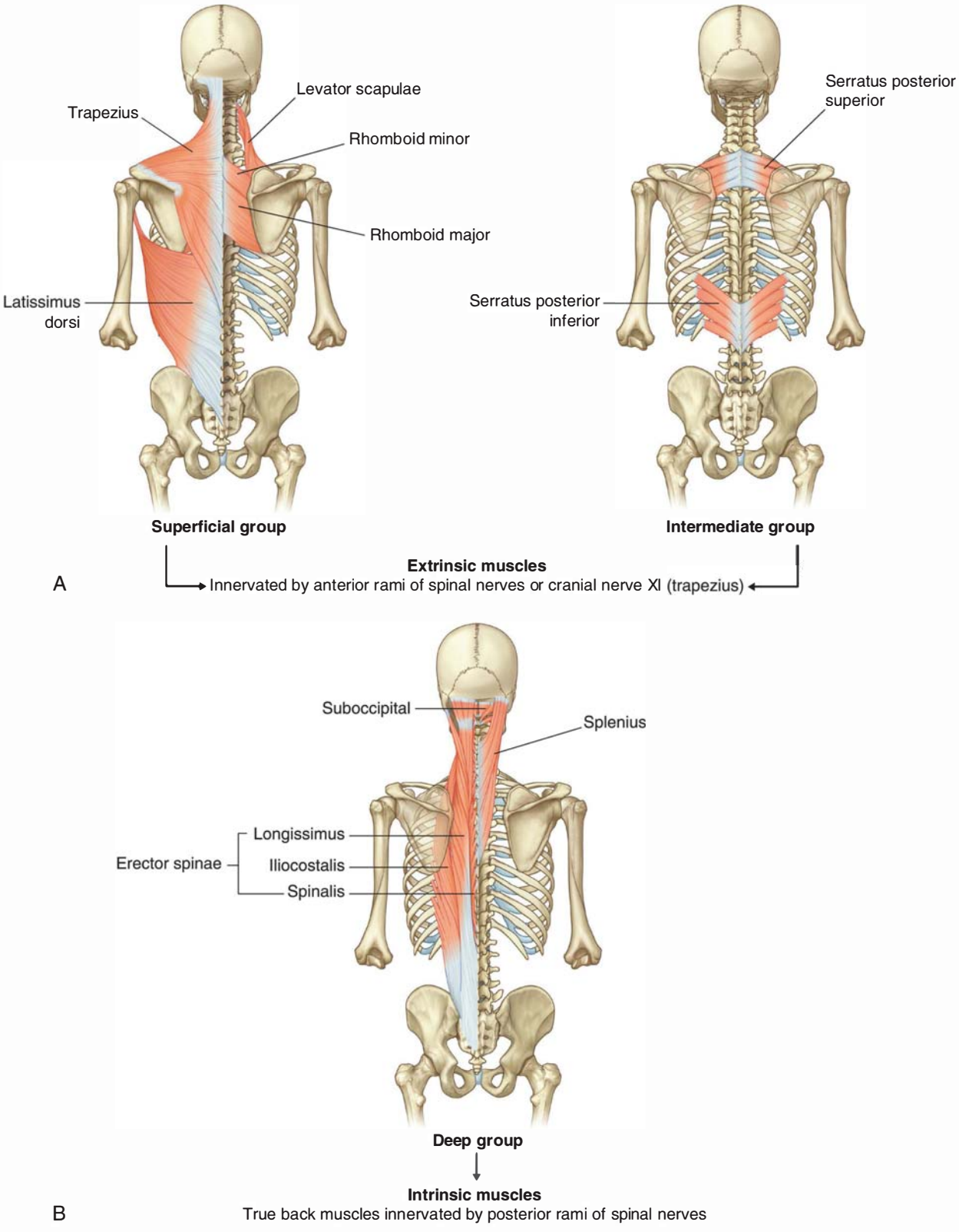


Fig. 2.7 Back muscles. **A.** Extrinsic muscles. **B.** Intrinsic muscles.

SUBASH KC/NMC-15TH/2014

Vertebral canal

The spinal cord lies within a bony canal formed by adjacent vertebrae and soft tissue elements (the vertebral canal) (Fig. 2.8):

- The anterior wall is formed by the vertebral bodies of the vertebrae, intervertebral discs, and associated ligaments.
- The lateral walls and roof are formed by the vertebral arches and ligaments.

Within the vertebral canal, the spinal cord is surrounded by a series of three connective tissue membranes (the meninges):

- The pia mater is the innermost membrane and is intimately associated with the surface of the spinal cord.
- The second membrane, the arachnoid mater, is separated from the pia by the subarachnoid space, which contains cerebrospinal fluid.
- The thickest and most external of the membranes, the dura mater, lies directly against, but is not attached to, the arachnoid mater.

In the vertebral canal, the dura mater is separated from surrounding bone by an extradural (epidural) space containing loose connective tissue, fat, and a venous plexus.

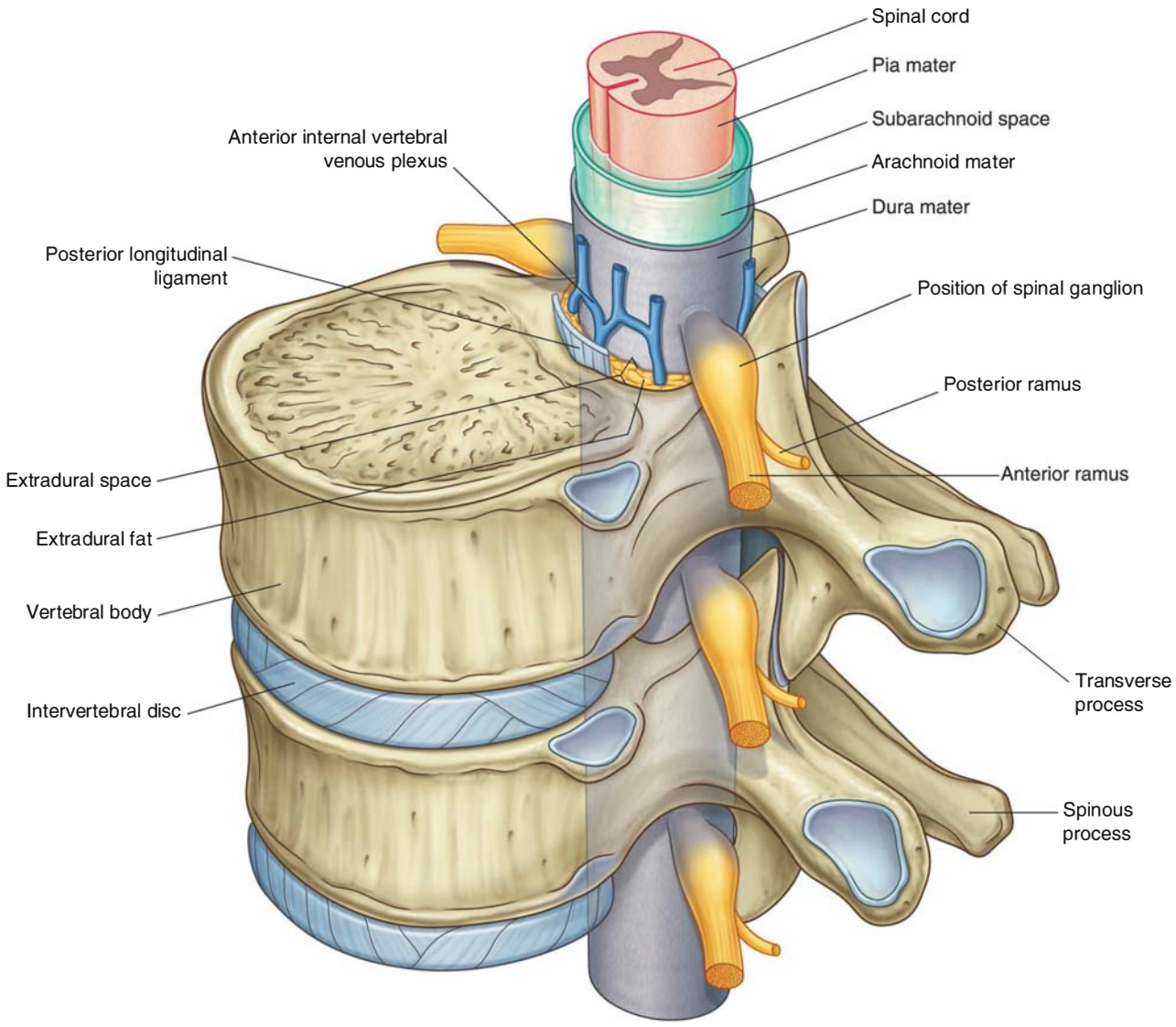


Fig. 2.8 Vertebral canal.

SUBASH KC/NMC-15TH/2014

Spinal nerves

The 31 pairs of spinal nerves are segmental in distribution and emerge from the vertebral canal between the pedicles of adjacent vertebrae. There are eight pairs of cervical nerves (C1 to C8), twelve thoracic (T1 to T12), five lumbar (L1 to L5), five sacral (S1 to S5), and one coccygeal (Co). Each nerve is attached to the spinal cord by a posterior root and an anterior root (Fig. 2.9).

After exiting the vertebral canal, each spinal nerve branches into:

- a posterior ramus—collectively, the small posterior rami innervate the back; and
- an anterior ramus—the much larger anterior rami innervate most other regions of the body except the head, which is innervated predominantly, but not exclusively, by cranial nerves.

The anterior rami form the major somatic plexuses (cervical, brachial, lumbar, and sacral) of the body. Major visceral components of the PNS (sympathetic trunk and prevertebral plexus) of the body are also associated mainly with the anterior rami of spinal nerves.

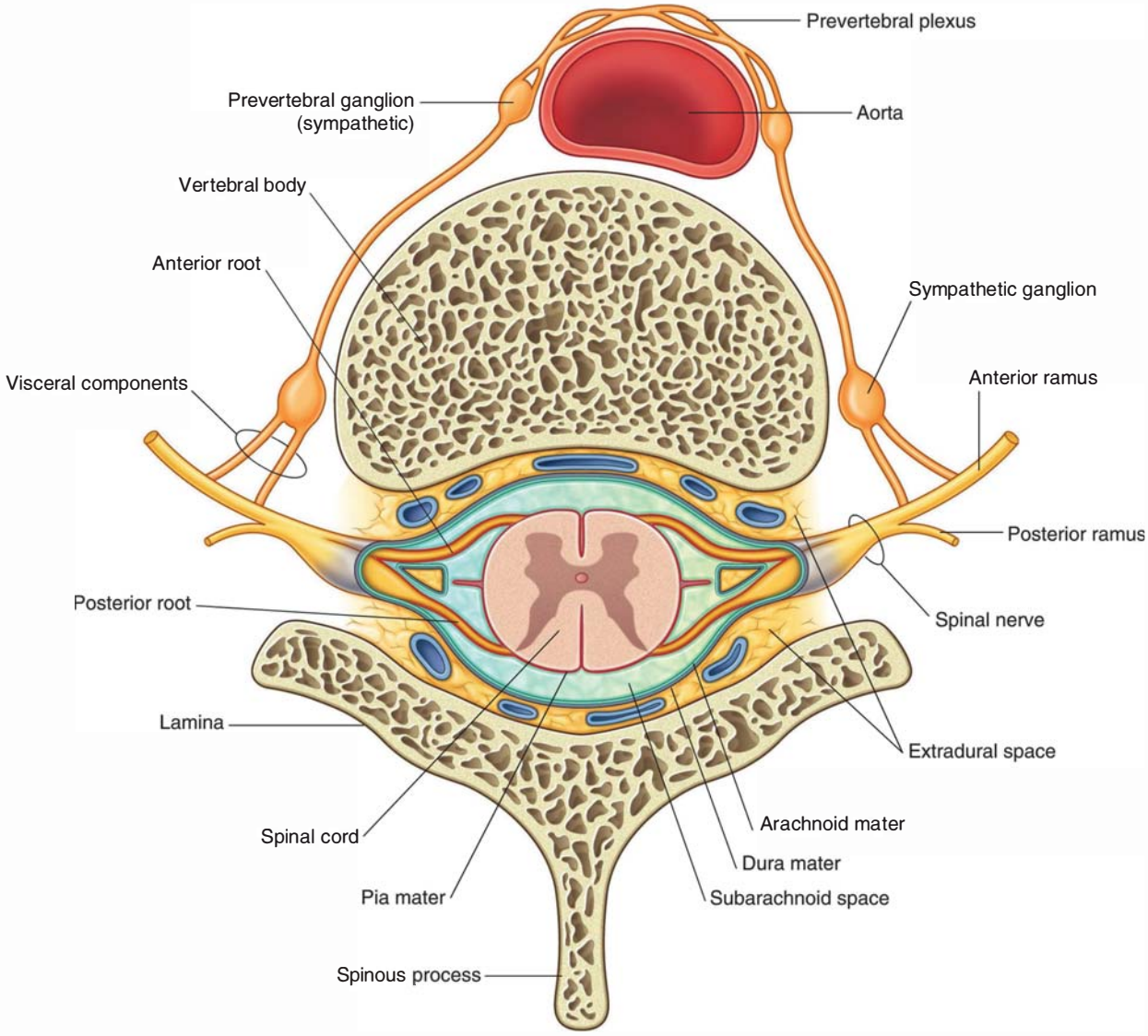


Fig. 2.9 Spinal nerves (transverse section).

SUBASH KC/NMC-15TH/2014

RELATIONSHIP TO OTHER REGIONS

Head

Cervical regions of the back constitute the skeletal and much of the muscular framework of the neck, which in turn supports and moves the head (Fig. 2.10).

The brain and cranial meninges are continuous with the spinal cord meninges at the foramen magnum of the skull. The paired vertebral arteries ascend, one on each side, through foramina in the transverse processes of cervical vertebrae and pass through the foramen magnum to participate, with the internal carotid arteries, in supplying blood to the brain.

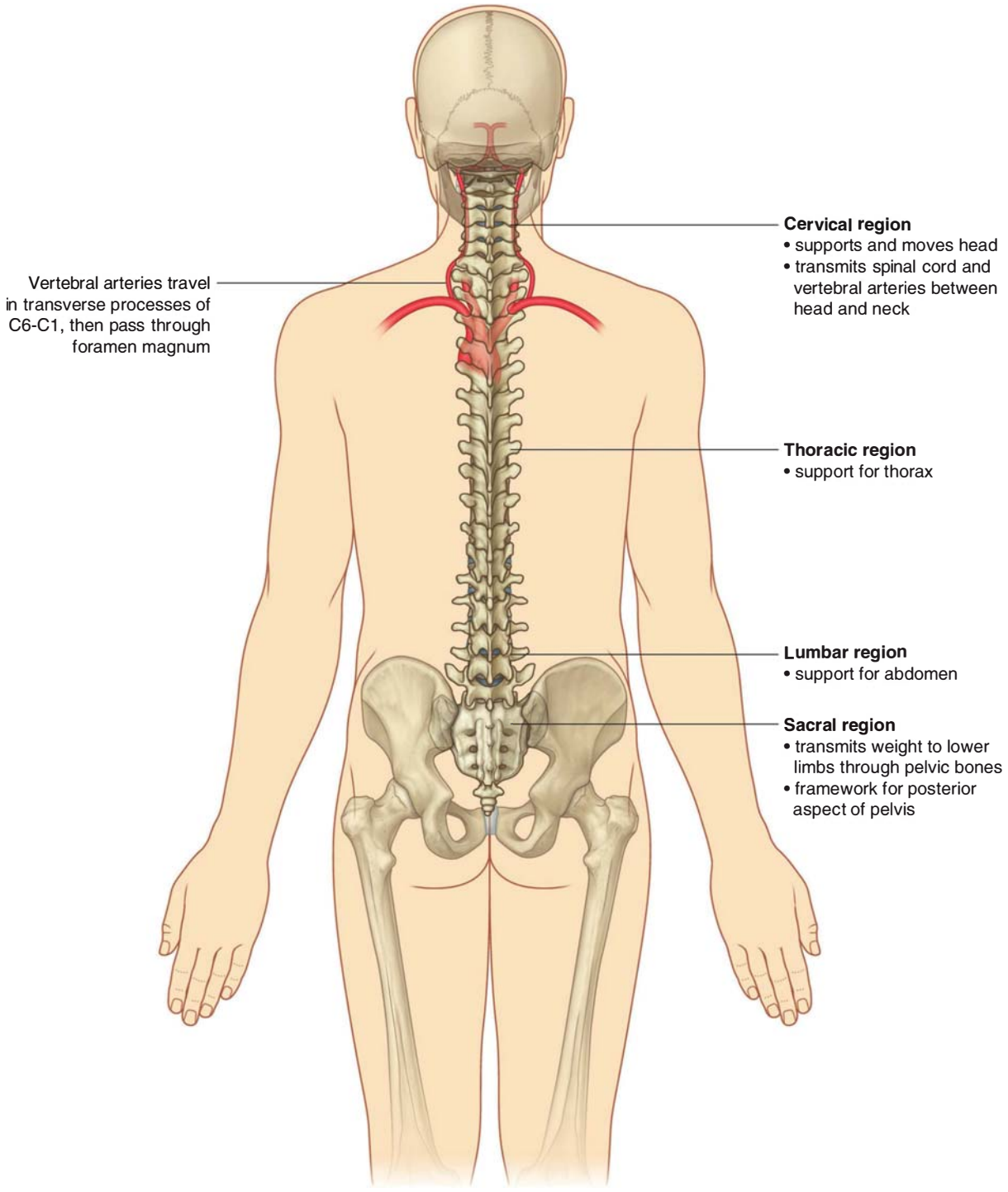


Fig. 2.10 Relationships of the back to other regions.

SUBASH KC/NMC-15TH/2014

Thorax, abdomen, and pelvis

The different regions of the vertebral column contribute to the skeletal framework of the thorax, abdomen, and pelvis (Fig. 2.10). In addition to providing support for each of these parts of the body, the vertebrae provide attachments for muscles and fascia, and articulation sites for other bones. The anterior rami of spinal nerves associated with the thorax, abdomen, and pelvis pass into these parts of the body from the back.

Limbs

The bones of the back provide extensive attachments for muscles associated with anchoring and moving the upper limbs on the trunk. This is less true of the lower limbs, which are firmly anchored to the vertebral column through articulation of the pelvic bones with the sacrum. The upper and lower limbs are innervated by anterior rami of spinal nerves that emerge from cervical and lumbosacral levels, respectively, of the vertebral column.

KEY FEATURES

Long vertebral column and short spinal cord

During development, the vertebral column grows much faster than the spinal cord. As a result, the spinal cord does not extend the entire length of the vertebral canal (Fig. 2.11).

In the adult, the spinal cord typically ends between vertebrae LI and LII, although it can end as high as vertebra TXII and as low as the disc between vertebrae LII and LIII.

Spinal nerves originate from the spinal cord at increasingly oblique angles from vertebrae CI to Co, and the nerve roots pass in the vertebral canal for increasingly longer distances. Their spinal cord level of origin therefore becomes increasingly dissociated from their vertebral column level of exit. This is particularly evident for lumbar and sacral spinal nerves.

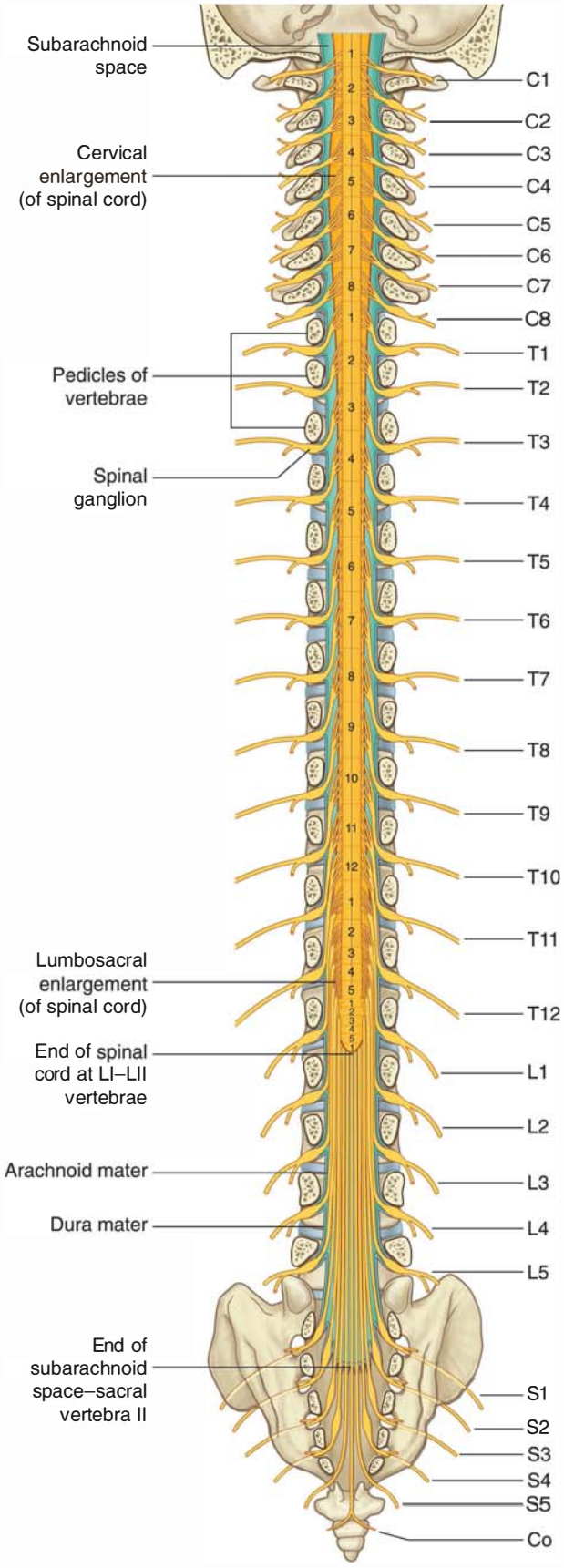


Fig. 2.11 Vertebral canal, spinal cord, and spinal nerves.

SUBASH KC/NMC-15TH/2014

Intervertebral foramina and spinal nerves

Each spinal nerve exits the vertebral canal laterally through an intervertebral foramen (Fig. 2.12). The foramen is formed between adjacent vertebral arches and is closely related to intervertebral joints:

- The superior and inferior margins are formed by notches in adjacent pedicles.
- The posterior margin is formed by the articular processes of the vertebral arches and the associated joint.
- The anterior border is formed by the intervertebral disc between the vertebral bodies of the adjacent vertebrae.

Any pathology that occludes or reduces the size of an intervertebral foramen, such as bone loss, herniation of the intervertebral disc, or dislocation of the zygapophysial joint (the joint between the articular processes), can affect the function of the associated spinal nerve.

Innervation of the back

Posterior branches of spinal nerves innervate the intrinsic muscles of the back and adjacent skin. The cutaneous distribution of these posterior rami extends into the gluteal region of the lower limb and the posterior aspect of the head. Parts of dermatomes innervated by the posterior rami of spinal nerves are shown in Fig. 2.13.

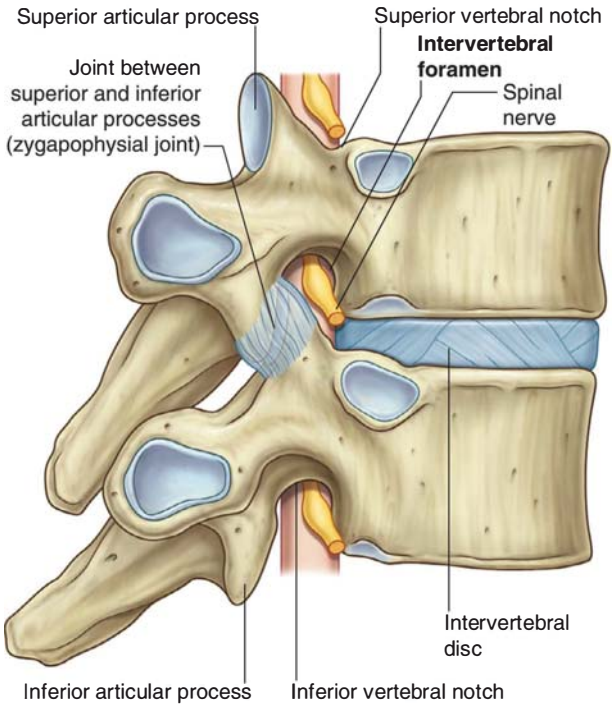
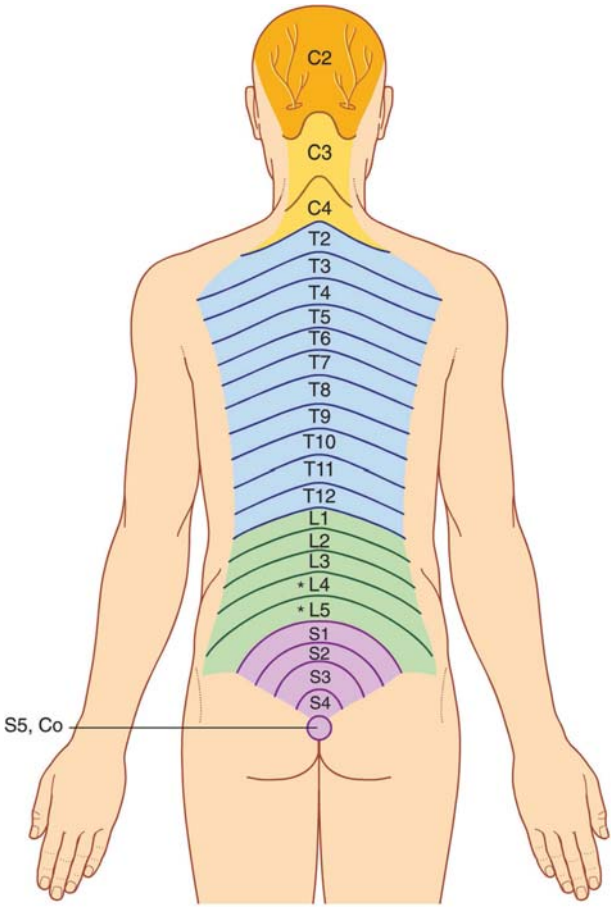


Fig. 2.12 Intervertebral foramina.



*The dorsal rami of L4 and L5 may not have cutaneous branches and may therefore not be represented as dermatomes on the back

Fig. 2.13 Dermatomes innervated by posterior rami of spinal nerves.

SUBASH KC/NMC-15TH/2014

Regional anatomy

SKELETAL FRAMEWORK

Skeletal components of the back consist mainly of the vertebrae and associated intervertebral discs. The skull, scapulae, pelvic bones, and ribs also contribute to the bony framework of the back and provide sites for muscle attachment.

Vertebrae

There are approximately 33 vertebrae, which are subdivided into five groups based on morphology and location (Fig. 2.14):

- The seven cervical vertebrae between the thorax and skull are characterized mainly by their small size and the presence of a foramen in each transverse process (Figs. 2.14 and 2.15).

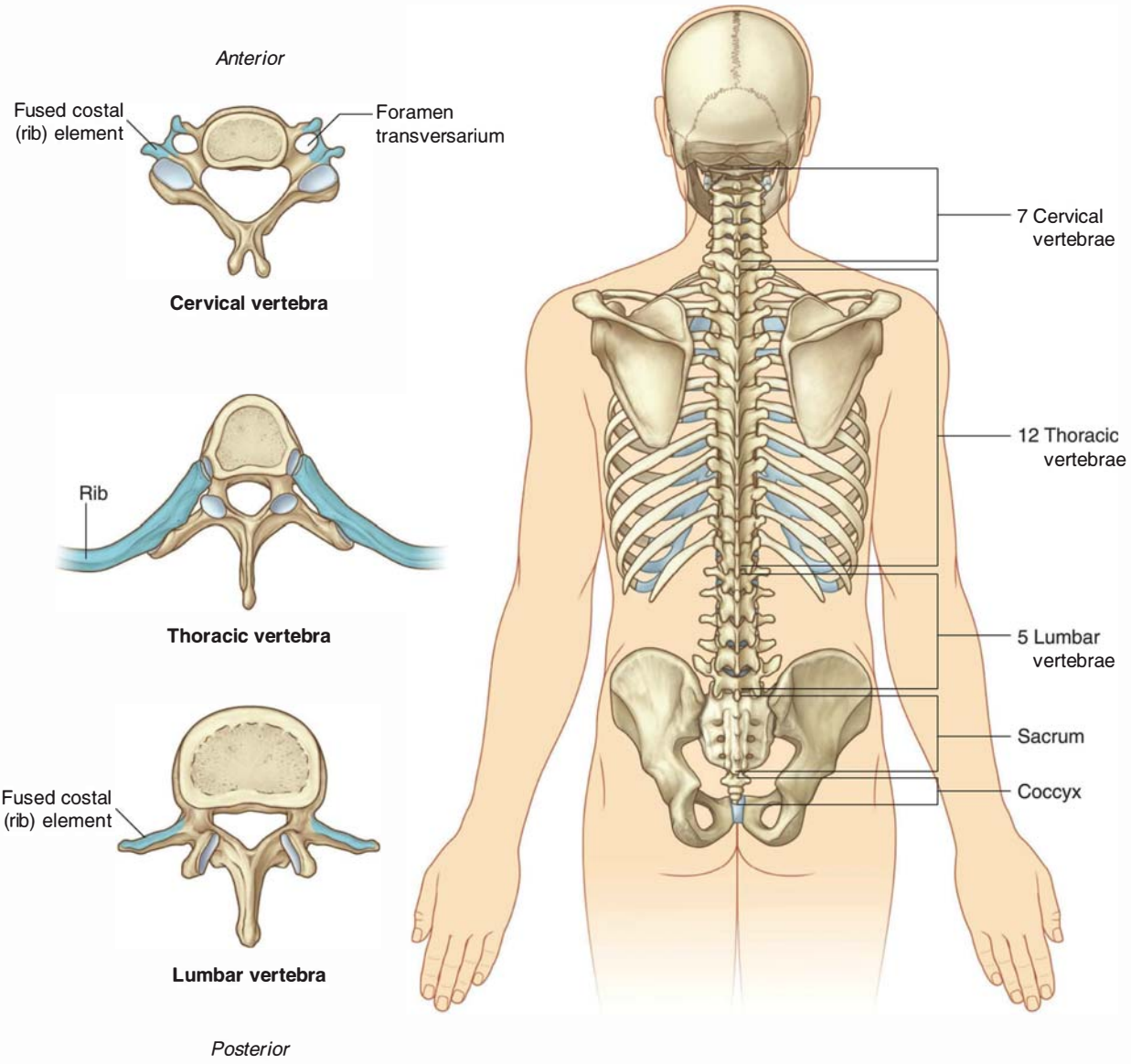


Fig. 2.14 Vertebrae.

SUBASH KC/NMC-15TH/2014

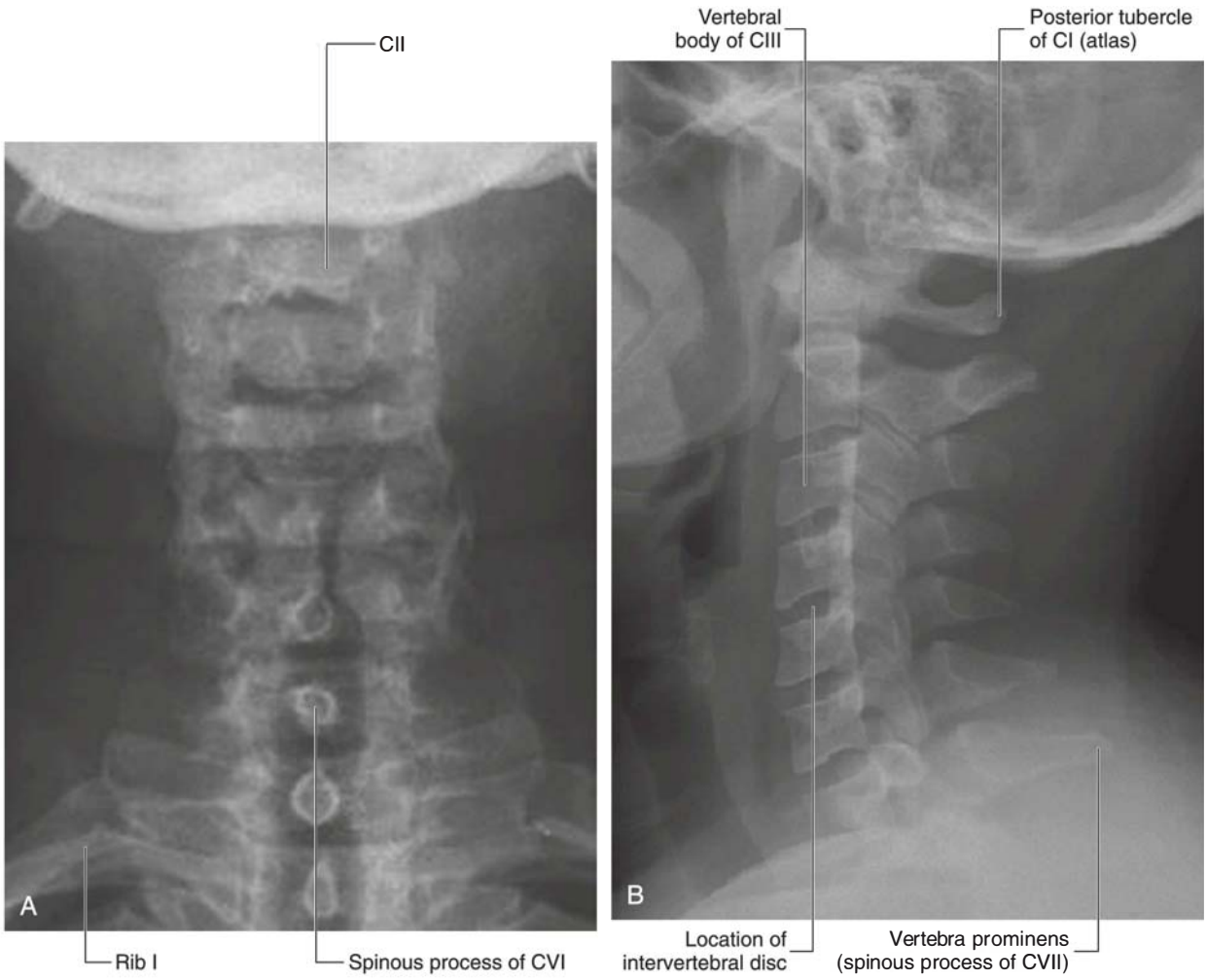


Fig. 2.15 Radiograph of cervical region of vertebral column. **A.** Anteroposterior view. **B.** Lateral view.

SUBASH KC/NMC-15TH/2014

- The 12 thoracic vertebrae are characterized by their articulated ribs (Figs. 2.14 and 2.16); although all vertebrae have rib elements, these elements are small and are incorporated into the transverse processes in regions other than the thorax; but in the thorax, the ribs are separate bones and articulate via synovial joints with the vertebral bodies and transverse processes of the associated vertebrae.
- Inferior to the thoracic vertebrae are five lumbar vertebrae, which form the skeletal support for the posterior abdominal wall and are characterized by their large size (Figs. 2.14 and 2.17).
- Next are five sacral vertebrae fused into one single bone called the sacrum, which articulates on each side with a pelvic bone and is a component of the pelvic wall.
- Inferior to the sacrum is a variable number, usually four, of coccygeal vertebrae, which fuse into a single small triangular bone called the coccyx.

In the embryo, the vertebrae are formed intersegmentally from cells called sclerotomes, which originate from adjacent somites (Fig. 2.18). Each vertebra is derived from the cranial parts of the two somites below, one on each side, and the caudal parts of the two somites above. The

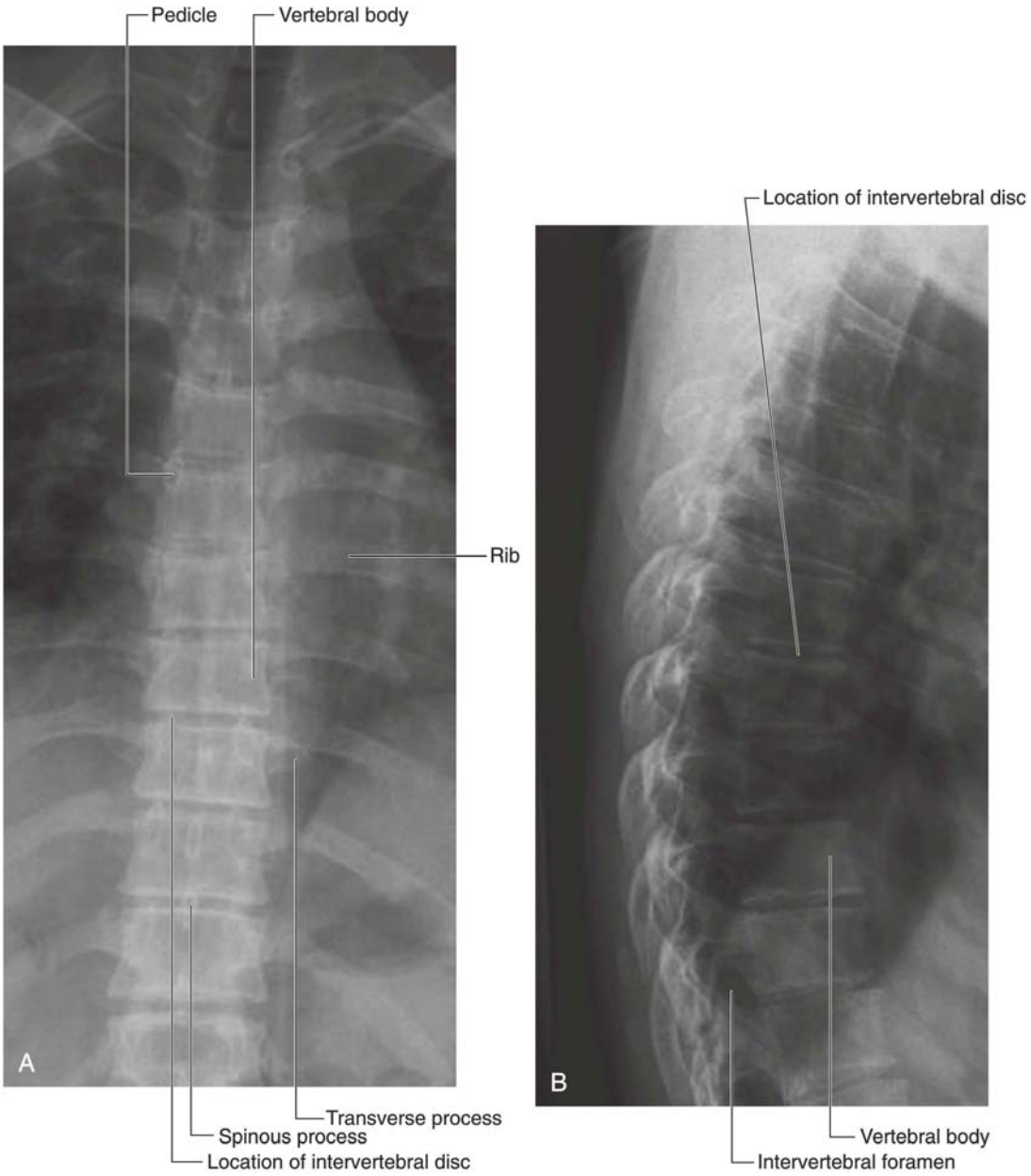


Fig. 2.16 Radiograph of thoracic region of vertebral column. **A.** Anteroposterior view. **B.** Lateral view.

SUBASH KC/NMC-15TH/2014

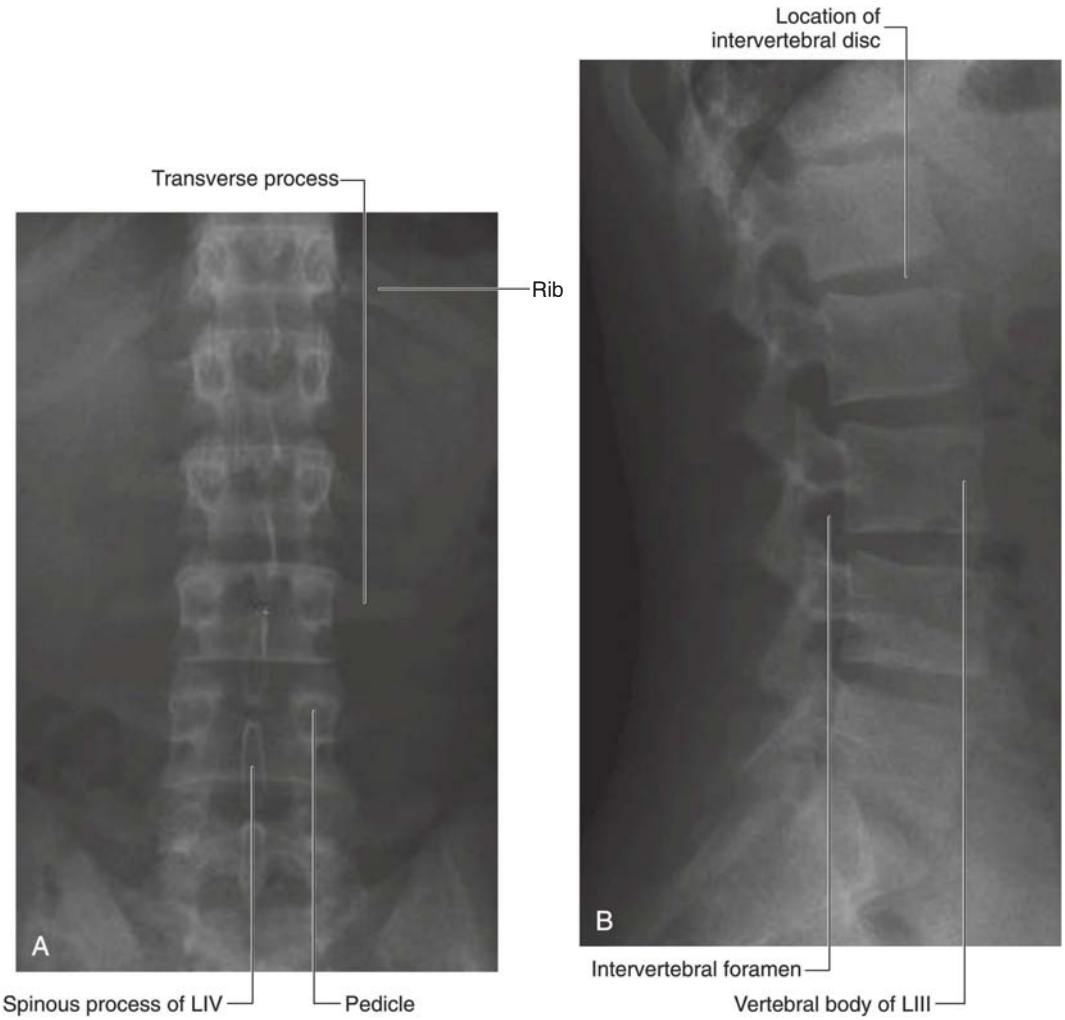


Fig. 2.17 Radiograph of lumbar region of vertebral column. **A.** Anteroposterior view. **B.** Lateral view.

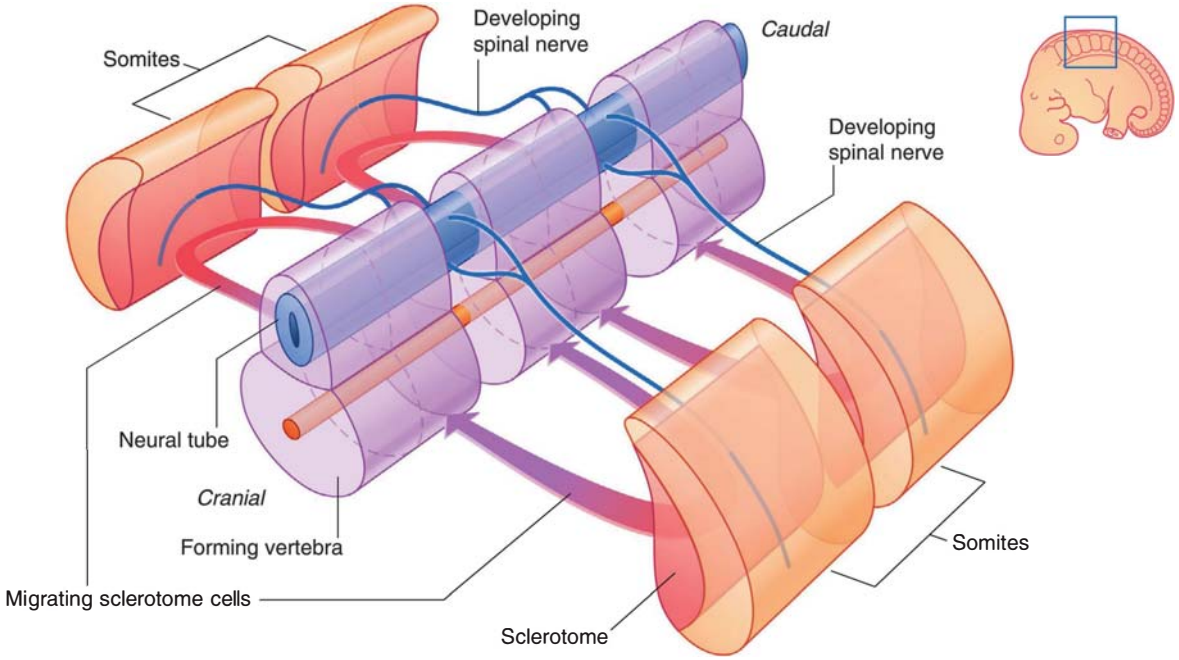


Fig. 2.18 Development of the vertebrae.

SUBASH KC/NMC-15TH/2014

spinal nerves develop segmentally and pass between the forming vertebrae.

Typical vertebra

A typical vertebra consists of a vertebral body and a posterior vertebral arch (Fig. 2.19). Extending from the vertebral arch are a number of processes for muscle attachment and articulation with adjacent bone.

The **vertebral body** is the weight-bearing part of the vertebra and is linked to adjacent vertebral bodies by intervertebral discs and ligaments. The size of vertebral bodies increases inferiorly as the amount of weight supported increases.

The **vertebral arch** forms the lateral and posterior parts of the vertebral foramen.

The vertebral foramina of all the vertebrae together form the **vertebral canal**, which contains and protects the spinal cord. Superiorly, the vertebral canal is continuous, through the foramen magnum of the skull, with the cranial cavity of the head.

The vertebral arch of each vertebra consists of pedicles and laminae (Fig. 2.19):

- The two **pedicles** are bony pillars that attach the vertebral arch to the vertebral body.
- The two **laminae** are flat sheets of bone that extend from each pedicle to meet in the midline and form the roof of the vertebral arch.

A **spinous process** projects posteriorly and inferiorly from the junction of the two laminae and is a site for muscle and ligament attachment.

A **transverse process** extends posterolaterally from the junction of the pedicle and lamina on each side and is a site for articulation with ribs in the thoracic region.

Also projecting from the region where the pedicles join the laminae are **superior** and **inferior articular processes** (Fig. 2.19), which articulate with the inferior and superior articular processes, respectively, of adjacent vertebrae.

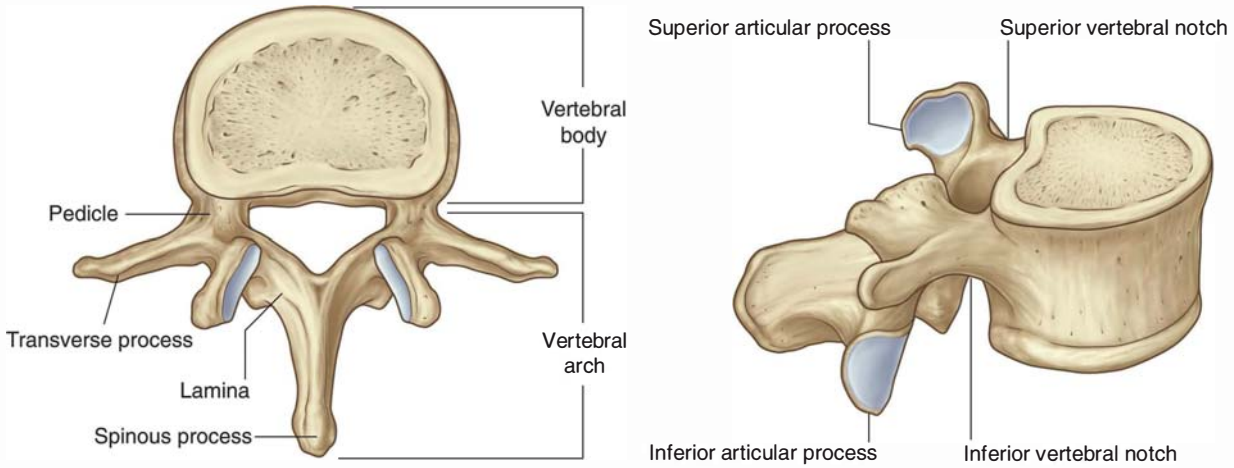


Fig. 2.19 Typical vertebra.

SUBASH KC/NMC-15TH/2014

Between the vertebral body and the origin of the articular processes, each pedicle is notched on its superior and inferior surfaces. These **superior** and **inferior vertebral notches** participate in forming intervertebral foramina.

Cervical vertebrae

The seven cervical vertebrae are characterized by their small size and by the presence of a foramen in each transverse process. A typical cervical vertebra has the following features (Fig. 2.20A):

- The vertebral body is short in height and square shaped when viewed from above and has a concave superior surface and a convex inferior surface.
- Each transverse process is trough shaped and perforated by a round **foramen transversarium**.
- The spinous process is short and bifid.
- The vertebral foramen is triangular.

The first and second cervical vertebrae—the atlas and axis—are specialized to accommodate movement of the head.

SUBASH KC/NMC-15TH/2014

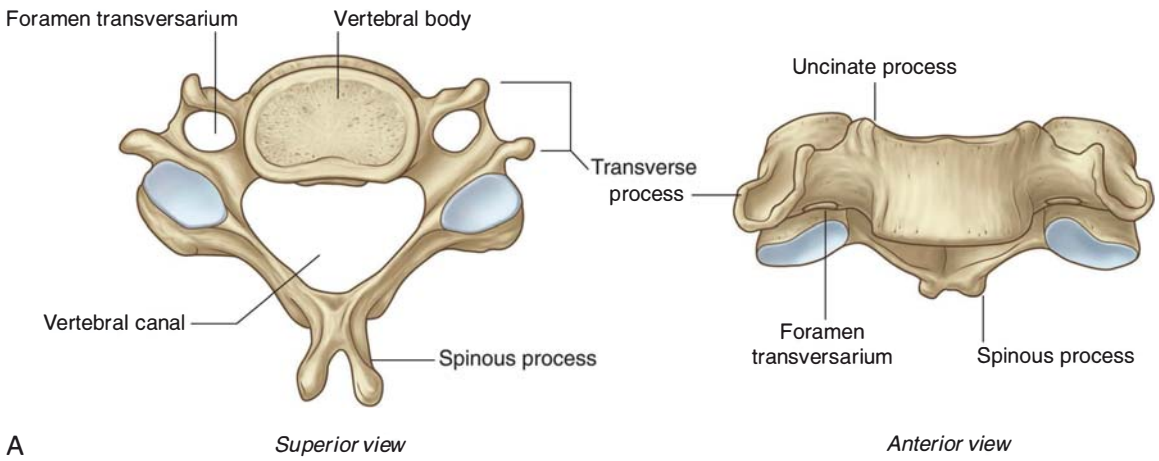


Fig. 2.20 Regional vertebrae. **A.** Typical cervical vertebra.

Continued

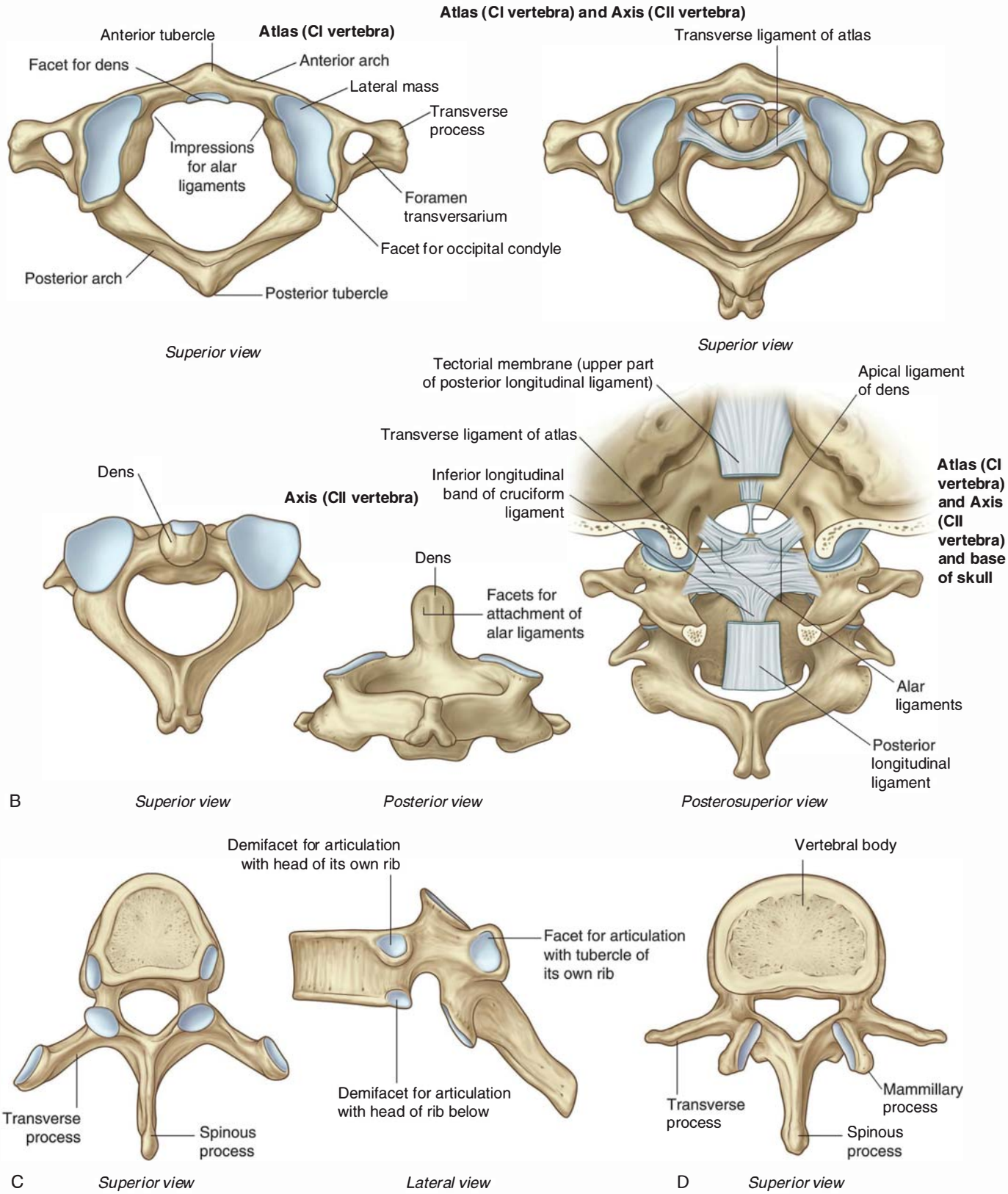


Fig. 2.20, cont'd **B.** Atlas and axis. **C.** Typical thoracic vertebra. **D.** Typical lumbar vertebra.

SUBASH KC/NMC-15TH/2014

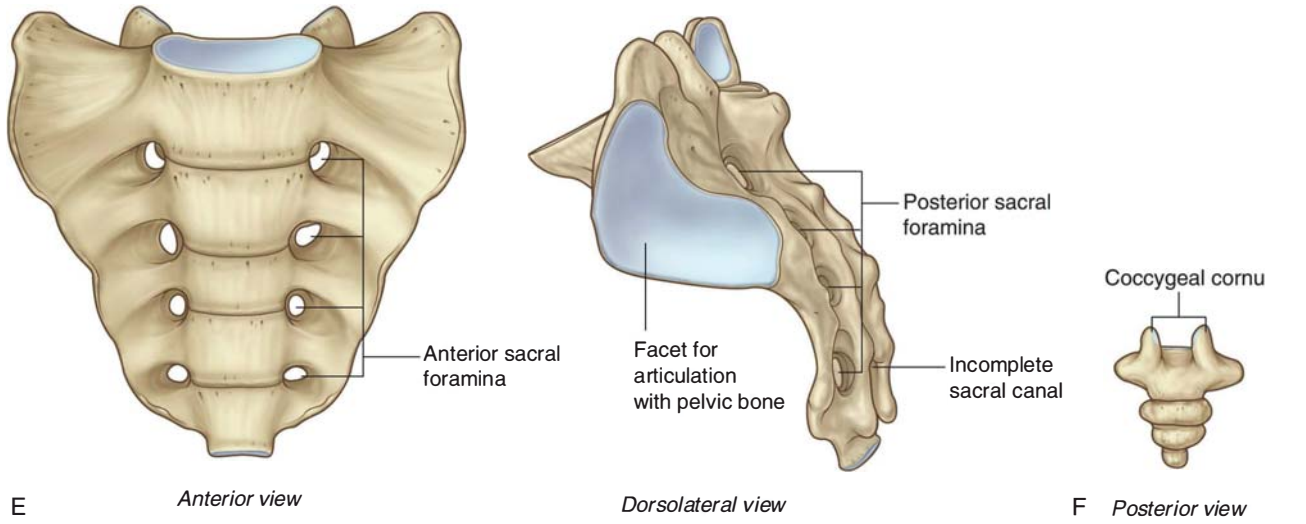


Fig. 2.20, cont'd E. Sacrum. **F.** Coccyx.

Atlas and axis

Vertebra CI (the **atlas**) articulates with the head (Fig. 2.21). Its major distinguishing feature is that it lacks a vertebral body (Fig. 2.20B). In fact, the vertebral body of CI fuses onto the body of CII during development to become the dens of CII. As a result, there is no intervertebral disc between CI and CII. When viewed from above, the atlas is ring shaped and composed of two **lateral masses** interconnected by an **anterior arch** and a **posterior arch**.

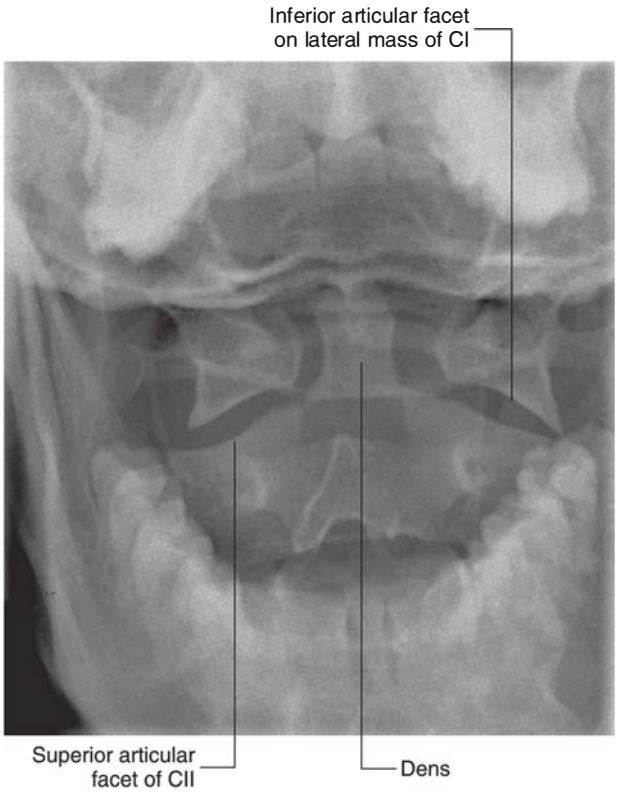


Fig. 2.21 Radiograph showing CI (atlas) and CII (axis) vertebrae. Open mouth, anteroposterior (odontoid peg) view.

Each lateral mass articulates above with an occipital **condyle** of the skull and below with the superior articular process of vertebra CII (the **axis**). The **superior articular surfaces** are bean shaped and concave, whereas the **inferior articular surfaces** are almost circular and flat.

The **atlanto-occipital joint** allows the head to nod up and down on the vertebral column.

The posterior surface of the anterior arch has an articular facet for the **dens**, which projects superiorly from the vertebral body of the axis. The dens is held in position by a strong **transverse ligament of atlas** posterior to it and spanning the distance between the oval attachment facets on the medial surfaces of the lateral masses of the atlas.

The dens acts as a pivot that allows the atlas and attached head to rotate on the axis, side to side.

The transverse processes of the atlas are large and protrude further laterally than those of the other cervical vertebrae and act as levers for muscle action, particularly for muscles that move the head at the **atlanto-axial joints**.

The axis is characterized by the large tooth-like dens, which extends superiorly from the vertebral body (Figs. 2.20B and 2.21). The anterior surface of the dens has an oval facet for articulation with the anterior arch of the atlas.

The two superolateral surfaces of the dens possess circular impressions that serve as attachment sites for strong alar ligaments, one on each side, which connect the dens to the medial surfaces of the occipital condyles. These **alar ligaments** check excessive rotation of the head and atlas relative to the axis.

SUBASH KC/NMC-15TH/2014

Thoracic vertebrae

The twelve thoracic vertebrae are all characterized by their articulation with ribs. A typical thoracic vertebra has two partial facets (superior and inferior costal facets) on each side of the vertebral body for articulation with the head of its own rib and the head of the rib below (Fig. 2.20C). The superior costal facet is much larger than the inferior costal facet.

Each transverse process also has a facet (transverse costal facet) for articulation with the tubercle of its own rib. The vertebral body of the vertebra is somewhat heart shaped when viewed from above, and the vertebral foramen is circular.

Lumbar vertebrae

The five lumbar vertebrae are distinguished from vertebrae in other regions by their large size (Fig. 2.20D). Also, they lack facets for articulation with ribs. The transverse processes are generally thin and long, with the exception of those on vertebra LV, which are massive and somewhat cone shaped for the attachment of **iliolumbar ligaments** to connect the transverse processes to the pelvic bones.

The vertebral body of a typical lumbar vertebra is cylindrical and the vertebral foramen is triangular in shape and larger than in the thoracic vertebrae.

Sacrum

The sacrum is a single bone that represents the five fused sacral vertebrae (Fig. 2.20E). It is triangular in shape with the **apex** pointed inferiorly, and is curved so that it has a concave anterior surface and a correspondingly convex posterior surface. It articulates above with vertebra LV and below with the coccyx. It has two large L-shaped facets, one on each lateral surface, for articulation with the pelvic bones.

The posterior surface of the sacrum has four pairs of posterior sacral foramina, and the anterior surface has four pairs of anterior sacral foramina for the passage of the posterior and anterior rami, respectively, of S1 to S4 spinal nerves.

The posterior wall of the vertebral canal may be incomplete near the inferior end of the sacrum.

Coccyx

The coccyx is a small triangular bone that articulates with the inferior end of the sacrum and represents three to four fused coccygeal vertebrae (Fig. 2.20F). It is characterized by its small size and by the absence of vertebral arches and therefore a vertebral canal.

Intervertebral foramina

Intervertebral foramina are formed on each side between adjacent parts of vertebrae and associated intervertebral discs (Fig. 2.22). The foramina allow structures, such as spinal nerves and blood vessels, to pass in and out of the vertebral canal.

An intervertebral foramen is formed by the inferior vertebral notch on the pedicle of the vertebra above and the superior vertebral notch on the pedicle of the vertebra below. The foramen is bordered:

- posteriorly by the zygapophysial joint between the articular processes of the two vertebrae, and

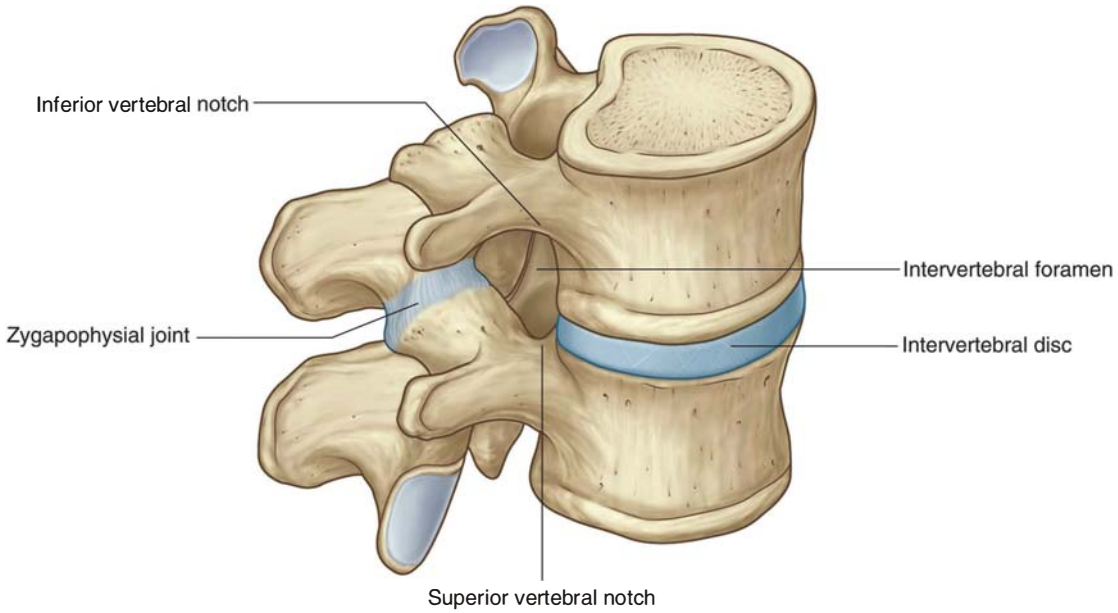


Fig. 2.22 Intervertebral foramen.

SUBASH KC/NMC-15TH/2014

- anteriorly by the intervertebral disc and adjacent vertebral bodies.

Each intervertebral foramen is a confined space surrounded by bone and ligament, and by joints. Pathology in any of these structures, and in the surrounding muscles, can affect structures within the foramen.

Posterior spaces between vertebral arches

In most regions of the vertebral column, the laminae and spinous processes of adjacent vertebrae overlap to form a reasonably complete bony dorsal wall for the vertebral canal. However, in the lumbar region, large gaps exist between the posterior components of adjacent vertebral arches (Fig. 2.23). These gaps between adjacent laminae and spinous processes become increasingly wide from vertebra LI to vertebra LV. The spaces can be widened further by flexion of the vertebral column. These gaps allow relatively easy access to the vertebral canal for clinical procedures.

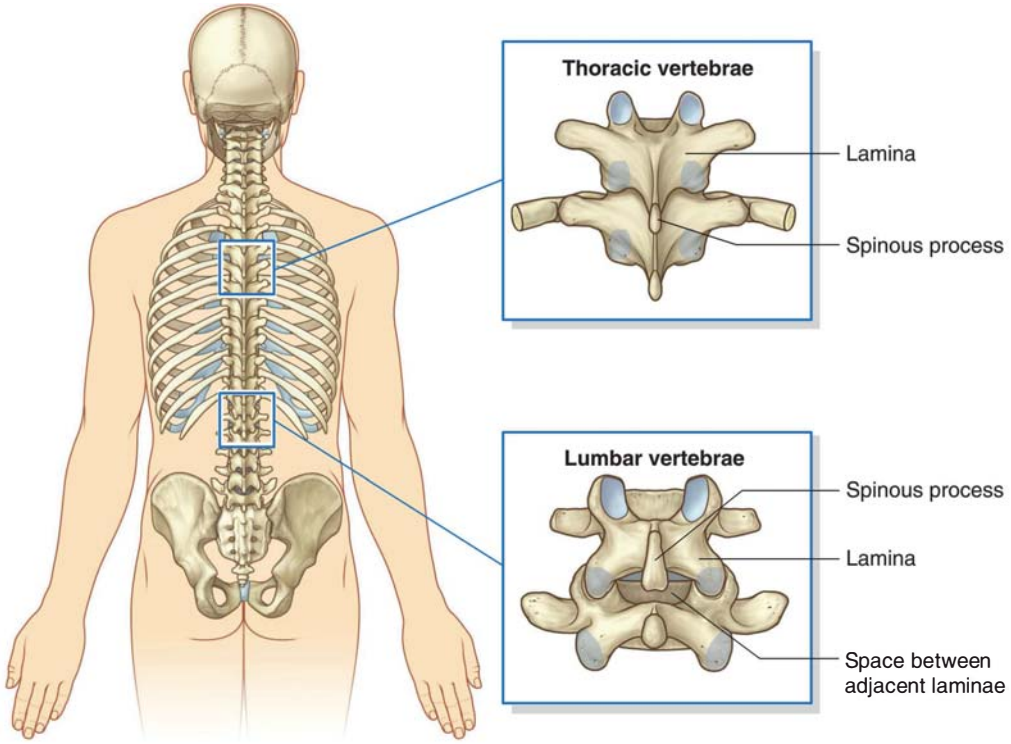


Fig. 2.23 Spaces between adjacent vertebral arches in the lumbar region.

SUBASH KC/NMC-15TH/2014

In the clinic

Spina bifida

Spina bifida is a disorder in which the two sides of vertebral arches, usually in lower vertebrae, fail to fuse during development, resulting in an "open" vertebral canal (Fig. 2.24). There are two types of spina bifida.

- The commonest type is spina bifida occulta, in which there is a defect in the vertebral arch of LV or SI. This defect occurs in as many as 10% of individuals and results in failure of the posterior arch to fuse in the midline. Clinically, the patient is asymptomatic, although physical examination may reveal a tuft of hair over the spinous processes.
- The more severe form of spina bifida involves complete failure of fusion of the posterior arch at the lumbosacral junction, with a large outpouching of the meninges. This may contain cerebrospinal fluid (a **meningocele**) or a portion of the spinal cord (a **myelomeningocele**). These abnormalities may result in a variety of neurological deficits, including problems with walking and bladder function.

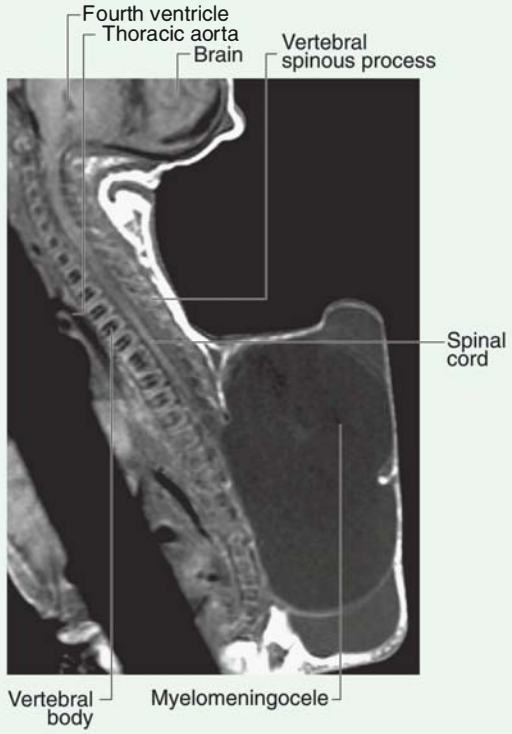


Fig. 2.24 T1-weighted MR image in the sagittal plane demonstrating a lumbosacral myelomeningocele. There is an absence of laminae and spinous processes in the lumbosacral region.

In the clinic

Vertebroplasty

Vertebroplasty is a new technique in which the body of a vertebra can be filled with bone cement (typically methyl methacrylate). The indications for the technique include vertebral body collapse and pain from the vertebral body, which may be secondary to tumor infiltration. The procedure is most commonly performed for osteoporotic wedge fractures, which are a considerable cause of morbidity and pain in older patients.

Osteoporotic wedge fractures typically occur in the thoracolumbar region, and the approach to performing vertebroplasty is novel and relatively straightforward. The procedure is performed under sedation or light general anesthetic. Using X-ray guidance the pedicle is identified on the anteroposterior image. A metal cannula is placed through the pedicle into the vertebral body. Liquid bone cement is injected via the cannula into the vertebral body (see Fig. 1.18, p. 17). The function of the bone cement is two-fold. First, it increases the strength of the vertebral body and prevents further loss of height. Furthermore, as the bone cement sets, there is a degree of heat generated that is believed to disrupt pain nerve endings.

SUBASH KC/NMC-15TH/2014

In the clinic

Scoliosis

Scoliosis is an abnormal lateral curvature of the vertebral column (Fig. 2.25).

A true scoliosis involves not only the curvature (right- or left-sided) but also a rotational element of one vertebra upon another.

The commonest types of scoliosis are those for which we have little understanding about how or why they occur and are termed idiopathic scoliosis. These are never present at birth and tend to occur in either the infantile, juvenile, or adolescent age groups. The vertebral bodies and posterior elements (pedicles and laminae) are normal in these patients.

When a scoliosis is present from birth (congenital scoliosis) it is usually associated with other developmental abnormalities. In these patients, there is a strong association with other abnormalities of the chest wall, genitourinary tract, and heart disease. This group of patients needs careful evaluation by many specialists.

A rare but important group of scoliosis is that in which the muscle is abnormal. Muscular dystrophy is the commonest example. The abnormal muscle does not retain the normal alignment of the vertebral column, and curvature develops as a result. A muscle biopsy is needed to make the diagnosis.

Other disorders that can produce scoliosis include bone tumors, spinal cord tumors, and localized disc protrusions.

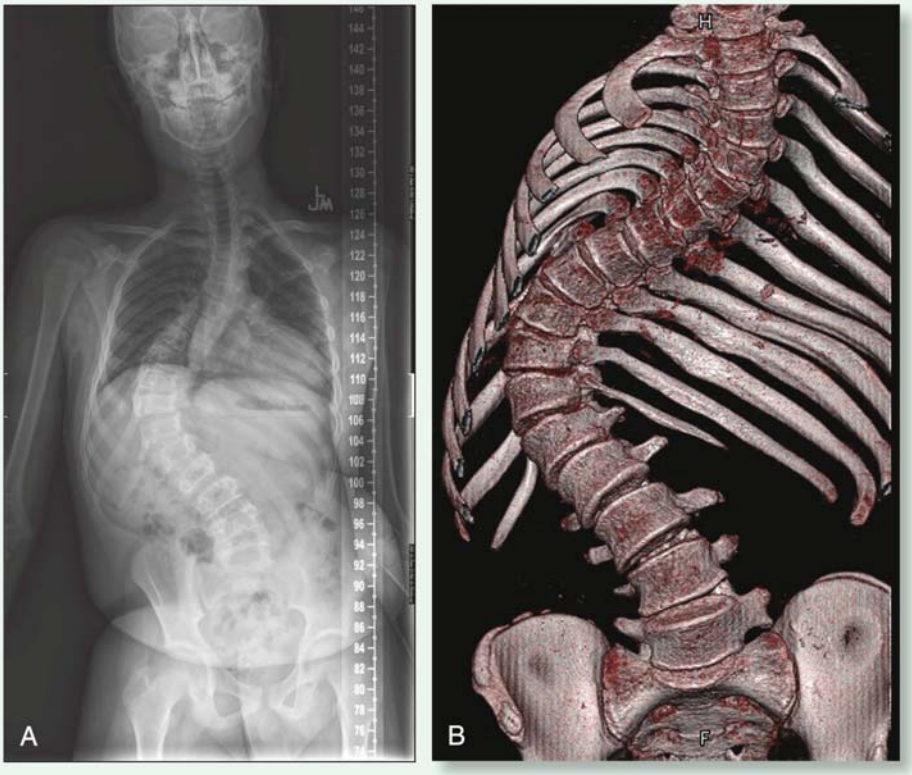


Fig. 2.25 Severe scoliosis. **A.** Radiograph, anteroposterior view. **B.** Volume-rendered CT, anterior view.

SUBASH KC/NMC-15TH/2014

In the clinic

Kyphosis

Kyphosis is abnormal curvature of the vertebral column in the thoracic region, producing a "hunchback" deformity. This condition occurs in certain disease states, the most dramatic of which is usually secondary to tuberculosis infection of a thoracic vertebral body, where the kyphosis becomes angulated at the site of the lesion. This produces the **gibbus deformity,** a deformity that was prevalent before the use of antituberculous medication.

Lordosis

Lordosis is abnormal curvature of the vertebral column in the lumbar region, producing a swayback deformity.

In the clinic

Variation in vertebral numbers

There are usually seven cervical vertebrae, although in certain diseases these may be fused. Fusion of cervical vertebrae (Fig. 2.26A) can be associated with other abnormalities, for example Klippel-Feil syndrome, in which there is fusion of vertebrae CI and CII or CV and CVI, and may be associated with a high-riding scapula (Sprengel's shoulder) and cardiac abnormalities.

Variations in the number of thoracic vertebrae also are well described.

One of the commonest abnormalities in the lumbar vertebrae is a partial fusion of vertebra LV with the sacrum (sacralization of the lumbar vertebra). Partial separation of vertebra SI from the sacrum (lumbarization of first sacral vertebra) may also occur (Fig. 2.26B).

A hemivertebra occurs when a vertebra develops only on one side (Fig. 2.26B).

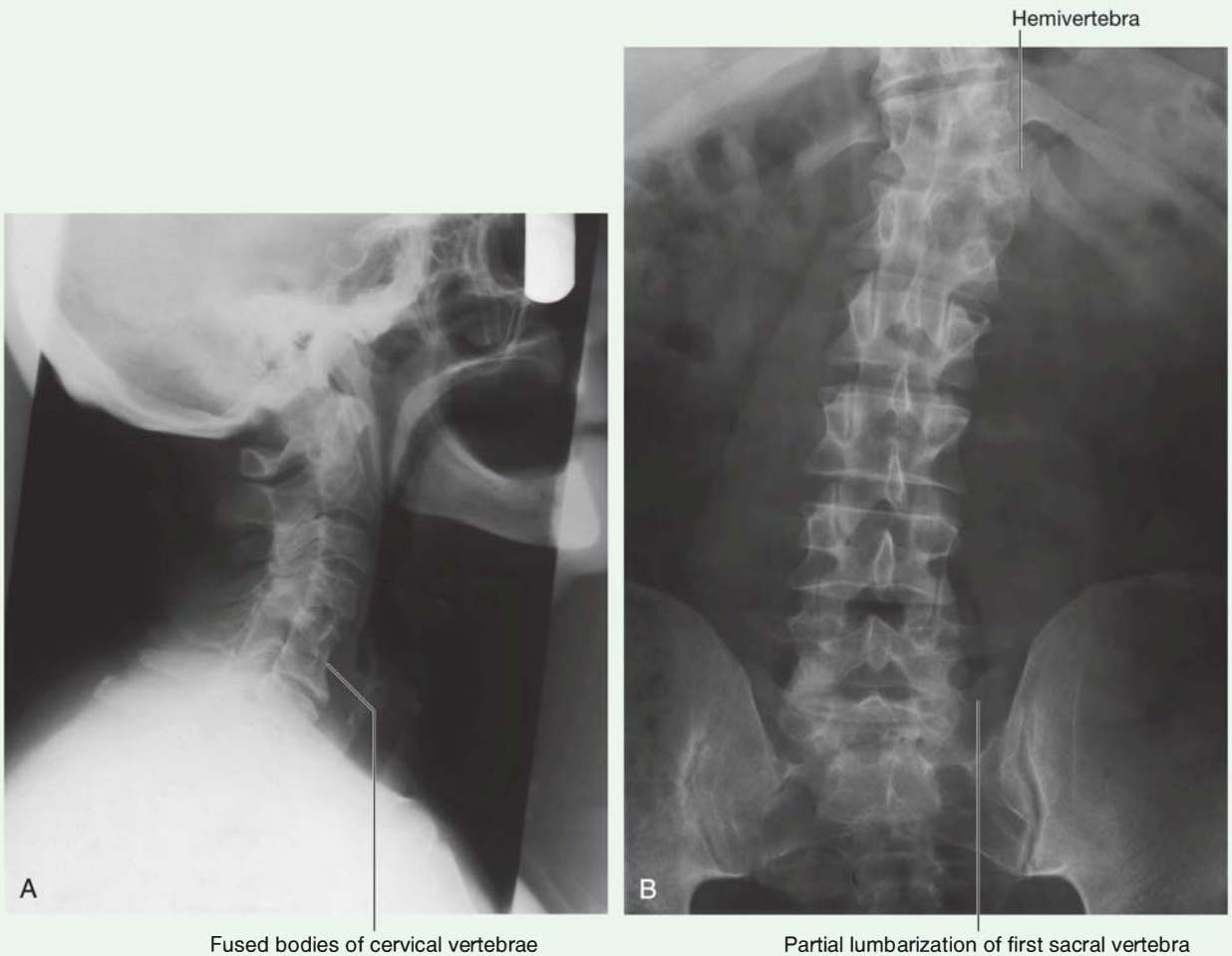


Fig. 2.26 Variations in vertebral number. **A.** Fused vertebral bodies of cervical vertebrae. **B.** Hemivertebra.

SUBASH KC/NMC-15TH/2014

In the clinic

The vertebrae and cancer

The vertebrae are common sites for metastatic disease (secondary spread of cancer cells). When cancer cells grow within the vertebral bodies and the posterior elements, they destroy the mechanical properties of the bone. A minor injury may therefore lead to vertebral collapse. Importantly, vertebrae that contain extensive metastatic disease may extrude fragments of tumor into the **vertebral canal**, compressing nerves and the spinal cord.

In the clinic

Osteoporosis

Osteoporosis is a pathophysiologic condition in which bone quality is normal but the quantity of bone is deficient. It is a metabolic bone disorder that commonly occurs in women in their 50s and 60s and in men in their 70s.

Many factors influence the development of osteoporosis, including genetic predetermination, level of activity and nutritional status, and, in particular, estrogen levels in women.

Typical complications of osteoporosis include "crush" vertebral body fractures, distal fractures of the radius, and hip fractures.

With increasing age and poor-quality bone, patients are more susceptible to fracture. Healing tends to be impaired in these elderly patients, who consequently require long hospital stays and prolonged rehabilitation.

Patients likely to develop osteoporosis can be identified by dual-photon X-ray absorptiometry (DXA) scanning. Low-dose X-rays are passed through the bone, and by counting the number of photons detected and knowing the dose given, the number of X-rays absorbed by the bone can be calculated. The amount of X-ray absorption can be directly correlated with the bone mass, and this can be used to predict whether a patient is at risk for osteoporotic fractures.

SUBASH KC/NMC-15TH/2014

JOINTS

Joints between vertebrae in the back

The two major types of joints between vertebrae are:

- symphyses between vertebral bodies (Fig. 2.27), and
- synovial joints between articular processes (Fig. 2.28).

A typical vertebra has a total of six joints with adjacent vertebrae: four synovial joints (two above and two below) and two symphyses (one above and one below). Each symphysis includes an intervertebral disc.

Although the movement between any two vertebrae is limited, the summation of movement among all vertebrae results in a large range of movement by the vertebral column.

Movements by the vertebral column include flexion, extension, lateral flexion, rotation, and circumduction.

Movements by vertebrae in a specific region (cervical, thoracic, and lumbar) are determined by the shape and orientation of joint surfaces on the articular processes and on the vertebral bodies.

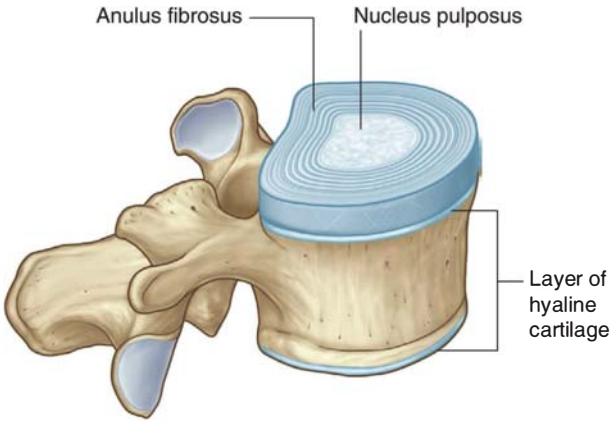


Fig. 2.27 Intervertebral joints.

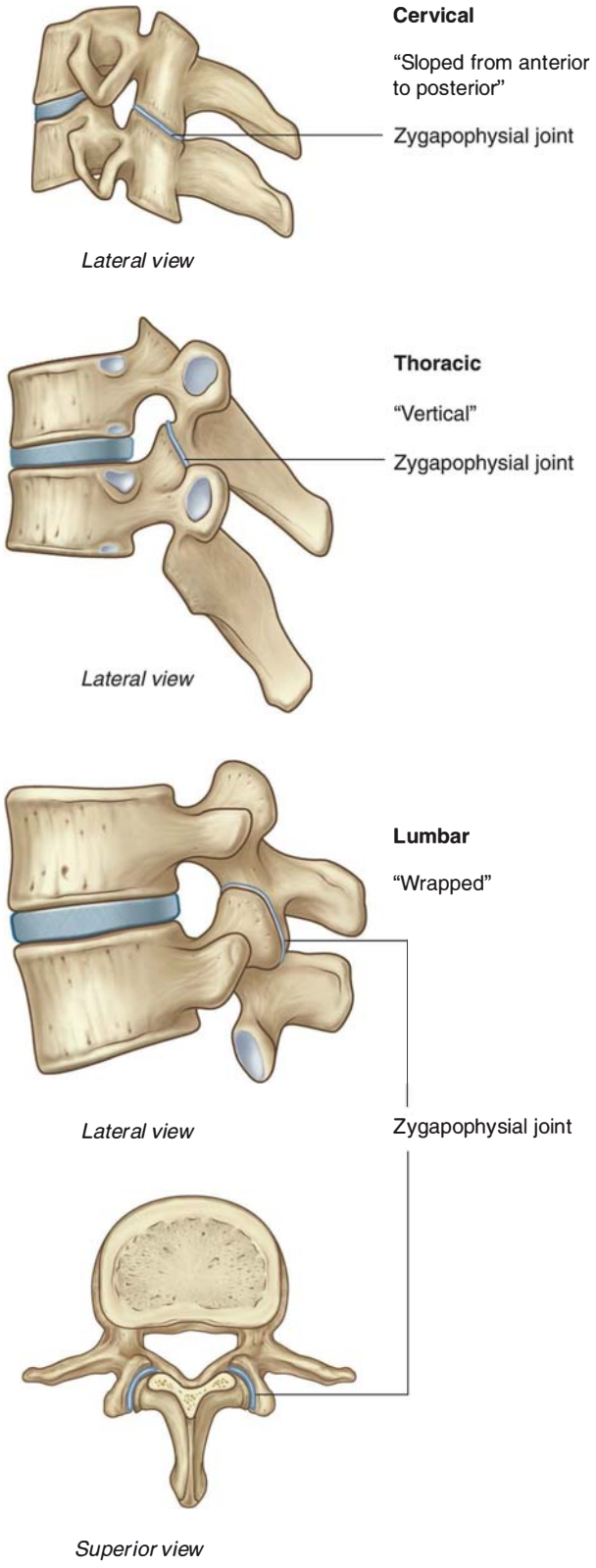


Fig. 2.28 Zygapophysial joints.

Symphyses between vertebral bodies (intervertebral discs)

The symphysis between adjacent vertebral bodies is formed by a layer of hyaline cartilage on each vertebral body and an intervertebral disc, which lies between the layers.

The **intervertebral disc** consists of an outer anulus fibrosus, which surrounds a central nucleus pulposus (Fig. 2.27).

- The **anulus fibrosus** consists of an outer ring of collagen surrounding a wider zone of fibrocartilage arranged in a lamellar configuration. This arrangement of fibers limits rotation between vertebrae.
- The **nucleus pulposus** fills the center of the intervertebral disc, is gelatinous, and absorbs compression forces between vertebrae.

Degenerative changes in the anulus fibrosus can lead to herniation of the nucleus pulposus. Posterolateral herniation can impinge on the roots of a spinal nerve in the intervertebral foramen.

Joints between vertebral arches (zygapophysial joints)

The synovial joints between superior and inferior articular processes on adjacent vertebrae are the zygapophysial joints (Fig. 2.28). A thin articular capsule attached to the margins of the articular facets encloses each joint.

In cervical regions, the zygapophysial joints slope inferiorly from anterior to posterior. This orientation facilitates flexion and extension. In thoracic regions, the joints are oriented vertically and limit flexion and extension, but facilitate rotation. In lumbar regions, the joint surfaces are curved and adjacent processes interlock, thereby limiting range of movement, though flexion and extension are still major movements in the lumbar region.

"Uncovertebral" joints

The lateral margins of the upper surfaces of typical cervical vertebrae are elevated into crests or lips termed uncinate processes. These may articulate with the body of the vertebra above to form small "uncovertebral" synovial joints (Fig. 2.29).

SUBASH KC/NMC-15TH/2014

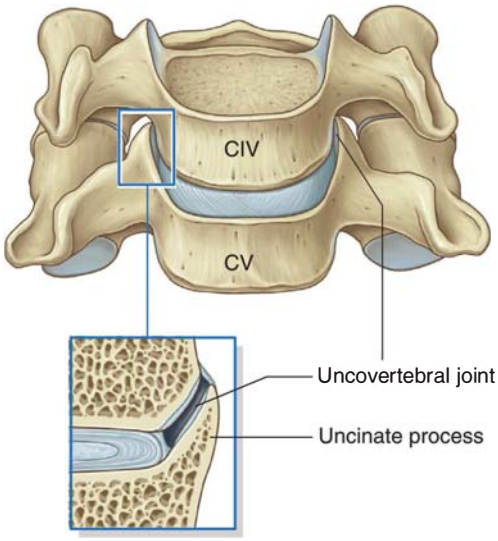


Fig. 2.29 Uncovertebral joint.

In the clinic

Back pain

Back pain is an extremely common disorder. It can be related to mechanical problems or to disc protrusion impinging on a nerve. In cases involving discs, it may be necessary to operate and remove the disc that is pressing on the nerve.

Not infrequently, patients complain of pain and no immediate cause is found; the pain is therefore attributed to mechanical discomfort, which may be caused by degenerative disease. One of the treatments is to pass a needle into the facet joint and inject it with local anesthetic and corticosteroid.

In the clinic

Herniation of intervertebral discs

The discs between the vertebrae are made up of a central portion (the nucleus pulposus) and a complex series of fibrous rings (anulus fibrosus). A tear can occur within the anulus fibrosus through which the material of the nucleus pulposus can track. After a period of time, this material may track into the vertebral canal or into the intervertebral foramen to impinge on neural structures (Fig. 2.30). This is a common cause of back pain. A disc may protrude posteriorly to directly impinge on the cord or the roots of the lumbar nerves, depending on the level, or may protrude posterolaterally adjacent to the pedicle and impinge on the descending root.

In cervical regions of the vertebral column, cervical disc protrusions often become ossified and are termed disc osteophyte bars.

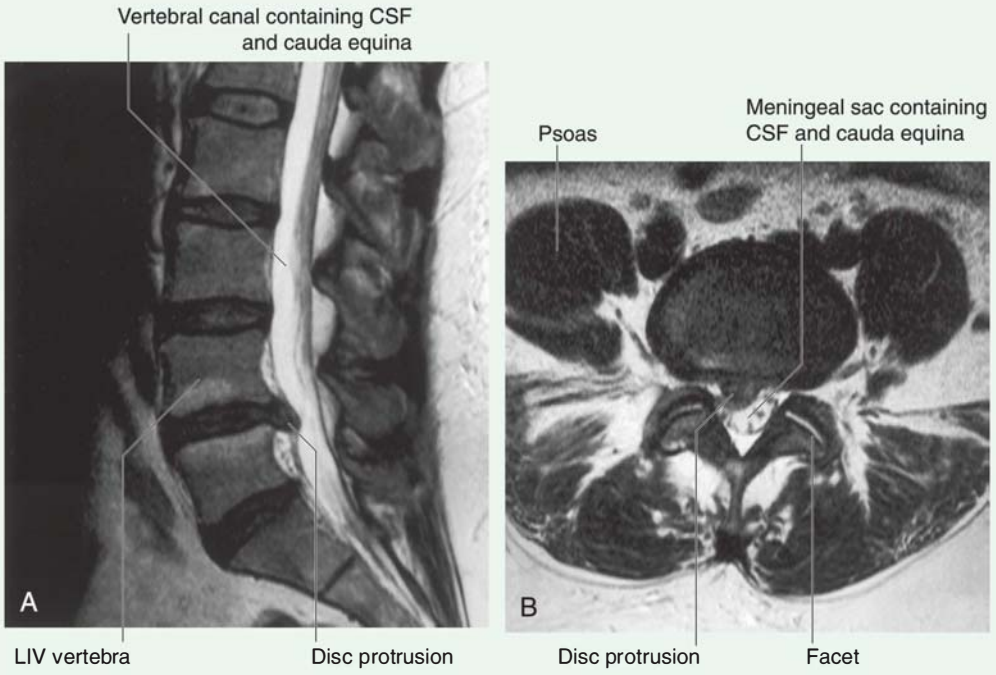


Fig. 2.30 Disc protrusion. T2-weighted magnetic resonance images of the lumbar region of the vertebral column. **A.** Sagittal plane. **B.** Axial plane.

SUBASH KC/NMC-15TH/2014

In the clinic

Joint diseases

Some diseases have a predilection for synovial joints rather than symphyses. A typical example is rheumatoid arthritis, which primarily affects synovial joints and synovial bursae, resulting in destruction of the joint and its lining. Symphyses are usually preserved.

LIGAMENTS

Joints between vertebrae are reinforced and supported by numerous ligaments, which pass between vertebral bodies and interconnect components of the vertebral arches.

Anterior and posterior longitudinal ligaments

The anterior and posterior longitudinal ligaments are on the anterior and posterior surfaces of the vertebral bodies and extend along most of the vertebral column (Fig. 2.31).

The **anterior longitudinal ligament** is attached superiorly to the base of the skull and extends inferiorly to attach to the anterior surface of the sacrum. Along its length it is attached to the vertebral bodies and intervertebral discs.

The **posterior longitudinal ligament** is on the posterior surfaces of the vertebral bodies and lines the anterior surface of the vertebral canal. Like the anterior longitudinal ligament, it is attached along its length to the vertebral bodies and intervertebral discs. The upper part of the posterior longitudinal ligament that connects CII to the intracranial aspect of the base of the skull is termed the **tectorial membrane** (see Fig. 2.20B).

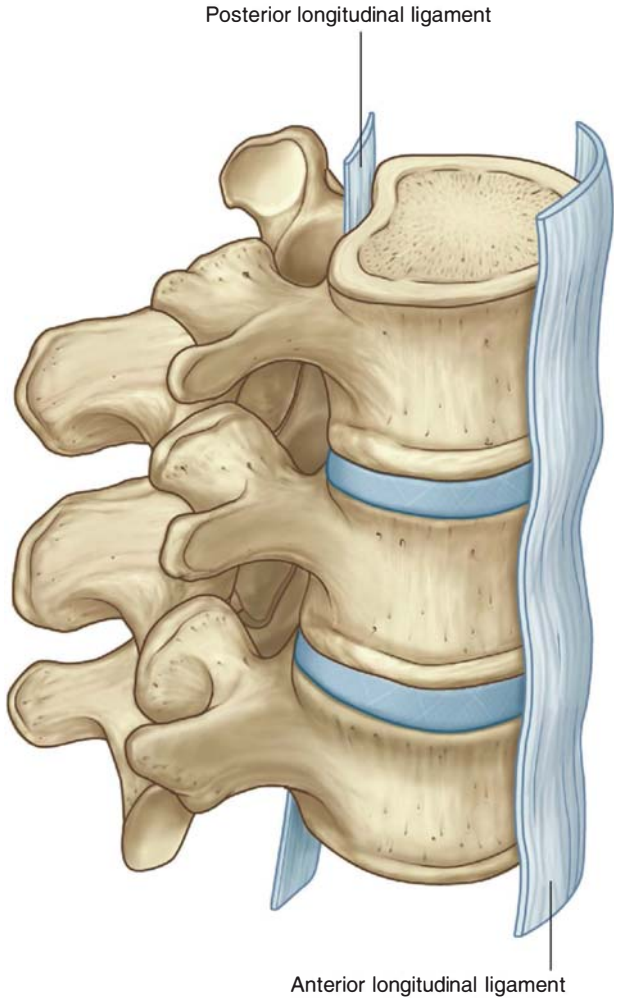


Fig. 2.31 Anterior and posterior longitudinal ligaments of vertebral column.

Ligamenta flava

The **ligamenta flava**, on each side, pass between the laminae of adjacent vertebrae (Fig. 2.32). These thin, broad ligaments consist predominantly of elastic tissue and form part of the posterior surface of the vertebral

SUBASH KC/NMC-15TH/2014

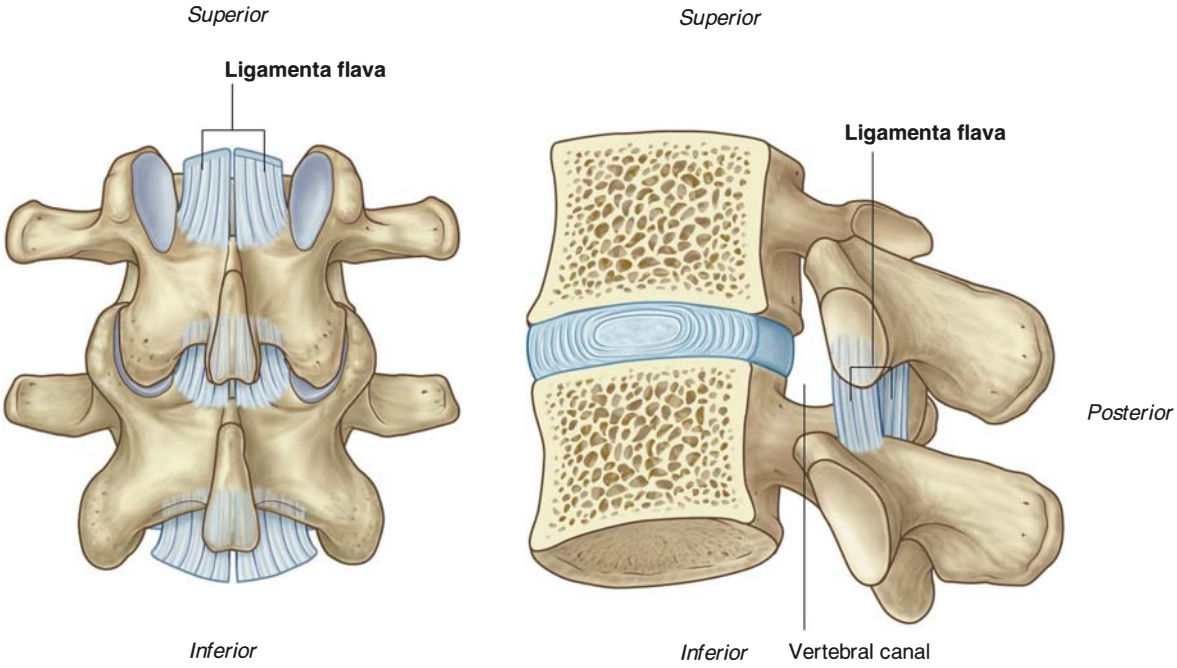


Fig. 2.32 Ligamenta flava.

canal. Each ligamentum flavum runs between the posterior surface of the lamina on the vertebra below to the anterior surface of the lamina of the vertebra above. The ligamenta flava resist separation of the laminae in flexion and assist in extension back to the anatomical position.

Supraspinous ligament and ligamentum nuchae

The supraspinous ligament connects and passes along the tips of the vertebral spinous processes from vertebra CVII to the sacrum (Fig. 2.33). From vertebra CVII to the skull, the ligament becomes structurally distinct from more caudal parts of the ligament and is called the ligamentum nuchae.

The **ligamentum nuchae** is a triangular, sheet-like structure in the median sagittal plane:

- The base of the triangle is attached to the skull, from the external occipital protuberance to the foramen magnum.
- The apex is attached to the tip of the spinous process of vertebra CVII.
- The deep side of the triangle is attached to the posterior tubercle of vertebra CI and the spinous processes of the other cervical vertebrae.

The ligamentum nuchae supports the head. It resists flexion and facilitates returning the head to the anatomical

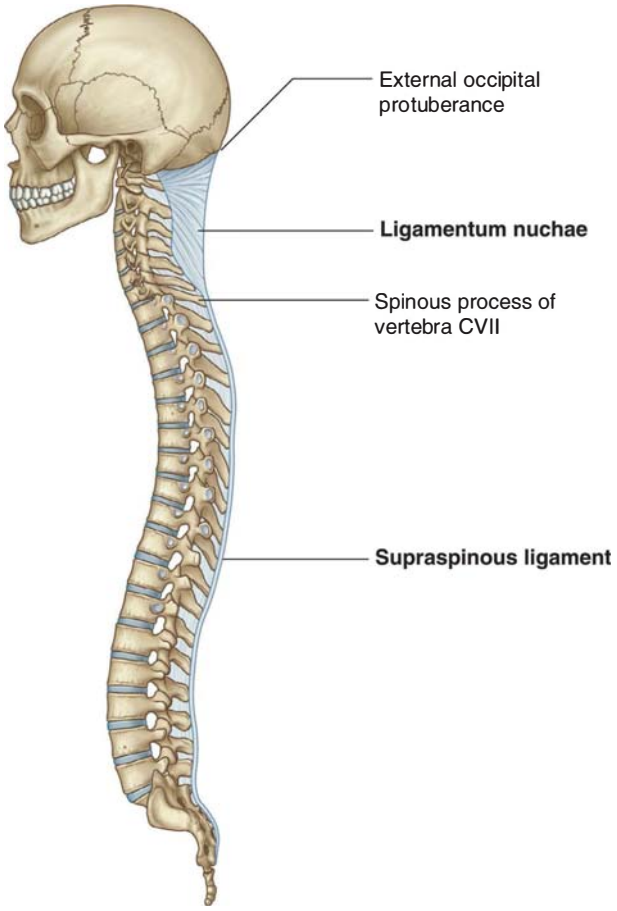


Fig. 2.33 Supraspinous ligament and ligamentum nuchae.

SUBASH KC/NMC-15TH/2014

position. The broad lateral surfaces and the posterior edge of the ligament provide attachment for adjacent muscles.

Interspinous ligaments

Interspinous ligaments pass between adjacent vertebral spinous processes (Fig. 2.34). They attach from the base to the apex of each spinous process and blend with the supraspinous ligament posteriorly and the ligamenta flava anteriorly on each side.

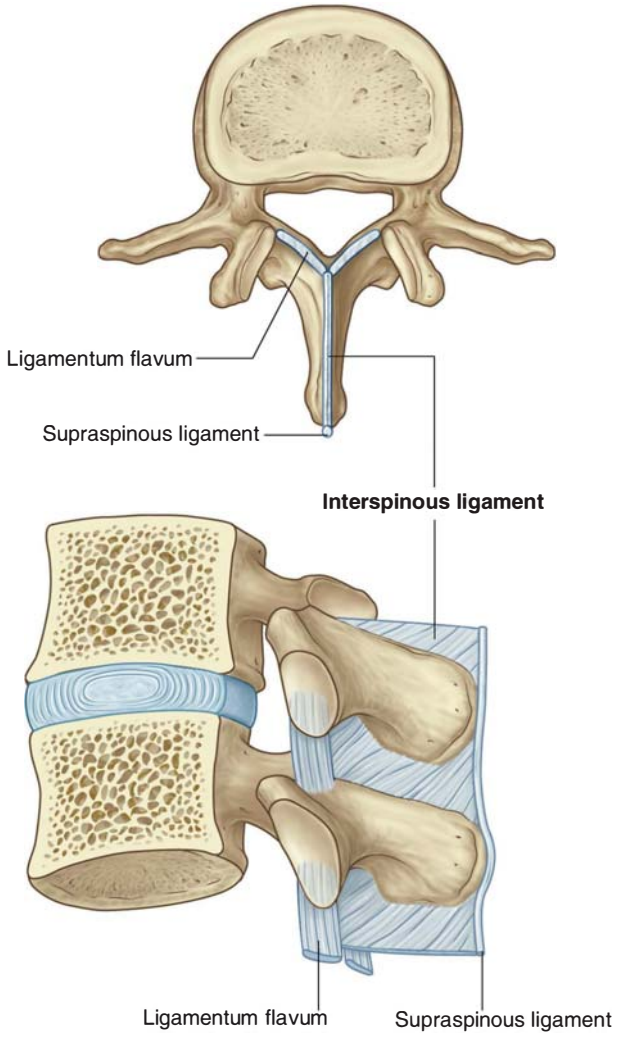


Fig. 2.34 Interspinous ligaments.

In the clinic

Ligamenta flava

The ligamenta flava are important structures within the vertebral canal. In degenerative conditions of the vertebral column, the ligamenta flava may hypertrophy. This is often associated with hypertrophy and arthritic change of the zygapophysial joints. In combination, zygapophysial joint hypertrophy, ligamenta flava hypertrophy, and a mild disc protrusion can reduce the dimensions of the vertebral canal, producing the syndrome of spinal stenosis.

In the clinic

Vertebral fractures

Vertebral fractures can occur anywhere along the vertebral column. In most instances, the fracture will heal under appropriate circumstances. At the time of injury, it is not the fracture itself but related damage to the contents of the vertebral canal and the surrounding tissues that determines the severity of the patient's condition.

Vertebral column stability is divided into three arbitrary clinical "columns": the **anterior column** consists of the vertebral bodies and the anterior longitudinal ligament; the **middle column** comprises the vertebral body and the posterior longitudinal ligament; and the **posterior column** is made up of the ligamenta flava, interspinous ligaments, supraspinous ligaments, and the ligamentum nuchae in the cervical vertebral column.

Destruction of one of the clinical columns is usually a stable injury requiring little more than rest and appropriate analgesia. Disruption of two columns is highly likely to be unstable and requires fixation and immobilization. A three-column spinal injury usually results in a significant neurological event and requires fixation to prevent further extension of the neurological defect and to create vertebral column stability.

At the craniocervical junction, a complex series of ligaments create stability. If the traumatic incident disrupts craniocervical stability, the chances of a significant spinal cord injury are extremely high. The consequences are quadriplegia. In addition, respiratory function may be compromised by paralysis of the phrenic nerve (which arises from spinal nerves C3 to C5), and severe hypotension (low blood pressure) may result from central disruption of the sympathetic part of the autonomic division of the nervous system.

Mid and lower cervical vertebral column disruption may produce a range of complex neurological problems

SUBASH KC/NMC-15TH/2014

In the clinic—cont'd

involving the upper and lower limbs, although below the level of C5, respiratory function is unlikely to be compromised.

Lumbar vertebral column injuries are rare. When they occur, they usually involve significant force. Knowing that a significant force is required to fracture a vertebra, one must assess the abdominal organs and the rest of the axial skeleton for further fractures and visceral rupture.

Vertebral injuries may also involve the soft tissues and supporting structures between the vertebrae. Typical examples of this are the unifacetal and bifacetal cervical vertebral dislocations that occur in hyperflexion injuries.

Pars interarticularis fractures

The pars interarticularis is a clinical term to describe the specific region of a vertebra between the superior and inferior facet (zygapophysial) joints (Fig. 2.35A). This region is susceptible to trauma, especially in athletes.

If a fracture occurs around the pars interarticularis, the vertebral body may slip anteriorly and compress the vertebral canal.

The most common sites for pars interarticularis fractures are the LIV and LV levels (Fig. 2.35B). (Clinicians often refer to parts of the back in shorthand terms that are not strictly anatomical; for example, facet joints and apophyseal joints are terms used instead of zygapophysial joints, and spinal column is used instead of vertebral column.)

It is possible for a vertebra to slip anteriorly upon its inferior counterpart without a pars interarticularis fracture. Usually this is related to abnormal anatomy of the facet joints, facet joint degenerative change. This disorder is termed **spondylolisthesis.**



Fig. 2.35 Radiograph of lumbar region of vertebral column, oblique view ("Scottie dog"). **A.** Normal radiograph of lumbar region of vertebral column, oblique view. In this view, the transverse process (nose), pedicle (eye), superior articular process (ear), inferior articular process (front leg), and pars interarticularis (neck) resemble a dog. A fracture of the pars interarticularis is visible as a break in the neck of the dog, or the appearance of a collar. **B.** Fracture of pars interarticularis.

SUBASH KC/NMC-15TH/2014

In the clinic

Surgical procedures on the back

Discectomy/laminectomy

A prolapsed intervertebral disc may impinge upon the meningeal (thecal) sac, cord, and most commonly the nerve root, producing symptoms attributable to that level. In some instances the disc protrusion will undergo a degree of involution that may allow symptoms to resolve without intervention. In some instances pain, loss of function, and failure to resolve may require surgery to remove the disc protrusion.

It is of the utmost importance that the level of the disc protrusion is identified before surgery. This may require MRI scanning and on-table fluoroscopy to prevent operating on the wrong level. A midline approach to the right or to the left of the spinous processes will depend upon the most prominent site of the disc bulge. In some instances removal of the lamina will increase the potential space and may relieve symptoms. Some surgeons perform a small fenestration (windowing) within the ligamentum flavum. This provides access to the canal. The meningeal sac and its contents are gently retracted, exposing the nerve root and the offending disc. The disc is dissected free, removing its effect on the nerve root and the canal.

Spinal Fusion

Spinal fusion is performed when it is necessary to fuse one vertebra with the corresponding superior or inferior vertebra, and in some instances multilevel fusion may be necessary. Indications are varied, though they include stabilization after fracture, stabilization related to tumor infiltration, and stabilization when mechanical pain is produced either from the disc or from the posterior elements.

There are a number of surgical methods in which a fusion can be performed, through either a posterior approach and fusing the posterior elements, an anterior approach by removal of the disc and either disc replacement or anterior fusion, or in some instances a 360° fusion where the posterior elements and the vertebral bodies are fused.

BACK MUSCULATURE

Muscles of the back are organized into superficial, intermediate, and deep groups.

Muscles in the superficial and intermediate groups are extrinsic muscles because they originate embryologically from locations other than the back. They are innervated by anterior rami of spinal nerves:

- The superficial group consists of muscles related to and involved in movements of the upper limb.
- The intermediate group consists of muscles attached to the ribs and may serve a respiratory function.

Muscles of the deep group are intrinsic muscles because they develop in the back. They are innervated by posterior rami of spinal nerves and are directly related to movements of the vertebral column and head.

Superficial group of back muscles

The muscles in the superficial group are immediately deep to the skin and superficial fascia (Figs. 2.36 to 2.39). They attach the superior part of the appendicular skeleton (clavicle, scapula, and humerus) to the axial skeleton (skull, ribs, and vertebral column). Because these muscles are primarily involved with movements of this part of the appendicular skeleton, they are sometimes referred to as the **appendicular group**.

Muscles in the superficial group include the trapezius, latissimus dorsi, rhomboid major, rhomboid minor, and levator scapulae. The rhomboid major, rhomboid minor, and levator scapulae muscles are located deep to the trapezius muscle in the superior part of the back.

SUBASH KC/NMC-15TH/2014

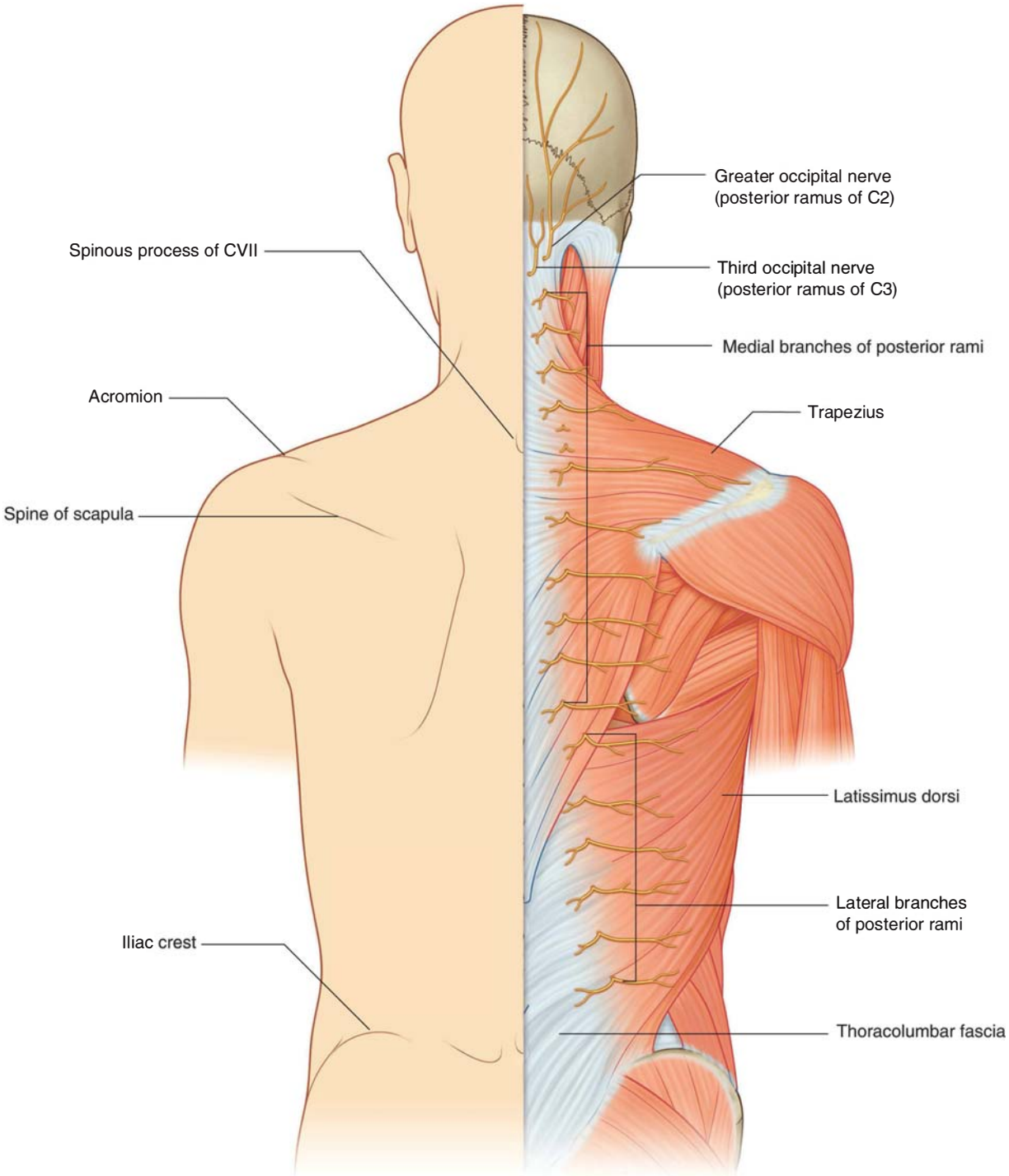


Fig. 2.36 Superficial group of back muscles—trapezius and latissimus dorsi.

SUBASH KC/NMC-15TH/2014

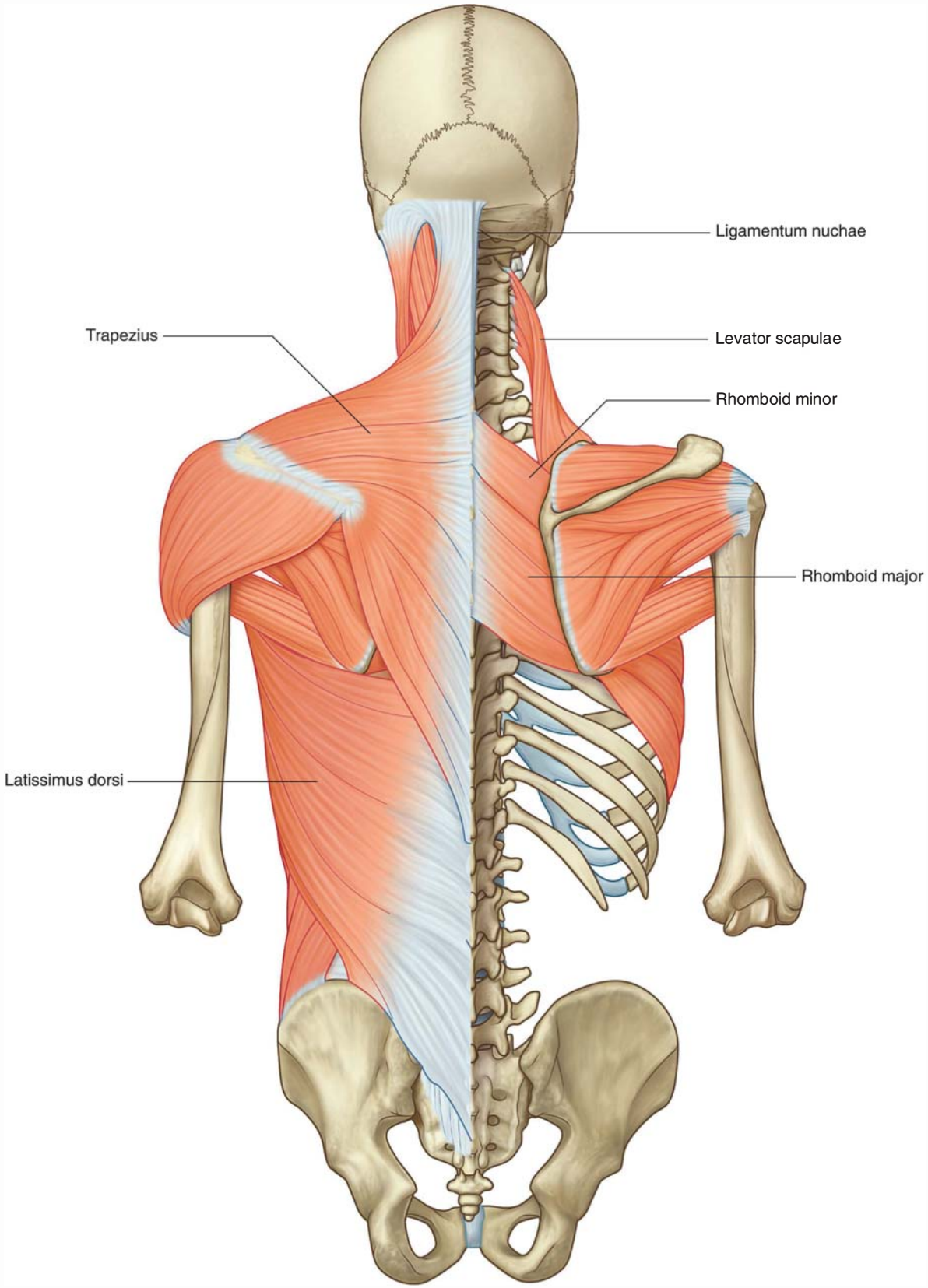


Fig. 2.37 Superficial group of back muscles—trapezius and latissimus dorsi, with rhomboid major, rhomboid minor, and levator scapulae located deep to trapezius in the superior part of the back.

SUBASH KC/NMC-15TH/2014

Trapezius

Each **trapezius** muscle is flat and triangular, with the base of the triangle situated along the vertebral column (the muscle's origin) and the apex pointing toward the tip of the shoulder (the muscle's insertion) (Fig. 2.37 and Table 2.1). The muscles on both sides together form a trapezoid.

The superior fibers of the trapezius, from the skull and upper portion of the vertebral column, descend to attach to the lateral third of the clavicle and to the acromion of the scapula. Contraction of these fibers elevates the scapula. In addition, the superior and inferior fibers work together to rotate the lateral aspect of the scapula upward, which needs to occur when raising the upper limb above the head.

Motor innervation of the trapezius is by the accessory nerve [XI], which descends from the neck onto the deep surface of the muscle (Fig. 2.38). Proprioceptive fibers from the trapezius pass in the branches of the cervical plexus and enter the spinal cord at spinal cord levels C3 and C4.

The blood supply to the trapezius is from the superficial branch of the transverse cervical artery, the acromial branch of the suprascapular artery, and the dorsal branches of posterior intercostal arteries.

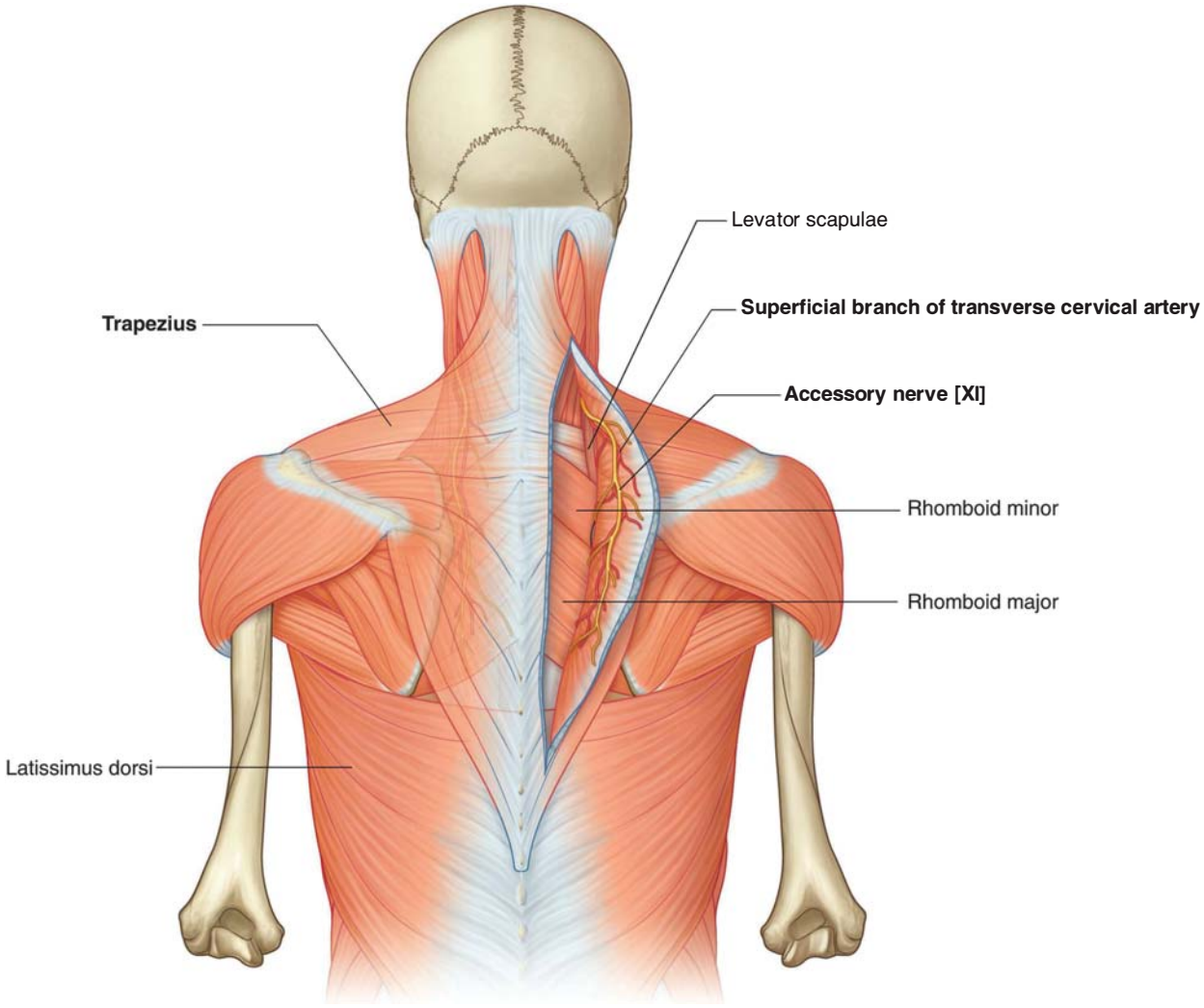


Fig. 2.38 Innervation and blood supply of trapezius.

SUBASH KC/NMC-15TH/2014

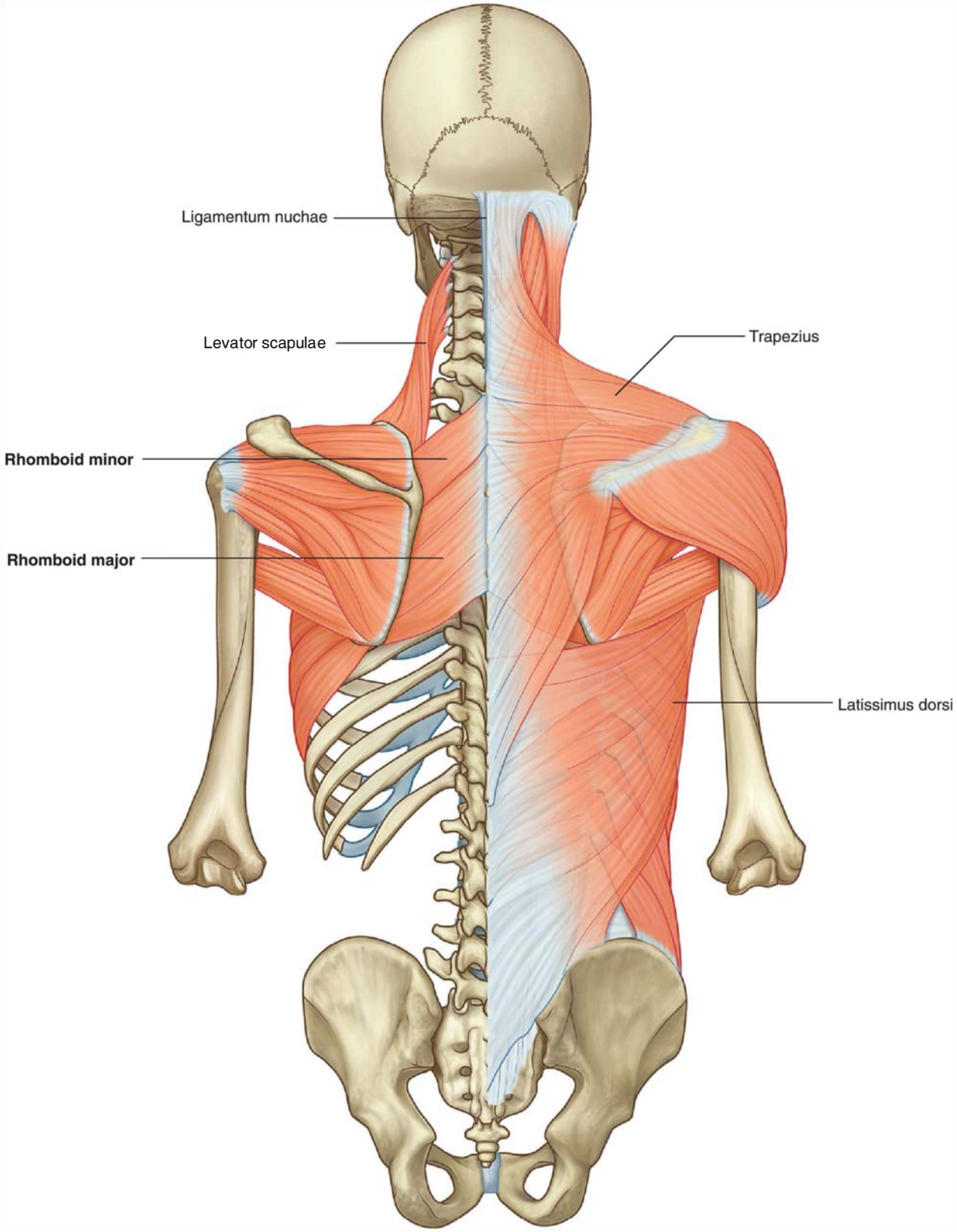


Fig. 2.39 Rhomboid muscles and levator scapulae.

SUBASH KC/NMC-15TH/2014

Table 2.1 Superficial (appendicular) group of back muscles

Muscle	Origin	Insertion	Innervation	Function
Trapezius	Superior nuchal line, external occipital protuberance, ligamentum nuchae, spinous processes of CVII to TXII	Lateral one third of clavicle, acromion, spine of scapula	Motor—accessory nerve [XI]; proprioception—C3 and C4	Assists in rotating the scapula during abduction of humerus above horizontal; upper fibers elevate, middle fibers adduct, and lower fibers depress scapula
Latissimus dorsi	Spinous processes of TVII to LV and sacrum, iliac crest, ribs X to XII	Floor of intertubercular sulcus of humerus	Thoracodorsal nerve (C6 to C8)	Extends, adducts, and medially rotates humerus
Levator scapulae	Transverse processes of CI to CIV	Upper portion medial border of scapula	C3 to C4 and dorsal scapular nerve (C4, C5)	Elevates scapula
Rhomboid major	Spinous processes of TII to TV	Medial border of scapula between spine and inferior angle	Dorsal scapular nerve (C4, C5)	Retracts (adducts) and elevates scapula
Rhomboid minor	Lower portion of ligamentum nuchae, spinous processes of CVII and TI	Medial border of scapula at spine of scapula	Dorsal scapular nerve (C4, C5)	Retracts (adducts) and elevates scapula

SUBASH KC/NMC-15TH/2014

Latissimus dorsi

Latissimus dorsi is a large, flat triangular muscle that begins in the lower portion of the back and tapers as it ascends to a narrow tendon that attaches to the humerus anteriorly (Figs. 2.36 to 2.39 and Table 2.1). As a result, movements associated with this muscle include extension, adduction, and medial rotation of the upper limb. The latissimus dorsi can also depress the shoulder, preventing its upward movement.

The thoracodorsal nerve of the brachial plexus innervates the latissimus dorsi muscle. Associated with this nerve is the thoracodorsal artery, which is the primary blood supply of the muscle. Additional small arteries come from dorsal branches of posterior intercostal and lumbar arteries.

Levator scapulae

Levator scapulae is a slender muscle that descends from the transverse processes of the upper cervical vertebrae to the upper portion of the scapula on its medial border at the superior angle (Fig. 2.37 and 2.39, and Table 2.1). It elevates the scapula and may assist other muscles in rotating the lateral aspect of the scapula inferiorly.

The levator scapulae is innervated by branches from the anterior rami of spinal nerves C3 and C4 and the dorsal scapular nerve, and its arterial supply consists of branches primarily from the transverse and ascending cervical arteries.

Rhomboid minor and rhomboid major

The two rhomboid muscles are inferior to levator scapulae (Fig. 2.39 and Table 2.1). **Rhomboid minor** is superior to rhomboid major, and is a small, cylindrical muscle that arises from the ligamentum nuchae of the neck and the spinous processes of vertebrae CVII and TI and attaches to the medial scapular border opposite the root of the spine of the scapula.

The larger **rhomboid major** originates from the spinous processes of the upper thoracic vertebrae and attaches to the medial scapular border inferior to rhomboid minor.

The two rhomboid muscles work together to retract or pull the scapula toward the vertebral column. With other

muscles they may also rotate the lateral aspect of the scapula inferiorly.

The dorsal scapular nerve, a branch of the brachial plexus, innervates both rhomboid muscles (Fig. 2.40).

Intermediate group of back muscles

The muscles in the intermediate group of back muscles consist of two thin muscular sheets in the superior and inferior regions of the back, immediately deep to the muscles in the superficial group (Fig. 2.41 and Table 2.2). Fibers from these two serratus posterior muscles (**serratus posterior superior** and **serratus posterior inferior**) pass obliquely outward from the vertebral column to attach to the ribs. This positioning suggests a respiratory function, and at times, these muscles have been referred to as the respiratory group.

Serratus posterior superior is deep to the rhomboid muscles, whereas serratus posterior inferior is deep to the latissimus dorsi. Both serratus posterior muscles are attached to the vertebral column and associated structures medially, and either descend (the fibers of the serratus posterior superior) or ascend (the fibers of the serratus posterior inferior) to attach to the ribs. These two muscles therefore elevate and depress the ribs.

The serratus posterior muscles are innervated by segmental branches of anterior rami of intercostal nerves. Their vascular supply is provided by a similar segmental pattern through the intercostal arteries.

SUBASH KC/NMC-15TH/2014

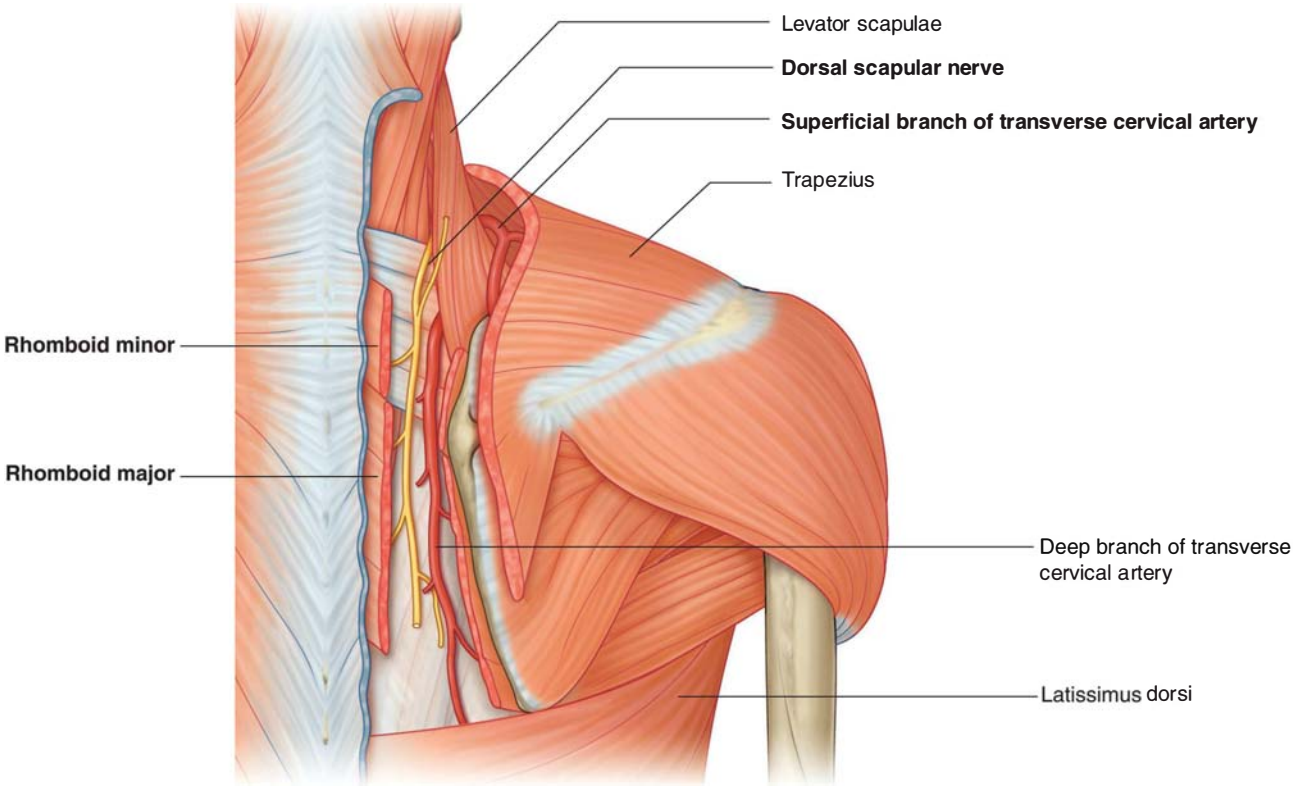


Fig. 2.40 Innervation and blood supply of the rhomboid muscles.

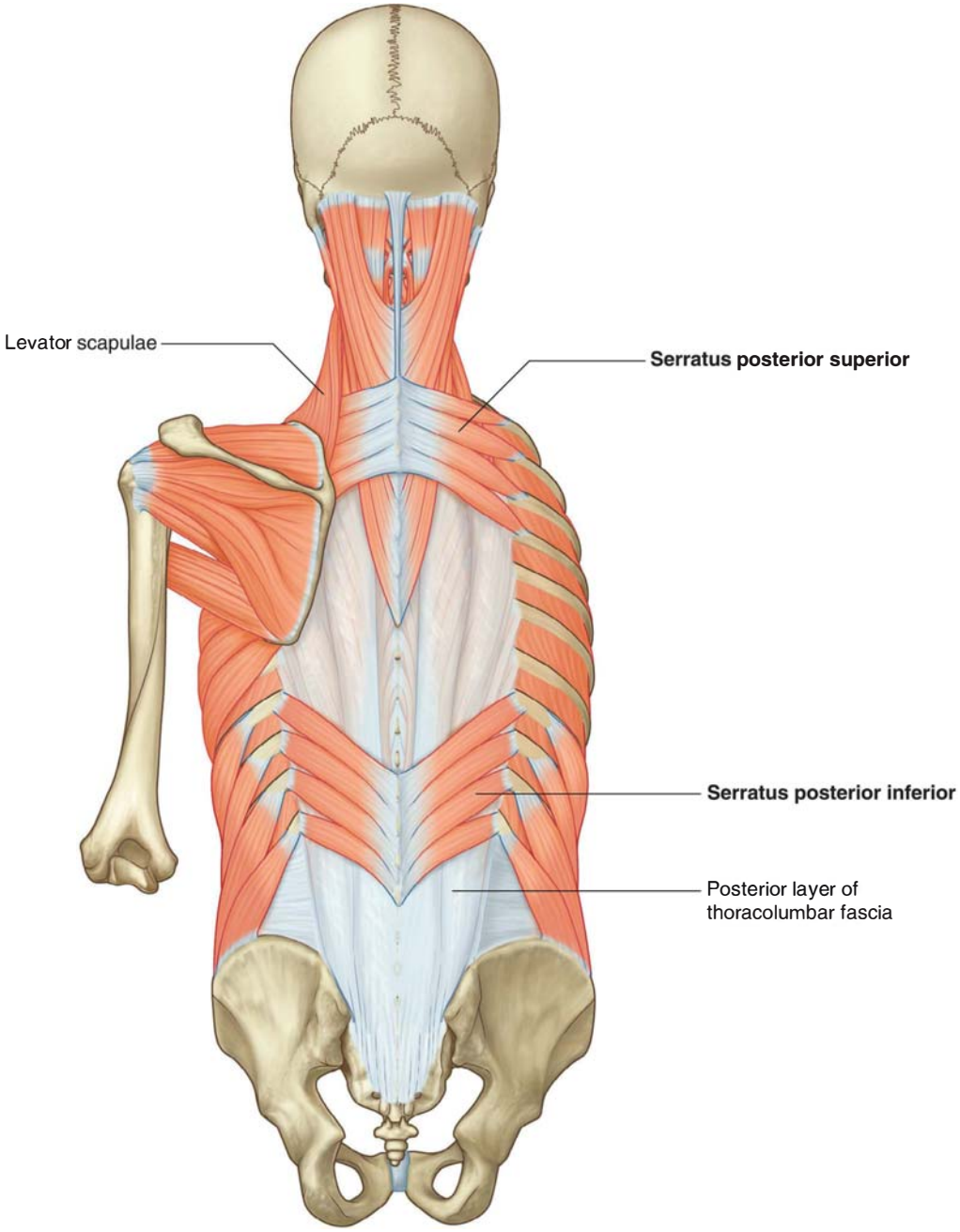


Fig. 2.41 Intermediate group of back muscles—serratus posterior muscles.

Table 2.2 Intermediate (respiratory) group of back muscles

Muscle	Origin	Insertion	Innervation	Function
Serratus posterior superior	Lower portion of ligamentum nuchae, spinous processes of CVII to TIII, and supraspinous ligaments	Upper border of ribs II to V just lateral to their angles	Anterior rami of upper thoracic nerves (T2 to T5)	Elevates ribs II to V
Serratus posterior inferior	Spinous processes of TXI to LIII and supraspinous ligaments	Lower border of ribs IX to XII just lateral to their angles	Anterior rami of lower thoracic nerves (T9 to T12)	Depresses ribs IX to XII and may prevent lower ribs from being elevated when the diaphragm contracts

SUBASH KC/NMC-15TH/2014

Deep group of back muscles

The deep or intrinsic muscles of the back extend from the pelvis to the skull and are innervated by segmental branches of the posterior rami of spinal nerves. They include:

- the extensors and rotators of the head and neck—the splenius capitis and cervicis (spinotransversales muscles),
- the extensors and rotators of the vertebral column—the erector spinae and transversospinales, and
- the short segmental muscles—the interspinales and intertransversarii.

The vascular supply to this deep group of muscles is through branches of the vertebral, deep cervical, occipital, transverse cervical, posterior intercostal, subcostal, lumbar, and lateral sacral arteries.

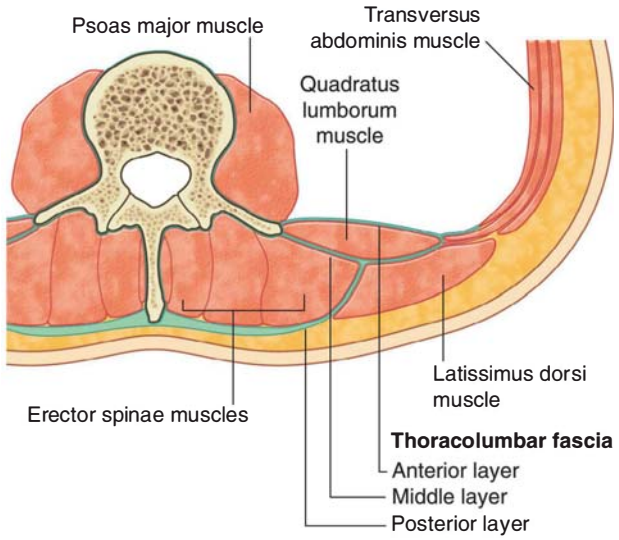


Fig. 2.42 Thoracolumbar fascia and the deep back muscles (transverse section).

Thoracolumbar fascia

The **thoracolumbar fascia** covers the deep muscles of the back and trunk (Fig. 2.42). This fascial layer is critical to the overall organization and integrity of the region:

- Superiorly, it passes anteriorly to the serratus posterior muscle and is continuous with deep fascia in the neck.
- In the thoracic region, it covers the deep muscles and separates them from the muscles in the superficial and intermediate groups.
- Medially, it attaches to the spinous processes of the thoracic vertebrae and, laterally, to the angles of the ribs.

The medial attachments of the latissimus dorsi and serratus posterior inferior muscles blend into the thoracolumbar fascia. In the lumbar region, the thoracolumbar fascia consists of three layers:

- The posterior layer is thick and is attached to the spinous processes of the lumbar vertebrae and sacral vertebrae and to the supraspinous ligament—from these attachments, it extends laterally to cover the erector spinae.
- The middle layer is attached medially to the tips of the transverse processes of the lumbar vertebrae and intertransverse ligaments—inferiorly, it is attached to the iliac crest and, superiorly, to the lower border of rib XII.
- The anterior layer covers the anterior surface of the quadratus lumborum muscle (a muscle of the posterior abdominal wall) and is attached medially to the transverse processes of the lumbar vertebrae—inferiorly, it is attached to the iliac crest and, superiorly, it forms the lateral arcuate ligament for attachment of the diaphragm.

The posterior and middle layers of the thoracolumbar fascia come together at the lateral margin of the erector spinae (Fig. 2.42). At the lateral border of the quadratus lumborum, the anterior layer joins them and forms the aponeurotic origin for the transversus abdominis muscle of the abdominal wall.

Spinotransversales muscles

The two spinotransversales muscles run from the spinous processes and ligamentum nuchae upward and laterally (Fig. 2.43 and Table 2.3):

- The splenius capitis is a broad muscle attached to the occipital bone and mastoid process of the temporal bone.
- The splenius cervicis is a narrow muscle attached to the transverse processes of the upper cervical vertebrae.

Together the spinotransversales muscles draw the head backward, extending the neck. Individually, each muscle rotates the head to one side—the same side as the contracting muscle.

SUBASH KC/NMC-15TH/2014

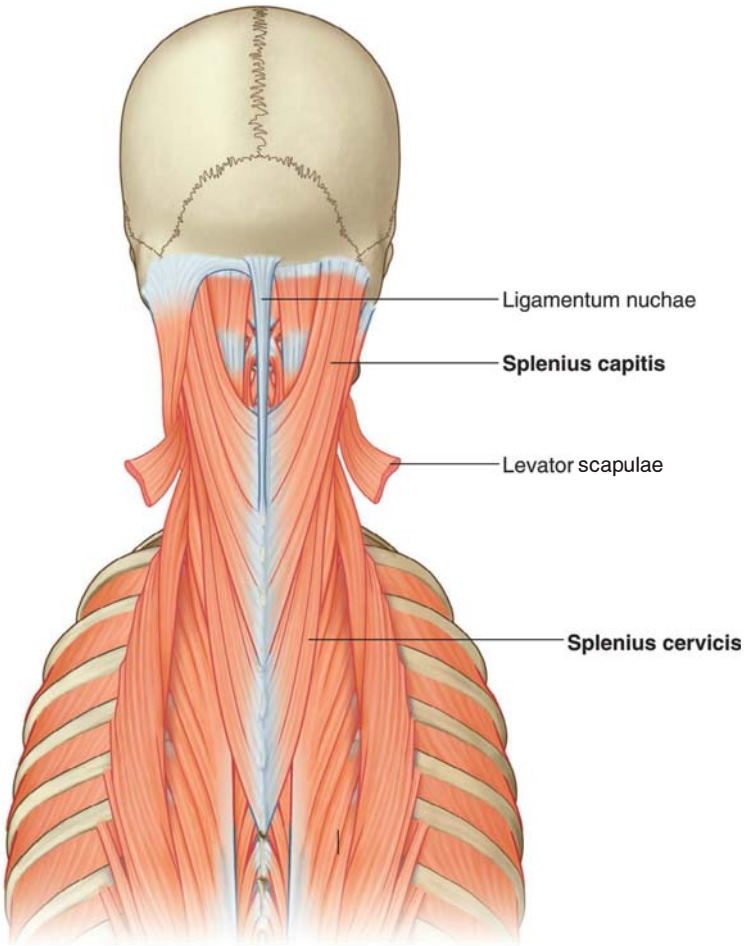


Fig. 2.43 Deep group of back muscles—spinotransversales muscles (splenius capitis and splenius cervicis).

Table 2.3 Spinotransversales muscles

Muscle	Origin	Insertion	Innervation	Function
Splenius capitis	Lower half of ligamentum nuchae, spinous processes of CVII to TIV	Mastoid process, skull below lateral one third of superior nuchal line	Posterior rami of middle cervical nerves	Together—draw head backward, extending neck; individually—draw and rotate head to one side (turn face to same side)
Splenius cervicis	Spinous processes of TIII to TVI	Transverse processes of CI to CIII	Posterior rami of lower cervical nerves	Together—extend neck; individually—draw and rotate head to one side (turn face to same side)

Erector spinae muscles

The erector spinae is the largest group of intrinsic back muscles. The muscles lie posterolaterally to the vertebral column between the spinous processes medially and the angles of the ribs laterally. They are covered in the thoracic and lumbar regions by thoracolumbar fascia and the serratus posterior inferior, rhomboid, and splenius muscles. The mass arises from a broad, thick tendon attached to the sacrum, the spinous processes of the lumbar and lower thoracic vertebrae, and the iliac crest (Fig. 2.44 and Table 2.4). It divides in the upper lumbar region into three vertical columns of muscle, each of which is further subdivided regionally (lumborum, thoracis, cervicis, and capitis), depending on where the muscles attach superiorly.

SUBASH KC/NMC-15TH/2014

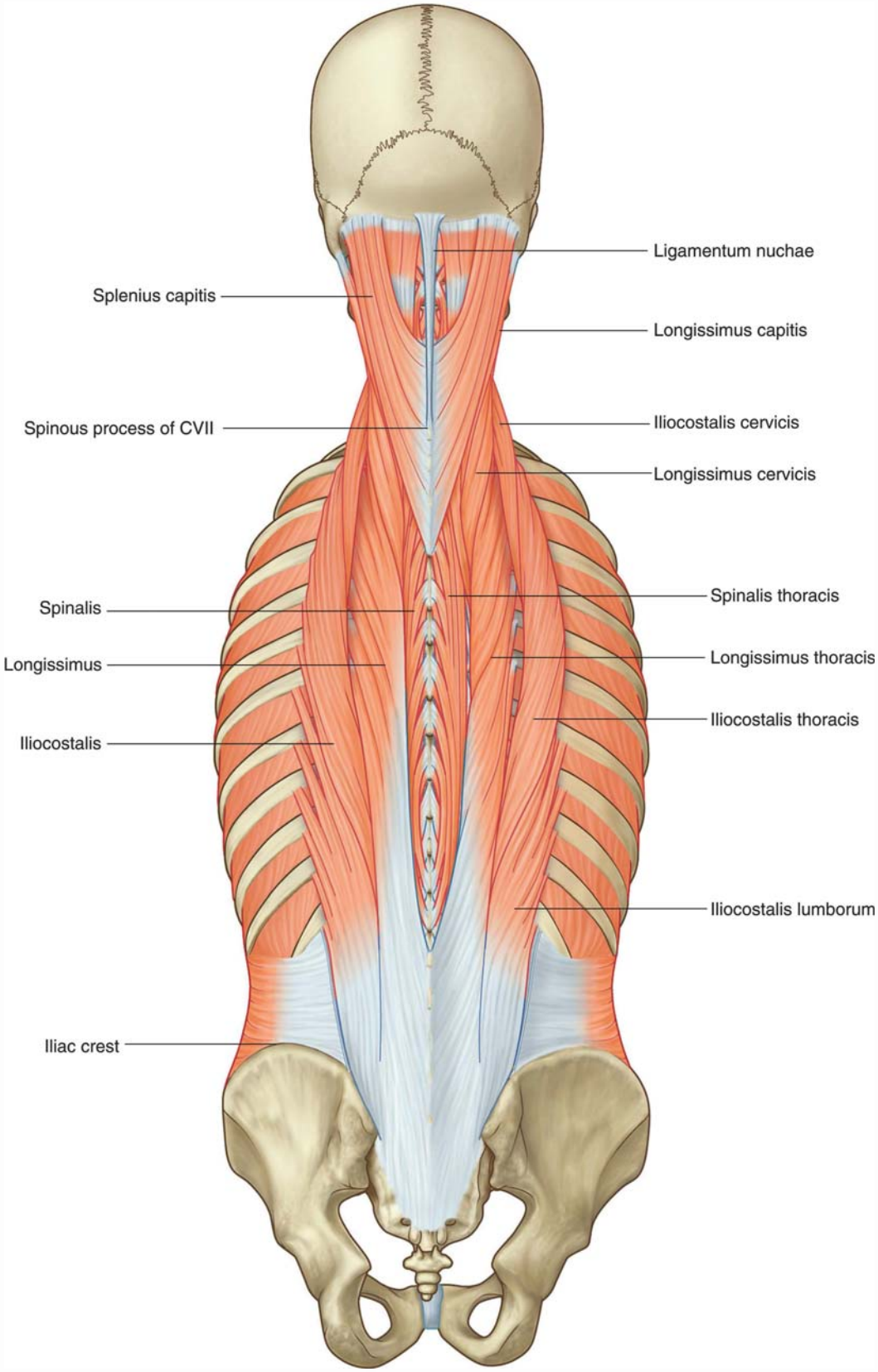


Fig. 2.44 Deep group of back muscles—erector spinae muscles.

SUBASH KC/NMC-15TH/2014

Table 2.4 Erector spinae group of back muscles

Muscle	Origin	Insertion
Iliocostalis lumborum	Sacrum, spinous processes of lumbar and lower two thoracic vertebrae and their supraspinous ligaments, and the iliac crest	Angles of the lower six or seven ribs
Iliocostalis thoracis	Angles of the lower six ribs	Angles of the upper six ribs and the transverse process of CVII
Iliocostalis cervicis	Angles of ribs III to VI	Transverse processes of CIV to CVI
Longissimus thoracis	Blends with iliocostalis in lumbar region and is attached to transverse processes of lumbar vertebrae	Transverse processes of all thoracic vertebrae and just lateral to the tubercles of the lower nine or ten ribs
Longissimus cervicis	Transverse processes of upper four or five thoracic vertebrae	Transverse processes of CII to CVI
Longissimus capitis	Transverse processes of upper four or five thoracic vertebrae and articular processes of lower three or four cervical vertebrae	Posterior margin of the mastoid process
Spinalis thoracis	Spinous processes of TX or TXI to LII	Spinous processes of TI to TVIII (varies)
Spinalis cervicis	Lower part of ligamentum nuchae and spinous process of CVII (sometimes TI to TII)	Spinous process of CII (axis)
Spinalis capitis	Usually blends with semispinalis capitis	With semispinalis capitis

SUBASH KC/NMC-15TH/2014

- The outer or most laterally placed column of the erector spinae muscles is the **iliocostalis**, which is associated with the costal elements and passes from the common tendon of origin to multiple insertions into the angles of the ribs and the transverse processes of the lower cervical vertebrae.
- The middle or intermediate column is the **longissimus**, which is the largest of the erector spinae subdivision extending from the common tendon of origin to the base of the skull. Throughout this vast expanse, the lateral positioning of the longissimus muscle is in the area of the transverse processes of the various vertebrae.
- The most medial muscle column is the **spinalis**, which is the smallest of the subdivisions and interconnects the spinous processes of adjacent vertebrae. The spinalis is most constant in the thoracic region and is generally absent in the cervical region. It is associated with a deeper muscle (the semispinalis capitis) as the erector spinae group approaches the skull.

The muscles in the erector spinae group are the primary extensors of the vertebral column and head. Acting bilaterally, they straighten the back, returning it to the upright position from a flexed position, and pull the head posteriorly. They also participate in controlling vertebral column flexion by contracting and relaxing in a coordinated fashion. Acting unilaterally, they bend the vertebral column laterally. In addition, unilateral contractions of muscles attached to the head turn the head to the actively contracting side.

Transversospinales muscles

The transversospinales muscles run obliquely upward and medially from transverse processes to spinous processes, filling the groove between these two vertebral projections (Fig. 2.45 and Table 2.5). They are deep to the erector spinae and consist of three major subgroups—the semispinalis, multifidus, and rotatores muscles.

- The **semispinalis** muscles are the most superficial collection of muscle fibers in the transversospinales group. These muscles begin in the lower thoracic region and end by attaching to the skull, crossing between four and six vertebrae from their point of origin to point of attachment. Semispinalis muscles are found in the thoracic and cervical regions, and attach to the occipital bone at the base of the skull.
- Deep to the semispinalis is the second group of muscles, the **multifidus**. Muscles in this group span the length of the vertebral column, passing from a lateral point of origin upward and medially to attach to spinous processes and spanning between two and four vertebrae. The multifidus muscles are present throughout the length of the vertebral column but are best developed in the lumbar region.
- The small **rotatores** muscles are the deepest of the transversospinales group. They are present throughout the length of the vertebral column but are best developed in the thoracic region. Their fibers pass upward and medially from transverse processes to spinous processes crossing two vertebrae (long rotators) or attaching to an adjacent vertebra (short rotators).

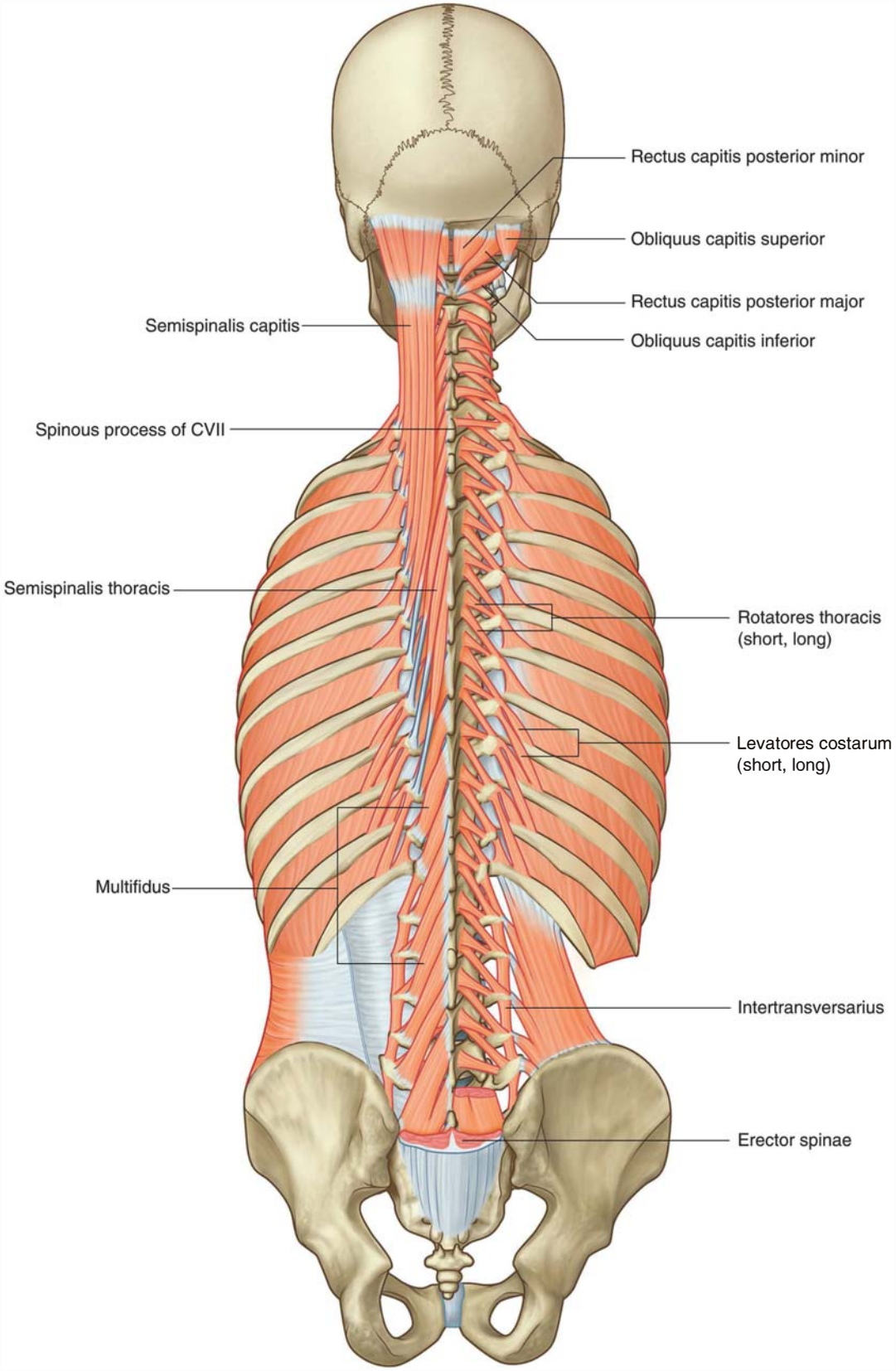


Fig. 2.45 Deep group of back muscles—transversospinales and segmental muscles.

SUBASH KC/NMC-15TH/2014

Table 2.5 Transversospinales group of back muscles

Muscle	Origin	Insertion
Semispinalis thoracis	Transverse processes of TVI to TX	Spinous processes of upper four thoracic and lower two cervical vertebrae
Semispinalis cervicis	Transverse processes of upper five or six thoracic vertebrae	Spinous processes of CII (axis) to CV
Semispinalis capitis	Transverse processes of TI to TVI (or TVII) and CVII and articular processes of CIV to CVI	Medial area between the superior and inferior nuchal lines of occipital bone
Multifidus	Sacrum, origin of erector spinae, posterior superior iliac spine, mammillary processes of lumbar vertebrae, transverse processes of thoracic vertebrae, and articular processes of lower four cervical vertebrae	Base of spinous processes of all vertebrae from LV to CII (axis)
Rotatores lumborum	Transverse processes of lumbar vertebrae	Spinous processes of lumbar vertebrae
Rotatores thoracis	Transverse processes of thoracic vertebrae	Spinous processes of thoracic vertebrae
Rotatores cervicis	Articular processes of cervical vertebrae	Spinous processes of cervical vertebrae

When muscles in the transversospinales group contract bilaterally, they extend the vertebral column, an action similar to that of the erector spinae group. However, when muscles on only one side contract, they pull the spinous processes toward the transverse processes on that side, causing the trunk to turn or rotate in the opposite direction.

One muscle in the transversospinales group, the **semispinalis capitis**, has a unique action because it attaches to the skull. Contracting bilaterally, this muscle pulls the head posteriorly, whereas unilateral contraction pulls the head posteriorly and turns it, causing the chin to move superiorly and turn toward the side of the contracting muscle. These actions are similar to those of the upper erector spinae.

Segmental muscles

The two groups of segmental muscles (Fig. 2.45 and Table 2.6) are deeply placed in the back and innervated by posterior rami of spinal nerves.

- The first group of segmental muscles are the **levatores costarum** muscles, which arise from the transverse processes of vertebrae CVII and TI to TXI. They have an oblique lateral and downward direction and insert into the rib below the vertebra of origin in the area of the tubercle. Contraction elevates the ribs.
- The second group of segmental muscles are the true segmental muscles of the back—the **interspinales**, which pass between adjacent spinous processes, and the **intertransversarii,** which pass between adjacent transverse processes. These postural muscles stabilize adjoining vertebrae during movements of the vertebral column to allow more effective action of the large muscle groups.

Suboccipital muscles

A small group of deep muscles in the upper cervical region at the base of the occipital bone move the head. They connect vertebra CI (the atlas) to vertebra CII (the axis) and connect both vertebrae to the base of the skull. Because of their location they are sometimes referred to as suboccipital muscles (Figs. 2.45 and 2.46 and Table 2.7). They include, on each side:

Table 2.6 Segmental back muscles

Muscle	Origin	Insertion	Function
Levatores costarum	Short paired muscles arising from transverse processes of CVII to TXI	The rib below vertebra of origin near tubercle	Contraction elevates rib
Interspinales	Short paired muscles attached to the spinous processes of contiguous vertebrae, one on each side of the interspinous ligament		Postural muscles that stabilize adjoining vertebrae during movements of vertebral column
Intertransversarii	Small muscles between the transverse processes of contiguous vertebrae		Postural muscles that stabilize adjoining vertebrae during movements of vertebral column

SUBASH KC/NMC-15TH/2014

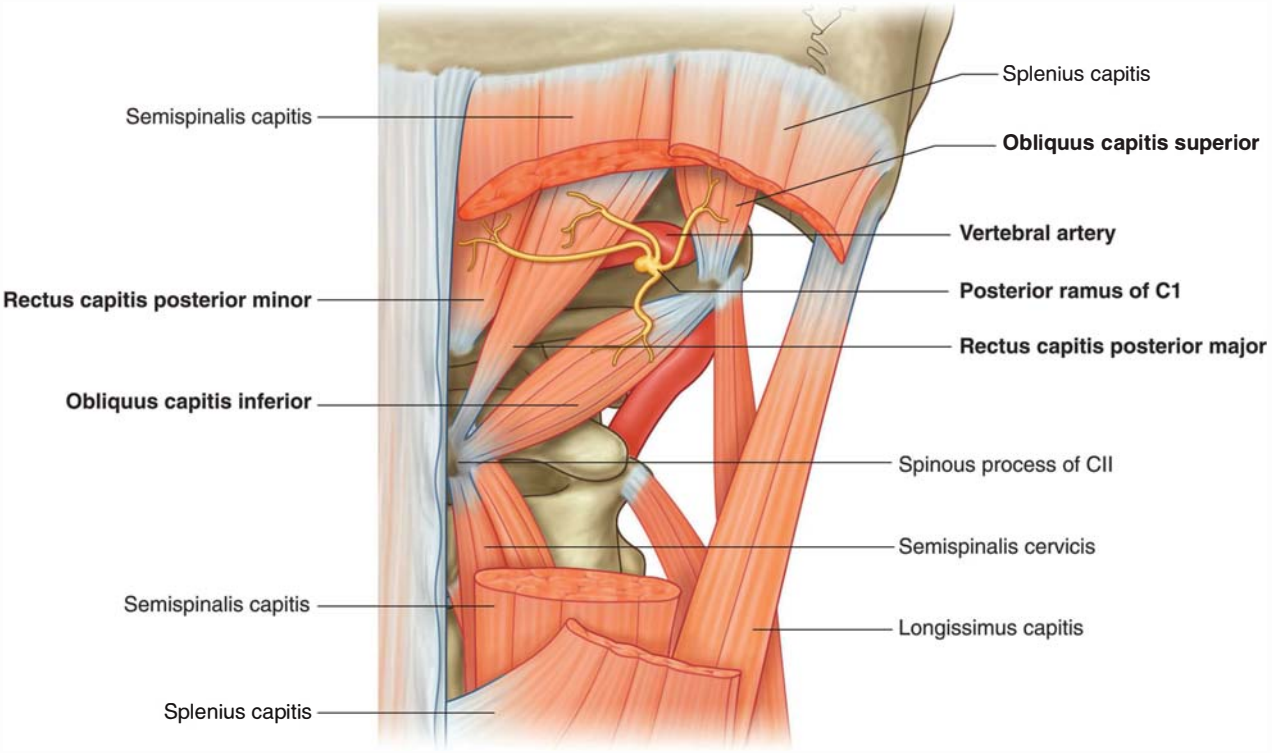


Fig. 2.46 Deep group of back muscles—suboccipital muscles. This also shows the borders of the suboccipital triangle.

Table 2.7 Suboccipital group of back muscles

Muscle	Origin	Insertion	Innervation	Function
Rectus capitis posterior major	Spinous process of axis (CII)	Lateral portion of occipital bone below inferior nuchal line	Posterior ramus of C1	Extension of head; rotation of face to same side as muscle
Rectus capitis posterior minor	Posterior tubercle of atlas (CI)	Medial portion of occipital bone below inferior nuchal line	Posterior ramus of C1	Extension of head
Obliquus capitis superior	Transverse process of atlas (CI)	Occipital bone between superior and inferior nuchal lines	Posterior ramus of C1	Extension of head and bends it to same side
Obliquus capitis inferior	Spinous process of axis (CII)	Transverse process of atlas (CI)	Posterior ramus of C1	Rotation of face to same side

- **rectus capitis posterior major,**
- **rectus capitis posterior minor,**
- **obliquus capitis inferior, and**
- **obliquus capitis superior.**

Contraction of the suboccipital muscles extends the head at the atlanto-axial joint.

The suboccipital muscles are innervated by the posterior ramus of the first cervical nerve, which enters the area between the vertebral artery and the posterior arch of the atlas (Fig. 2.46). The vascular supply to the muscles in this area is from branches of the vertebral and occipital arteries.

The suboccipital muscles form the boundaries of the **suboccipital triangle**, an area that contains several important structures (Fig. 2.46):

- The rectus capitis posterior major muscle forms the medial border of the triangle.
- The obliquus capitis superior muscle forms the lateral border.
- The obliquus capitis inferior muscle forms the inferior border.

SUBASH KC/NMC-15TH/2014

The contents of the area outlined by these muscles are the posterior ramus of C1, the vertebral artery, and associated veins.

In the clinic

Nerve injuries affecting superficial back muscles

Weakness in the trapezius, caused by an interruption of the accessory nerve [XI], may appear as drooping of the shoulder, inability to raise the arm above the head because of impaired rotation of the scapula, or weakness in attempting to raise the shoulder (i.e., shrug the shoulder against resistance).

A weakness in, or an inability to use, the latissimus dorsi, resulting from an injury to the thoracodorsal nerve, diminishes the capacity to pull the body upward while climbing or doing a pull-up.

An injury to the dorsal scapular nerve, which innervates the rhomboids, may result in a lateral shift in the position of the scapula on the affected side (i.e., the normal position of the scapula is lost because of the affected muscle's inability to prevent antagonistic muscles from pulling the scapula laterally).

SPINAL CORD

The spinal cord extends from the foramen magnum to approximately the level of the disc between vertebrae LI and LII in adults, although it can end as high as vertebra TXII or as low as the disc between vertebrae LII and LIII (Fig. 2.47). In neonates, the spinal cord extends approximately to vertebra LIII but can reach as low as vertebra LIV. The distal end of the cord (the **conus medullaris**) is cone shaped. A fine filament of connective tissue (the pial part of the **filum terminale**) continues inferiorly from the apex of the **conus medullaris**.

The spinal cord is not uniform in diameter along its length. It has two major swellings or enlargements in regions associated with the origin of spinal nerves that innervate the upper and lower limbs. A **cervical enlargement** occurs in the region associated with the origins of spinal nerves C5 to T1, which innervate the upper limbs. A lumbosacral enlargement occurs in the region associated with the origins of spinal nerves L1 to S3, which innervate the lower limbs.

The external surface of the spinal cord is marked by a number of fissures and sulci (Fig. 2.48):

- The **anterior median fissure** extends the length of the anterior surface.

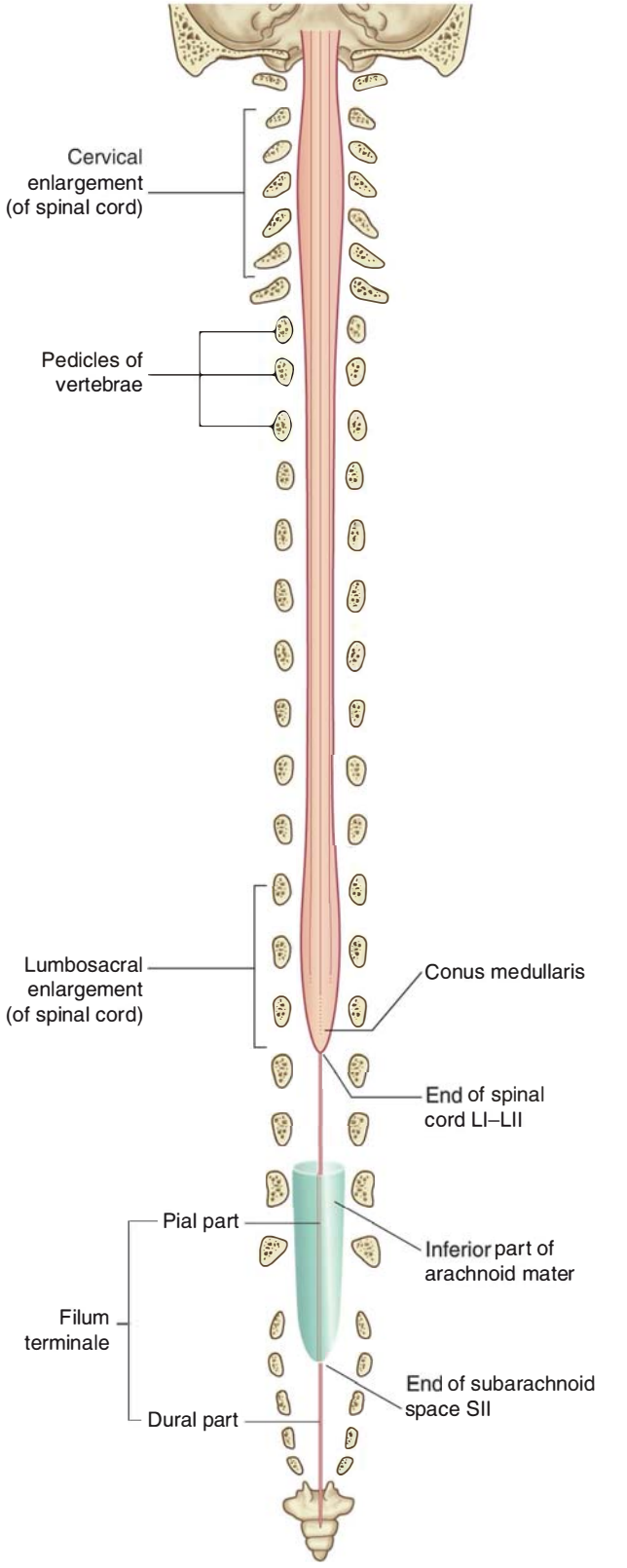


Fig. 2.47 Spinal cord.

SUBASH KC/NMC-15TH/2014

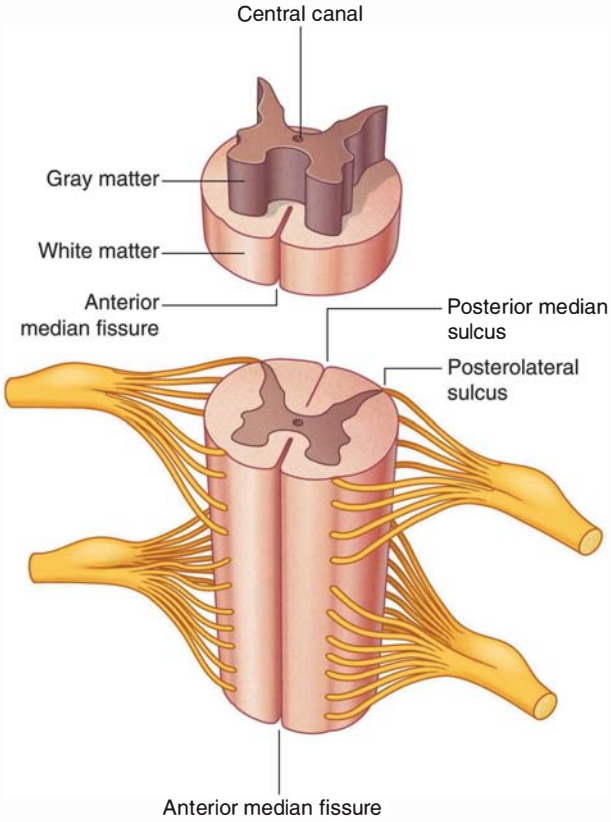


Fig. 2.48 Features of the spinal cord.

- The **posterior median sulcus** extends along the posterior surface.
- The **posterolateral sulcus** on each side of the posterior surface marks where the posterior rootlets of spinal nerves enter the cord.

Internally, the cord has a small central canal surrounded by gray and white matter:

- The gray matter is rich in nerve cell bodies, which form longitudinal columns along the cord, and in cross section these columns form a characteristic H-shaped appearance in the central regions of the cord.
- The white matter surrounds the gray matter and is rich in nerve cell processes, which form large bundles or tracts that ascend and descend in the cord to other spinal cord levels or carry information to and from the brain.

Vasculature

Arteries

The arterial supply to the spinal cord comes from two sources (Fig. 2.49). It consists of:

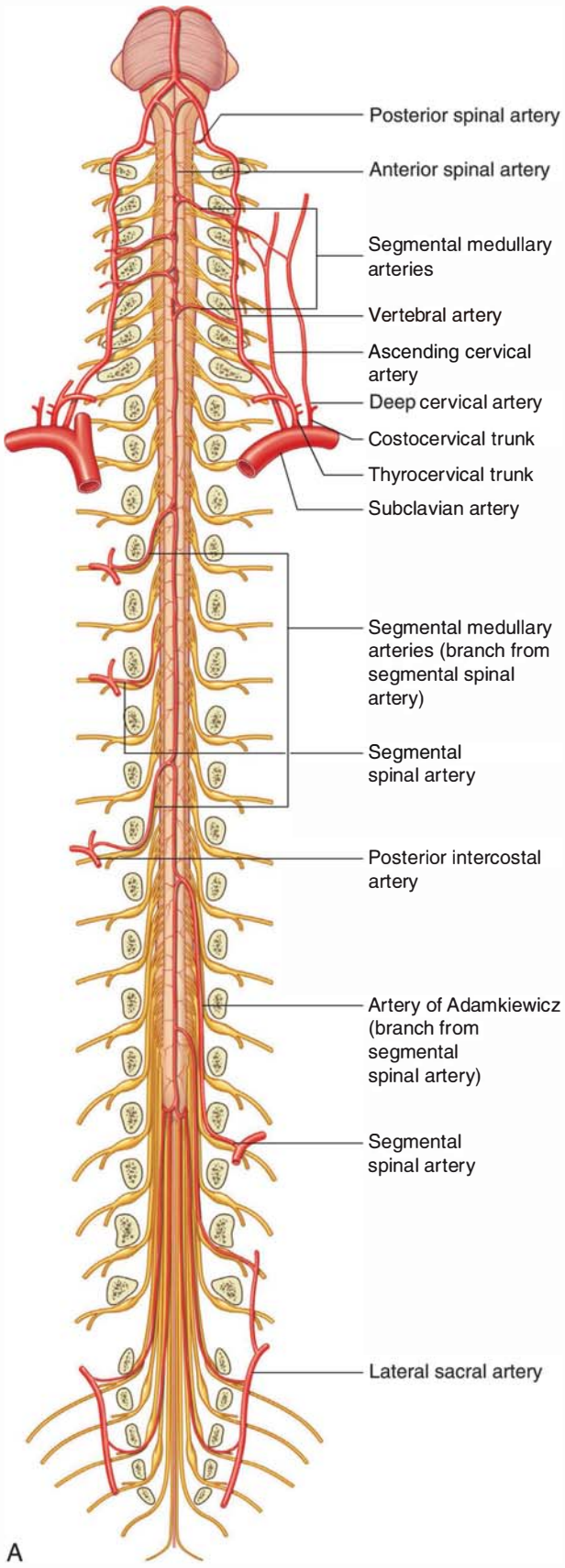


Fig. 2.49 Arteries that supply the spinal cord. **A.** Anterior view of spinal cord (not all segmental spinal arteries are shown).

SUBASH KC/NMC-15TH/2014

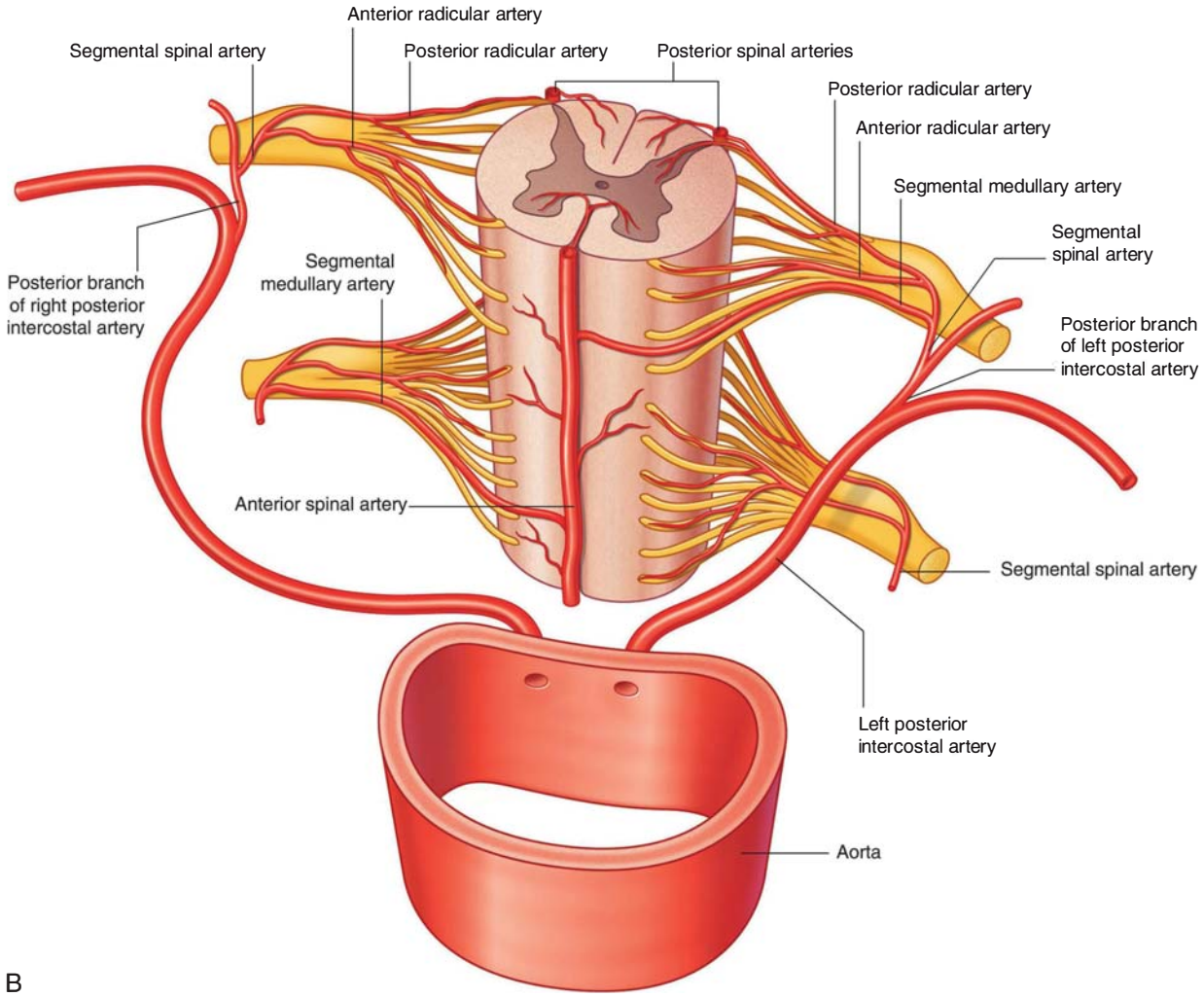


Fig. 2.49, cont'd B. Segmental supply of spinal cord.

- longitudinally oriented vessels, arising superior to the cervical portion of the cord, which descend on the surface of the cord; and
- feeder arteries that enter the vertebral canal through the intervertebral foramina at every level; these feeder vessels, or **segmental spinal arteries**, arise predominantly from the vertebral and deep cervical arteries in the neck, the posterior intercostal arteries in the thorax, and the lumbar arteries in the abdomen.

After entering an intervertebral foramen, the segmental spinal arteries give rise to **anterior** and **posterior radicular arteries** (Fig. 2.49). This occurs at every vertebral level. The radicular arteries follow, and supply, the anterior and posterior roots. At various vertebral levels, the **segmental spinal arteries** also give off **segmental medullary arteries** (Fig. 2.49). These vessels pass directly to the longitudinally oriented vessels, reinforcing these.

The longitudinal vessels consist of:

- a single **anterior spinal artery**, which originates within the cranial cavity as the union of two vessels that arise from the vertebral arteries—the resulting single anterior spinal artery passes inferiorly, approximately parallel to the anterior median fissure, along the surface of the spinal cord; and
- two **posterior spinal arteries**, which also originate in the cranial cavity, usually arising directly from a terminal branch of each vertebral artery (the posterior inferior cerebellar artery)—the right and left posterior spinal arteries descend along the spinal cord, each as two branches that bracket the posterolateral sulcus and the connection of posterior roots with the spinal cord.

The anterior and posterior spinal arteries are reinforced along their length by eight to ten segmental medullary

SUBASH KC/NMC-15TH/2014

arteries (Fig. 2.49). The largest of these is the **arteria radicularis magna** or the **artery of Adamkiewicz** (Fig. 2.49). This vessel arises in the lower thoracic or upper lumbar region, usually on the left side, and reinforces the arterial supply to the lower portion of the spinal cord, including the lumbar enlargement.

Veins

Veins that drain the spinal cord form a number of longitudinal channels (Fig. 2.50):

- Two pairs of veins on each side bracket the connections of the posterior and anterior roots to the cord.
- One midline channel parallels the anterior median fissure.
- One midline channel passes along the posterior median sulcus.

These longitudinal channels drain into an extensive internal vertebral plexus in the extradural (epidural) space of the vertebral canal, which then drains into segmentally arranged vessels that connect with major systemic veins, such as the azygos system in the thorax. The internal vertebral plexus also communicates with intracranial veins.

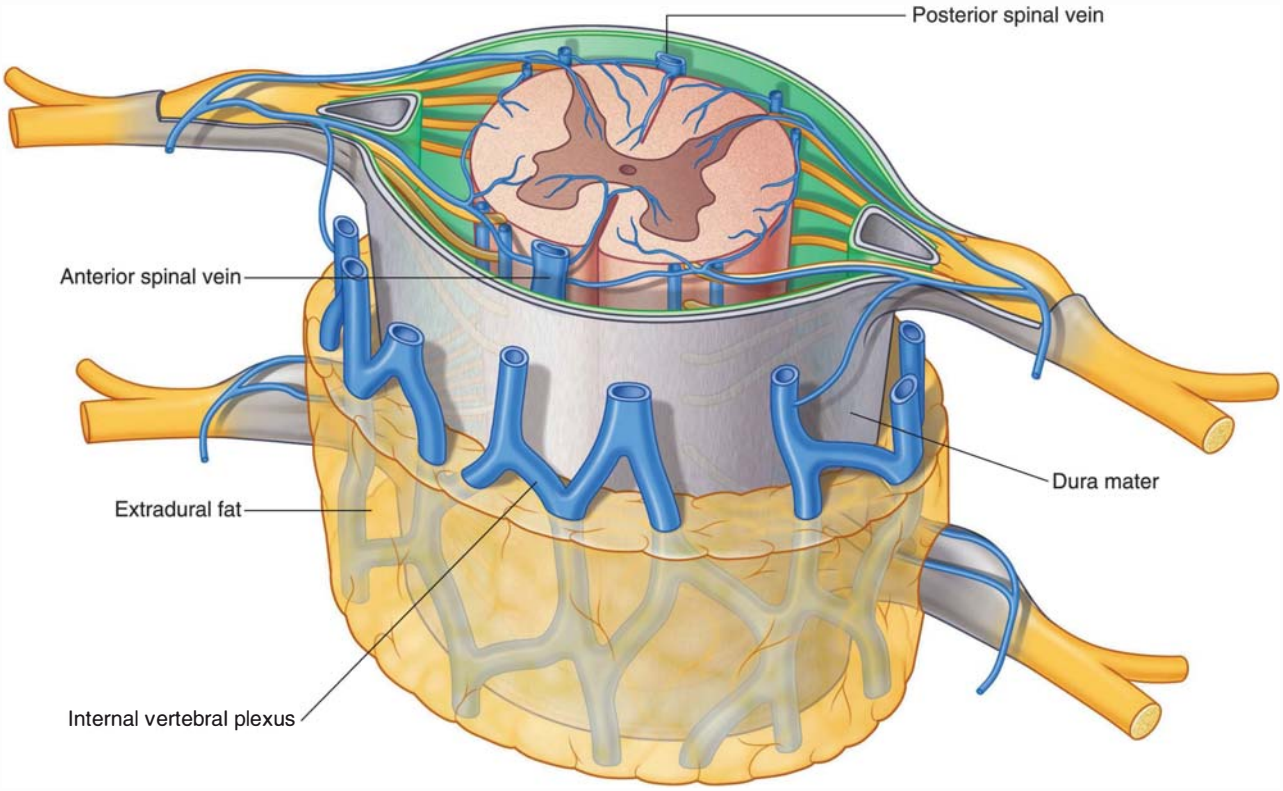


Fig. 2.50 Veins that drain the spinal cord.

SUBASH KC/NMC-15TH/2014

Meninges

Spinal dura mater

The **spinal dura mater** is the outermost meningeal membrane and is separated from the bones forming the vertebral canal by an extradural space (Fig. 2.51). Superiorly, it is continuous with the inner meningeal layer of cranial dura mater at the foramen magnum of the skull. Inferiorly, the dural sac dramatically narrows at the level of the lower border of vertebra SII and forms an investing sheath for the pial part of the filum terminale of the spinal cord. This terminal cord-like extension of dura mater (the dural part of the filum terminale) attaches to the posterior surface of the vertebral bodies of the coccyx.

As spinal nerves and their roots pass laterally, they are surrounded by tubular sleeves of dura mater, which merge with and become part of the outer covering (epineurium) of the nerves.

Arachnoid mater

The **arachnoid mater** is a thin delicate membrane against, but not adherent to, the deep surface of the dura mater (Fig. 2.51). It is separated from the pia mater by the subarachnoid space. The arachnoid mater ends at the level of vertebra SII (see Fig. 2.47).

Subarachnoid space

The subarachnoid space between the arachnoid and pia mater contains CSF (Fig. 2.51). The subarachnoid space around the spinal cord is continuous at the foramen magnum with the subarachnoid space surrounding the brain. Inferiorly, the subarachnoid space terminates at approximately the level of the lower border of vertebra SII (see Fig. 2.47).

Delicate strands of tissue (**arachnoid trabeculae**) are continuous with the arachnoid mater on one side and the pia mater on the other; they span the subarachnoid space and interconnect the two adjacent membranes. Large blood vessels are suspended in the subarachnoid space by similar strands of material, which expand over the vessels to form a continuous external coat.

The subarachnoid space extends further inferiorly than the spinal cord. The spinal cord ends at approximately the disc between vertebrae LI and LII, whereas the

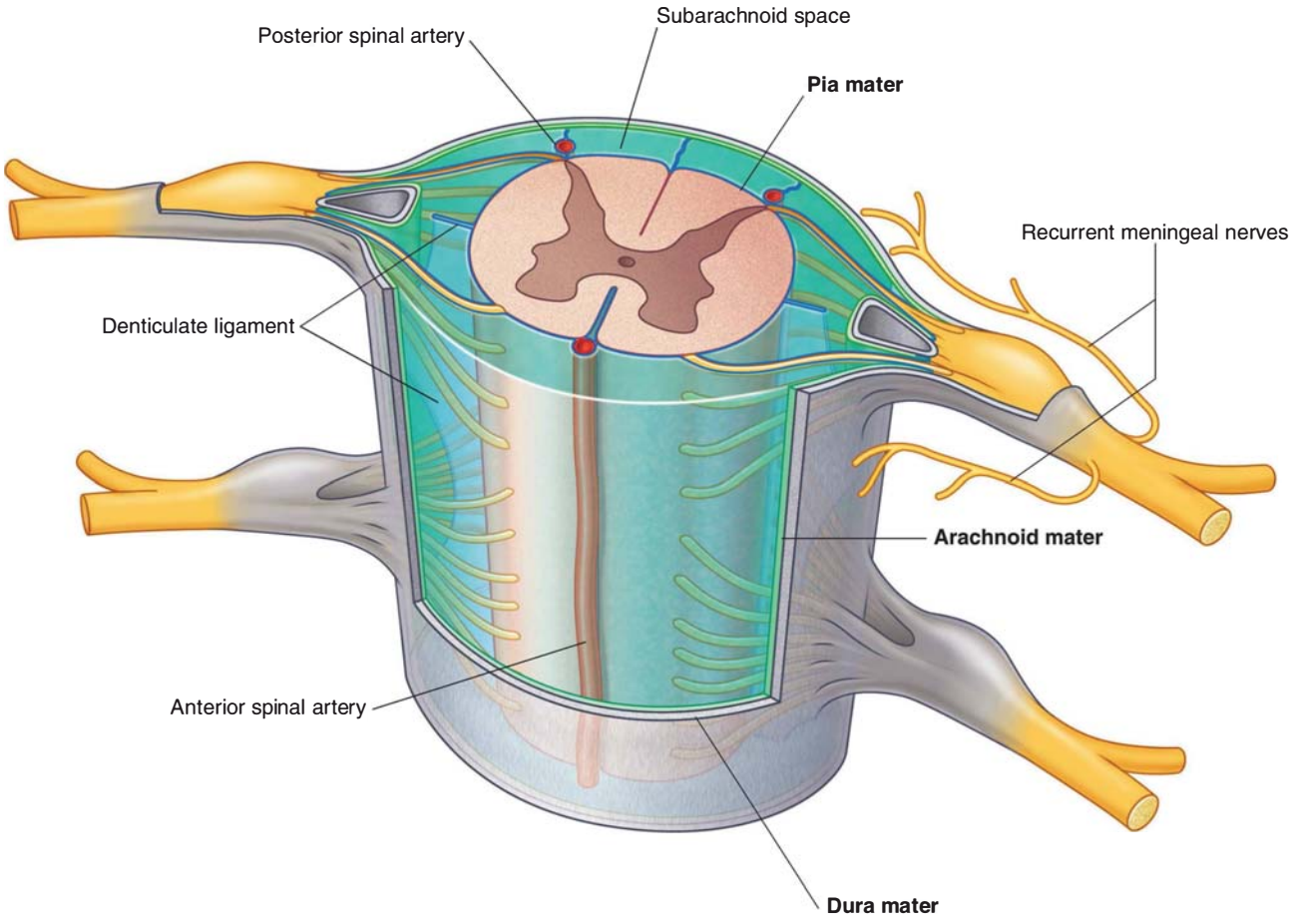


Fig. 2.51 Meninges.

SUBASH KC/NMC-15TH/2014

subarachnoid space extends to approximately the lower border of vertebra SII (see Fig. 2.47). The subarachnoid space is largest in the region inferior to the terminal end of the spinal cord, where it surrounds the cauda equina. As a consequence, CSF can be withdrawn from the subarachnoid space in the lower lumbar region without endangering the spinal cord.

Pia mater

The spinal pia mater is a vascular membrane that firmly adheres to the surface of the spinal cord (Fig. 2.51). It extends into the anterior median fissure and reflects as sleeve-like coatings onto posterior and anterior rootlets and roots as they cross the subarachnoid space. As the roots exit the space, the sleeve-like coatings reflect onto the arachnoid mater.

On each side of the spinal cord, a longitudinally oriented sheet of pia mater (the **denticulate ligament**) extends laterally from the cord toward the arachnoid and dura mater (Fig. 2.51).

- Medially, each denticulate ligament is attached to the spinal cord in a plane that lies between the origins of the posterior and anterior rootlets.
- Laterally, each denticulate ligament forms a series of triangular extensions along its free border, with the apex of each extension being anchored through the arachnoid mater to the dura mater.

The lateral attachments of the denticulate ligaments generally occur between the exit points of adjacent posterior and anterior rootlets. The ligaments function to position the spinal cord in the center of the subarachnoid space.

Arrangement of structures in the vertebral canal

The vertebral canal is bordered:

- anteriorly by the bodies of the vertebrae, intervertebral discs, and posterior longitudinal ligament (Fig. 2.52);
- laterally, on each side by the pedicles and intervertebral foramina; and
- posteriorly by the laminae and ligamenta flava, and in the median plane the roots of the interspinous ligaments and vertebral spinous processes.

Between the walls of the vertebral canal and the dural sac is an extradural space containing a vertebral plexus of veins embedded in fatty connective tissue.

The vertebral spinous processes can be palpated through the skin in the midline in thoracic and lumbar regions of the back. Between the skin and spinous processes is a layer of superficial fascia. In lumbar regions, the adjacent spinous processes and the associated laminae on either side of the midline do not overlap, resulting in gaps between adjacent vertebral arches.

When carrying out a lumbar puncture (spinal tap), the needle passes between adjacent vertebral spinous processes, through the supraspinous and interspinous ligaments, and enters the extradural space. The needle continues through the dura and arachnoid mater and enters the subarachnoid space, which contains CSF.

SUBASH KC/NMC-15TH/2014

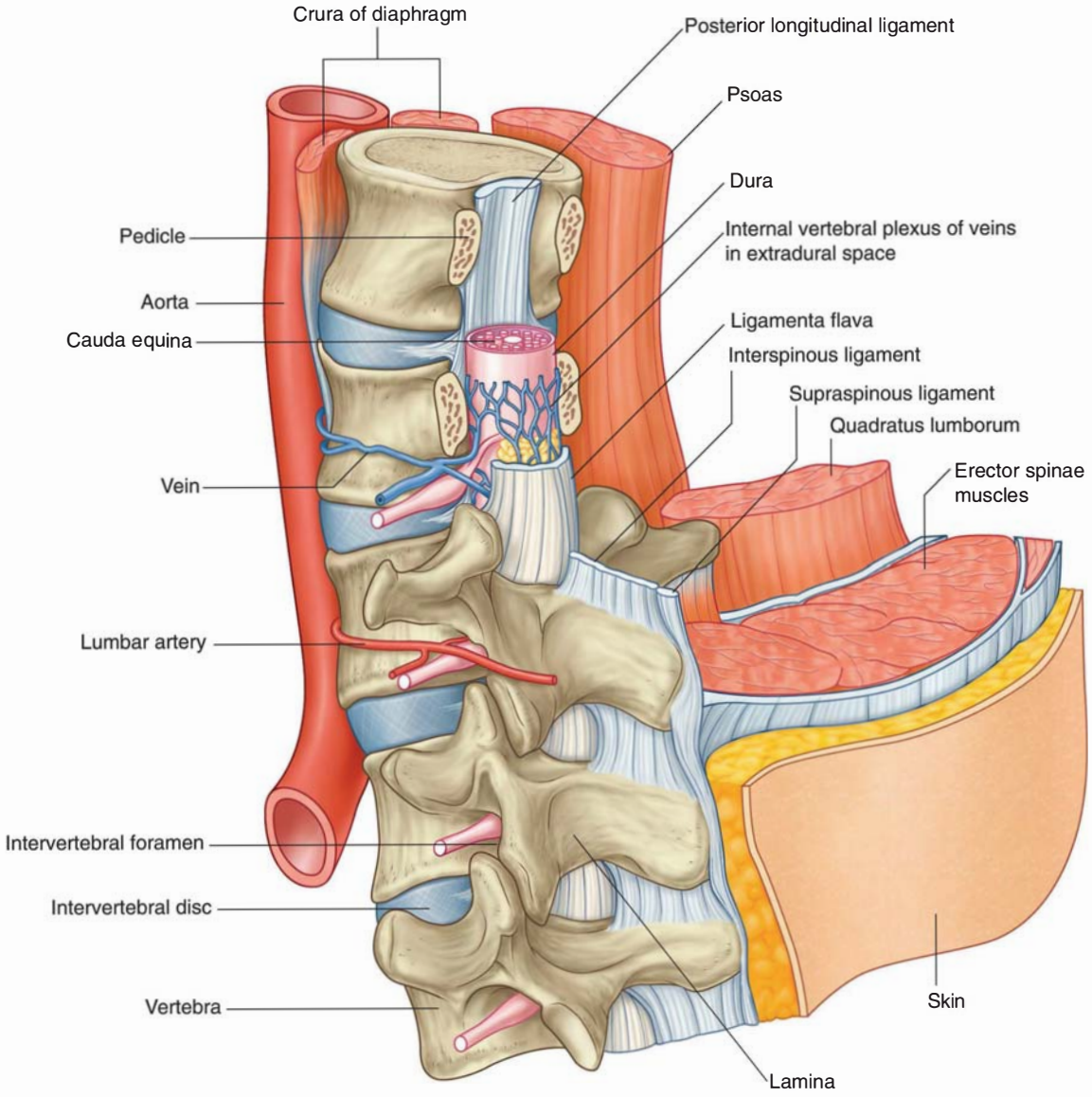


Fig. 2.52 Arrangement of structures in the vertebral canal and the back (lumbar region).

SUBASH KC/NMC-15TH/2014

In the clinic

Lumbar cerebrospinal fluid tap

A lumbar tap (puncture) is carried out to obtain a sample of CSF for examination. In addition, passage of a needle or conduit into the subarachnoid space (CSF space) is used to inject antibiotics, chemotherapeutic agents, and anesthetics.

The lumbar region is an ideal site to access the subarachnoid space because the spinal cord terminates around the level of the disc between vertebrae LI and LII in the adult. The subarachnoid space extends to the region of the lower border of the SII vertebra. There is therefore a large CSF-filled space containing lumbar and sacral nerve roots but no spinal cord.

Depending on the clinician's preference, the patient is placed in the lateral or prone position. A needle is passed in the midline in between the spinous processes into the extradural space. Further advancement punctures the dura and arachnoid mater to enter the subarachnoid space. Most needles push the roots away from the tip without causing the patient any symptoms. Once the needle is in the subarachnoid space, fluid can be aspirated. In some situations, it is important to measure CSF pressure.

Local anesthetics can be injected into the extradural space or the subarachnoid space to anesthetize the sacral and lumbar nerve roots. Such anesthesia is useful for operations on the pelvis and the legs, which can then be carried out without the need for general anesthesia. When procedures are carried out, the patient must be in the erect position and not lying on his or her side or in the head-down position. If a patient lies on his or her side, the anesthesia is likely to be unilateral. If the patient is placed in the head-down position, the anesthetic can pass cranially and potentially depress respiration.

In some instances, anesthesiologists choose to carry out **extradural anesthesia.** A needle is placed through the skin, supraspinous ligament, interspinous ligament, and ligamenta flava into the areolar tissue and fat around the dura mater. Anesthetic agent is introduced and diffuses around the vertebral canal to anesthetize the exiting nerve roots and diffuse into the subarachnoid space.

Spinal nerves

Each spinal nerve is connected to the spinal cord by posterior and anterior roots (Fig. 2.53):

- The **posterior root** contains the processes of sensory neurons carrying information to the CNS—the cell bodies of the sensory neurons, which are derived embryologically from neural crest cells, are clustered in a **spinal ganglion** at the distal end of the posterior root, usually in the intervertebral foramen.
- The **anterior root** contains motor nerve fibers, which carry signals away from the CNS—the cell bodies of the primary motor neurons are in anterior regions of the spinal cord.

Medially, the posterior and anterior roots divide into rootlets, which attach to the spinal cord.

A **spinal segment** is the area of the spinal cord that gives rise to the **posterior** and **anterior rootlets**, which will form a single pair of spinal nerves. Laterally, the posterior and anterior roots on each side join to form a spinal nerve.

Each spinal nerve divides, as it emerges from an intervertebral foramen, into two major branches: a small posterior ramus and a much larger anterior ramus (Fig. 2.53):

- The **posterior rami** innervate only intrinsic back muscles (the epaxial muscles) and an associated narrow strip of skin on the back.
- The **anterior rami** innervate most other skeletal muscles (the hypaxial muscles) of the body, including those of the limbs and trunk, and most remaining areas of the skin, except for certain regions of the head.

SUBASH KC/NMC-15TH/2014

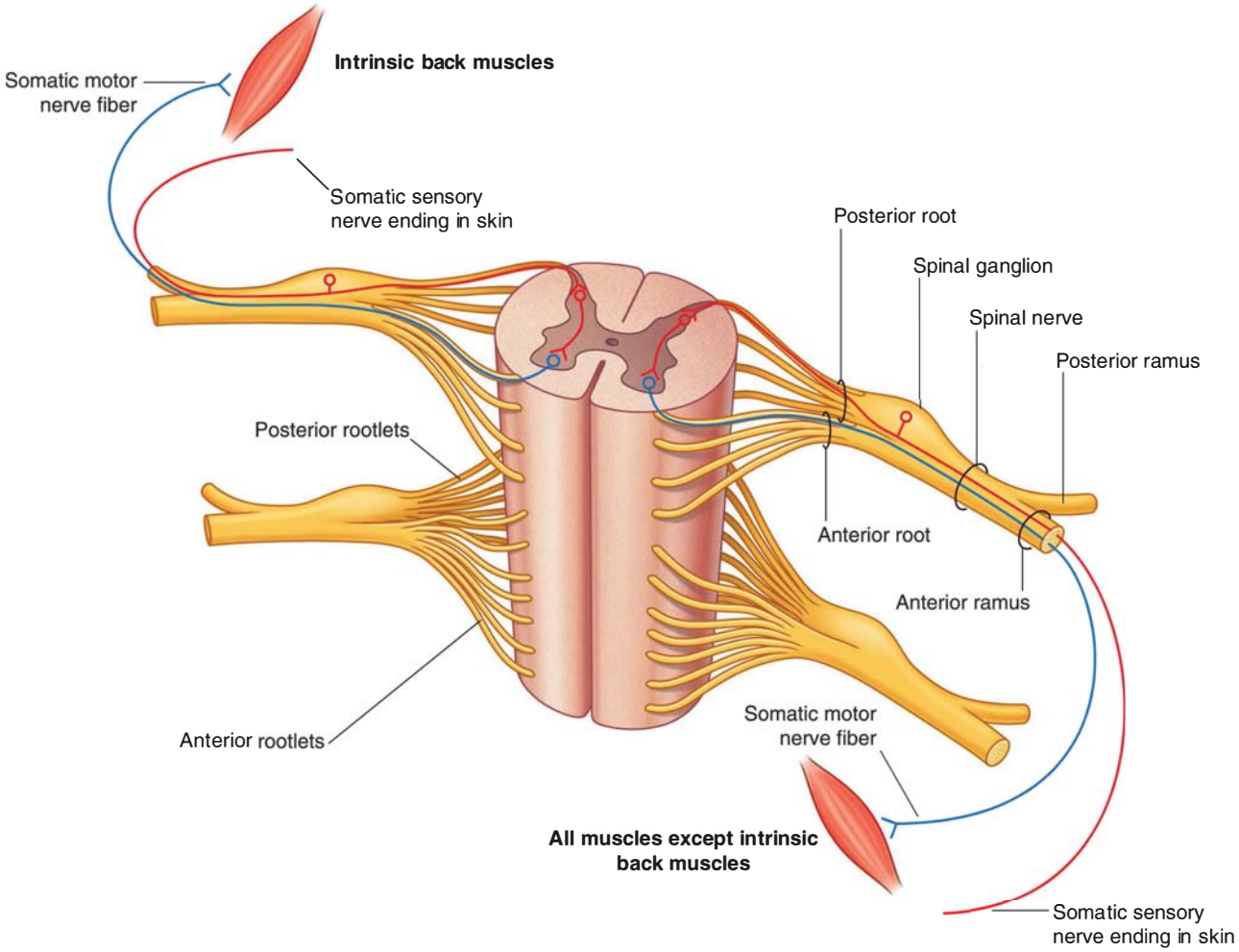


Fig. 2.53 Basic organization of a spinal nerve.

Near the point of division into anterior and posterior rami, each spinal nerve gives rise to two to four small recurrent meningeal (sinuvertebral) nerves (see Fig. 2.51). These nerves reenter the intervertebral foramen to supply dura, ligaments, intervertebral discs, and blood vessels.

All major somatic plexuses (cervical, brachial, lumbar, and sacral) are formed by anterior rami.

Because the spinal cord is much shorter than the vertebral column, the roots of spinal nerves become longer and pass more obliquely from the cervical to coccygeal regions of the vertebral canal (Fig. 2.54).

In adults, the spinal cord terminates at a level approximately between vertebrae LI and LII, but this can range between vertebra TXII and the disc between vertebrae LII and LIII. Consequently, posterior and anterior roots forming spinal nerves emerging between vertebrae in the lower regions of the vertebral column are connected to the spinal cord at higher vertebral levels.

Below the end of the spinal cord, the posterior and anterior roots of lumbar, sacral, and coccygeal nerves pass inferiorly to reach their exit points from the vertebral canal. This terminal cluster of roots is the cauda equina.

SUBASH KC/NMC-15TH/2014

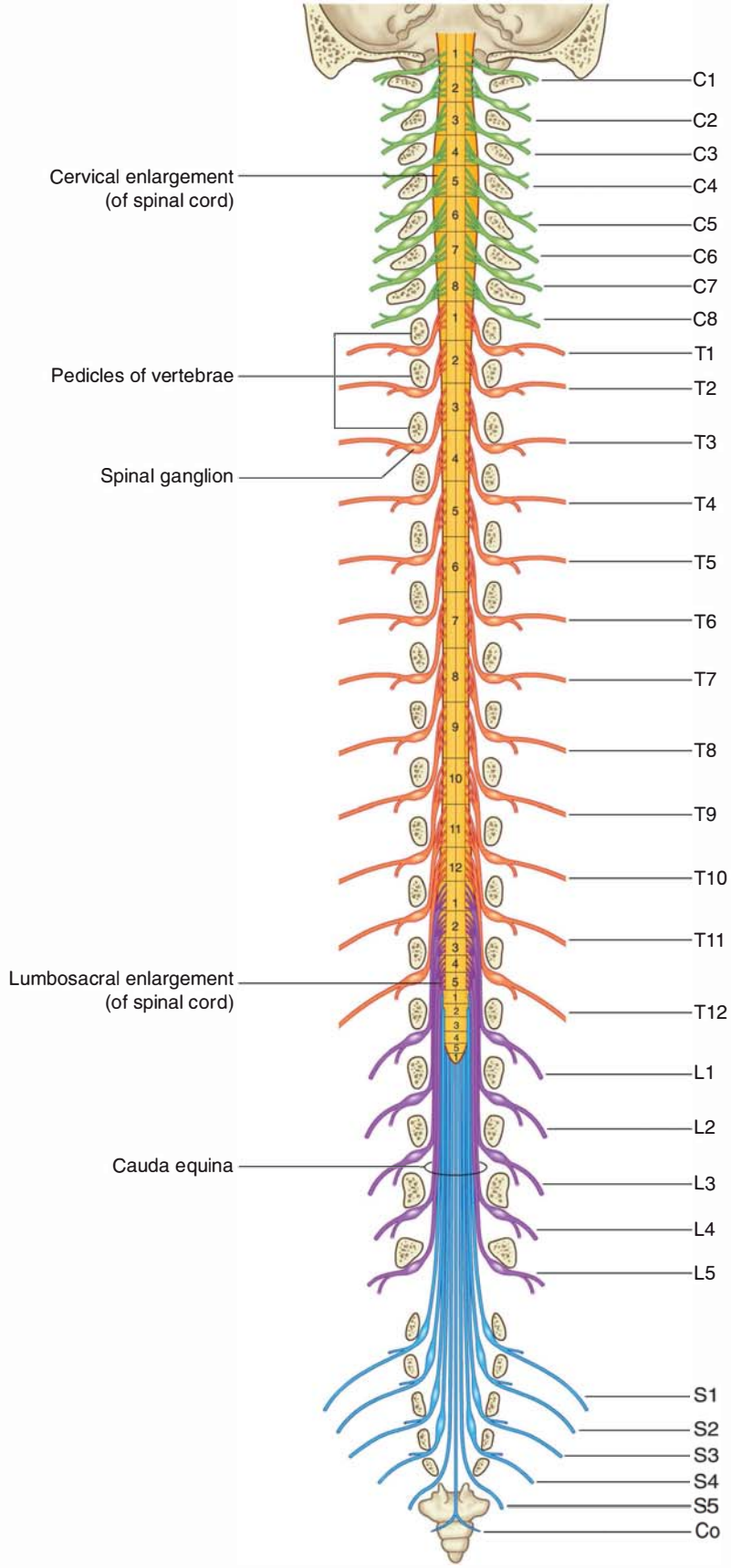


Fig. 2.54 Course of spinal nerves in the vertebral canal.

SUBASH KC/NMC-15TH/2014

Nomenclature of spinal nerves

There are approximately 31 pairs of spinal nerves (Fig. 2.54), named according to their position with respect to associated vertebrae:

- eight cervical nerves—C1 to C8,
- twelve thoracic nerves—T1 to T12,
- five lumbar nerves—L1 to L5,
- five sacral nerves—S1 to S5,
- one coccygeal nerve—Co.

The first cervical nerve (C1) emerges from the vertebral canal between the skull and vertebra CI (Fig. 2.55). Therefore cervical nerves C2 to C7 also emerge from the vertebral canal above their respective vertebrae. Because there are only seven cervical vertebrae, C8 emerges between vertebrae CVII and TI. As a consequence, all remaining spinal nerves, beginning with T1, emerge from the vertebral canal below their respective vertebrae.

In the clinic

Herpes zoster

Herpes zoster is the virus that produces chickenpox in children. In some patients the virus remains dormant in the cells of the spinal ganglia. Under certain circumstances, the virus becomes activated and travels along the neuronal bundles to the areas supplied by that nerve (the dermatome). A rash ensues, which is characteristically exquisitely painful. Importantly, this typical dermatomal distribution is characteristic of this disorder.

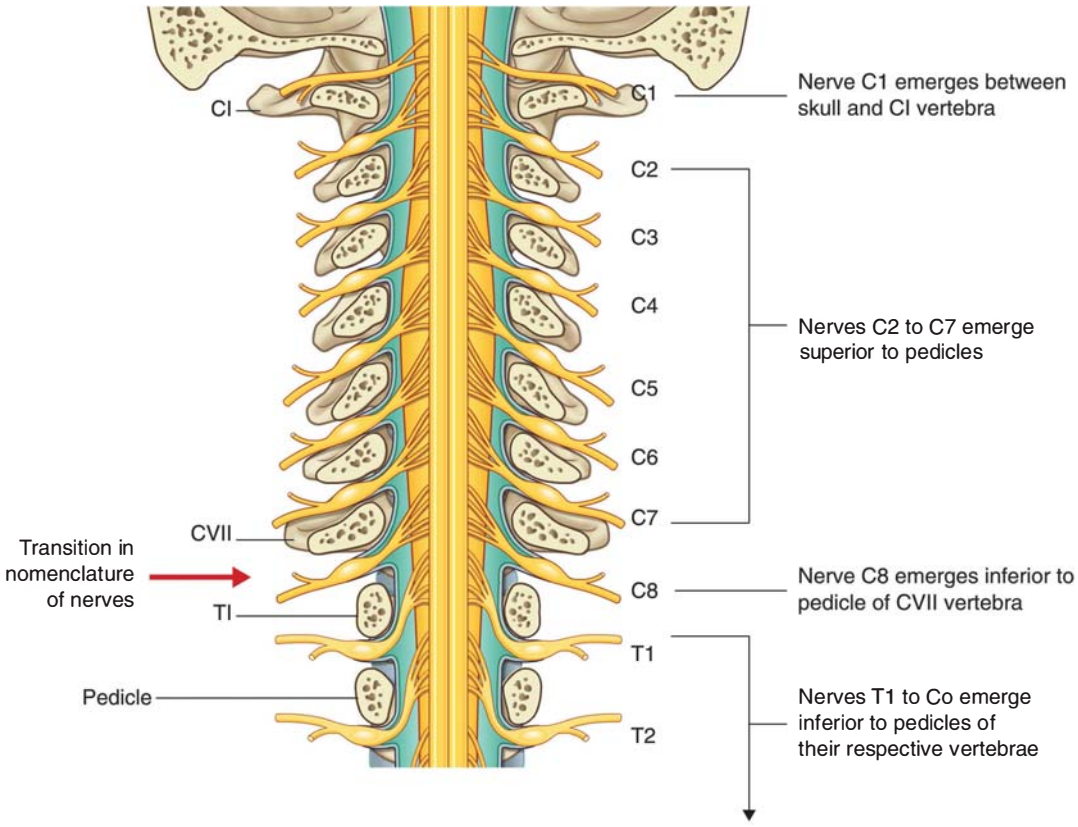


Fig. 2.55 Nomenclature of the spinal nerves.

SUBASH KC/NMC-15TH/2014

In the clinic

Back pain—alternative explanations

Back pain is an extremely common condition affecting almost all individuals at some stage during their life. It is of key clinical importance to identify whether the back pain relates to the vertebral column and its attachments or relates to other structures.

The failure to consider other potential structures that may produce back pain can lead to significant mortality and morbidity. Pain may refer to the back from a number of organs situated in the retroperitoneum. Pancreatic pain in particular refers to the back and may be associated with pancreatic cancer and pancreatitis. Renal pain, which may be produced by stones in the renal collecting system or renal tumors, also typically refers to the back. More often than not this is usually unilateral; however, it can produce central posterior back pain. Enlarged lymph nodes in the pre- and para-aortic region may produce central posterior back pain and may be a sign of solid tumor malignancy, infection, or Hodgkin's lymphoma. An enlarging abdominal aorta (abdominal aortic aneurysm) may cause back pain as it enlarges without rupture. Therefore it is critical to think of this structure as a potential cause of back pain, because treatment will be lifesaving. Moreover, a ruptured abdominal aortic aneurysm may also cause acute back pain in the first instance.

In all patients back pain requires careful assessment not only of the vertebral column but also of the chest and abdomen in order not to miss other important anatomical structures that may produce signs and symptoms radiating to the back.

SUBASH KC/NMC-15TH/2014

Surface anatomy

Back surface anatomy

Surface features of the back are used to locate muscle groups for testing peripheral nerves, to determine regions of the vertebral column, and to estimate the approximate position of the inferior end of the spinal cord. They are also used to locate organs that occur posteriorly in the thorax and abdomen.

Absence of lateral curvatures

When viewed from behind, the normal vertebral column has no lateral curvatures. The vertical skin furrow between muscle masses on either side of the midline is straight (Fig. 2.56).



Fig. 2.56 Normal appearance of the back. **A.** In women. **B.** In men.

SUBASH KC/NMC-15TH/2014

Primary and secondary curvatures in the sagittal plane

When viewed from the side, the normal vertebral column has primary curvatures in the thoracic and sacral/coccygeal regions and secondary curvatures in the cervical and lumbar regions (Fig. 2.57). The primary curvatures are concave anteriorly. The secondary curvatures are concave posteriorly.

Useful nonvertebral skeletal landmarks

A number of readily palpable bony features provide useful landmarks for defining muscles and for locating structures associated with the vertebral column. Among these features are the external occipital protuberance, the scapula, and the iliac crest (Fig. 2.58).

The external occipital protuberance is palpable in the midline at the back of the head just superior to the hairline.

The spine, medial border, and inferior angle of the scapula are often visible and are easily palpable.

The iliac crest is palpable along its entire length, from the anterior superior iliac spine at the lower lateral margin of the anterior abdominal wall to the posterior superior iliac spine near the base of the back. The position of the posterior superior iliac spine is often visible as a "sacral dimple" just lateral to the midline.

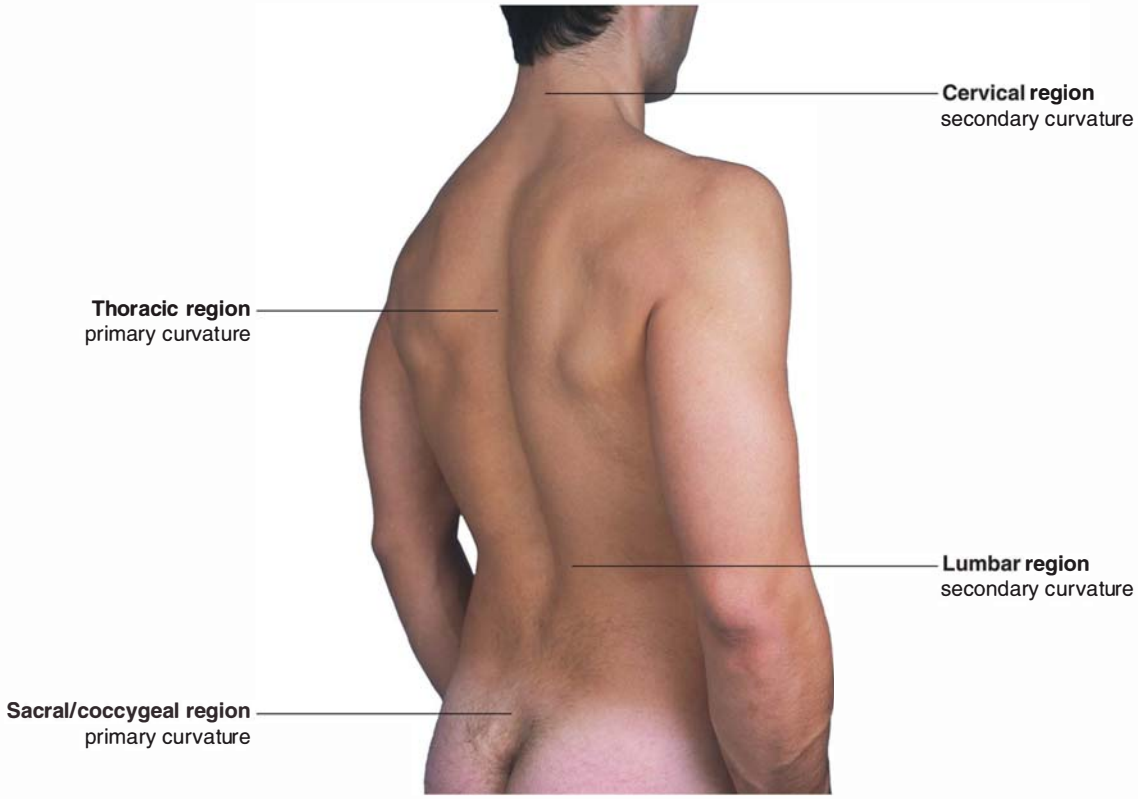


Fig. 2.57 Normal curvatures of the vertebral column.

SUBASH KC/NMC-15TH/2014

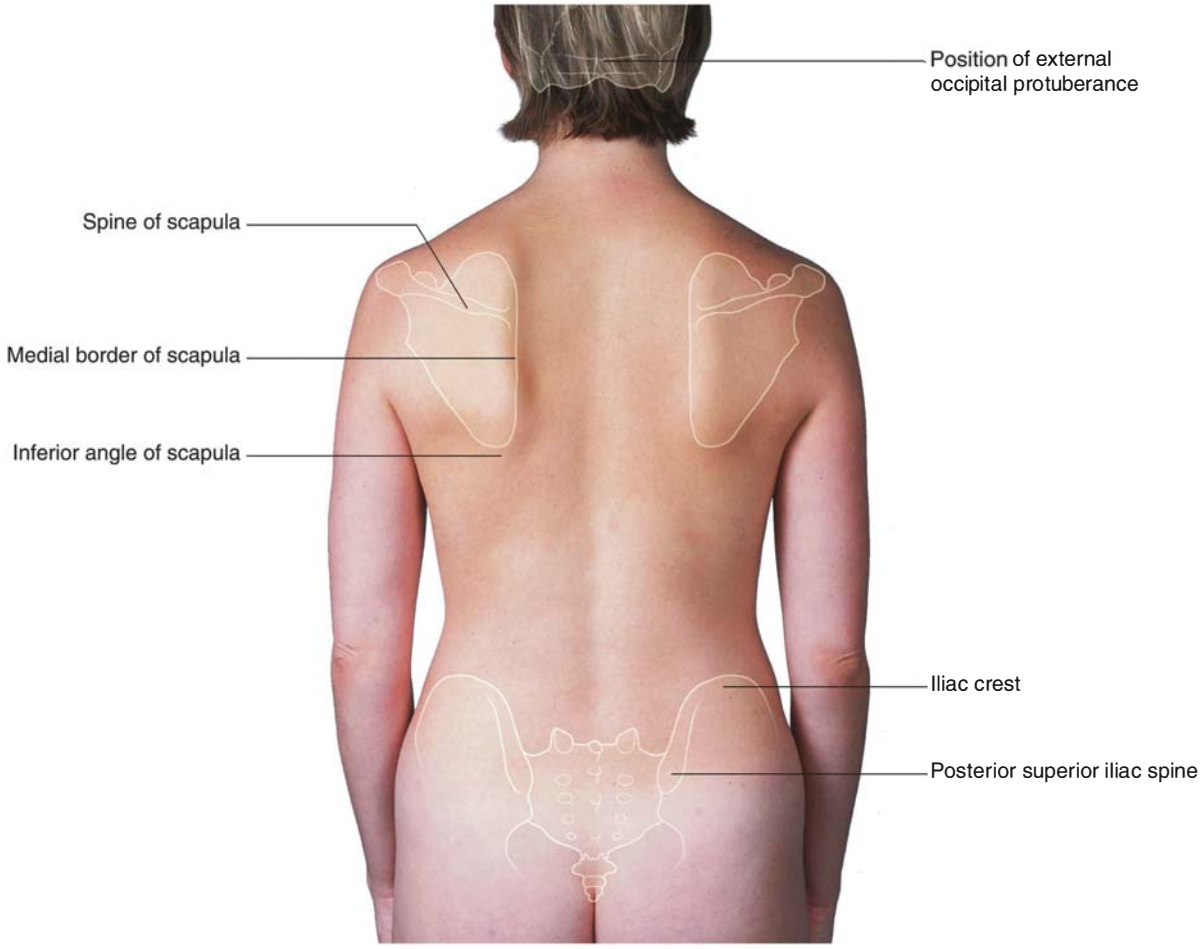


Fig. 2.58 Back of a woman with major palpable bony landmarks indicated.

SUBASH KC/NMC-15TH/2014

How to identify specific vertebral spinous processes

Identification of vertebral spinous processes (Fig. 2.59A) can be used to differentiate between regions of the vertebral column and facilitate visualizing the position of deeper structures, such as the inferior ends of the spinal cord and subarachnoid space.

The spinous process of vertebra CII can be identified through deep palpation as the most superior bony protuberance in the midline inferior to the skull.

Most of the other spinous processes, except for that of vertebra CVII, are not readily palpable because they are obscured by soft tissue.

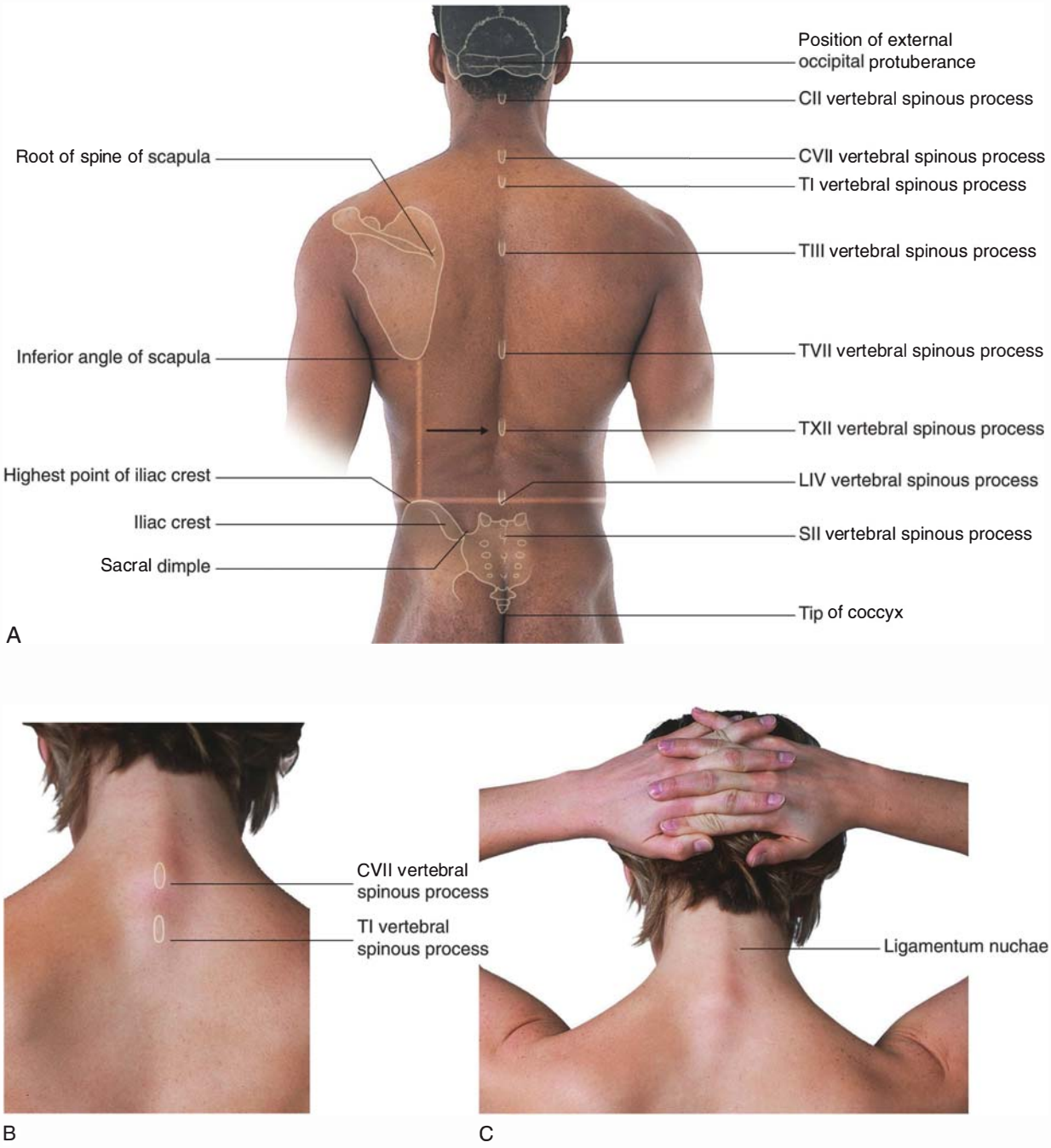


Fig. 2.59 The back with the positions of vertebral spinous processes and associated structures indicated. **A.** In a man. **B.** In a woman with neck flexed. The prominent CVII and TI vertebral spinous processes are labeled. **C.** In a woman with neck flexed to accentuate the ligamentum nuchae.

SUBASH KC/NMC-15TH/2014

The spinous process of CVII is usually visible as a prominent eminence in the midline at the base of the neck (Fig. 2.59B), particularly when the neck is flexed.

Extending between CVII and the external occipital protuberance of the skull is the ligamentum nuchae, which is readily apparent as a longitudinal ridge when the neck is flexed (Fig. 2.59C).

Inferior to the spinous process of CVII is the spinous process of TI, which is also usually visible as a midline protuberance. Often it is more prominent than the spinous process of CVII (Fig. 2.59A,B).

The root of the spine of the scapula is at the same level as the spinous process of vertebra TIII, and the inferior angle of the scapula is level with the spinous process of vertebra TVII (Fig. 2.59A).

The spinous process of vertebra TXII is level with the midpoint of a vertical line between the inferior angle of the scapula and the iliac crest (Fig. 2.59A).

A horizontal line between the highest point of the iliac crest on each side crosses through the spinous process of vertebra LIV. The LIII and LV vertebral spinous processes can be palpated above and below the LIV spinous process, respectively (Fig. 2.59A).

The sacral dimples that mark the position of the posterior superior iliac spine are level with the SII vertebral spinous process (Fig. 2.59A).

The tip of the coccyx is palpable at the base of the vertebral column between the gluteal masses (Fig. 2.59A).

The tips of the vertebral spinous processes do not always lie in the same horizontal plane as their corresponding vertebral bodies. In thoracic regions, the spinous processes are long and sharply sloped downward so that their tips lie at the level of the vertebral body below. In other words, the tip of the TIII vertebral spinous process lies at vertebral level TIV.

In lumbar and sacral regions, the spinous processes are generally shorter and less sloped than in thoracic regions, and their palpable tips more closely reflect the position of their corresponding vertebral bodies. As a consequence, the palpable end of the spinous process of vertebra LIV lies at approximately the LIV vertebral level.

Visualizing the inferior ends of the spinal cord and subarachnoid space

The spinal cord does not occupy the entire length of the vertebral canal. Normally in adults, it terminates at the level of the disc between vertebrae LI and LII; however, it may end as high as TXII or as low as the disc between vertebrae LII and LIII. The subarachnoid space ends at approximately the level of vertebra SII (Fig. 2.60A).

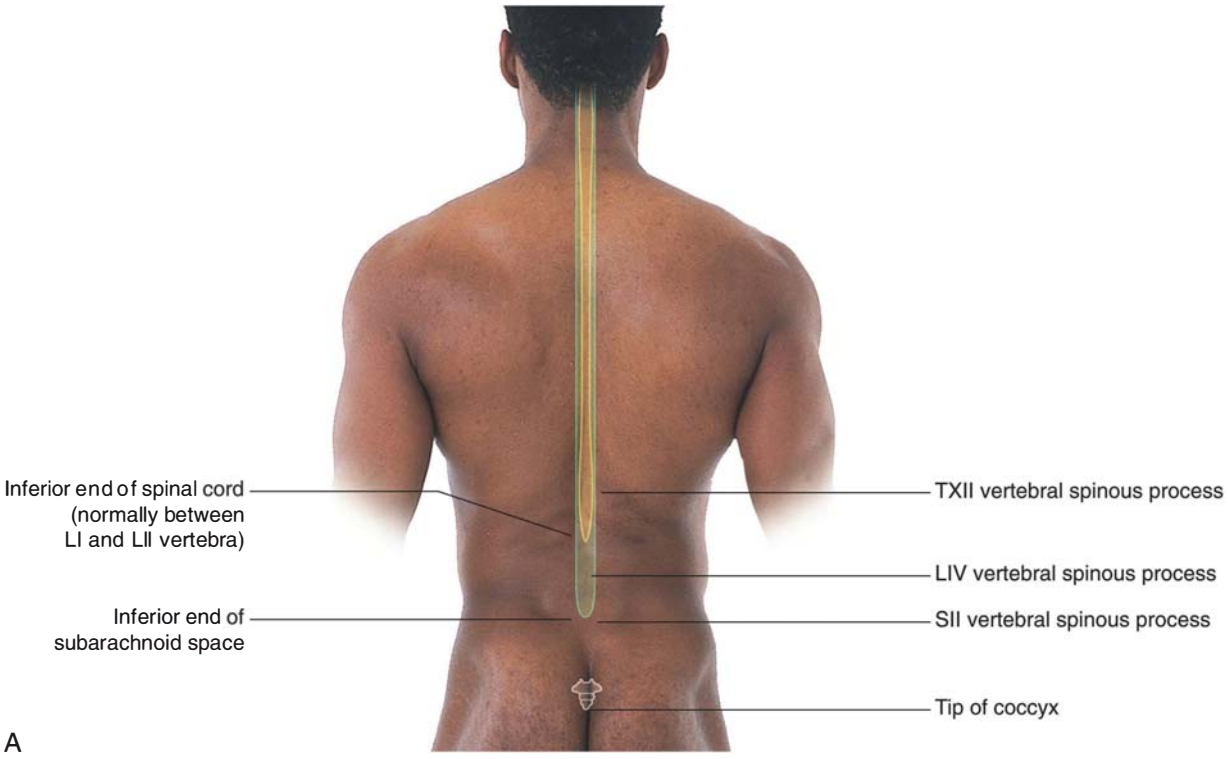


Fig. 2.60 Back with the ends of the spinal cord and subarachnoid space indicated. **A.** In a man.

Continued

SUBASH KC/NMC-15TH/2014

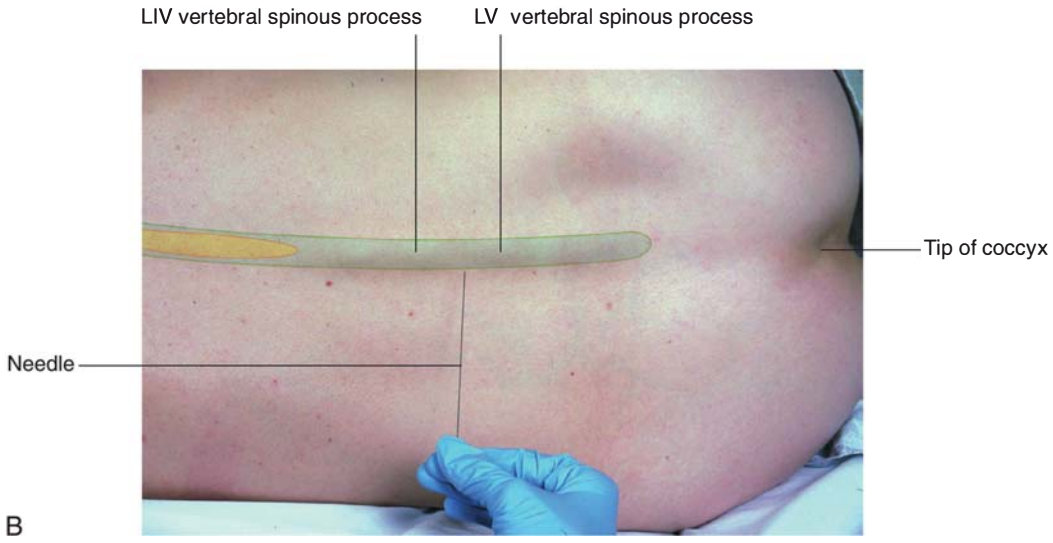


Fig. 2.60, cont'd Back with the ends of the spinal cord and subarachnoid space indicated. **B.** In a woman lying on her side in a fetal position, which accentuates the lumbar vertebral spinous processes and opens the spaces between adjacent vertebral arches. Cerebrospinal fluid can be withdrawn from the subarachnoid space in lower lumbar regions without endangering the spinal cord.

SUBASH KC/NMC-15TH/2014

Because the subarachnoid space can be accessed in the lower lumbar region without endangering the spinal cord, it is important to be able to identify the position of the lumbar vertebral spinous processes. The LIV vertebral spinous process is level with a horizontal line between the highest points on the iliac crests. In the lumbar region, the palpable ends of the vertebral spinous processes lie opposite their corresponding vertebral bodies. The subarachnoid space can be accessed between vertebral levels LIII and LIV and between LIV and LV without endangering the spinal cord (Fig. 2.60B). The subarachnoid space ends at vertebral level SII, which is level with the sacral dimples marking the posterior superior iliac spines.

Identifying major muscles

A number of intrinsic and extrinsic muscles of the back can readily be observed and palpated. The largest of these are the trapezius and latissimus dorsi muscles (Fig. 2.61A and 2.61B). Retracting the scapulae toward the midline can accentuate the rhomboid muscles (Fig. 2.61C), which lie deep to the trapezius muscle. The erector spinae muscles are visible as two longitudinal columns separated by a furrow in the midline (Fig. 2.61A).

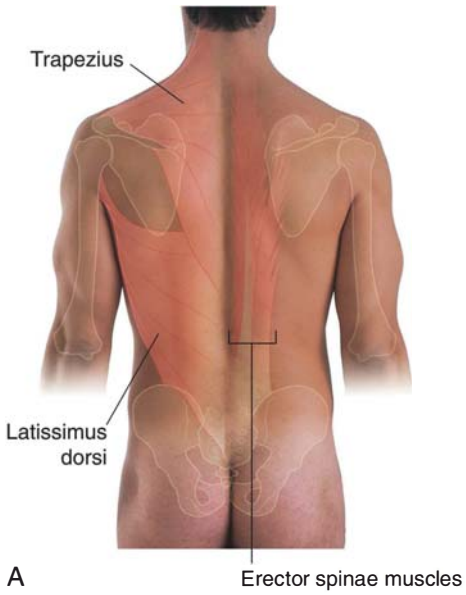
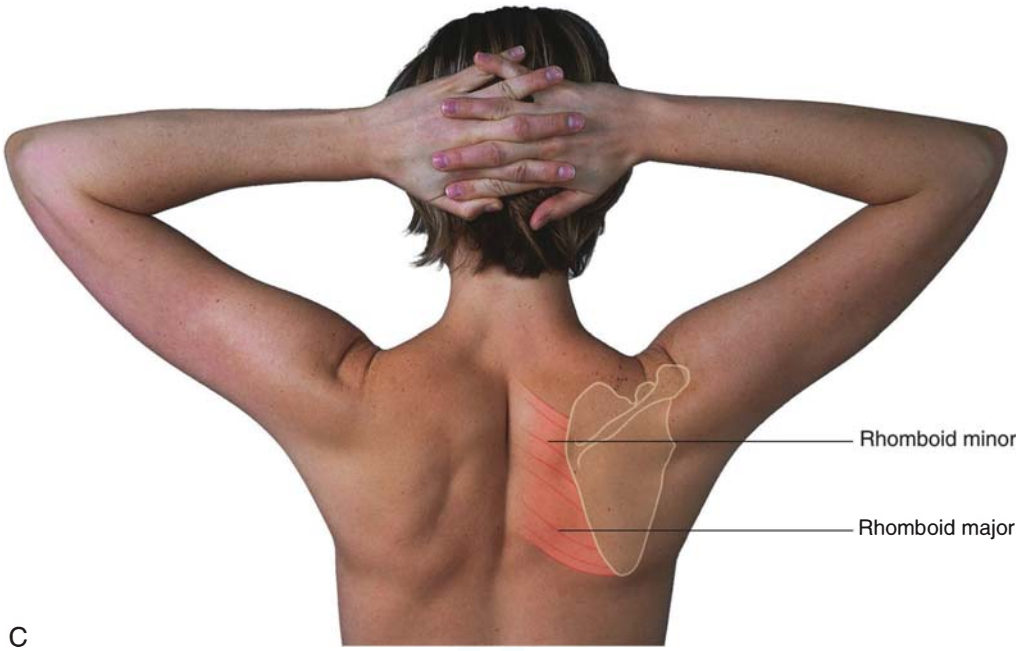


Fig. 2.61 Back muscles. **A.** In a man with latissimus dorsi, trapezius, and erector spinae muscles outlined.



B



C

Fig. 2.61, cont'd Back muscles. **B.** In a man with arms abducted to accentuate the lateral margins of the latissimus dorsi muscles. **C.** In a woman with scapulae externally rotated and forcibly retracted to accentuate the rhomboid muscles.

SUBASH KC/NMC-15TH/2014

Clinical cases

Case 1

SCIATICA VERSUS LUMBAGO

A 50-year-old woman visited her local family practitioner with severe lower back pain radiating into her right buttock.

Low back pain is a common problem in family practice.

Of the many common causes of low back pain some need to be identified early to commence appropriate treatment. The common causes include an anular disc tear, a disc prolapse that impinges directly on a nerve root, spinal stenosis, and mechanical zygapophysial joint pain. Overall, the main causes can be distilled into three central groups: mechanical back pain, degenerative joint disease, and neuronal compression.

Sciatica and lumbago are not the same. **Lumbago** is a generic term referring to low back pain. **Sciatica** is a name given to pain in the area of distribution of the sciatic nerve (L4 to S3), which is commonly felt in the buttock and over the posterolateral aspects of the leg.

Case 2

CERVICAL SPINAL CORD INJURY

A 45-year-old man was involved in a serious car accident. On examination he had a severe injury to the cervical region of his vertebral column with damage to the spinal cord. In fact, his breathing became erratic and stopped.

If the cervical spinal cord injury is above the level of C5, breathing is likely to stop. The phrenic nerve takes origin from C3, C4, and C5 and supplies the diaphragm. Breathing may not cease immediately if the lesion is just below C5, but does so as the cord becomes edematous and damage progresses superiorly. In addition, some respiratory and ventilatory exchange may occur by using neck muscles plus the sternocleidomastoid and trapezius muscles, which are innervated by the accessory nerve [XI].

The patient was unable to sense or move his upper and lower limbs.

The patient has paralysis of the upper and lower limbs and is therefore quadriplegic. If breathing is unaffected, the lesion is below the level of C5 or at the level of C5. The nerve supply to the upper limbs is via the brachial plexus, which begins at the C5 level. The site of the spinal cord injury is at or above the C5 level.

It is important to remember that although the cord has been transected in the cervical region, the cord below this level is intact. Reflex activity may therefore occur below the injury, but communication with the brain is lost.

SUBASH KC/NMC-15TH/2014

Case 3

PSOAS ABSCESS

A 25-year-old woman complained of increasing lumbar back pain. Over the ensuing weeks she was noted to have an enlarging lump in the right groin, which was mildly tender to touch. On direct questioning, the patient also complained of a productive cough with sputum containing mucus and blood, and she had a mild temperature.

The chest radiograph revealed a cavitating apical lung mass, which explains the pulmonary history.

Given the age of the patient a primary lung cancer is unlikely. The hemoptysis (coughing up blood in the sputum) and the rest of the history suggest the patient has a lung infection. Given the chest radiographic findings of a cavity in the apex of the lung, a diagnosis of tuberculosis (TB) was made. This was confirmed by bronchoscopy and aspiration of pus, which was cultured.

During the patient's pulmonary infection, the tuberculous bacillus had spread via the blood to vertebra L1. The bone destruction began in the cancellous bone of the vertebral body close to the intervertebral discs. This disease progressed and eroded into the intervertebral disc, which became infected. The disc was destroyed, and the infected disc material extruded around the disc anteriorly and passed into the psoas muscle sheath. This is not an uncommon finding for a tuberculous infection of the lumbar portion of the vertebral column.

As the infection progressed, the pus spread within the psoas muscle sheath beneath the inguinal ligament to produce a hard mass in the groin. This is a typical finding for a psoas abscess.

Fortunately for the patient, there was no evidence of any damage within the vertebral canal.

The patient underwent a radiologically guided drainage of the psoas abscess and was treated for over 6 months with a long-term antibiotic regimen. She made an excellent recovery with no further symptoms, although the cavities within the lungs remain. It healed with sclerosis.

Case 4

DISSECTING THORACIC ANEURYSM

A 72-year-old fit and healthy man was brought to the emergency department with severe back pain beginning at the level of the shoulder blades and extending to the midlumbar region. The pain was of relatively acute onset and was continuous. The patient was able to walk to the gurney as he entered the ambulance; however, at the emergency department the patient complained of inability to use both legs.

The attending physician examined the back thoroughly and found no significant abnormality. He noted that there was reduced sensation in both legs, and there was virtually no power in extensor or flexor groups. The patient was tachycardic, which was believed to be due to pain, and the blood pressure obtained in the ambulance measured 120/80 mm Hg. It was noted that the patient's current blood pressure was 80/40 mm Hg; however, the patient did not complain of typical clinical symptoms of hypotension.

On first inspection, it is difficult to "add up" these clinical symptoms and signs. In essence we have a progressive paraplegia associated with severe back pain and an anomaly in blood pressure measurements, which are not compatible with the clinical state of the patient.

It was deduced that the blood pressure measurements were obtained in different arms, and both were reassessed.

The blood pressure measurements were true. In the right arm the blood pressure measured 120/80 mm Hg and in the left arm the blood pressure measured 80/40 mm Hg. This would imply a deficiency of blood to the left arm.

The patient was transferred from the emergency department to the CT scanner, and a scan was performed that included the chest, abdomen, and pelvis.

The CT scan demonstrated a dissecting thoracic aortic aneurysm. Aortic dissection occurs when the tunica intima and part of the tunica media of the wall of the aorta become separated from the remainder of the tunica media and the tunica adventitia of the aorta wall. This produces a false lumen. Blood passes not only in the true aortic lumen but also through a small hole into the wall of the aorta and into the false lumen. It often reenters the true aortic lumen inferiorly. This produces two

(continues)

SUBASH KC/NMC-15TH/2014

Case 4 *(continued)*

channels through which blood may flow. The process of the aortic dissection produces considerable pain for the patient and is usually of rapid onset. Typically the pain is felt between the shoulder blades and radiating into the back, and although the pain is not from the back musculature or the vertebral column, careful consideration of structures other than the back should always be sought.

The difference in the blood pressure between the two arms indicates the level at which the dissection has begun. The "point of entry" is proximal to the left subclavian artery. At this level a small flap has been created, which limits the blood flow to the left upper limb, giving the low blood pressure recording. The brachiocephalic trunk has not been affected by the aortic dissection, and hence blood flow remains appropriate to the right upper limb.

The paraplegia was caused by ischemia to the spinal cord.

The blood supply to the spinal cord is from a single anterior spinal artery and two posterior spinal arteries. These arteries are fed via segmental spinal arteries at every vertebral level. There are a number of reinforcing arteries (segmental medullary arteries) along the length of the spinal cord—the largest of which is the artery of Adamkiewicz. This artery of Adamkiewicz, a segmental medullary artery, typically arises from the lower thoracic or upper lumbar region, and unfortunately during this patient's aortic dissection, the origin of this vessel was disrupted. This produces acute spinal cord ischemia and has produced the paraplegia in the patient.

Unfortunately, the dissection extended, the aorta ruptured, and the patient succumbed.

SUBASH KC/NMC-15TH/2014

Case 5

SACRAL TUMOR

A 55-year-old woman came to her physician with sensory alteration in the right gluteal (buttock) region and in the intergluteal (natal) cleft. Examination also demonstrated low-grade weakness of the muscles of the foot and subtle weakness of the extensor hallucis longus, extensor digitorum longus, and fibularis tertius on the right. The patient also complained of some mild pain symptoms posteriorly in the right gluteal region.

A lesion was postulated in the left sacrum.

Pain in the right sacro-iliac region could easily be attributed to the sacro-iliac joint, which is often very sensitive to pain. The weakness of the intrinsic muscles of the foot and the extensor hallucis longus, extensor digitorum longus, and fibularis tertius muscles raises the possibility of an abnormality affecting the nerves exiting the sacrum and possibly the lumbosacral junction. The altered sensation around the gluteal region toward the anus would also support these anatomical localizing features.

An X-ray was obtained of the pelvis.

The X-ray appeared on first inspection unremarkable. However, the patient underwent further investigation, including CT and MRI, which demonstrated a large destructive lesion involving the whole of the left sacrum extending into the anterior sacral foramina at the S1, S2, and S3 levels. Interestingly, plain radiographs of the sacrum may often appear normal on first inspection, and further imaging should always be sought in patients with a suspected sacral abnormality.

The lesion was expansile and lytic.

Most bony metastases are typically nonexpansile. They may well erode the bone, producing lytic type of lesions, or may become very sclerotic (prostate metastases and breast metastases). From time to time we see a mixed pattern of lytic and sclerotic.

There are a number of uncommon instances in which certain metastases are expansile and lytic. These typically occur in renal metastases and may be seen in multiple myeloma. The anatomical importance of these specific tumors is that they often expand and impinge upon other structures. The expansile nature of this patient's tumor within the sacrum was the cause for compression of the sacral nerve roots, producing her symptoms.

The patient underwent a course of radiotherapy, had the renal tumor excised, and is currently undergoing a course of chemoimmunotherapy.

3 Thorax

ADDITIONAL LEARNING RESOURCES for Chapter 3, *Thorax*, on STUDENT CONSULT (www.studentconsult.com):

- Image Library—illustrations of thoracic anatomy, Chapter 3
- Self-Assessment—National Board style multiple-choice questions, Chapter 3
- Short Questions—these are questions requiring short responses, Chapter 3
- Interactive Surface Anatomy—interactive surface animations, Chapter 3
- Medical Clinical Case Studies, Chapter 3
  - Cardiac tamponade
  - Patent ductus arteriosus
  - Subclavian steal syndrome
  - Sinus of Valsalva aneurysm

Free Online Anatomy and Embryology Self-Study Course

- Anatomy modules 4 through 9
- Embryology modules 61 through 64

Conceptual overview 123

General description 123
Functions 124
- Breathing 124
- Protection of vital organs 124
- Conduit 124

Component parts 124
- Thoracic wall 124
- Superior thoracic aperture 126
- Inferior thoracic aperture 126
- Diaphragm 127
- Mediastinum 128
- Pleural cavities 128

Relationship to other regions 130
- Neck 130
- Upper limb 130
- Abdomen 130
- Breast 131

Key features 132
- Vertebral level TIV/V 132
- Venous shunts from left to right 132
- Segmental neurovascular supply of thoracic wall 134
- Sympathetic system 136
- Flexible wall and inferior thoracic aperture 136
- Innervation of the diaphragm 138

Regional anatomy 139

Pectoral region 139
- Breast 139
- Muscles of the pectoral region 142

Thoracic wall 143
- Skeletal framework 143
- Intercostal spaces 150

Diaphragm 161
- Arterial supply 162
- Venous drainage 162
- Innervation 162

SUBASH KC/NMC-15TH/2014

Movements of the thoracic wall and diaphragm during breathing 162
Pleural cavities 162
Pleura 163
Lungs 167
Mediastinum 180
Middle mediastinum 180
Superior mediastinum 210
Posterior mediastinum 222
Anterior mediastinum 230

Surface anatomy 231

Thorax surface anatomy 231
How to count ribs 231
Surface anatomy of the breast in women 232
Visualizing structures at the TIV/V vertebral level 232
Visualizing structures in the superior mediastinum 234
Visualizing the margins of the heart 235
Where to listen for heart sounds 236
Visualizing the pleural cavities and lungs, pleural recesses, and lung lobes and fissures 236
Where to listen for lung sounds 238

Clinical cases 241

SUBASH KC/NMC-15TH/2014

Conceptual overview

GENERAL DESCRIPTION

The **thorax** is an irregularly shaped cylinder with a narrow opening (superior thoracic aperture) superiorly and a relatively large opening (inferior thoracic aperture) inferiorly (Fig. 3.1). The superior thoracic aperture is open, allowing continuity with the neck; the inferior thoracic aperture is closed by the diaphragm.

The musculoskeletal wall of the thorax is flexible and consists of segmentally arranged vertebrae, ribs, and muscles and the sternum.

The **thoracic cavity** enclosed by the thoracic wall and the diaphragm is subdivided into three major compartments:

- a left and a right pleural cavity, each surrounding a lung, and
- the mediastinum.

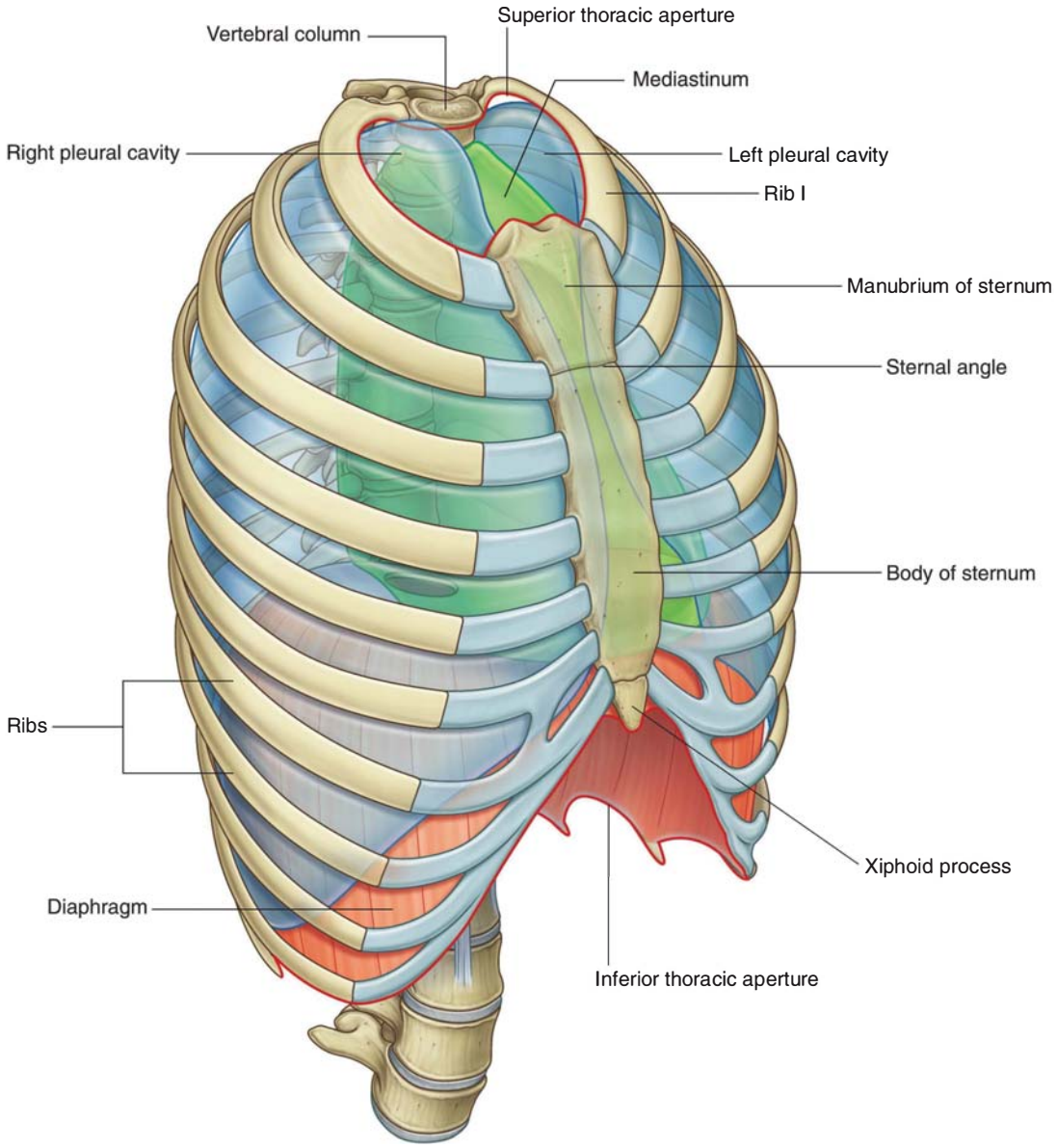


Fig. 3.1 Thoracic wall and cavity.

SUBASH KC/NMC-15TH/2014

The mediastinum is a thick, flexible soft tissue partition oriented longitudinally in a median sagittal position. It contains the heart, esophagus, trachea, major nerves, and major systemic blood vessels.

The pleural cavities are completely separated from each other by the mediastinum. Therefore, abnormal events in one pleural cavity do not necessarily affect the other cavity. This also means that the mediastinum can be entered surgically without opening the pleural cavities.

Another important feature of the pleural cavities is that they extend above the level of rib I. The apex of each lung actually extends into the root of the neck. As a consequence, abnormal events in the root of the neck can involve the adjacent pleura and lung, and events in the adjacent pleura and lung can involve the root of the neck.

FUNCTIONS

Breathing

One of the most important functions of the thorax is breathing. The thorax not only contains the lungs but also provides the machinery necessary—the diaphragm, thoracic wall, and ribs—for effectively moving air into and out of the lungs.

Up and down movements of the diaphragm and changes in the lateral and anterior dimensions of the thoracic wall, caused by movements of the ribs, alter the volume of the thoracic cavity and are key elements in breathing.

Protection of vital organs

The thorax houses and protects the heart, lungs, and great vessels. Because of the domed shape of the diaphragm, the thoracic wall also offers protection to some important abdominal viscera.

Much of the liver lies under the right dome of the diaphragm, and the stomach and spleen lie under the left. The posterior aspects of the superior poles of the kidneys lie on the diaphragm and are anterior to rib XII, on the right, and to ribs XI and XII, on the left.

Conduit

The mediastinum acts as a conduit for structures that pass completely through the thorax from one body region to another and for structures that connect organs in the thorax to other body regions.

The esophagus, vagus nerves, and thoracic duct pass through the mediastinum as they course between the abdomen and neck.

The phrenic nerves, which originate in the neck, also pass through the mediastinum to penetrate and supply the diaphragm.

Other structures such as the trachea, thoracic aorta, and superior vena cava course within the mediastinum en route to and from major visceral organs in the thorax.

COMPONENT PARTS

Thoracic wall

The **thoracic wall** consists of skeletal elements and muscles (Fig. 3.1):

- Posteriorly, it is made up of twelve thoracic vertebrae and their intervening intervertebral discs;
- Laterally, the wall is formed by **ribs** (twelve on each side) and three layers of flat muscles, which span the intercostal spaces between adjacent ribs, move the ribs, and provide support for the intercostal spaces;
- Anteriorly, the wall is made up of the **sternum**, which consists of the manubrium of sternum, body of sternum, and xiphoid process.

The manubrium of sternum, angled posteriorly on the body of sternum at the manubriosternal joint, forms the sternal angle, which is a major surface landmark used by clinicians in performing physical examinations of the thorax.

The anterior (distal) end of each rib is composed of costal cartilage, which contributes to the mobility and elasticity of the wall.

SUBASH KC/NMC-15TH/2014

All ribs articulate with thoracic vertebrae posteriorly. Most ribs (from rib II to IX) have three articulations with the vertebral column. The head of each rib articulates with the body of its own vertebra and with the body of the vertebra above (Fig. 3.2). As these ribs curve posteriorly, each also articulates with the transverse process of its vertebra.

Anteriorly, the costal cartilages of ribs I to VII articulate with the sternum.

The costal cartilages of ribs VIII to X articulate with the inferior margins of the costal cartilages above them. Ribs XI and XII are called floating ribs because they do not articulate with other ribs, costal cartilages, or the sternum. Their costal cartilages are small, only covering their tips.

The skeletal framework of the thoracic wall provides extensive attachment sites for muscles of the neck, abdomen, back, and upper limbs.

A number of these muscles attach to ribs and function as accessory respiratory muscles; some of them also stabilize the position of the first and last ribs.

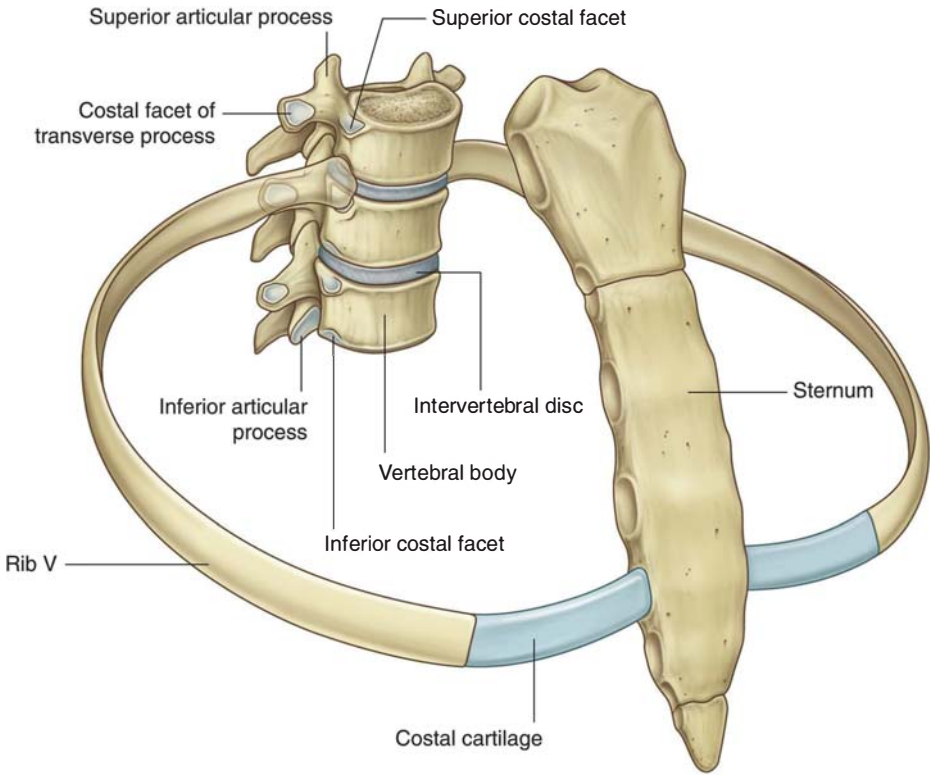


Fig. 3.2 Joints between ribs and vertebrae.

SUBASH KC/NMC-15TH/2014

Superior thoracic aperture

Completely surrounded by skeletal elements, the **superior thoracic aperture** consists of the body of vertebra TI posteriorly, the medial margin of rib I on each side, and the manubrium anteriorly.

The superior margin of the manubrium is in approximately the same horizontal plane as the intervertebral disc between vertebrae TII and TIII.

The first ribs slope inferiorly from their posterior articulation with vertebra TI to their anterior attachment to the manubrium. Consequently, the plane of the superior thoracic aperture is at an oblique angle, facing somewhat anteriorly.

At the superior thoracic aperture, the superior aspects of the pleural cavities, which surround the lungs, lie on either side of the entrance to the mediastinum (Fig. 3.3).

Structures that pass between the upper limb and thorax pass over rib I and the superior part of the pleural cavity as they enter and leave the mediastinum. Structures that pass between the neck and head and the thorax pass more vertically through the superior thoracic aperture.

Inferior thoracic aperture

The **inferior thoracic aperture** is large and expandable. Bone, cartilage, and ligaments form its margin (Fig. 3.4A).

The inferior thoracic aperture is closed by the diaphragm, and structures passing between the abdomen and thorax pierce or pass posteriorly to the diaphragm.

Skeletal elements of the inferior thoracic aperture are:

- the body of vertebra TXII posteriorly,
- rib XII and the distal end of rib XI posterolaterally,
- the distal cartilaginous ends of ribs VII to X, which unite to form the costal margin anterolaterally, and
- the xiphoid process anteriorly.

The joint between the costal margin and sternum lies roughly in the same horizontal plane as the intervertebral disc between vertebrae TIX and TX. In other words, the posterior margin of the inferior thoracic aperture is inferior to the anterior margin.

When viewed anteriorly, the inferior thoracic aperture is tilted superiorly.

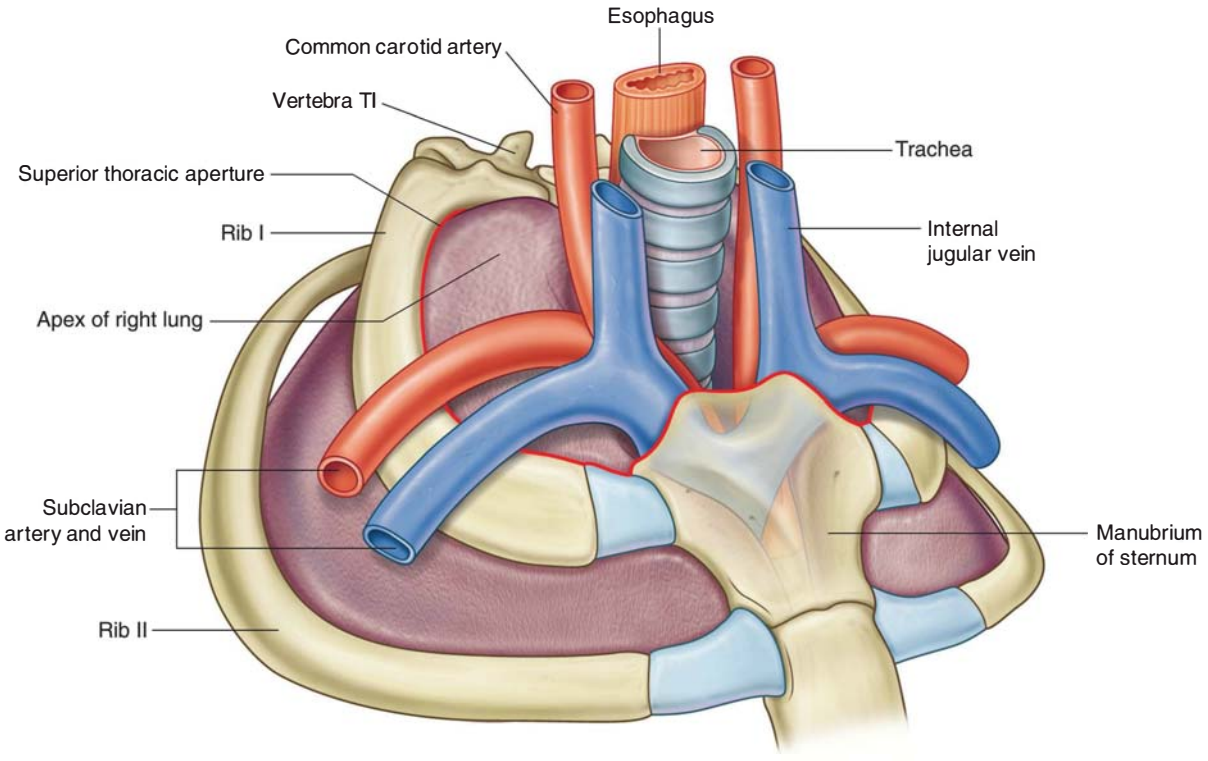


Fig. 3.3 Superior thoracic aperture.

SUBASH KC/NMC-15TH/2014

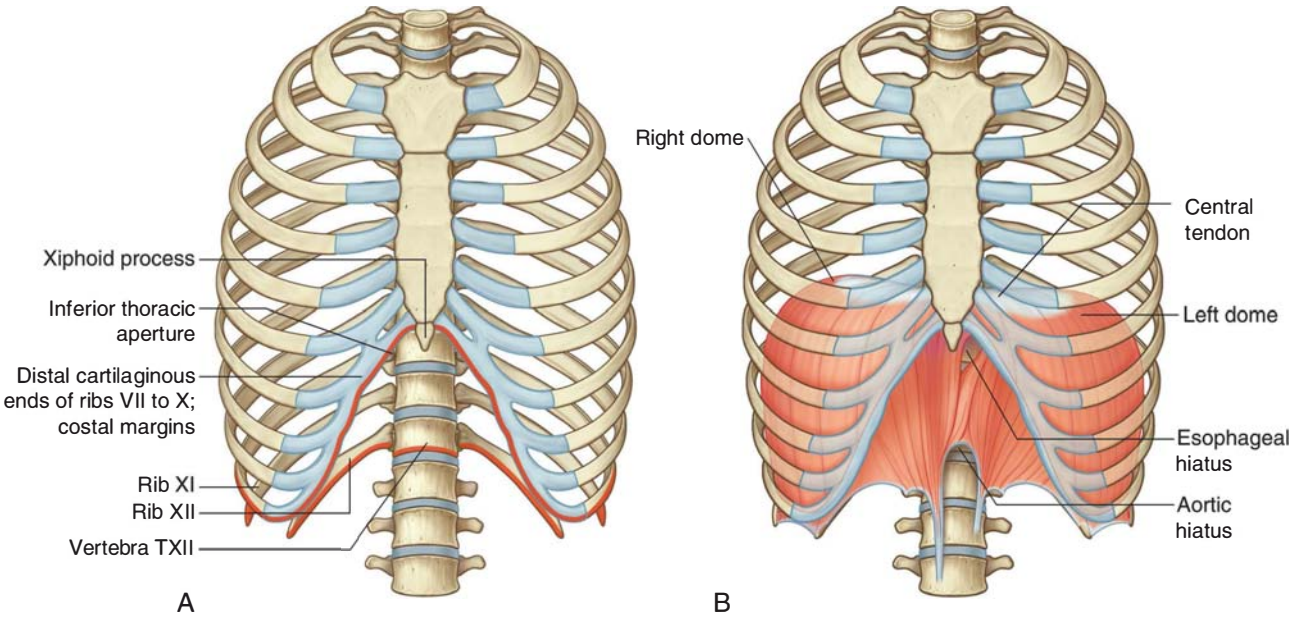


Fig. 3.4 A. Inferior thoracic aperture. **B.** Diaphragm.

Diaphragm

The musculotendinous **diaphragm** seals the inferior thoracic aperture (Fig. 3.4B).

Generally, muscle fibers of the diaphragm arise radially, from the margins of the inferior thoracic aperture, and converge into a large central tendon.

Because of the oblique angle of the inferior thoracic aperture, the posterior attachment of the diaphragm is inferior to the anterior attachment.

The diaphragm is not flat; rather, it "balloons" superiorly, on both the right and left sides, to form domes. The right dome is higher than the left, reaching as far as rib V.

As the diaphragm contracts, the height of the domes decreases and the volume of the thorax increases.

The esophagus and inferior vena cava penetrate the diaphragm; the aorta passes posterior to the diaphragm.

SUBASH KC/NMC-15TH/2014

Mediastinum

The **mediastinum** is a thick midline partition that extends from the sternum anteriorly to the thoracic vertebrae posteriorly, and from the superior thoracic aperture to the inferior thoracic aperture.

A horizontal plane passing through the sternal angle and the intervertebral disc between vertebrae TIV and TV separates the mediastinum into superior and inferior parts (Fig. 3.5). The inferior part is further subdivided by the pericardium, which encloses the pericardial cavity surrounding the heart. The pericardium and heart constitute the middle mediastinum.

The anterior mediastinum lies between the sternum and the pericardium; the posterior mediastinum lies between the pericardium and thoracic vertebrae.

Pleural cavities

The two pleural cavities are situated on either side of the mediastinum (Fig. 3.6).

Each **pleural cavity** is completely lined by a mesothelial membrane called the pleura.

During development, the lungs grow out of the mediastinum, becoming surrounded by the pleural cavities. As a result, the outer surface of each organ is covered by pleura.

Each lung remains attached to the mediastinum by a root formed by the airway, pulmonary blood vessels, lymphatic tissues, and nerves.

The pleura lining the walls of the cavity is the parietal pleura, whereas that reflected from the mediastinum at the roots and onto the surfaces of the lungs is the visceral pleura. Only a potential space normally exists between the visceral pleura covering lung and the parietal pleura lining the wall of the thoracic cavity.

The lung does not completely fill the potential space of the pleural cavity, resulting in recesses, which do not contain lung and are important for accommodating changes in lung volume during breathing. The costodiaphragmatic recess, which is the largest and clinically most important recess, lies inferiorly between the thoracic wall and diaphragm.

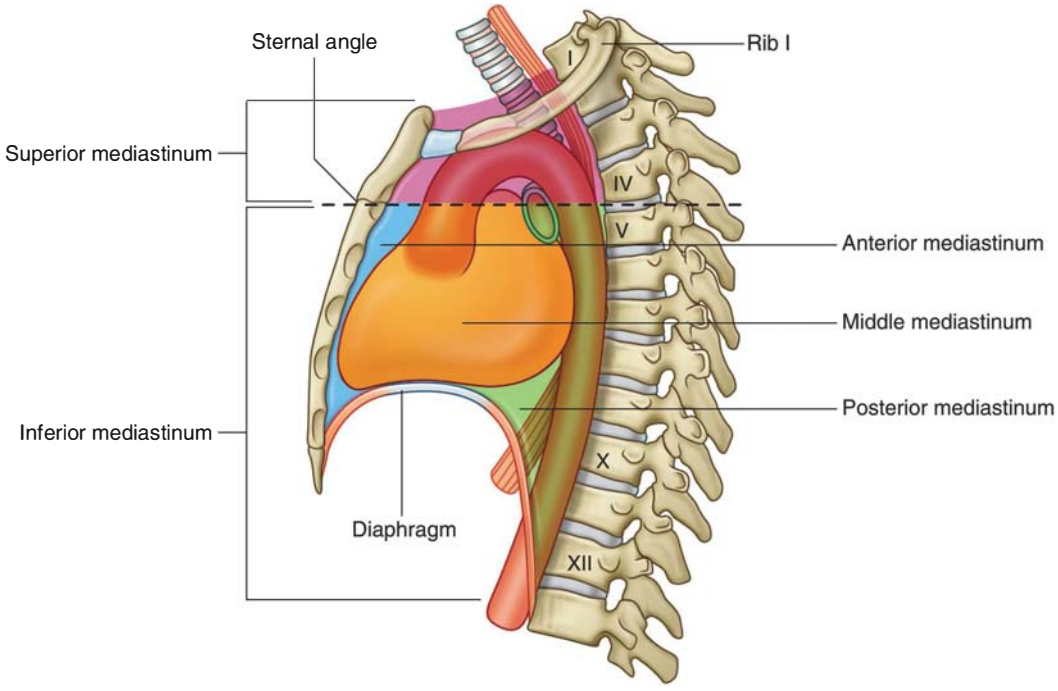


Fig. 3.5 Subdivisions of the mediastinum.

SUBASH KC/NMC-15TH/2014

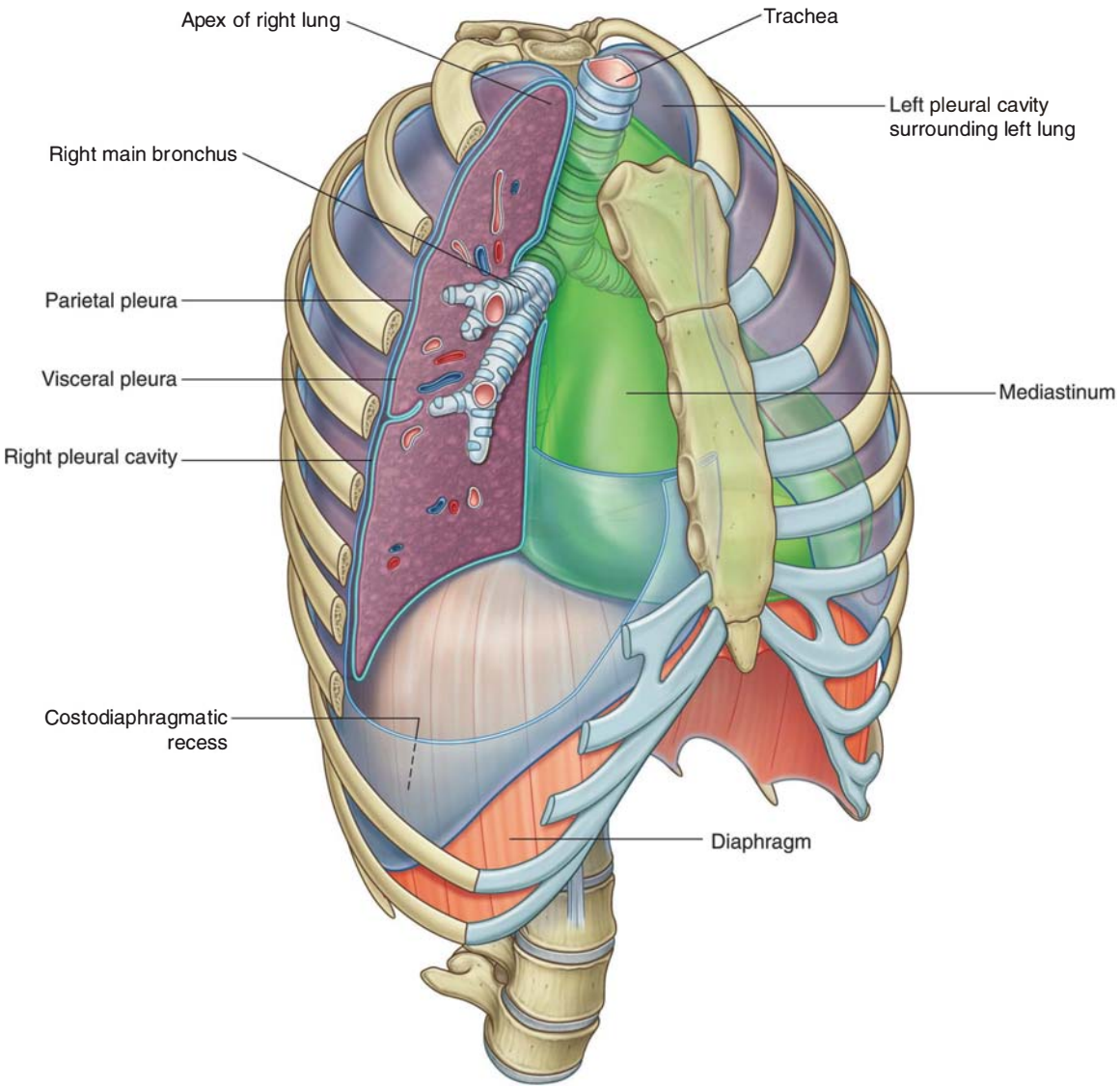


Fig. 3.6 Pleural cavities.

SUBASH KC/NMC-15TH/2014

RELATIONSHIP TO OTHER REGIONS

Neck

The superior thoracic aperture opens directly into the root of the neck (Fig. 3.7).

The superior aspect of each pleural cavity extends approximately 2–3 cm above rib I and the costal cartilage into the neck. Between these pleural extensions, major visceral structures pass between the neck and superior mediastinum. In the midline, the trachea lies immediately anterior to the esophagus. Major blood vessels and nerves pass in and out of the thorax at the superior thoracic aperture anteriorly and laterally to these structures.

Upper limb

An **axillary inlet**, or gateway to the upper limb, lies on each side of the superior thoracic aperture. These two axillary inlets and the superior thoracic aperture communicate superiorly with the root of the neck (Fig. 3.7).

Each axillary inlet is formed by:

- the superior margin of the scapula posteriorly,
- the clavicle anteriorly, and
- the lateral margin of rib I medially.

The apex of each triangular inlet is directed laterally and is formed by the medial margin of the coracoid process, which extends anteriorly from the superior margin of the scapula.

The base of the axillary inlet's triangular opening is the lateral margin of rib I.

Large blood vessels passing between the axillary inlet and superior thoracic aperture do so by passing over rib I.

Proximal parts of the brachial plexus also pass between the neck and upper limb by passing through the axillary inlet.

Abdomen

The diaphragm separates the thorax from the abdomen. Structures that pass between the thorax and abdomen either penetrate the diaphragm or pass posteriorly to it (Fig. 3.8):

- The inferior vena cava pierces the **central tendon of the diaphragm** to enter the right side of the mediastinum near vertebral level TVIII.
- The esophagus penetrates the muscular part of the diaphragm to leave the mediastinum and enter the abdomen just to the left of the midline at vertebral level TX.

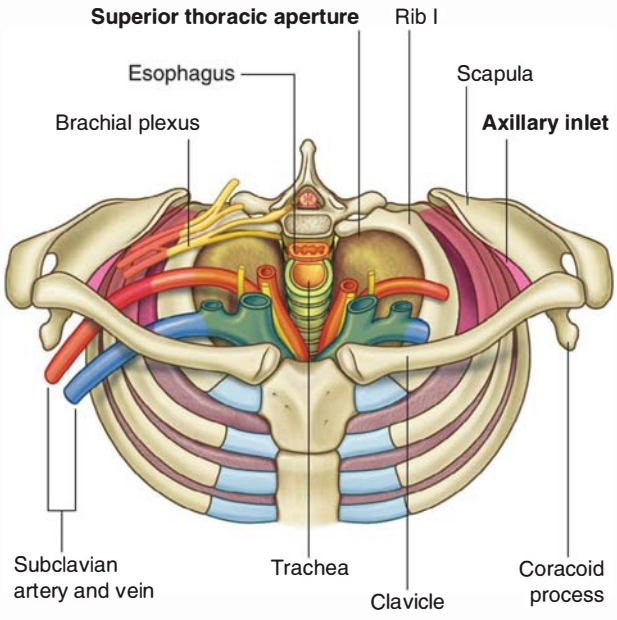


Fig. 3.7 Superior thoracic aperture and axillary inlet.

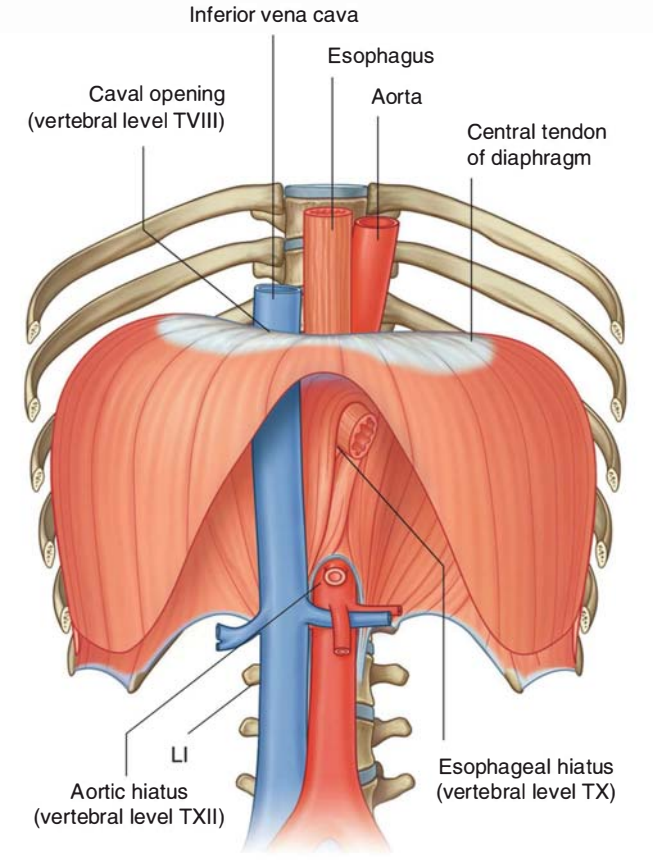


Fig. 3.8 Major structures passing between abdomen and thorax.

SUBASH KC/NMC-15TH/2014

- The aorta passes posteriorly to the diaphragm at the midline at vertebral level TXII.
- Numerous other structures that pass between the thorax and abdomen pass through or posterior to the diaphragm.

Breast

The breasts, consisting of secretory glands, superficial fascia, and overlying skin, are in the **pectoral region** on each side of the anterior thoracic wall (Fig. 3.9).

Vessels, lymphatics, and nerves associated with the breast are as follows:

- Branches from the internal thoracic arteries and veins perforate the anterior chest wall on each side of the sternum to supply anterior aspects of the thoracic wall. Those branches associated mainly with the second to fourth intercostal spaces also supply the anteromedial parts of each breast.
- Lymphatic vessels from the medial part of the breast accompany the perforating arteries and drain into the parasternal nodes on the deep surface of the thoracic wall.
- Vessels and lymphatics associated with lateral parts of the breast emerge from or drain into the **axillary region** of the upper limb.
- Lateral and anterior branches of the fourth to sixth intercostal nerves carry general sensation from the skin of the breast.

SUBASH KC/NMC-15TH/2014

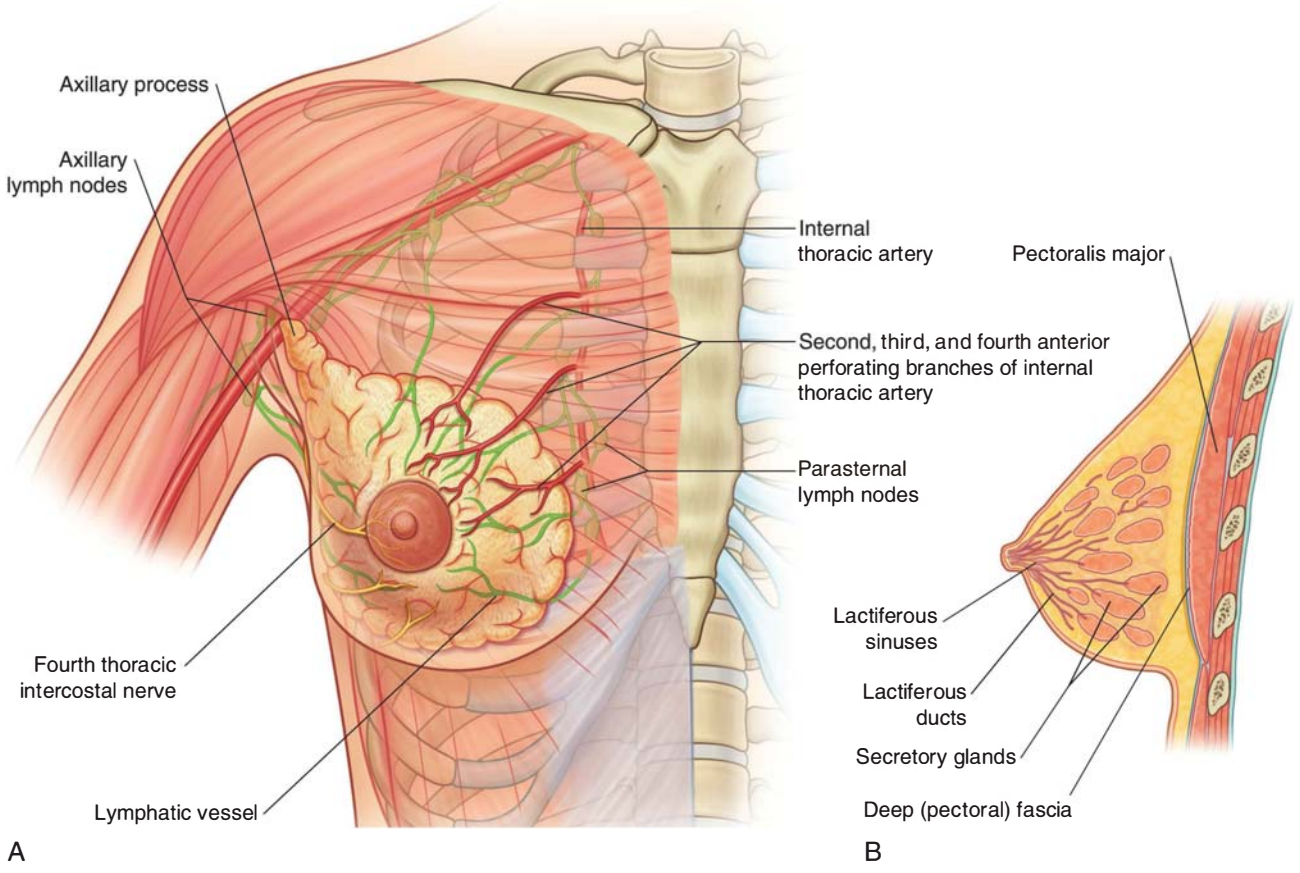


Fig. 3.9 Right breast.

KEY FEATURES

Vertebral level TIV/V

When working with patients, physicians use vertebral levels to determine the position of important anatomical structures within body regions.

The horizontal plane passing through the disc that separates thoracic vertebrae TIV and TV is one of the most significant planes in the body (Fig. 3.10) because it:

- passes through the sternal angle anteriorly, marking the position of the anterior articulation of the costal cartilage of rib II with the sternum. The sternal angle is used to find the position of rib II as a reference for counting ribs (because of the overlying clavicle, rib I is not palpable);
- separates the superior mediastinum from the inferior mediastinum and marks the position of the superior limit of the pericardium;
- marks where the arch of the aorta begins and ends;
- passes through the site where the superior vena cava penetrates the pericardium to enter the heart;
- is the level at which the trachea bifurcates into right and left main bronchi; and
- marks the superior limit of the pulmonary trunk.

Venous shunts from left to right

The **right atrium** is the chamber of the heart that receives deoxygenated blood returning from the body. It lies on the right side of the midline, and the two major veins, the superior and inferior venae cavae, that drain into it are also located on the right side of the body. This means that, to get to the right side of the body, all blood coming from the left side has to cross the midline. This left-to-right shunting is carried out by a number of important and, in some cases, very large veins, several of which are in the thorax (Fig. 3.11).

In adults, the left brachiocephalic vein crosses the midline immediately posterior to the manubrium and delivers blood from the left side of the head and neck, the left upper limb, and part of the left thoracic wall into the superior vena cava.

The hemiazygos and accessory hemiazygos veins drain posterior and lateral parts of the left thoracic wall, pass immediately anterior to the bodies of thoracic vertebrae, and flow into the azygos vein on the right side, which ultimately connects with the superior vena cava.

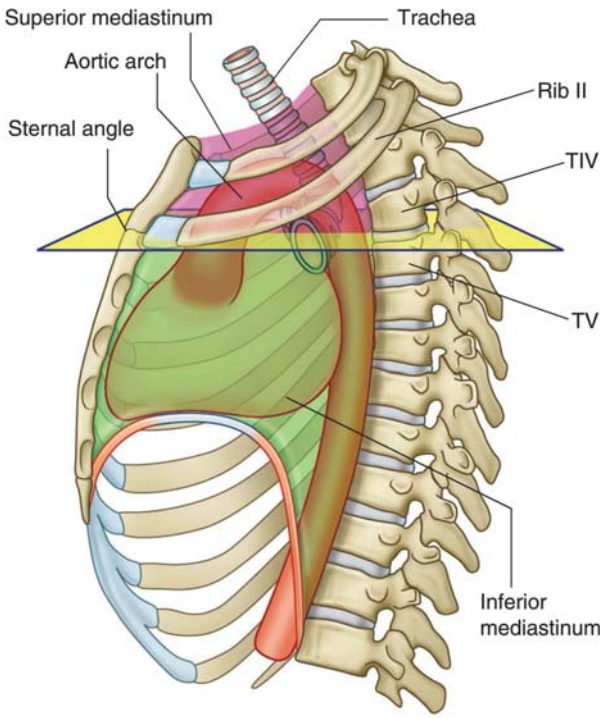


Fig. 3.10 Vertebral level TIV/V.

SUBASH KC/NMC-15TH/2014

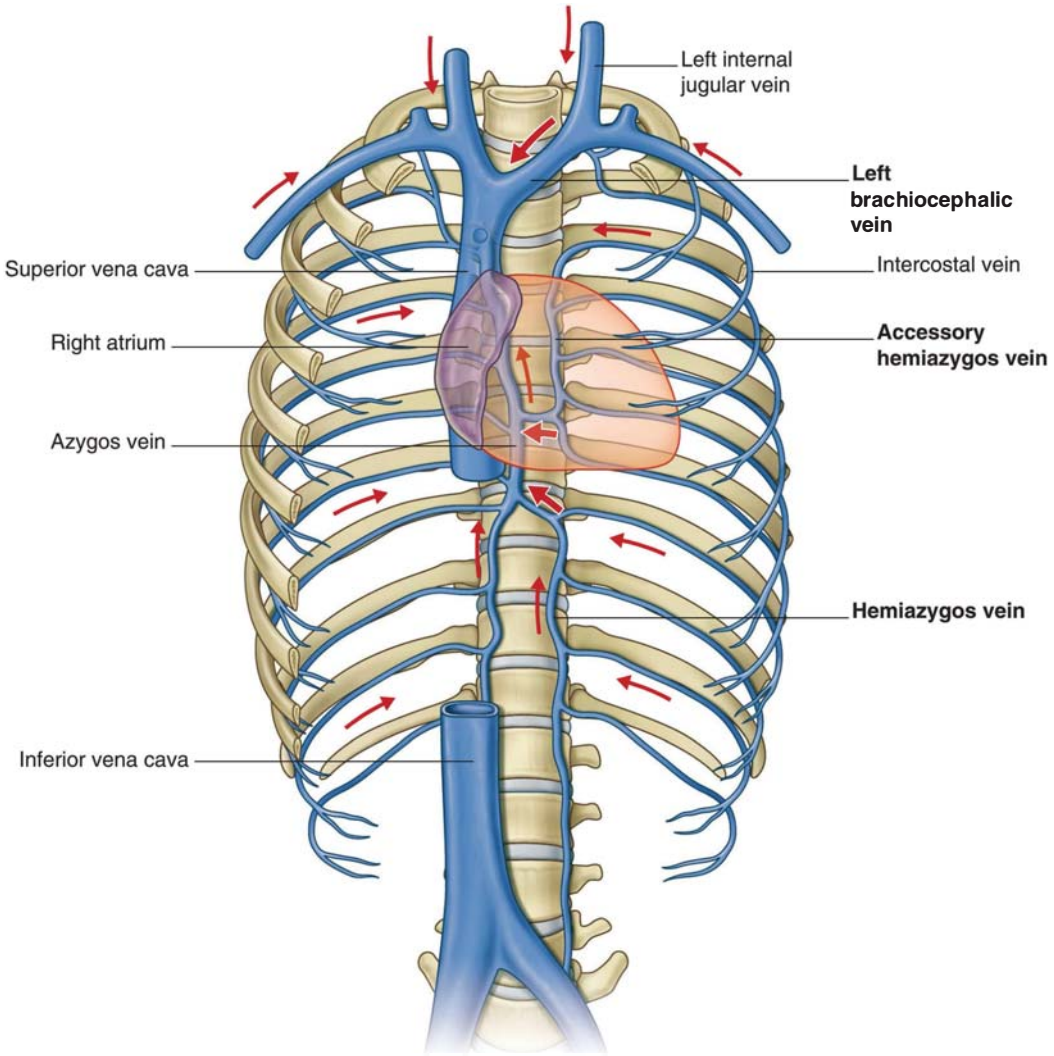


Fig. 3.11 Left-to-right venous shunts.

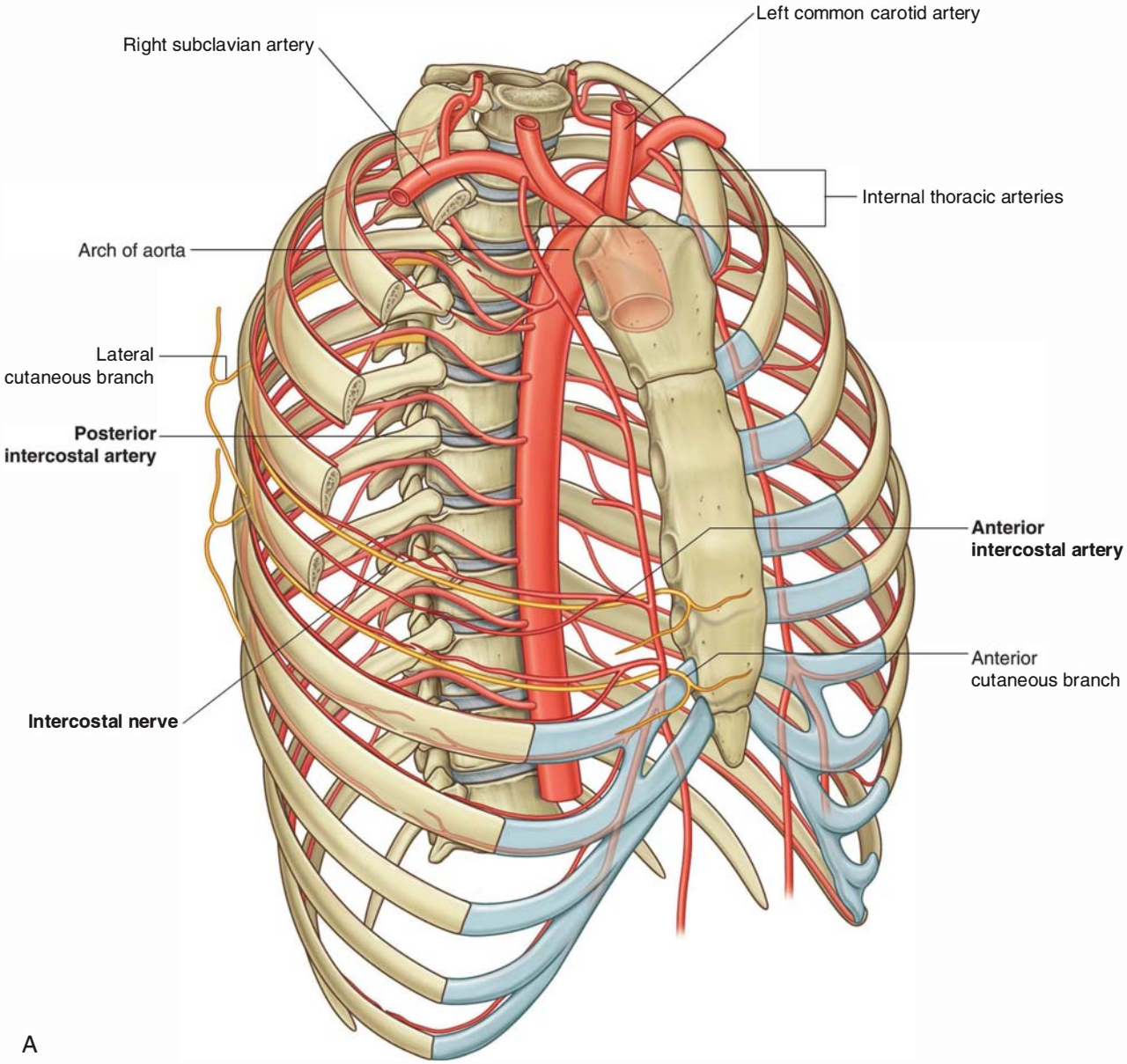
SUBASH KC/NMC-15TH/2014

Segmental neurovascular supply of thoracic wall

The arrangement of vessels and nerves that supply the thoracic wall reflects the segmental organization of the wall. Arteries to the wall arise from two sources:

- the thoracic aorta, which is in the posterior mediastinum, and
- a pair of vessels, the internal thoracic arteries, which run along the deep aspect of the anterior thoracic wall on either side of the sternum.

Posterior and anterior intercostal vessels branch segmentally from these arteries and pass laterally around the wall, mainly along the inferior margin of each rib (Fig. 3.12A). Running with these vessels are intercostal nerves (the anterior rami of thoracic spinal nerves), which innervate the wall, related parietal pleura, and associated



SUBASH KC/NMC-15TH/2014

Fig. 3.12 A. Segmental neurovascular supply of thoracic wall.

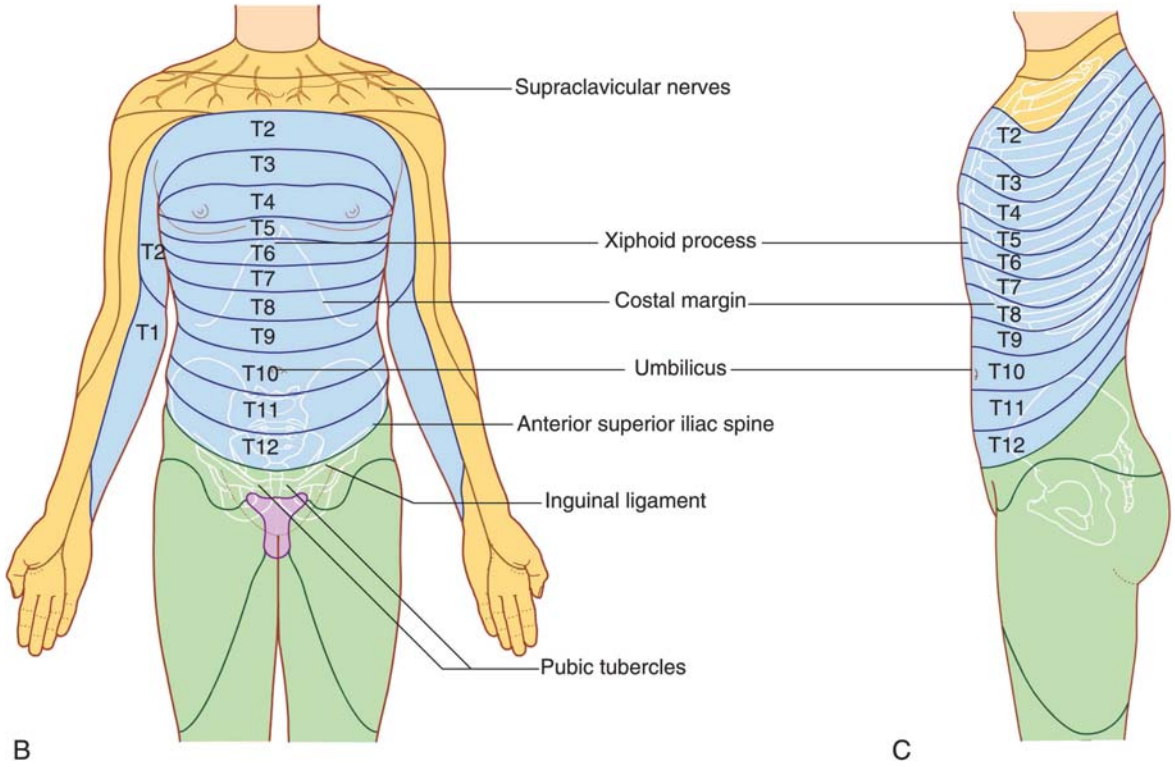


Fig. 3.12, cont'd B. Anterior view of thoracic dermatomes associated with thoracic spinal nerves. **C.** Lateral view of dermatomes associated with thoracic spinal nerves.

skin. The position of these nerves and vessels relative to the ribs must be considered when passing objects, such as chest tubes, through the thoracic wall.

Dermatomes of the thorax generally reflect the segmental organization of the thoracic spinal nerves (Fig. 3.12B). The exception occurs, anteriorly and superiorly, with the first thoracic dermatome, which is located mostly in the upper limb, and not on the trunk.

The anterosuperior region of the trunk receives branches from the anterior ramus of C4 via supraclavicular branches of the cervical plexus.

The highest thoracic dermatome on the anterior chest wall is T2, which also extends into the upper limb. In the midline, skin over the xiphoid process is innervated by T6.

Dermatomes of T7 to T12 follow the contour of the ribs onto the anterior abdominal wall (Fig. 3.12C).

SUBASH KC/NMC-15TH/2014

Sympathetic system

All preganglionic nerve fibers of the sympathetic system are carried out of the spinal cord in spinal nerves T1 to L2 (Fig. 3.13). This means that sympathetic fibers found anywhere in the body ultimately emerge from the spinal cord as components of these spinal nerves. Preganglionic sympathetic fibers destined for the head are carried out of the spinal cord in spinal nerve T1.

Flexible wall and inferior thoracic aperture

The thoracic wall is expandable because most ribs articulate with other components of the wall by true joints that allow movement, and because of the shape and orientation of the ribs (Fig. 3.14).

A rib's posterior attachment is superior to its anterior attachment. Therefore, when a rib is elevated, it moves the anterior thoracic wall forward relative to the posterior wall, which is fixed. In addition, the middle part of each rib is inferior to its two ends, so that when this region of the rib is elevated, it expands the thoracic wall laterally. Finally, because the diaphragm is muscular, it changes the volume of the thorax in the vertical direction.

Changes in the anterior, lateral, and vertical dimensions of the thoracic cavity are important for breathing.

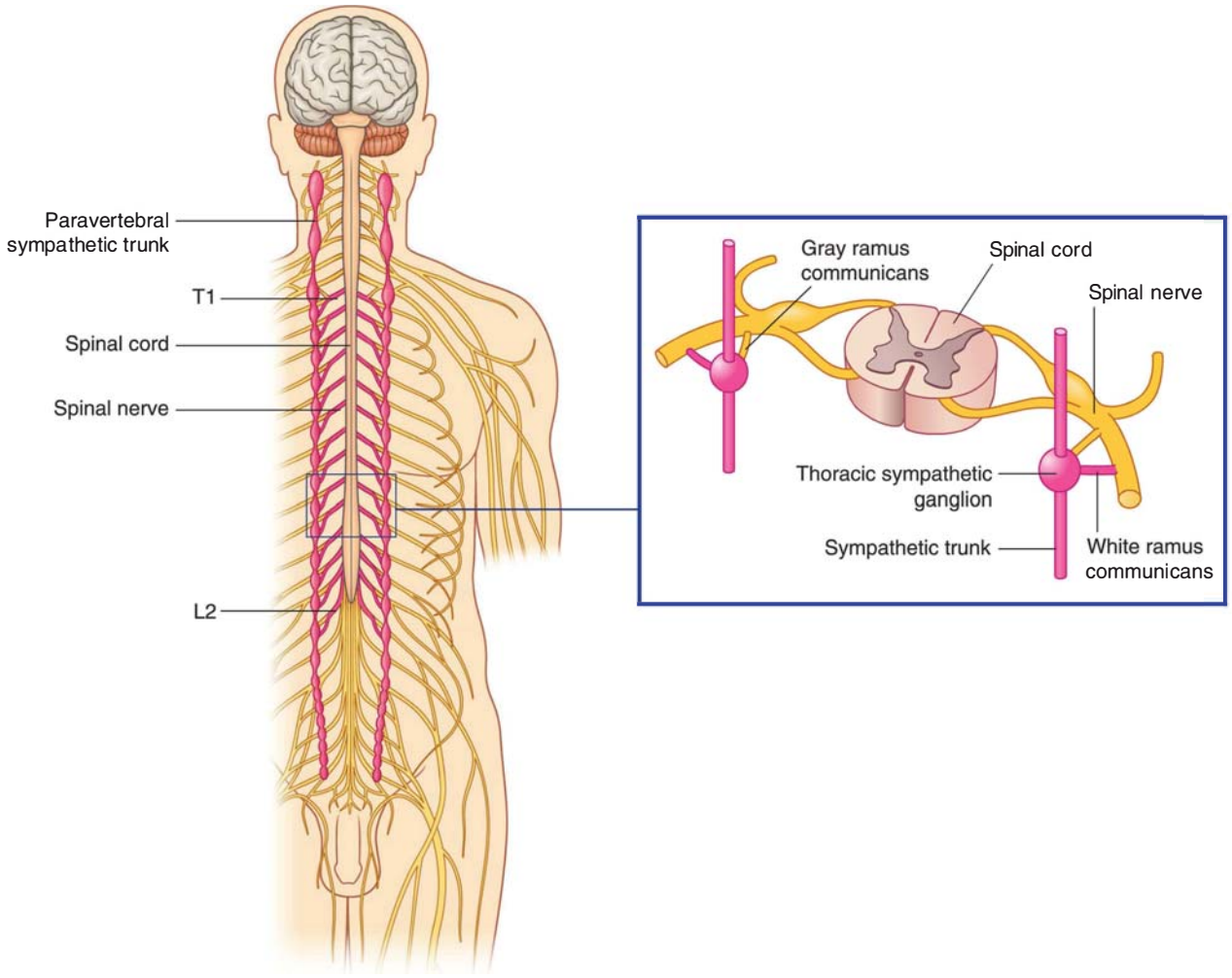


Fig. 3.13 Sympathetic trunks.

SUBASH KC/NMC-15TH/2014

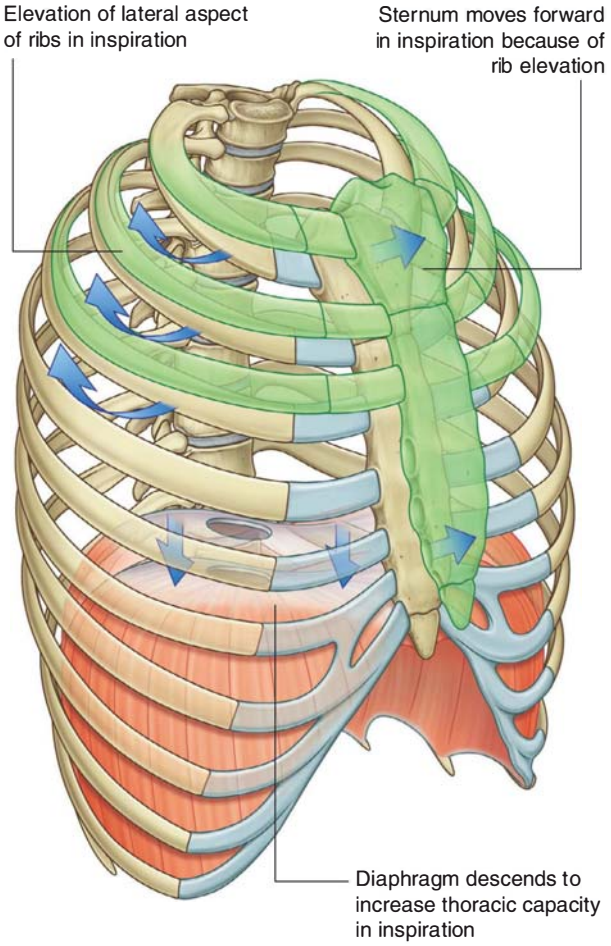


Fig. 3.14 Flexible thoracic wall and inferior thoracic aperture.

SUBASH KC/NMC-15TH/2014

Innervation of the diaphragm

The diaphragm is innervated by two phrenic nerves that originate, one on each side, as branches of the cervical plexus in the neck (Fig. 3.15). They arise from the anterior rami of cervical nerves C3, C4, and C5, with the major contribution coming from C4.

The **phrenic nerves** pass vertically through the neck, the superior thoracic aperture, and the mediastinum to supply motor innervation to the entire diaphragm, including the crura (muscular extensions that attach the diaphragm to the upper lumbar vertebrae). In the mediastinum, the phrenic nerves pass anteriorly to the roots of the lungs.

The tissues that initially give rise to the diaphragm are in an anterior position on the embryological disc before the head fold develops, which explains the cervical origin of the nerves that innervate the diaphragm. In other words, the tissue that gives rise to the diaphragm originates superior to the ultimate location of the diaphragm.

Spinal cord injuries below the level of the origin of the phrenic nerve do not affect movement of the diaphragm.

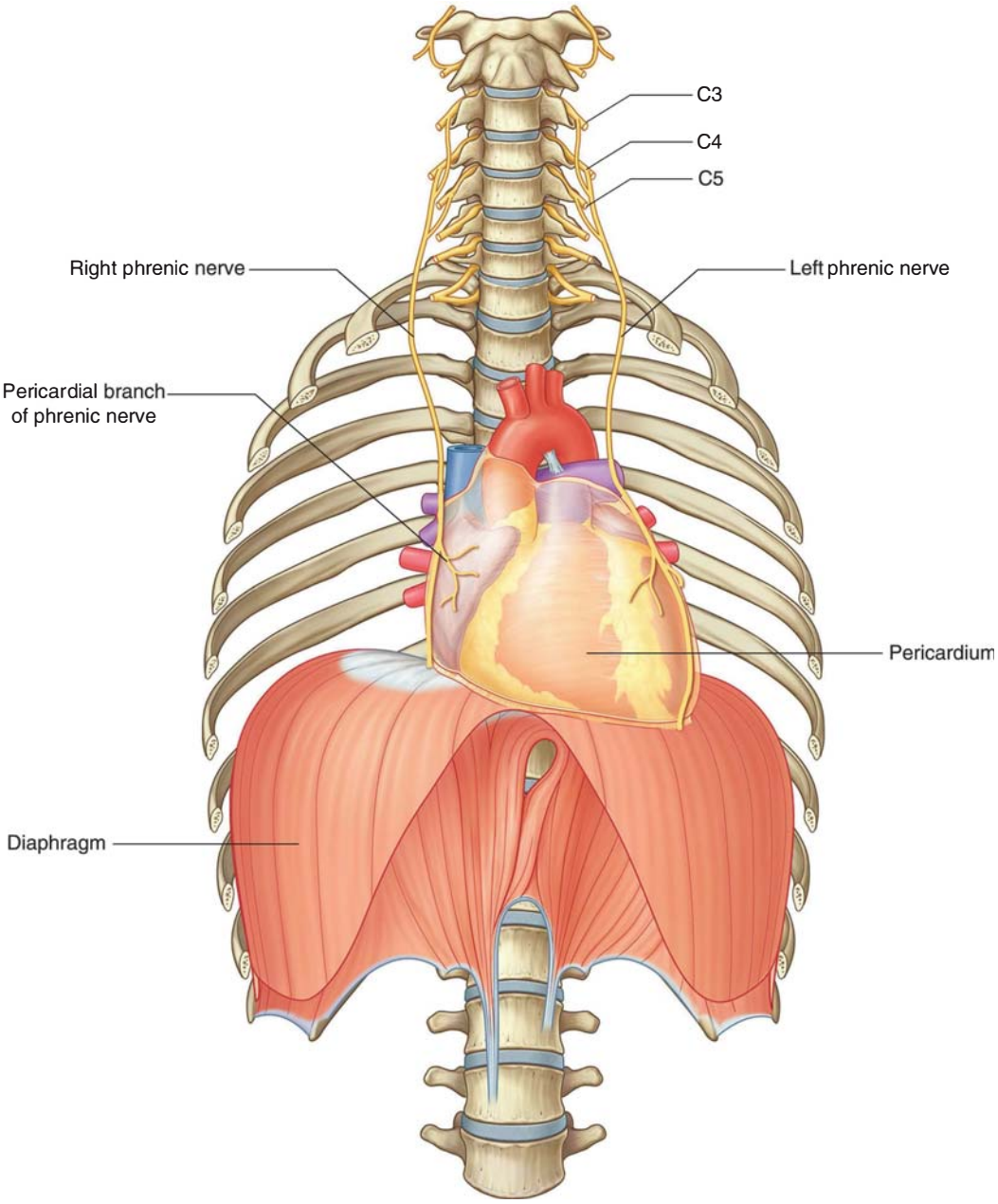


Fig. 3.15 Innervation of the diaphragm.

SUBASH KC/NMC-15TH/2014

Regional anatomy

The cylindrical thorax consists of:

- a wall,
- two pleural cavities,
- the lungs, and
- the mediastinum.

The thorax houses the heart and lungs, acts as a conduit for structures passing between the neck and the abdomen, and plays a principal role in breathing. In addition, the thoracic wall protects the heart and lungs and provides support for the upper limbs. Muscles anchored to the anterior thoracic wall provide some of this support, and together with their associated connective tissues, nerves, and vessels, and the overlying skin and superficial fascia, define the pectoral region.

PECTORAL REGION

The pectoral region is external to the anterior thoracic wall and anchors the upper limb to the trunk. It consists of:

- a superficial compartment containing skin, superficial fascia, and breasts; and
- a deep compartment containing muscles and associated structures.

Nerves, vessels, and lymphatics in the superficial compartment emerge from the thoracic wall, the axilla, and the neck.

Breast

The breasts consist of mammary glands and associated skin and connective tissues. The **mammary glands** are modified sweat glands in the superficial fascia anterior to the pectoral muscles and the anterior thoracic wall (Fig. 3.16).

The mammary glands consist of a series of ducts and associated secretory lobules. These converge to form 15 to 20 **lactiferous ducts**, which open independently onto the **nipple**. The nipple is surrounded by a circular pigmented area of skin termed the **areola**.

A well-developed, connective tissue stroma surrounds the ducts and lobules of the mammary gland. In certain regions, this condenses to form well-defined ligaments, the suspensory ligaments of breast, which are continuous with the dermis of the skin and support the breast. Carcinoma of the breast creates tension on these ligaments, causing pitting of the skin.

In nonlactating women, the predominant component of the breasts is fat, while glandular tissue is more abundant in lactating women.

The breast lies on deep fascia related to the pectoralis major muscle and other surrounding muscles. A layer of loose connective tissue (the **retromammary space**) separates the breast from the deep fascia and provides some degree of movement over underlying structures.

The base, or attached surface, of each breast extends vertically from ribs II to VI, and transversely from the sternum to as far laterally as the midaxillary line.

Arterial supply

The breast is related to the thoracic wall and to structures associated with the upper limb; therefore, vascular supply and drainage can occur by multiple routes (Fig. 3.16):

- laterally, vessels from the axillary artery—superior thoracic, thoraco-acromial, lateral thoracic, and subscapular arteries;
- medially, branches from the internal thoracic artery;
- the second to fourth intercostal arteries via branches that perforate the thoracic wall and overlying muscle.

Venous drainage

Veins draining the breast parallel the arteries and ultimately drain into the axillary, internal thoracic, and intercostal veins.

Innervation

Innervation of the breast is via anterior and lateral cutaneous branches of the second to sixth intercostal nerves. The nipple is innervated by the fourth intercostal nerve.

Lymphatic drainage

Lymphatic drainage of the breast is as follows:

- Approximately 75% is via lymphatic vessels that drain laterally and superiorly into **axillary nodes** (Fig. 3.16).
- Most of the remaining drainage is into parasternal nodes deep to the anterior thoracic wall and associated with the internal thoracic artery.
- Some drainage may occur via lymphatic vessels that follow the lateral branches of posterior intercostal arteries and connect with intercostal nodes situated near the heads and necks of ribs.

SUBASH KC/NMC-15TH/2014

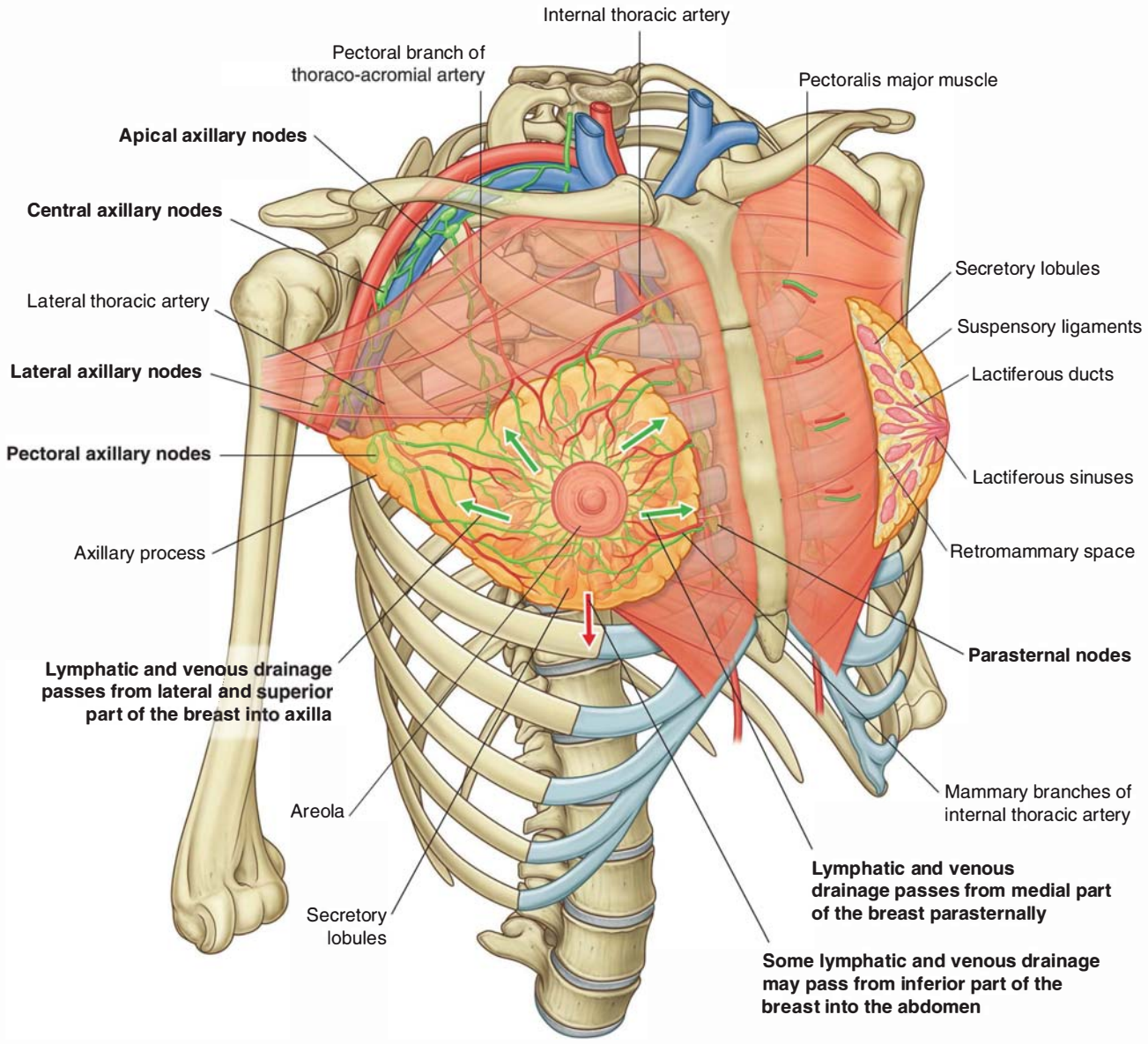


Fig. 3.16 Breasts.

SUBASH KC/NMC-15TH/2014

Axillary nodes drain into the subclavian trunks, parasternal nodes drain into the bronchomediastinal trunks, and intercostal nodes drain either into the thoracic duct or into the bronchomediastinal trunks.

Breast in men

The breast in men is rudimentary and consists only of small ducts, often composed of cords of cells, that normally do not extend beyond the areola. Breast cancer can occur in men.

In the clinic

Axillary tail of breast

It is important for clinicians to remember when evaluating the breast for pathology that the upper lateral region of the breast can project around the lateral margin of the pectoralis major muscle and into the axilla. This axillary process (axillary tail) may perforate deep fascia and extend as far superiorly as the apex of the axilla.

SUBASH KC/NMC-15TH/2014

In the clinic

Breast cancer

Breast cancer is one of the most common malignancies in women. It develops in the cells of the acini, lactiferous ducts, and lobules of the breast. Tumor growth and spread depends on the exact cellular site of origin of the cancer. These factors affect the response to surgery, chemotherapy, and radiotherapy. Breast tumors spread via the lymphatics and veins, or by direct invasion.

When a patient has a lump in the breast, a diagnosis of breast cancer is confirmed by a biopsy and histological evaluation. Once confirmed, the clinician must attempt to stage the tumor.

Staging the tumor means defining the:

- size of the primary tumor,
- exact site of the primary tumor,
- number and sites of lymph node spread, and
- organs to which the tumor may have spread.

Computed tomography (CT) scanning of the body may be carried out to look for any spread to the lungs (pulmonary metastases), liver (hepatic metastases), or bone (bony metastases).

Further imaging may include bone scanning using radioactive isotopes, which are avidly taken up by the tumor metastases in bone.

Lymph drainage of the breast is complex. Lymph vessels pass to axillary, supraclavicular, and parasternal nodes and may even pass to abdominal lymph nodes, as well as to the opposite breast. Containment of nodal metastatic breast cancer is therefore potentially difficult because it can spread through many lymph node groups.

Subcutaneous lymphatic obstruction and tumor growth pull on connective tissue ligaments in the breast, resulting in the appearance of an orange peel texture (**peau d'orange**) on the surface of the breast. Further subcutaneous spread can induce a rare manifestation of breast cancer that produces a hard, woody texture to the skin (**cancer en cuirasse**).

A mastectomy (surgical removal of the breast) involves excision of the breast tissue to the pectoralis major muscle and fascia. Within the axilla the breast tissue must be removed from the medial axillary wall. Closely applied to the medial axillary wall is the long thoracic nerve. Damage to this nerve can result in paralysis of the serratus anterior muscle, producing a characteristic "winged" scapula. It is also possible to damage the nerve to the latissimus dorsi muscle, and this may affect extension, medial rotation, and adduction of the humerus.

Muscles of the pectoral region

Each pectoral region contains the pectoralis major, pectoralis minor, and subclavius muscles (Fig. 3.17 and Table 3.1). All originate from the anterior thoracic wall and insert into bones of the upper limb.

Pectoralis major

The **pectoralis major** muscle is the largest and most superficial of the pectoral region muscles. It directly underlies the breast and is separated from it by deep fascia and the loose connective tissue of the retromammary space.

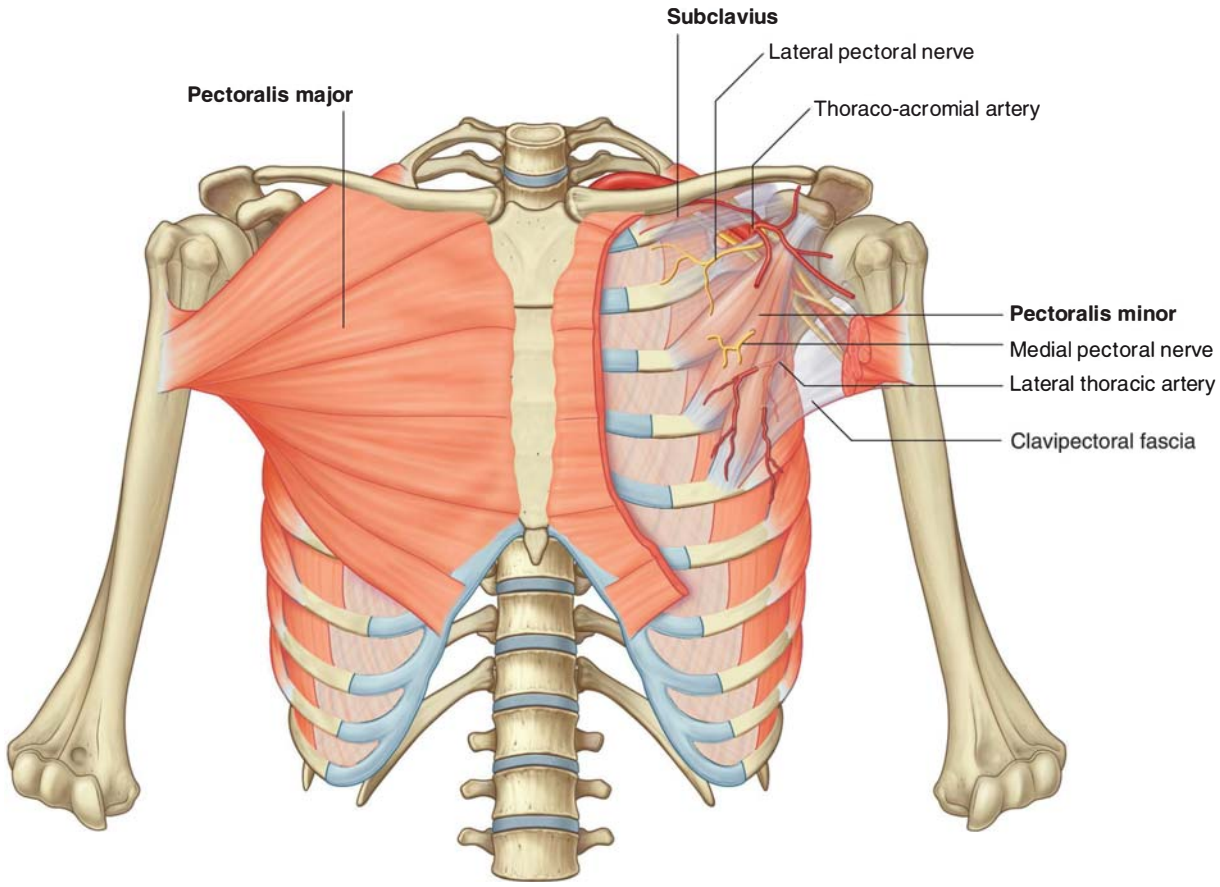


Fig. 3.17 Muscles and fascia of the pectoral region.

Table 3.1 Muscles of the pectoral region

Muscle	Origin	Insertion	Innervation	Function
Pectoralis major	Medial half of clavicle and anterior surface of sternum, first seven costal cartilages, aponeurosis of external oblique	Lateral lip of intertubercular sulcus of humerus	Medial and lateral pectoral nerves	Adduction, medial rotation, and flexion of the humerus at the shoulder joint
Subclavius	Rib I at junction between rib and costal cartilage	Groove on inferior surface of middle third of clavicle	Nerve to subclavius	Pulls clavicle medially to stabilize sternoclavicular joint; depresses tip of shoulder
Pectoralis minor	Anterior surfaces of the third, fourth, and fifth ribs, and deep fascia overlying the related intercostal spaces	Coracoid process of scapula	Medial pectoral nerves	Depresses tip of shoulder; protracts scapula

SUBASH KC/NMC-15TH/2014

The pectoralis major has a broad origin that includes the anterior surfaces of the medial half of the clavicle, the sternum, and related costal cartilages. The muscle fibers converge to form a flat tendon, which inserts into the lateral lip of the intertubercular sulcus of the humerus.

The pectoralis major adducts, flexes, and medially rotates the arm.

Subclavius and pectoralis minor muscles

The **subclavius** and **pectoralis minor muscles** underlie the pectoralis major:

- The subclavius is small and passes laterally from the anterior and medial part of rib I to the inferior surface of the clavicle.
- The pectoralis minor passes from the anterior surfaces of ribs III to V to the coracoid process of the scapula.

Both the subclavius and pectoralis minor pull the tip of the shoulder inferiorly.

A continuous layer of deep fascia, the **clavipectoral fascia**, encloses the subclavius and pectoralis minor and attaches to the clavicle above and to the floor of the axilla below.

The muscles of the pectoral region form the anterior wall of the axilla, a region between the upper limb and the neck through which all major structures pass. Nerves, vessels, and lymphatics that pass between the pectoral region and the axilla pass through the clavipectoral fascia between the subclavius and pectoralis minor or pass under the inferior margins of the pectoralis major and minor.

THORACIC WALL

The thoracic wall is segmental in design and composed of skeletal elements and muscles. It extends between:

- the superior thoracic aperture, bordered by vertebra TI, rib I, and the manubrium of the sternum; and
- the inferior thoracic aperture, bordered by vertebra TXII, rib XII, the end of rib XI, the costal margin, and the xiphoid process of the sternum.

Skeletal framework

The skeletal elements of the thoracic wall consist of the thoracic vertebrae, intervertebral discs, ribs, and sternum.

Thoracic vertebrae

There are twelve **thoracic vertebrae**, each of which is characterized by articulations with ribs.

Typical thoracic vertebra

A typical thoracic vertebra has a heart-shaped **vertebral body**, with roughly equal dimensions in the transverse and anteroposterior directions, and a long spinous process (Fig. 3.18). The **vertebral foramen** is generally circular and the **laminae** are broad and overlap with those of the vertebra below. The **superior articular processes** are flat, with their articular surfaces facing almost directly posteriorly, while the **inferior articular processes** project from the laminae and their articular facets face anteriorly.

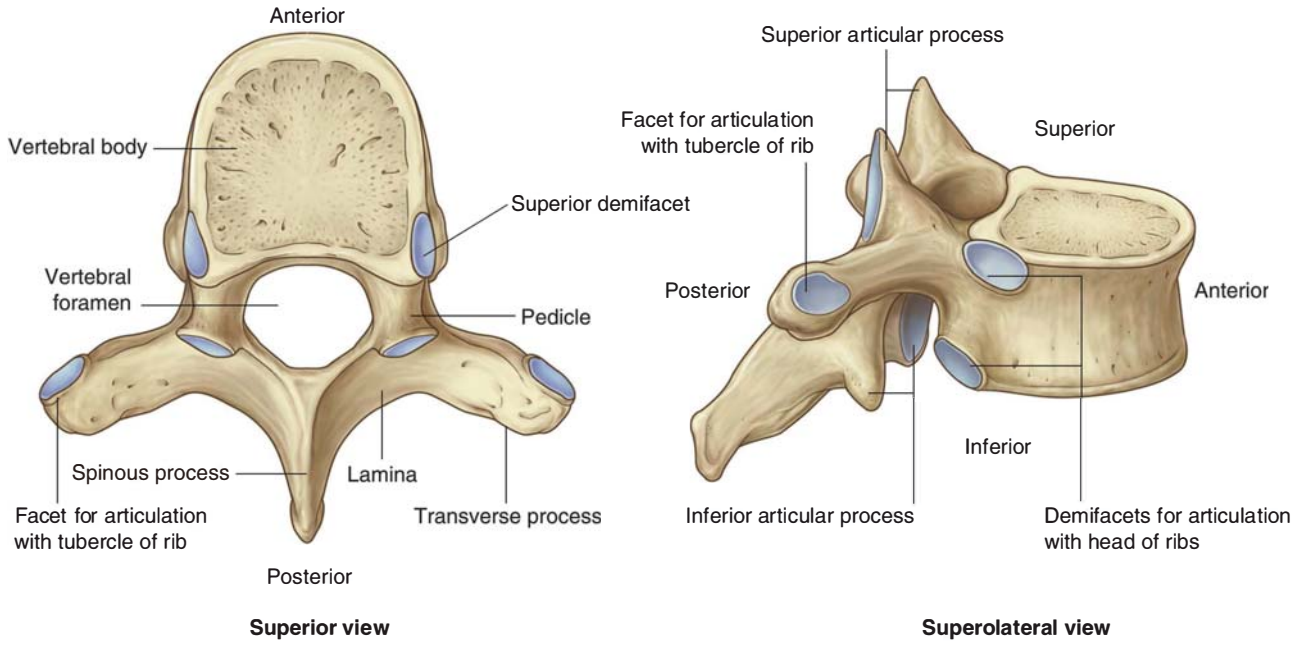


Fig. 3.18 Typical thoracic vertebra.

SUBASH KC/NMC-15TH/2014

The **transverse processes** are club shaped and project posterolaterally.

Articulation with ribs

A typical thoracic vertebra has three sites on each side for articulation with ribs.

- Two demifacets (i.e., partial facets) are located on the superior and inferior aspects of the body for articulation with corresponding sites on the heads of adjacent ribs. The **superior costal facet** articulates with part of the head of its own rib, and the **inferior costal facet** articulates with part of the head of the rib below.
- An oval facet (**transverse costal facet**) at the end of the transverse process articulates with the tubercle of its own rib.

Not all vertebrae articulate with ribs in the same fashion (Fig. 3.19):

- The superior costal facets on the body of vertebra TI are complete and articulate with a single facet on the head of its own rib—in other words, the head of rib I does not articulate with vertebra CVII.
- Similarly, vertebra TX (and often TIX) articulates only with its own ribs and therefore lacks inferior demifacets on the body.
- Vertebrae TXI and TXII articulate only with the heads of their own ribs—they lack transverse costal facets and have only a single complete facet on each side of their bodies.

Ribs

There are twelve pairs of ribs, each terminating anteriorly in a costal cartilage (Fig. 3.20).

Although all ribs articulate with the vertebral column, only the costal cartilages of the upper seven ribs, known as **true ribs**, articulate directly with the sternum. The remaining five pairs of ribs are **false ribs**:

- The costal cartilages of ribs VIII to X articulate anteriorly with the costal cartilages of the ribs above.
- Ribs XI and XII have no anterior connection with other ribs or with the sternum and are often called **floating ribs**.

A typical rib consists of a curved shaft with anterior and posterior ends (Fig. 3.21). The anterior end is continuous with its costal cartilage. The posterior end articulates with the vertebral column and is characterized by a head, neck, and tubercle.

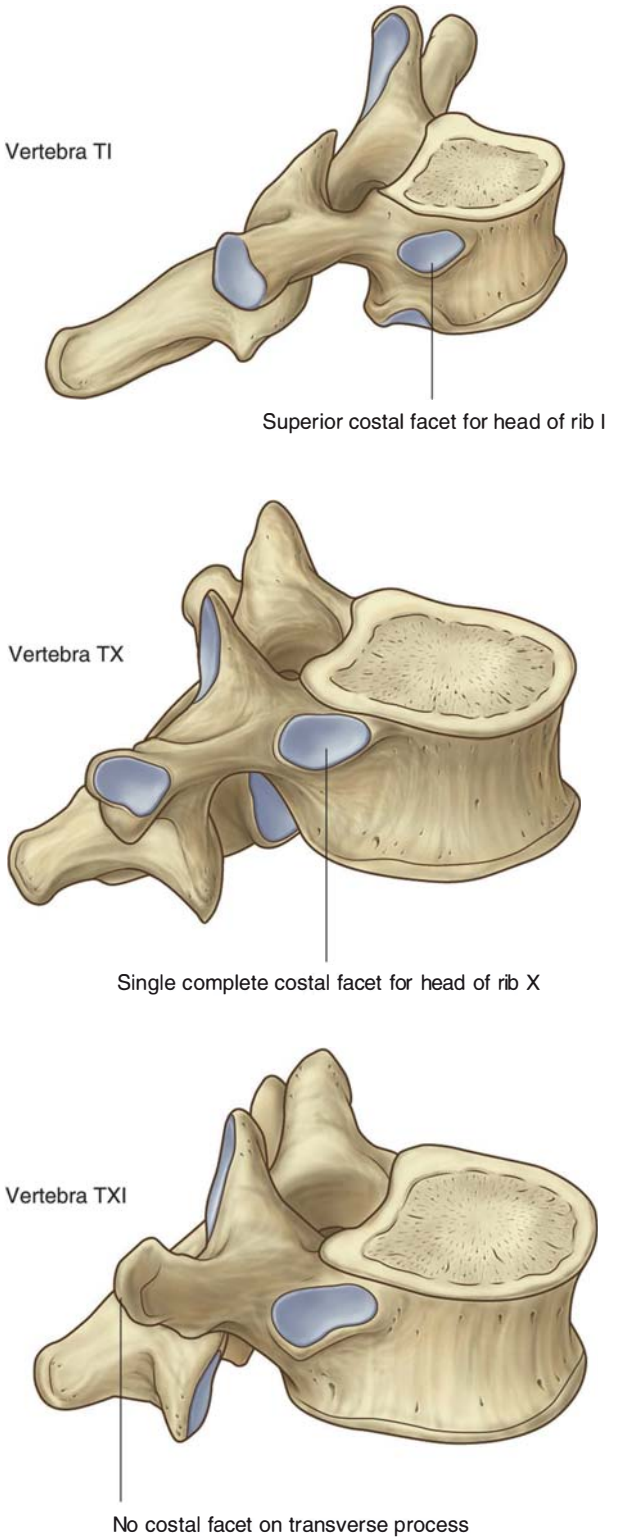


Fig. 3.19 Atypical thoracic vertebrae.

SUBASH KC/NMC-15TH/2014

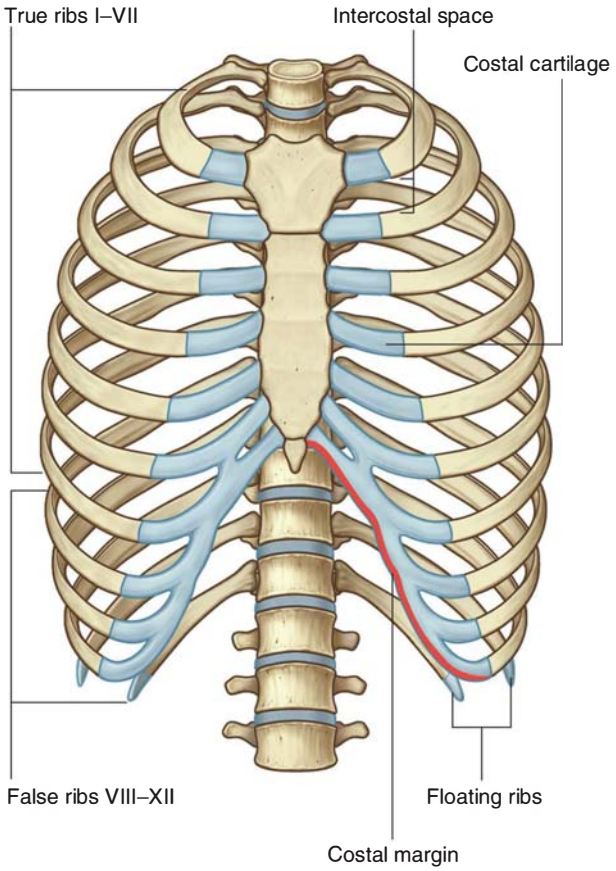


Fig. 3.20 Ribs.

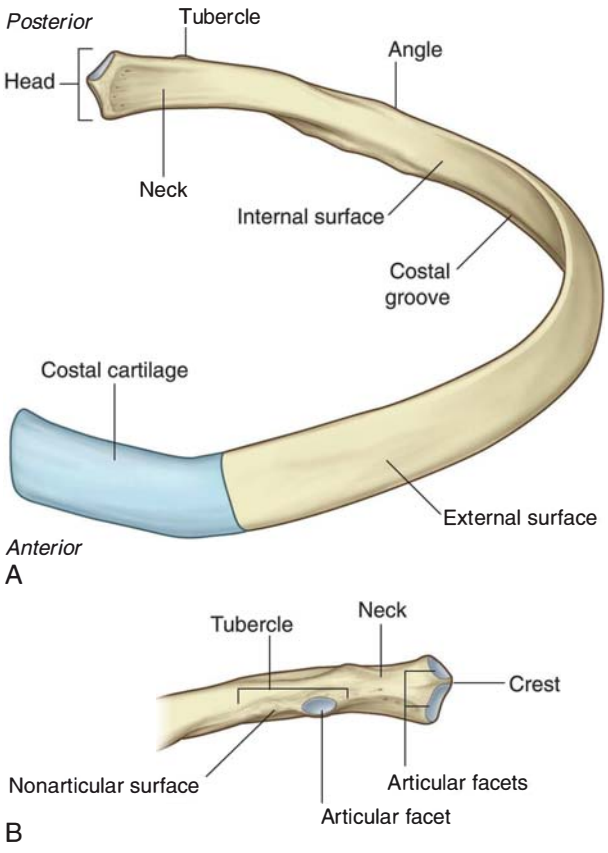


Fig. 3.21 A typical rib. **A.** Anterior view. **B.** Posterior view of proximal end of rib.

SUBASH KC/NMC-15TH/2014

The **head** is somewhat expanded and typically presents two articular surfaces separated by a **crest**. The smaller superior surface articulates with the inferior costal facet on the body of the vertebra above, whereas the larger inferior facet articulates with the superior costal facet of its own vertebra.

The **neck** is a short flat region of bone that separates the head from the tubercle.

The **tubercle** projects posteriorly from the junction of the neck with the shaft and consists of two regions, an articular part and a nonarticular part:

- The articular part is medial and has an oval facet for articulation with a corresponding facet on the transverse process of the associated vertebra.
- The raised nonarticular part is roughened by ligament attachments.

The shaft is generally thin and flat with internal and external surfaces.

The superior margin is smooth and rounded, whereas the inferior margin is sharp. The shaft bends forward just laterally to the tubercle at a site termed the **angle**. It also has a gentle twist around its longitudinal axis so that the external surface of the anterior part of the shaft faces somewhat superiorly relative to the posterior part. The inferior margin of the internal surface is marked by a distinct **costal groove**.

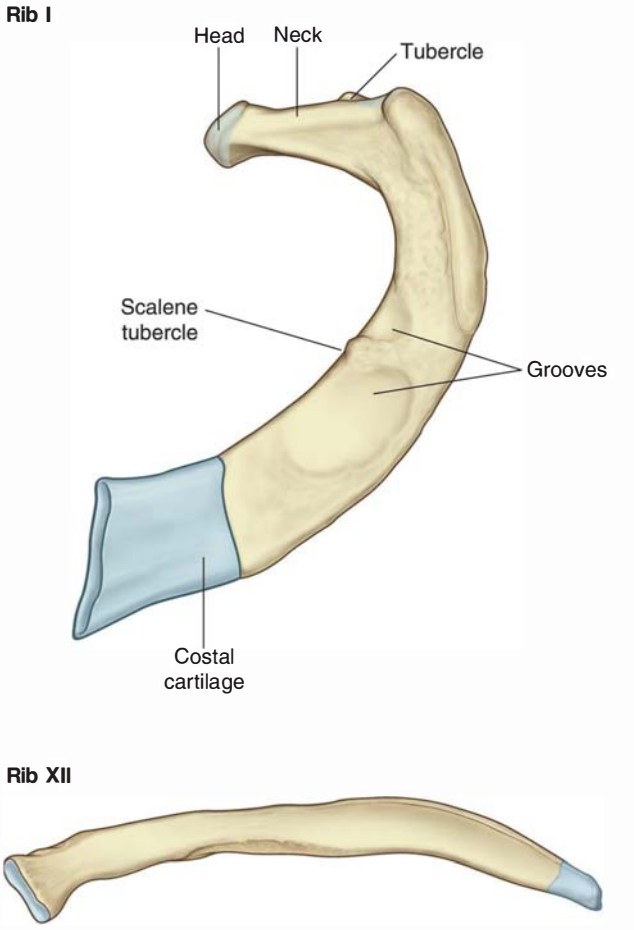


Fig. 3.22 Atypical ribs.

Distinct features of upper and lower ribs

The upper and lower ribs have distinct features (Fig. 3.22).

Rib I

Rib I is flat in the horizontal plane and has broad superior and inferior surfaces. From its articulation with vertebra TI, it slopes inferiorly to its attachment to the manubrium of the sternum. The head articulates only with the body of vertebra TI and therefore has only one articular surface. Like other ribs, the tubercle has a facet for articulation with the transverse process. The superior surface of the rib is characterized by a distinct tubercle, the **scalene tubercle**, which separates two smooth grooves that cross the rib approximately midway along the shaft. The anterior groove is caused by the subclavian vein, and the posterior groove is caused by the subclavian artery. Anterior and posterior to these grooves, the shaft is roughened by muscle and ligament attachments.

Rib II

Rib II, like rib I, is flat but twice as long. It articulates with the vertebral column in a way typical of most ribs.

Rib X

The head of **rib X** has a single facet for articulation with its own vertebra.

Ribs XI and XII

Ribs XI and **XII** articulate only with the bodies of their own vertebrae and have no tubercles or necks. Both ribs are short, have little curve, and are pointed anteriorly.

SUBASH KC/NMC-15TH/2014

Sternum

The adult **sternum** consists of three major elements: the broad and superiorly positioned manubrium of the sternum, the narrow and longitudinally oriented body of the sternum, and the small and inferiorly positioned xiphoid process (Fig. 3.23).

Manubrium of the sternum

The **manubrium of the sternum** forms part of the bony framework of the neck and the thorax.

The superior surface of the manubrium is expanded laterally and bears a distinct and palpable notch, the **jugular notch** (**suprasternal notch**), in the midline. On either side of this notch is a large oval fossa for articulation with the clavicle. Immediately inferior to this fossa, on each lateral surface of the manubrium, is a facet for the attachment of the first costal cartilage. At the lower end of the lateral border is a demifacet for articulation with the upper half of the anterior end of the second costal cartilage.

Body of the sternum

The **body of the sternum** is flat.

The anterior surface of the body of the sternum is often marked by transverse ridges that represent lines of fusion between the segmental elements called sternebrae, from which this part of the sternum arises embryologically.

The lateral margins of the body of the sternum have articular facets for costal cartilages. Superiorly, each lateral margin has a demifacet for articulation with the inferior aspect of the second costal cartilage. Inferior to this demifacet are four facets for articulation with the costal cartilages of ribs III to VI.

At the inferior end of the body of the sternum is a demifacet for articulation with the upper demifacet on the seventh costal cartilage. The inferior end of the body of the sternum is attached to the xiphoid process.

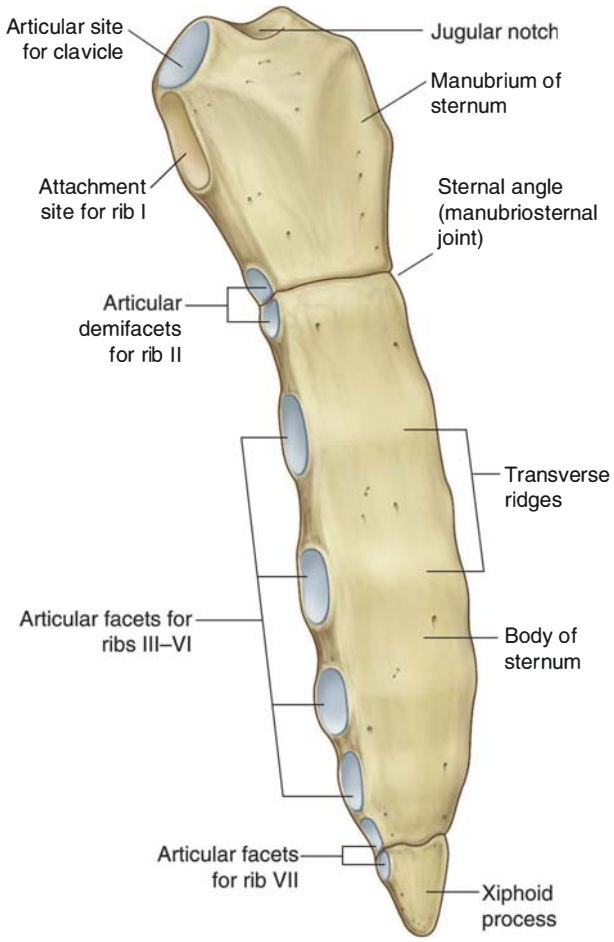


Fig. 3.23 Sternum.

SUBASH KC/NMC-15TH/2014

Xiphoid process

The **xiphoid process** is the smallest part of the sternum. Its shape is variable: it may be wide, thin, pointed, bifid, curved, or perforated. It begins as a cartilaginous structure, which becomes ossified in the adult. On each side of its upper lateral margin is a demifacet for articulation with the inferior end of the seventh costal cartilage.

Joints

Costovertebral joints

A typical rib articulates with:

- the bodies of adjacent vertebrae, forming a joint with the head of the rib; and
- the transverse process of its related vertebra, forming a **costotransverse joint** (Fig. 3.24).

Together, the costovertebral joints and related ligaments allow the necks of the ribs either to rotate around their longitudinal axes, which occurs mainly in the upper ribs, or to ascend and descend relative to the vertebral column, which occurs mainly in the lower ribs. The combined movements of all of the ribs on the vertebral column are essential for altering the volume of the thoracic cavity during breathing.

Joint with head of rib

The two facets on the head of the rib articulate with the superior facet on the body of its own vertebra and with the inferior facet on the body of the vertebra above. This joint is divided into two synovial compartments by an intra-articular ligament, which attaches the crest to the adjacent intervertebral disc and separates the two articular surfaces on the head of the rib. The two synovial compartments and the intervening ligament are surrounded by a single joint

SUBASH KC/NMC-15TH/2014

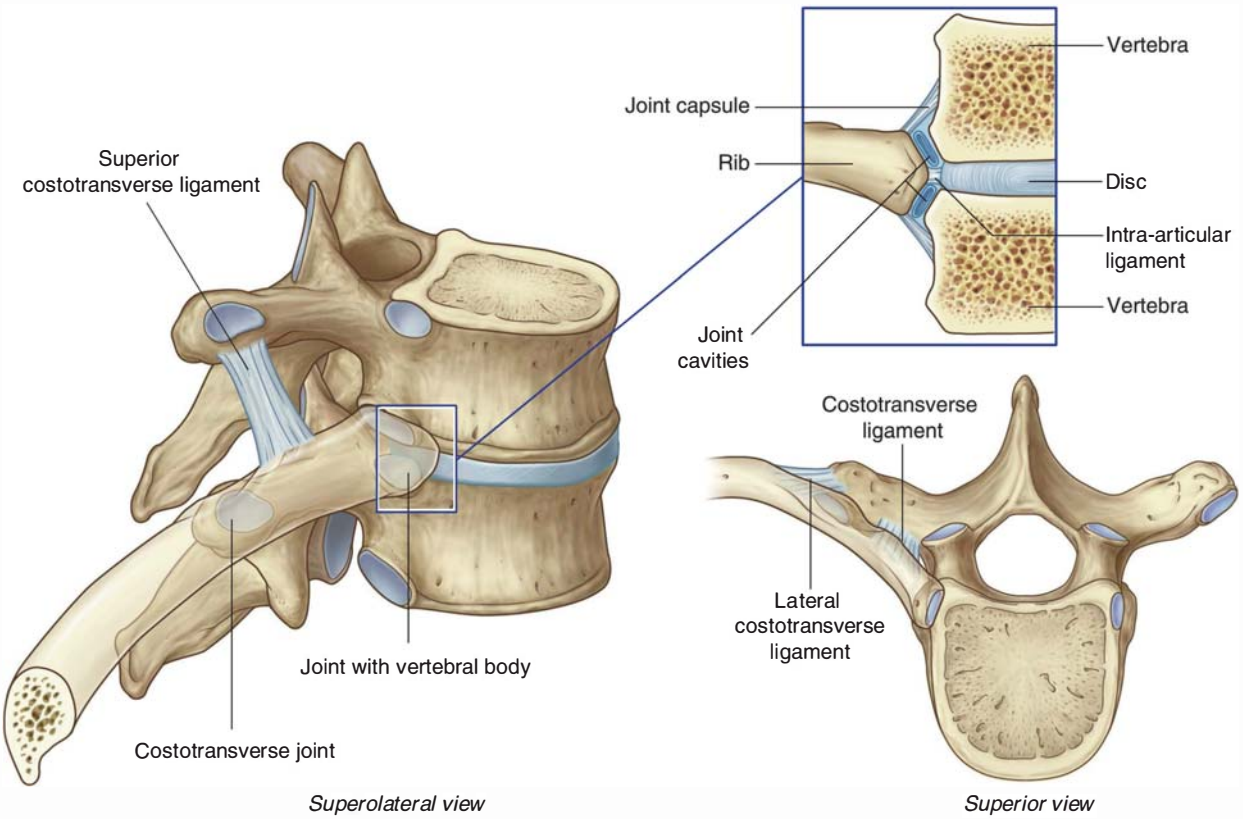


Fig. 3.24 Costovertebral joints.

capsule attached to the outer margins of the combined articular surfaces of the head and vertebral column.

Costotransverse joints

Costotransverse joints are synovial joints between the tubercle of a rib and the transverse process of the related vertebra (Fig. 3.24). The capsule surrounding each joint is thin. The joint is stabilized by two strong extracapsular ligaments that span the space between the transverse process and the rib on the medial and lateral sides of the joint:

- The **costotransverse ligament** is medial to the joint and attaches the neck of the rib to the transverse process.
- The **lateral costotransverse ligament** is lateral to the joint and attaches the tip of the transverse process to the roughened nonarticular part of the tubercle of the rib.

A third ligament, the **superior costotransverse ligament**, attaches the superior surface of the neck of the rib to the transverse process of the vertebra above.

Slight gliding movements occur at the costotransverse joints.

Sternocostal joints

The sternocostal joints are joints between the upper seven costal cartilages and the sternum (Fig. 3.25).

The joint between rib I and the manubrium is not synovial and consists of a fibrocartilaginous connection

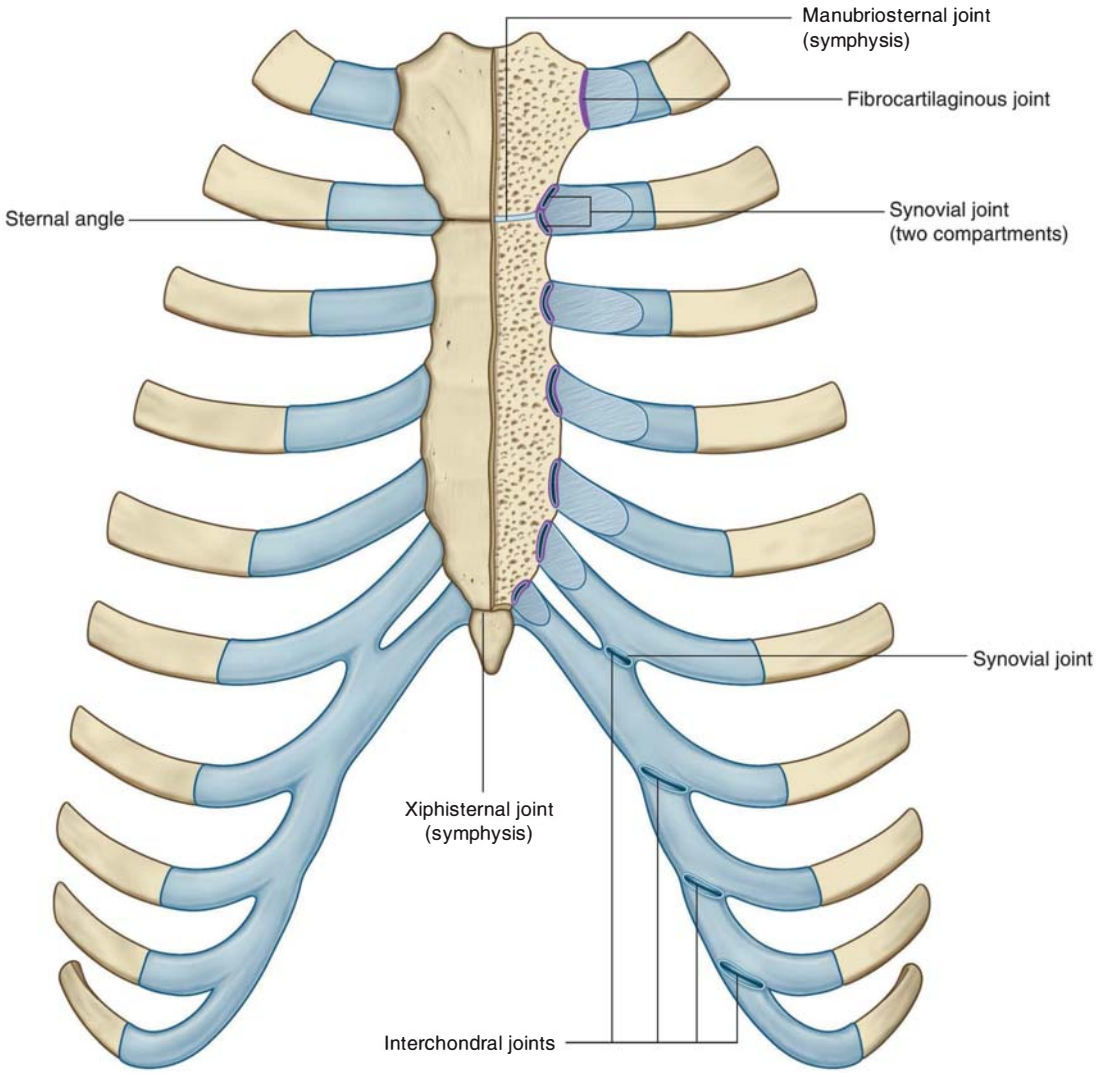


Fig. 3.25 Sternocostal joints.

SUBASH KC/NMC-15TH/2014

between the manubrium and the costal cartilage. The second to seventh joints are synovial and have thin capsules reinforced by surrounding sternocostal ligaments.

The joint between the second costal cartilage and the sternum is divided into two compartments by an intra-articular ligament. This ligament attaches the second costal cartilage to the junction of the manubrium and the body of the sternum.

Interchondral joints

Interchondral joints occur between the costal cartilages of adjacent ribs (Fig. 3.25), mainly between the costal cartilages of ribs VII to X, but may also involve the costal cartilages of ribs V and VI.

Interchondral joints provide indirect anchorage to the sternum and contribute to the formation of a smooth inferior costal margin. They are usually synovial, and the thin fibrous capsules are reinforced by interchondral ligaments.

Manubriosternal and xiphisternal joints

The joints between the manubrium and the body of the sternum and between the body of the sternum and the xiphoid process are usually symphyses (Fig. 3.25). Only slight angular movements occur between the manubrium and the body of the sternum during respiration. The joint between the body of the sternum and the xiphoid process often becomes ossified with age.

A clinically useful feature of the manubriosternal joint is that it can be palpated easily. This is because the manubrium normally angles posteriorly on the body of the sternum, forming a raised feature referred to as the sternal angle. This elevation marks the site of articulation of rib II with the sternum. Rib I is not palpable, because it lies inferior to the clavicle and is embedded in tissues at the base of the neck. Therefore, rib II is used as a reference for counting ribs and can be felt immediately lateral to the sternal angle.

In addition, the sternal angle lies on a horizontal plane that passes through the intervertebral disc between vertebrae TIV and TV (see Fig. 3.10). This plane separates the superior mediastinum from the inferior mediastinum and marks the superior border of the pericardium. The plane also passes through the end of the ascending aorta and the beginning of the arch of the aorta, the end of the arch of the aorta and the beginning of the thoracic aorta, and the bifurcation of the trachea, and just superior to the pulmonary trunk (see Fig. 3.79 and 3.86).

Intercostal spaces

Intercostal spaces lie between adjacent ribs and are filled by intercostal muscles (Fig. 3.26).

Intercostal nerves and associated major arteries and veins lie in the **costal groove** along the inferior margin of the superior rib and pass in the plane between the inner two layers of muscles.

In each space, the vein is the most superior structure and is therefore highest in the costal groove. The artery is inferior to the vein, and the nerve is inferior to the artery and often not protected by the groove. Therefore, the nerve is the structure most at risk when objects perforate the upper aspect of an intercostal space. Small collateral branches of the major intercostal nerves and vessels are often present superior to the inferior rib below.

Deep to the intercostal spaces and ribs, and separating these structures from the underlying pleura, is a layer of loose connective tissue, called **endothoracic fascia**, which contains variable amounts of fat.

Superficial to the spaces are deep fascia, superficial fascia, and skin. Muscles associated with the upper limbs and back overlie the spaces.

In the clinic

Cervical ribs

Cervical ribs are present in approximately 1% of the population.

A cervical rib is an accessory rib articulating with vertebra CVII; the anterior end attaches to the superior border of the anterior aspect of rib I.

Plain radiographs may demonstrate cervical ribs as small horn-like structures (see Fig. 3.106).

It is often not appreciated by clinicians that a fibrous band commonly extends from the anterior tip of the small cervical ribs to rib I, producing a "cervical band" that is not visualized on radiography. In patients with cervical ribs and cervical bands, structures that normally pass over rib I (see Fig. 3.7) are elevated by, and pass over, the cervical rib and band.

Clinically, "thoracic outlet syndrome" is used to describe symptoms resulting from abnormal compression of the brachial plexus of nerves as it passes over the first rib and through the axillary inlet into the upper limb. The anterior ramus of T1 passes superiorly out of the superior thoracic aperture to join and become part of the brachial plexus. The cervical band from a cervical rib is one cause of thoracic outlet syndrome by putting upward stresses on the lower parts of the brachial plexus as they pass over the first rib.

SUBASH KC/NMC-15TH/2014

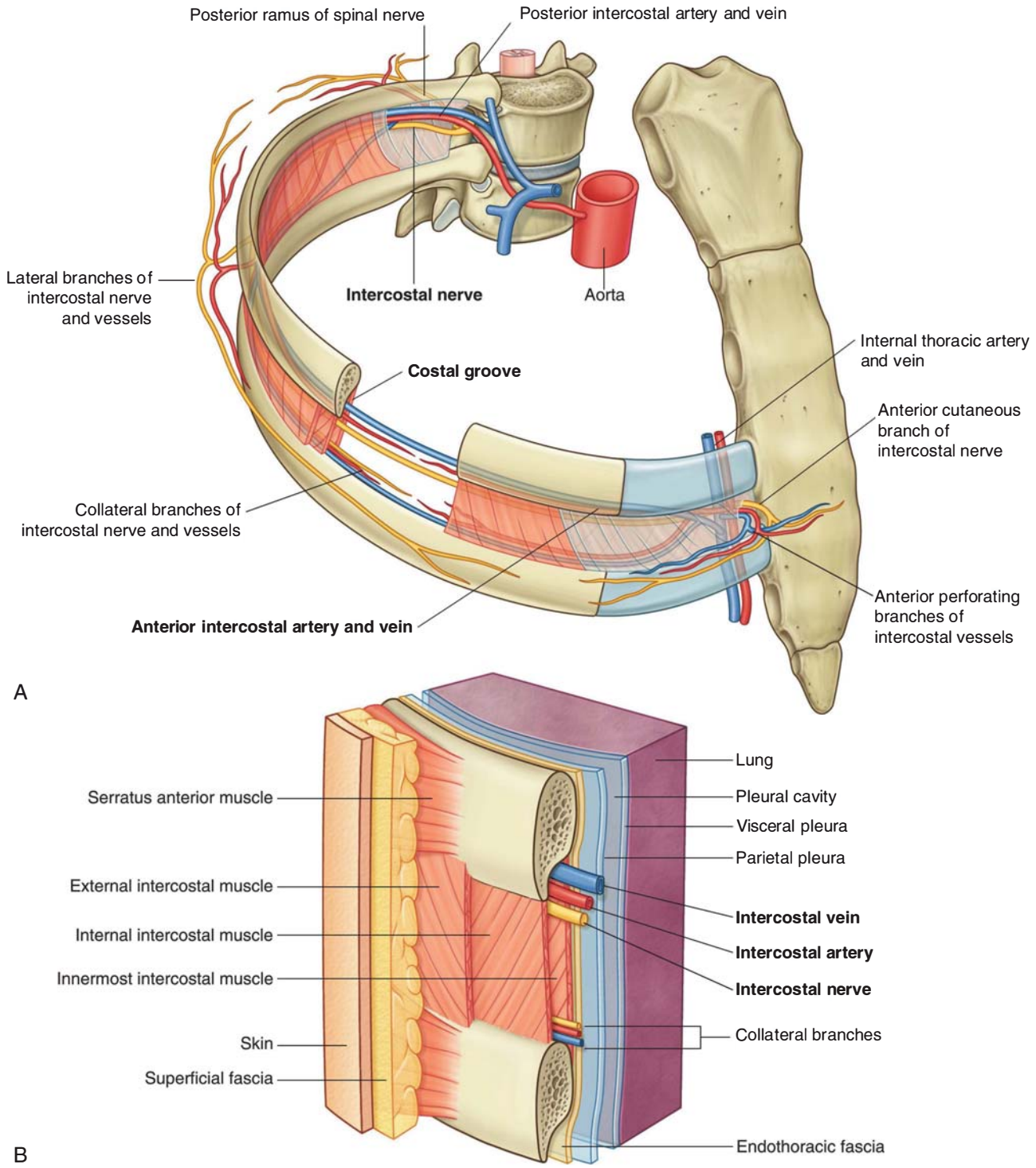


Fig. 3.26 Intercostal space. **A.** Anterolateral view. **B.** Details of an intercostal space and relationships.

SUBASH KC/NMC-15TH/2014

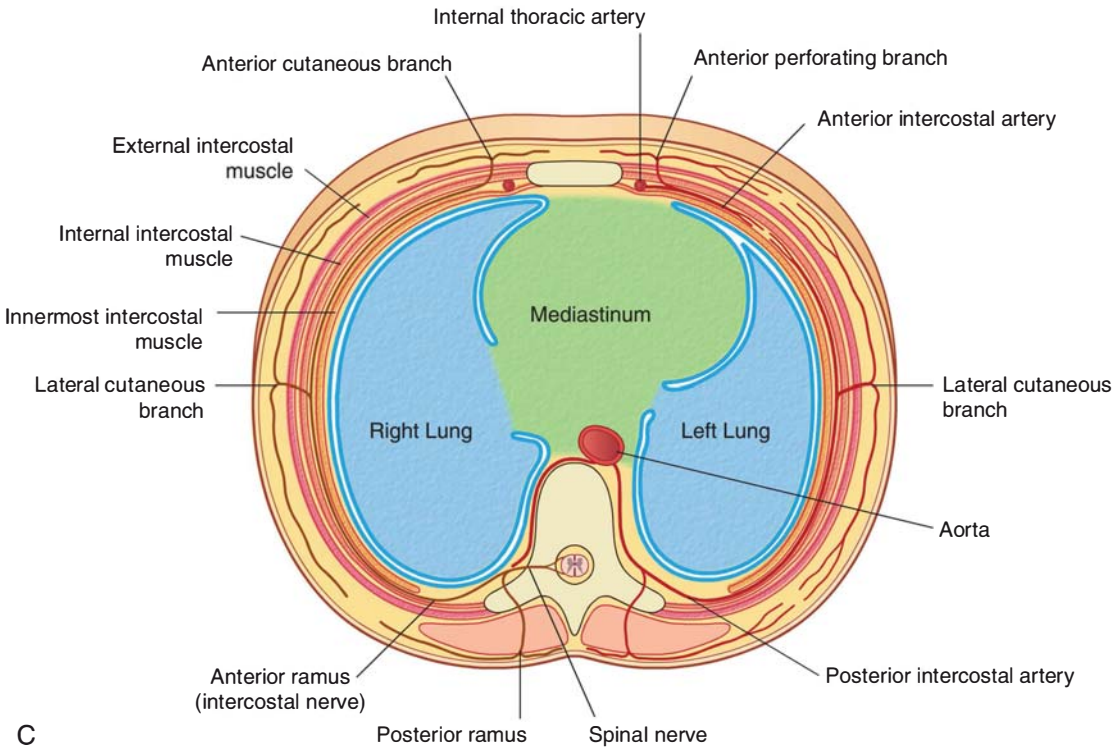


Fig. 3.26, cont'd Intercostal space. **C.** Transverse section.

SUBASH KC/NMC-15TH/2014

In the clinic

Collection of sternal bone marrow

The subcutaneous position of the sternum makes it possible to place a needle through the hard outer cortex into the internal (or medullary) cavity containing bone marrow. Once the needle is in this position, bone marrow can be aspirated. Evaluation of this material under the microscope helps clinicians diagnose certain blood diseases such as leukemia.

In the clinic

Rib fractures

Single rib fractures are of little consequence, though extremely painful.

After severe trauma, ribs may be broken in two or more places. If enough ribs are broken, a loose segment of chest wall, a flail segment (**flail chest**), is produced. When the patient takes a deep inspiration, the flail segment moves in the opposite direction to the chest wall, preventing full lung expansion and creating a paradoxically moving segment. If a large enough segment of chest wall is affected, ventilation may be impaired and assisted ventilation may be required until the ribs have healed.

Muscles

Muscles of the thoracic wall include those that fill and support the intercostal spaces, those that pass between the sternum and the ribs, and those that cross several ribs between costal attachments (Table 3.2).

The muscles of the thoracic wall, together with muscles between the vertebrae and ribs posteriorly (i.e., the **levatores costarum** and **serratus posterior superior** and **serratus posterior inferior** muscles) alter the position of the ribs and sternum and so change the thoracic volume during breathing. They also reinforce the thoracic wall.

Intercostal muscles

The **intercostal muscles** are three flat muscles found in each intercostal space that pass between adjacent ribs (Fig. 3.27). Individual muscles in this group are named according to their positions:

- The external intercostal muscles are the most superficial.
- The internal intercostal muscles are sandwiched between the external and innermost muscles.
- The innermost intercostal muscles are the deepest of the three muscles.

Table 3.2 Muscles of the thoracic wall

Muscle	Superior attachment	Inferior attachment	Innervation	Function
External intercostal	Inferior margin of rib above	Superior margin of rib below	Intercostal nerves; T1–T11	Most active during inspiration; supports intercostal space; moves ribs superiorly
Internal intercostal	Lateral edge of costal groove of rib above	Superior margin of rib below deep to the attachment of the related external intercostal	Intercostal nerves; T1–T11	Most active during expiration; supports intercostal space; moves ribs inferiorly
Innermost intercostal	Medial edge of costal groove of rib above	Internal aspect of superior margin of rib below	Intercostal nerves; T1–T11	Acts with internal intercostal muscles
Subcostales	Internal surface (near angle) of lower ribs	Internal surface of second or third rib below	Related intercostal nerves	May depress ribs
Transversus thoracis	Inferior margins and internal surfaces of costal cartilages of second to sixth ribs	Inferior aspect of deep surface of body of sternum, xiphoid process, and costal cartilages of ribs IV–VII	Related intercostal nerves	Depresses costal cartilages

SUBASH KC/NMC-15TH/2014

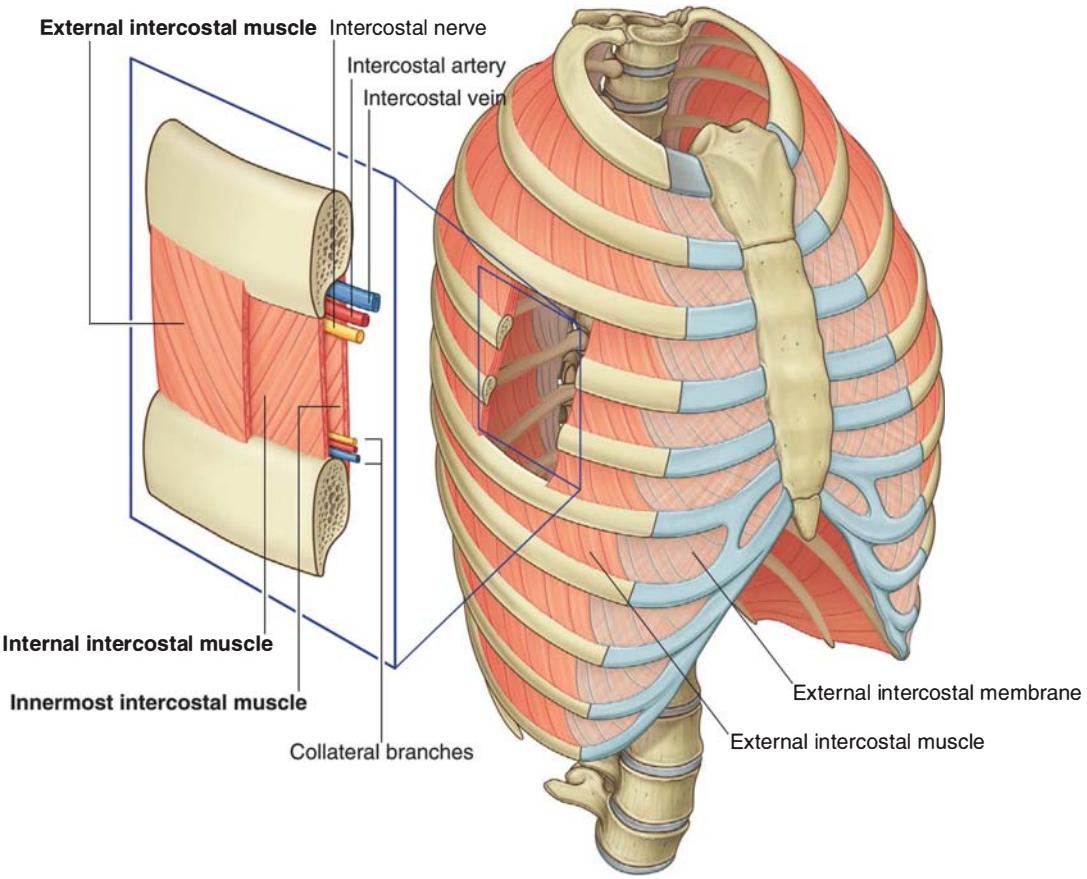


Fig. 3.27 Intercostal muscles.

The intercostal muscles are innervated by the related intercostal nerves. As a group, the intercostal muscles provide structural support for the intercostal spaces during breathing. They can also move the ribs.

External intercostal muscles

The eleven pairs of **external intercostal muscles** extend from the inferior margins (lateral edges of costal grooves) of the ribs above to the superior margins of the ribs below. When the thoracic wall is viewed from a lateral position, the muscle fibers pass obliquely anteroinferiorly (Fig. 3.27). The muscles extend around the thoracic wall from the regions of the tubercles of the ribs to the costal cartilages, where each layer continues as a thin connective tissue aponeurosis termed the **external intercostal membrane**. The external intercostal muscles are most active in inspiration.

Internal intercostal muscles

The eleven pairs of **internal intercostal muscles** pass between the most inferior lateral edge of the costal grooves of the ribs above, to the superior margins of the ribs below. They extend from parasternal regions, where the muscles course between adjacent costal cartilages, to the angle of the ribs posteriorly (Fig. 3.27). This layer continues medially toward the vertebral column, in each intercostal space, as the **internal intercostal membrane**. The muscle fibers pass in the opposite direction to those of the external intercostal muscles. When the thoracic wall is viewed from a lateral position, the muscle fibers pass obliquely posteroinferiorly. The internal intercostal muscles are most active during expiration.

Innermost intercostal muscles

The **innermost intercostal muscles** are the least distinct of the intercostal muscles, and the fibers have the same orientation as the internal intercostals (Fig. 3.27). These muscles are most evident in the lateral thoracic wall. They extend between the inner surfaces of adjacent ribs from the medial edge of the costal groove to the deep surface of the rib below. Importantly, the neurovascular bundles associated with the intercostal spaces pass around the thoracic wall in the costal grooves in a plane between the innermost and internal intercostal muscles.

Subcostales

The **subcostales** are in the same plane as the innermost intercostals, span multiple ribs, and are more numerous in lower regions of the posterior thoracic wall (Fig. 3.28A). They extend from the internal surfaces of one rib to the internal surface of the second (next) or third rib below. Their fibers parallel the course of the internal intercostal muscles and extend from the angle of the ribs to more medial positions on the ribs below.

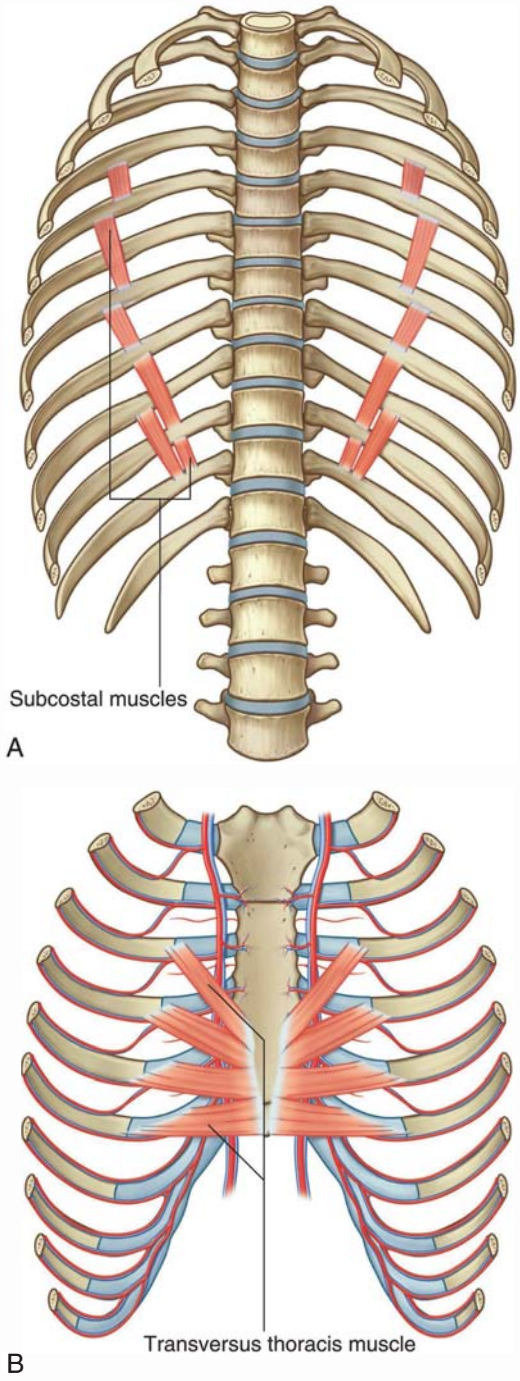


Fig. 3.28 **A.** Subcostal muscles. **B.** Transversus thoracis muscles.

Transversus thoracis muscles

The **transversus thoracis muscles** are found on the deep surface of the anterior thoracic wall (Fig. 3.28B) and in the same plane as the innermost intercostals.

SUBASH KC/NMC-15TH/2014

The transversus thoracis muscles originate from the posterior aspect of the xiphoid process, the inferior part of the body of the sternum, and the adjacent costal cartilages of the lower true ribs. They pass superiorly and laterally to insert into the lower borders of the costal cartilages of ribs III to VI. They most likely pull these latter elements inferiorly.

The transversus thoracis muscles lie deep to the internal thoracic vessels and secure these vessels to the wall.

Arterial supply

Vessels that supply the thoracic wall consist mainly of posterior and anterior intercostal arteries, which pass around the wall between adjacent ribs in intercostal spaces (Fig. 3.29). These arteries originate from the aorta and internal thoracic arteries, which in turn arise from the subclavian arteries in the root of the neck. Together, the intercostal arteries form a basket-like pattern of vascular supply around the thoracic wall.

Posterior intercostal arteries

Posterior intercostal arteries originate from vessels associated with the posterior thoracic wall. The upper two posterior intercostal arteries on each side are derived from the **supreme intercostal artery**, which descends into the thorax as a branch of the costocervical trunk in the neck. The **costocervical trunk** is a posterior branch of the subclavian artery (Fig. 3.29).

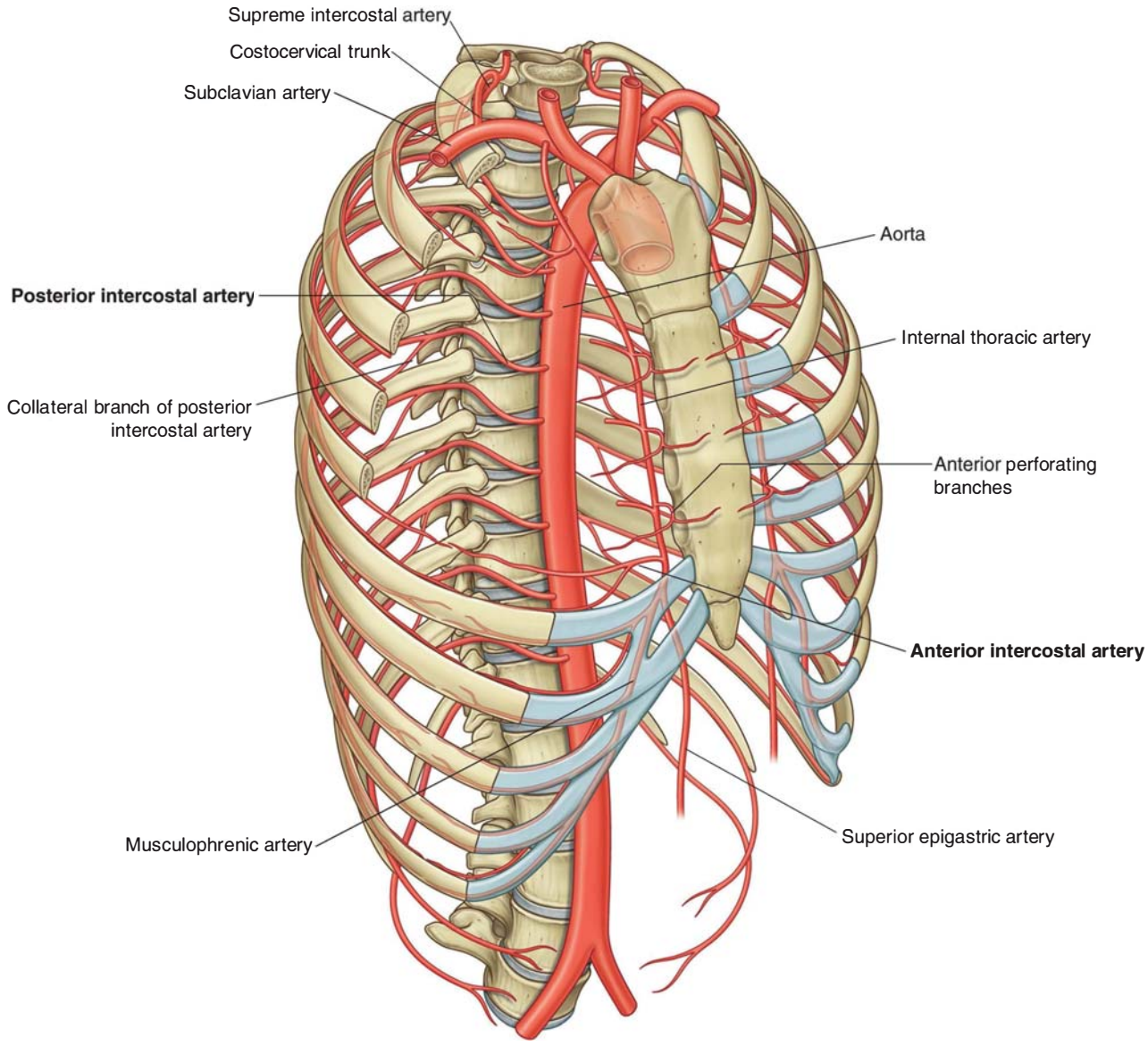


Fig. 3.29 Arteries of the thoracic wall.

SUBASH KC/NMC-15TH/2014

The remaining nine pairs of posterior intercostal arteries arise from the posterior surface of the thoracic aorta. Because the aorta is on the left side of the vertebral column, those posterior intercostal vessels passing to the right side of the thoracic wall cross the midline anterior to the bodies of the vertebrae and therefore are longer than the corresponding vessels on the left.

In addition to having numerous branches that supply various components of the wall, the posterior intercostal arteries have branches that accompany lateral cutaneous branches of the intercostal nerves to superficial regions.

Anterior intercostal arteries

The **anterior intercostal arteries** originate directly or indirectly as lateral branches from the internal thoracic arteries (Fig. 3.29).

Each **internal thoracic artery** arises as a major branch of the subclavian artery in the neck. It passes anteriorly over the cervical dome of the pleura and descends vertically through the superior thoracic aperture and along the deep aspect of the anterior thoracic wall. On each side, the internal thoracic artery lies posterior to the costal cartilages of the upper six ribs and about 1 cm lateral to the sternum. At approximately the level of the sixth intercostal space, it divides into two terminal branches:

- the **superior epigastric artery**, which continues inferiorly into the anterior abdominal wall (Fig. 3.29);
- the **musculophrenic artery**, which passes along the costal margin, goes through the diaphragm, and ends near the last intercostal space.

Anterior intercostal arteries that supply the upper six intercostal spaces arise as lateral branches from the internal thoracic artery, whereas those supplying the lower spaces arise from the musculophrenic artery.

In each intercostal space, the anterior intercostal arteries usually have two branches:

- One passes below the margin of the upper rib.
- The other passes above the margin of the lower rib and meets a collateral branch of the posterior intercostal artery.

The distributions of the anterior and posterior intercostal vessels overlap and can develop anastomotic connections. The anterior intercostal arteries are generally smaller than the posterior vessels.

In addition to anterior intercostal arteries and a number of other branches, the internal thoracic arteries give rise to perforating branches that pass directly forward between the costal cartilages to supply structures external to the thoracic wall. These vessels travel with the anterior cutaneous branches of the intercostal nerves.

Venous drainage

Venous drainage from the thoracic wall generally parallels the pattern of arterial supply (Fig. 3.30).

Centrally, the intercostal veins ultimately drain into the azygos system of veins or into **internal thoracic veins**, which connect with the **brachiocephalic veins** in the neck.

Often the upper posterior intercostal veins on the left side come together and form the **left superior intercostal vein**, which empties into the left brachiocephalic vein.

Similarly, the upper posterior intercostal veins on the right side may come together and form the **right superior intercostal vein**, which empties into the **azygos vein**.

SUBASH KC/NMC-15TH/2014

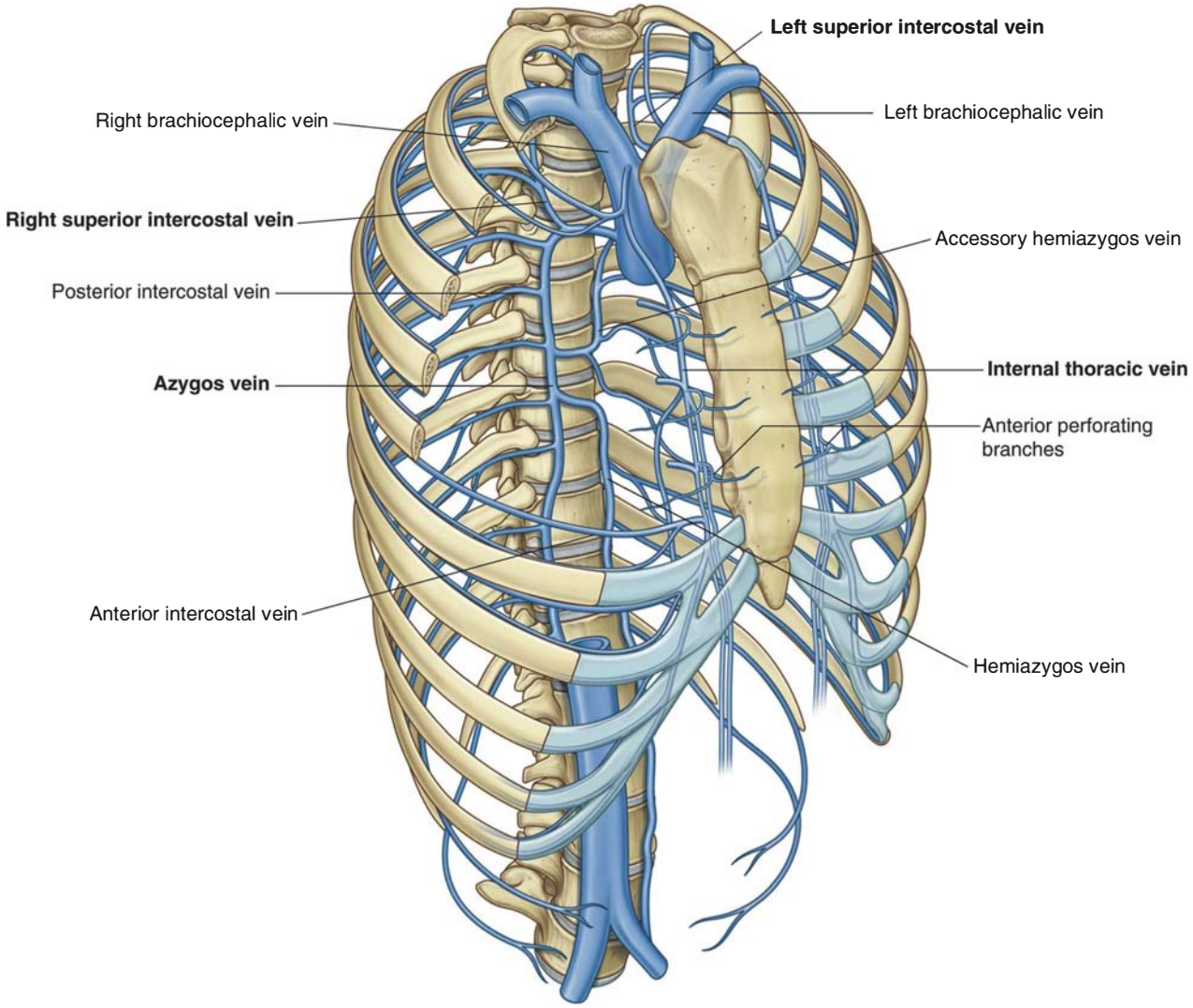


Fig. 3.30 Veins of the thoracic wall.

SUBASH KC/NMC-15TH/2014

Lymphatic drainage

Lymphatic vessels of the thoracic wall drain mainly into lymph nodes associated with the internal thoracic arteries (**parasternal nodes**), with the heads and necks of ribs (**intercostal nodes**), and with the diaphragm (**diaphragmatic nodes**) (Fig. 3.31). Diaphragmatic nodes are posterior to the xiphoid and at sites where the phrenic nerves penetrate the diaphragm. They also occur in regions where the diaphragm is attached to the vertebral column.

Parasternal nodes drain into **bronchomediastinal trunks**. Intercostal nodes in the upper thorax also drain into bronchomediastinal trunks, whereas intercostal nodes in the lower thorax drain into the **thoracic duct**.

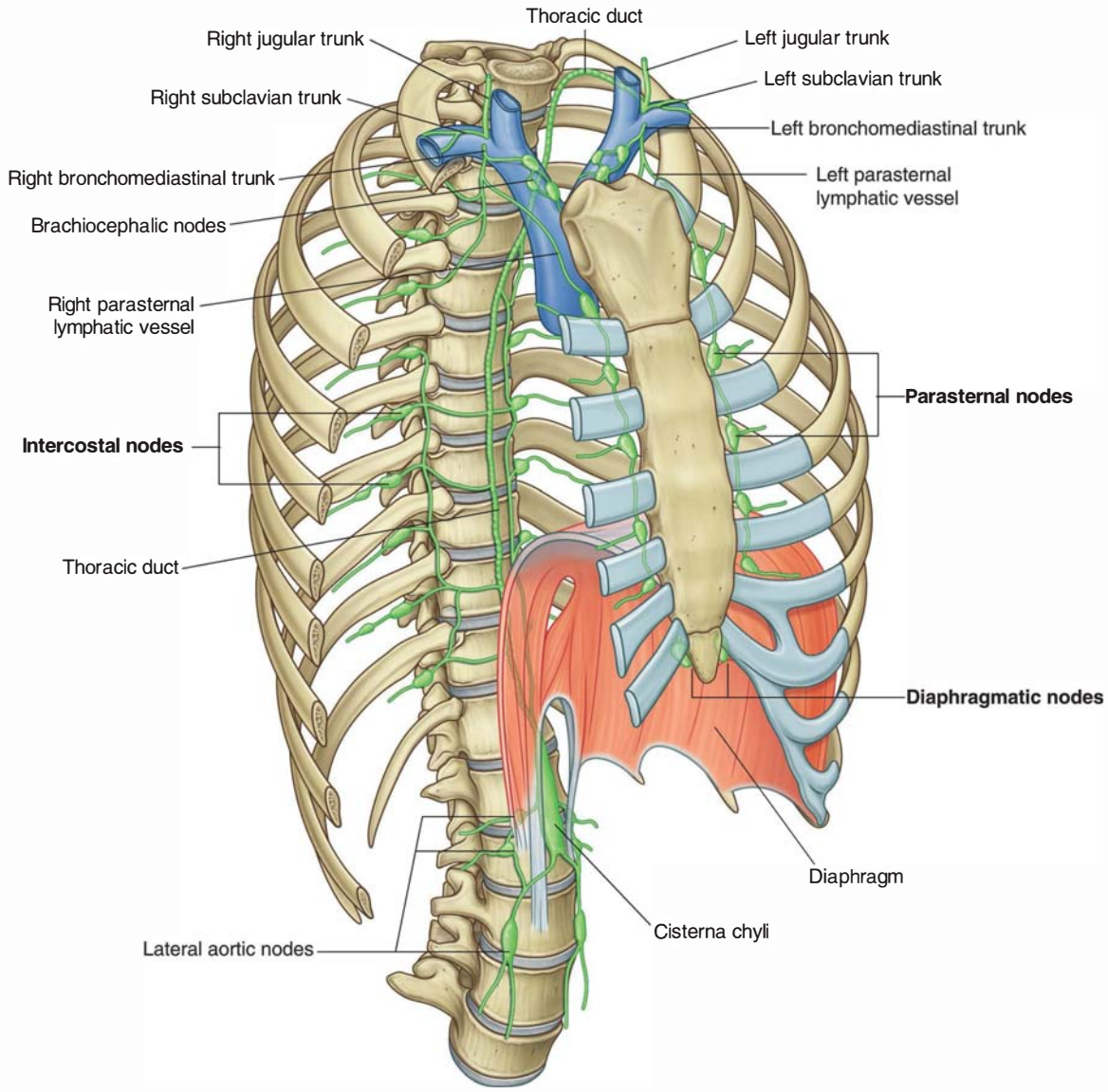
Nodes associated with the diaphragm interconnect with parasternal, prevertebral, and juxta-esophageal nodes, **brachiocephalic nodes** (anterior to the brachiocephalic veins in the superior mediastinum), and **lateral aortic/lumbar nodes** (in the abdomen).

Superficial regions of the thoracic wall drain mainly into **axillary lymph nodes** in the axilla or parasternal nodes.

Innervation

Intercostal nerves

Innervation of the thoracic wall is mainly by the **intercostal nerves**, which are the anterior rami of spinal nerves



SUBASH KC/NMC-15TH/2014

Fig. 3.31 Major lymphatic vessels and nodes of the thoracic wall.

T1 to T11 and lie in the intercostal spaces between adjacent ribs. The anterior ramus of spinal nerve T12 (the **subcostal nerve**) is inferior to rib XII (Fig. 3.32).

A typical intercostal nerve passes laterally around the thoracic wall in an intercostal space. The largest of the branches is the **lateral cutaneous branch**, which pierces the lateral thoracic wall and divides into an anterior branch and a posterior branch that innervate the overlying skin.

The intercostal nerves end as **anterior cutaneous branches**, which emerge either parasternally, between adjacent costal cartilages, or laterally to the midline, on the anterior abdominal wall, to supply the skin.

In addition to these major branches, small collateral branches can be found in the intercostal space running along the superior border of the lower rib.

In the thorax, the intercostal nerves carry:

- somatic motor innervation to the muscles of the thoracic wall (intercostal, subcostal, and transversus thoracis muscles),
- somatic sensory innervation from the skin and parietal pleura, and
- postganglionic sympathetic fibers to the periphery.

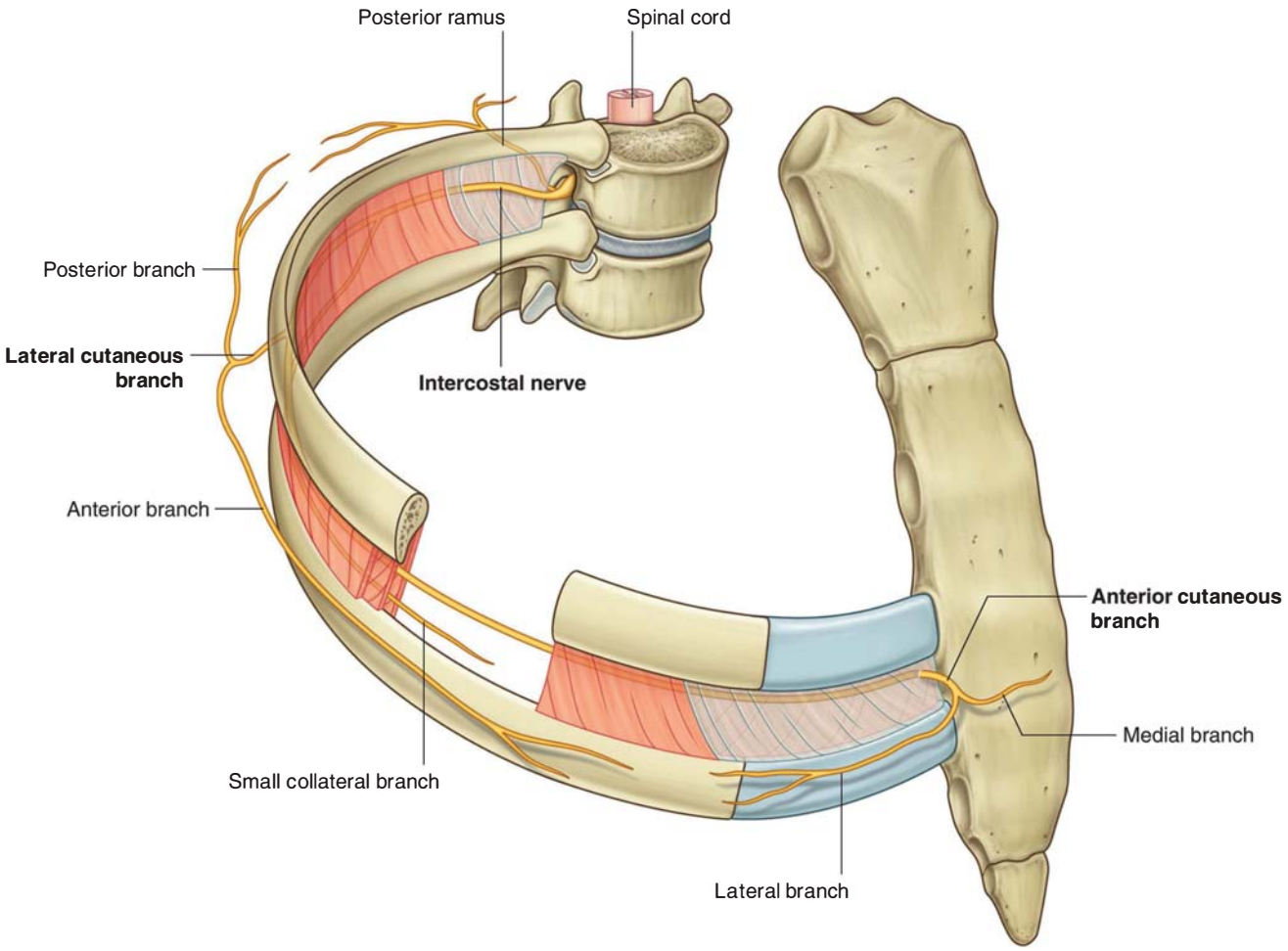


Fig. 3.32 Intercostal nerves.

SUBASH KC/NMC-15TH/2014

Sensory innervation of the skin overlying the upper thoracic wall is supplied by cutaneous branches (supraclavicular nerves), which descend from the cervical plexus in the neck.

In addition to innervating the thoracic wall, intercostal nerves innervate other regions:

- The anterior ramus of T1 contributes to the brachial plexus.
- The lateral cutaneous branch of the second intercostal nerve (the **intercostobrachial nerve**) contributes to cutaneous innervation of the medial surface of the upper arm.
- The lower intercostal nerves supply the muscles, skin, and peritoneum of the abdominal wall.

In the clinic

Thoracostomy (chest) tube insertion

Insertion of a chest tube is a commonly performed procedure and is indicated to relieve air or fluid trapped in the thorax between the lung and the chest wall (pleural cavity). This procedure is done for pneumothorax, hemothorax, hemopneumothorax, malignant pleural effusion empyema, hydrothorax, and chylothorax, and also after thoracic surgery.

The position of the thoracostomy tube should be between the anterior axillary and midaxillary anatomical lines from anterior to posterior and either the fourth or fifth intercostal space from cephalad to caudad. The position of the ribs in this region should be clearly marked. Anesthetic should be applied to the superior border of the rib and the inferior aspect of the intercostal space, including one rib and space above and one rib and space below. The neurovascular bundle runs in the neurovascular plane, which lies in the superior aspect of the intercostal space (just below the rib); hence, the reason for positioning the tube on the superior border of a rib (i.e., at the lowest position in the intercostal space).

In the clinic

Surgical access to the chest

A surgical access is potentially more challenging in the chest given the rigid nature of the thoracic cage. Moreover, access is also dependent upon the organ that is operated upon and its relationships to subdiaphragmatic structures and structures in the neck.

A standard incision site would include a median sternotomy to obtain access to the heart, including the coronary arteries and the cardiac valves. A left lateral thoracotomy or a right lateral thoracotomy is an incision through an intercostal space to access the lungs and the mediastinal structures.

Minimally invasive thoracic surgery (video-assisted thoracic surgery [VATS]) involves making small (1-cm) incisions in the intercostal spaces, placing a small camera on a telescope, and manipulating other instruments through additional small incisions. A number of procedures can be performed in this manner, including lobectomy, lung biopsy, and esophagectomy.

In the clinic

Intercostal nerve block

Local anesthesia of intercostal nerves produces excellent analgesia in patients with chest trauma and those patients requiring anesthesia for a thoracotomy, mastectomy, or upper abdominal surgical procedures.

The intercostal nerves are situated inferior to the rib borders in the neurovascular bundle. Each neurovascular bundle is situated deep to the external and internal intercostal muscle groups.

The nerve block may be undertaken using a "blind" technique or under direct imaging guidance.

The patient is placed in the appropriate position to access the rib. Typically, under ultrasound guidance, a needle may be advanced into the region of the subcostal groove, followed by an injection with local anesthesia. Depending on the type of local anesthetic, analgesia may be short- or long-acting.

Given the position of the neurovascular bundle and the subcostal groove, complications may include puncture of the parietal pleura and an ensuing pneumothorax. Bleeding may also occur if the artery or vein is damaged during the procedure.

SUBASH KC/NMC-15TH/2014

DIAPHRAGM

The **diaphragm** is a thin musculotendinous structure that fills the inferior thoracic aperture and separates the thoracic cavity from the abdominal cavity (Fig. 3.33 and see Chapter 4). It is attached peripherally to the:

- xiphoid process of the sternum,
- costal margin of the thoracic wall,
- ends of ribs XI and XII,
- ligaments that span across structures of the posterior abdominal wall, and
- vertebrae of the lumbar region.

From these peripheral attachments, muscle fibers converge to join the central tendon. The pericardium is attached to the middle part of the central tendon.

In the median sagittal plane, the diaphragm slopes inferiorly from its anterior attachment to the xiphoid, approximately at vertebral level TVIII/IX, to its posterior attachment to the **median arcuate ligament**, crossing anteriorly to the aorta at approximately vertebral level TXII.

Structures traveling between the thorax and abdomen pass through the diaphragm or between the diaphragm and its peripheral attachments:

- The inferior vena cava passes through the central tendon at approximately vertebral level TVIII.

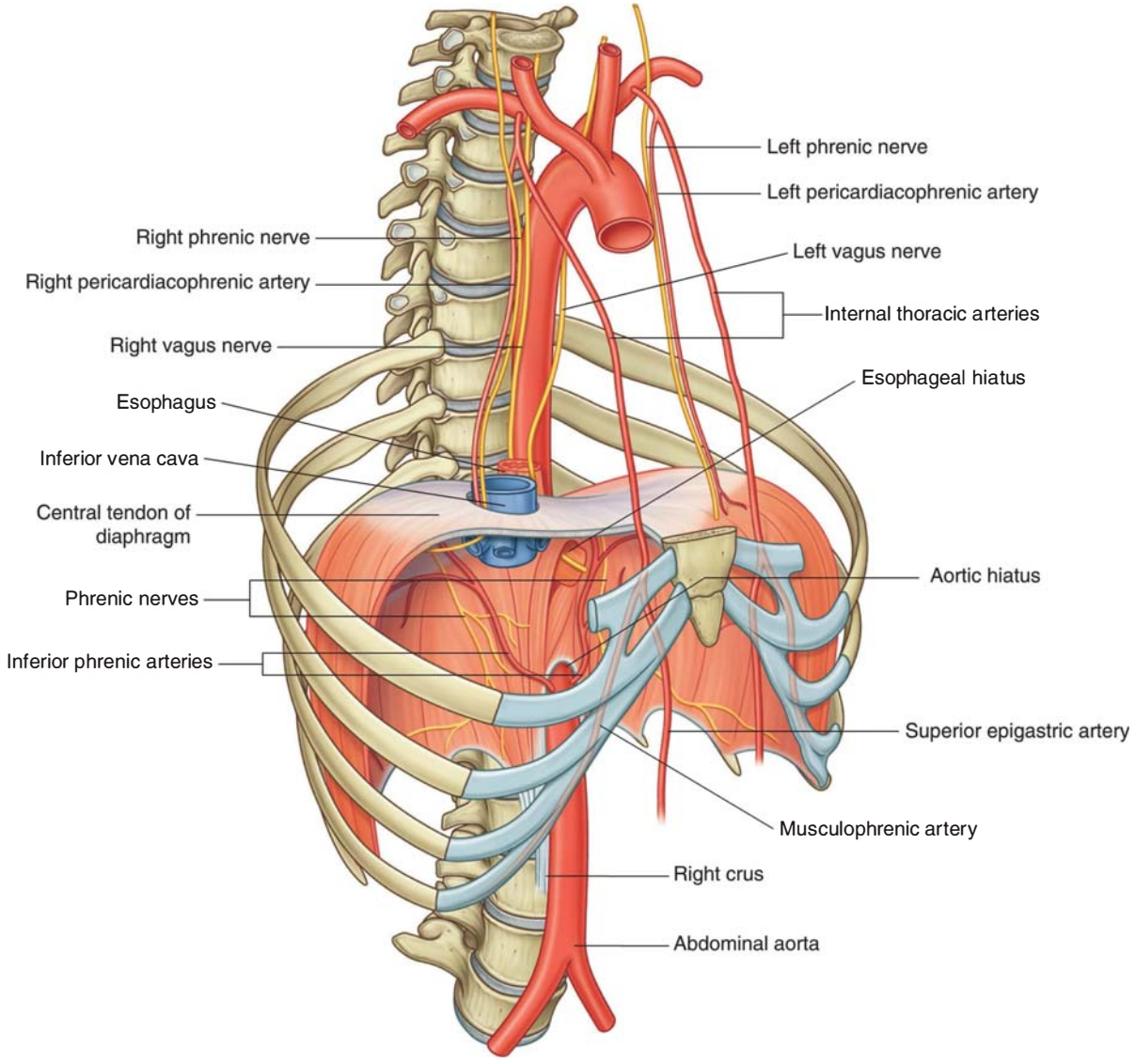


Fig. 3.33 Diaphragm.

SUBASH KC/NMC-15TH/2014

- The esophagus passes through the muscular part of the diaphragm, just to the left of midline, approximately at vertebral level TX.
- The vagus nerves pass through the diaphragm with the esophagus.
- The aorta passes behind the posterior attachment of the diaphragm at vertebral level TXII.
- The thoracic duct passes behind the diaphragm with the aorta.
- The azygos and hemiazygos veins may also pass through the aortic hiatus or through the crura of the diaphragm.

Other structures outside the posterior attachments of the diaphragm lateral to the aortic hiatus include the sympathetic trunks. The greater, lesser, and least splanchnic nerves penetrate the crura.

Arterial supply

The arterial supply to the diaphragm is from vessels that arise superiorly and inferiorly to it (see Fig. 3.33). From above, pericardiacophrenic and musculophrenic arteries supply the diaphragm. These vessels are branches of the internal thoracic arteries. **Superior phrenic arteries**, which arise directly from lower parts of the thoracic aorta, and small branches from intercostal arteries contribute to the supply. The largest arteries supplying the diaphragm arise from below it. These arteries are the **inferior phrenic arteries**, which branch directly from the abdominal aorta.

Venous drainage

Venous drainage of the diaphragm is by veins that generally parallel the arteries. The veins drain into:

- the brachiocephalic veins in the neck,
- the **azygos system of veins**, or
- abdominal veins (left suprarenal vein and inferior vena cava).

Innervation

The diaphragm is innervated by the **phrenic nerves** (C3, C4, and C5), which penetrate the diaphragm and innervate it from its abdominal surface.

Contraction of the domes of the diaphragm flattens the diaphragm, thereby increasing thoracic volume. Movements of the diaphragm are essential for normal breathing.

MOVEMENTS OF THE THORACIC WALL AND DIAPHRAGM DURING BREATHING

One of the principal functions of the thoracic wall and the diaphragm is to alter the volume of the thorax and thereby move air in and out of the lungs.

During breathing, the dimensions of the thorax change in the vertical, lateral, and anteroposterior directions. Elevation and depression of the diaphragm significantly alter the vertical dimensions of the thorax. Depression results when the muscle fibers of the diaphragm contract. Elevation occurs when the diaphragm relaxes.

Changes in the anteroposterior and lateral dimensions result from elevation and depression of the ribs (Fig. 3.34). The posterior ends of the ribs articulate with the vertebral column, whereas the anterior ends of most ribs articulate with the sternum or adjacent ribs.

Because the anterior ends of the ribs are inferior to the posterior ends, when the ribs are elevated, they move the sternum upward and forward. Also, the angle between the body of the sternum and the manubrium may become slightly less acute. When the ribs are depressed, the sternum moves downward and backward. This "pump handle" movement changes the dimensions of the thorax in the anteroposterior direction (Fig. 3.34A).

As well as the anterior ends of the ribs being lower than the posterior ends, the middles of the shafts tend to be lower than the two ends. When the shafts are elevated, the middles of the shafts move laterally. This "bucket handle" movement increases the lateral dimensions of the thorax (Fig. 3.34B).

Any muscles attaching to the ribs can potentially move one rib relative to another and therefore act as accessory respiratory muscles. Muscles in the neck and the abdomen can fix or alter the positions of upper and lower ribs.

PLEURAL CAVITIES

Two **pleural cavities**, one on either side of the mediastinum, surround the lungs (Fig. 3.35):

- Superiorly, they extend above rib I into the root of the neck.
- Inferiorly, they extend to a level just above the costal margin.
- The medial wall of each pleural cavity is the mediastinum.

SUBASH KC/NMC-15TH/2014

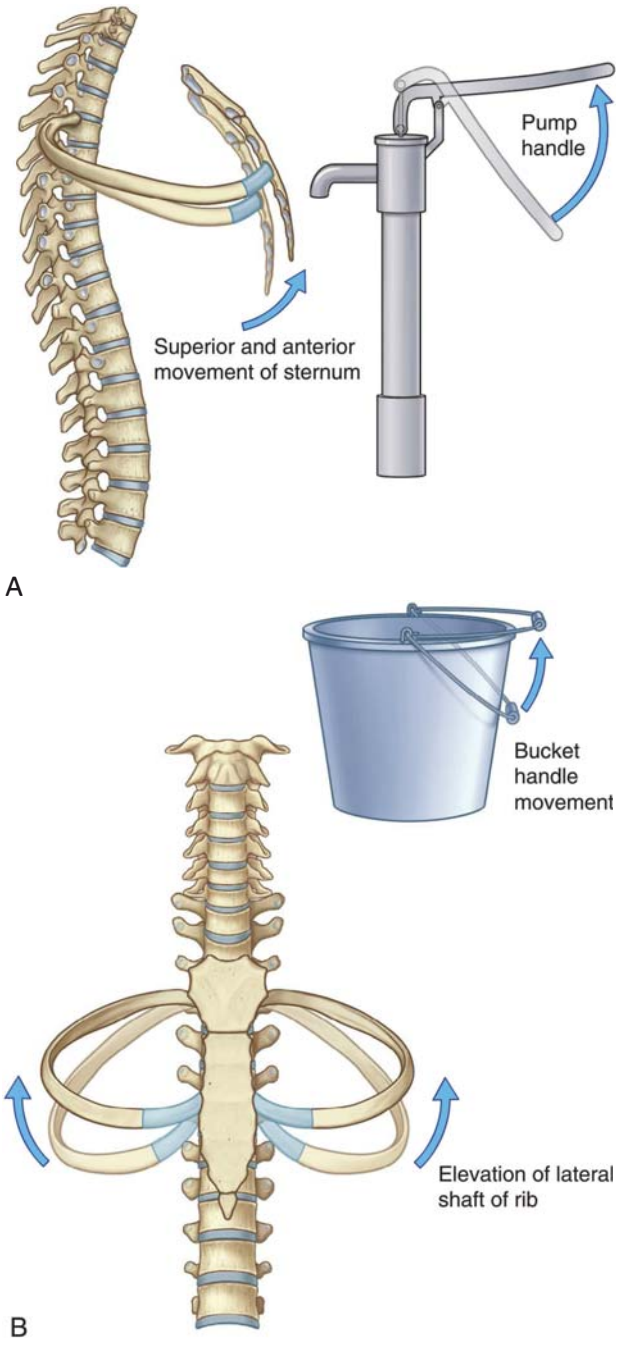


Fig. 3.34 Movement of thoracic wall during breathing. **A.** Pump handle movement of ribs and sternum. **B.** Bucket handle movement of ribs.

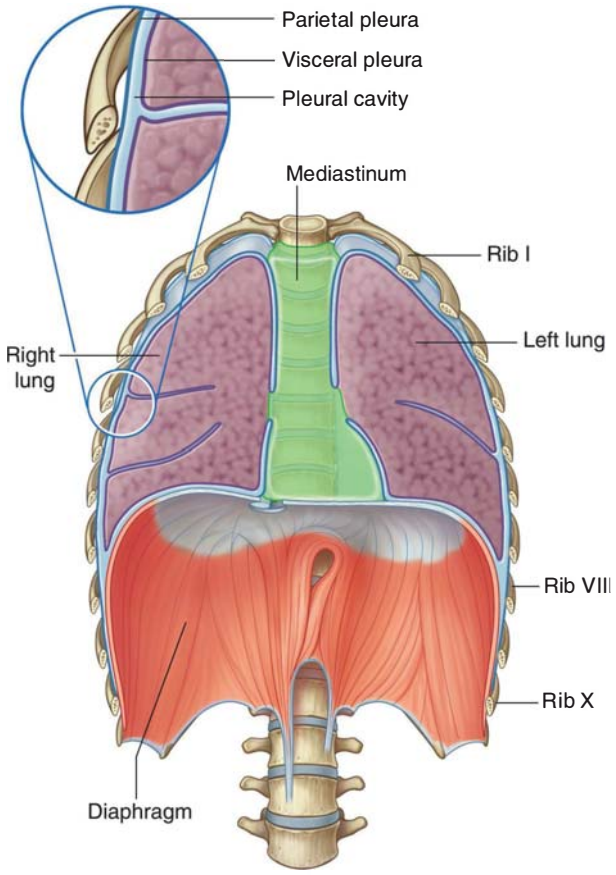


Fig. 3.35 Pleural cavities.

Pleura

Each pleural cavity is lined by a single layer of flat cells, mesothelium, and an associated layer of supporting connective tissue; together, they form the pleura.

The **pleura** is divided into two major types, based on location:

- Pleura associated with the walls of a pleural cavity is **parietal pleura** (Fig. 3.35).
- Pleura that reflects from the medial wall and onto the surface of the lung is **visceral pleura** (Fig. 3.35), which adheres to and covers the lung.

SUBASH KC/NMC-15TH/2014

Each pleural cavity is the potential space enclosed between the visceral and parietal pleurae. They normally contain only a very thin layer of serous fluid. As a result, the surface of the lung, which is covered by visceral pleura, directly opposes and freely slides over the parietal pleura attached to the wall.

Parietal pleura

The names given to the parietal pleura correspond to the parts of the wall with which they are associated (Fig. 3.36):

- Pleura related to the ribs and intercostal spaces is termed the **costal part**.
- Pleura covering the diaphragm is the **diaphragmatic part**.
- Pleura covering the mediastinum is the **mediastinal part**.
- The dome-shaped layer of parietal pleura lining the cervical extension of the pleural cavity is **cervical pleura** (**dome of pleura** or **pleural cupola**).

Covering the superior surface of the cervical pleura is a distinct dome-like layer of fascia, the **suprapleural membrane** (Fig. 3.36). This connective tissue membrane is attached laterally to the medial margin of the first rib and behind to the transverse process of vertebra CVII. Superiorly, the membrane receives muscle fibers from some of the

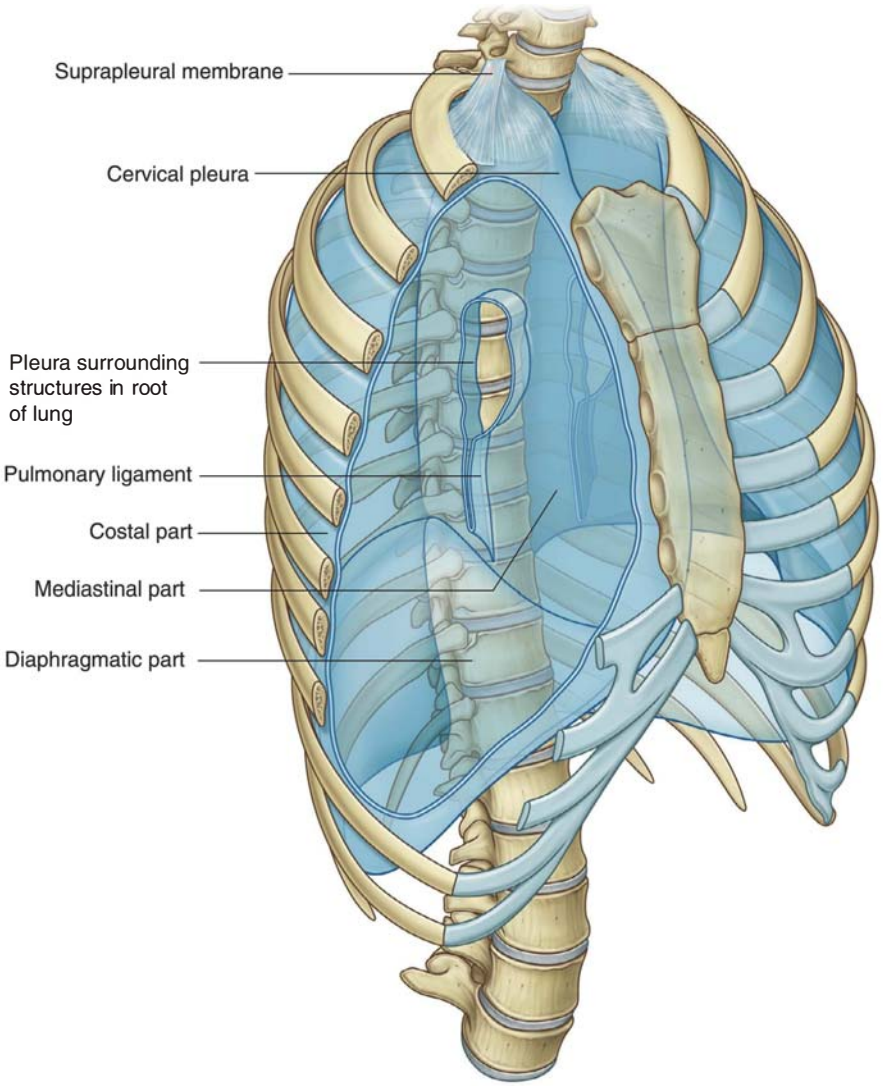


Fig. 3.36 Parietal pleura.

SUBASH KC/NMC-15TH/2014

deep muscles in the neck (scalene muscles) that function to keep the membrane taught. The suprapleural membrane provides apical support for the pleural cavity in the root of the neck.

In the region of vertebrae TV to TVII, the mediastinal pleura reflects off the mediastinum as a tubular, sleeve-like covering for structures (i.e., airway, vessels, nerves, lymphatics) that pass between the lung and mediastinum. This sleeve-like covering, and the structures it contains, forms the **root of the lung**. The root joins the medial surface of the lung at an area referred to as the **hilum of the lung**. Here, the mediastinal pleura is continuous with the visceral pleura.

The parietal pleural is innervated by somatic afferent fibers. The costal pleura is innervated by branches from the intercostal nerves, and pain would be felt in relation to the thoracic wall. The diaphragmatic pleura and the mediastinal pleura are innervated mainly by the phrenic nerves (originating at spinal cord levels C3, C4, and C5). Pain from these areas would refer to the C3, C4, and C5 dermatomes (lateral neck and the supraclavicular region of the shoulder).

Peripheral reflections

The peripheral reflections of parietal pleura mark the extent of the pleural cavities (Fig. 3.37).

Superiorly, the pleural cavity can project as much as 3–4 cm above the first costal cartilage but does not extend above the neck of rib I. This limitation is caused by the inferior slope of rib I to its articulation with the manubrium.

Anteriorly, the pleural cavities approach each other posterior to the upper part of the sternum. However, posterior to the lower part of the sternum, the parietal pleura does not come as close to the midline on the left side as it does on the right because the middle mediastinum, containing the pericardium and heart, bulges to the left.

Inferiorly, the costal pleura reflects onto the diaphragm above the costal margin. In the midclavicular line, the pleural cavity extends inferiorly to approximately rib VIII. In the midaxillary line, it extends to rib X. From this point, the inferior margin courses somewhat horizontally, crossing ribs XI and XII to reach vertebra TXII. From the midclavicular line to the vertebral column, the inferior

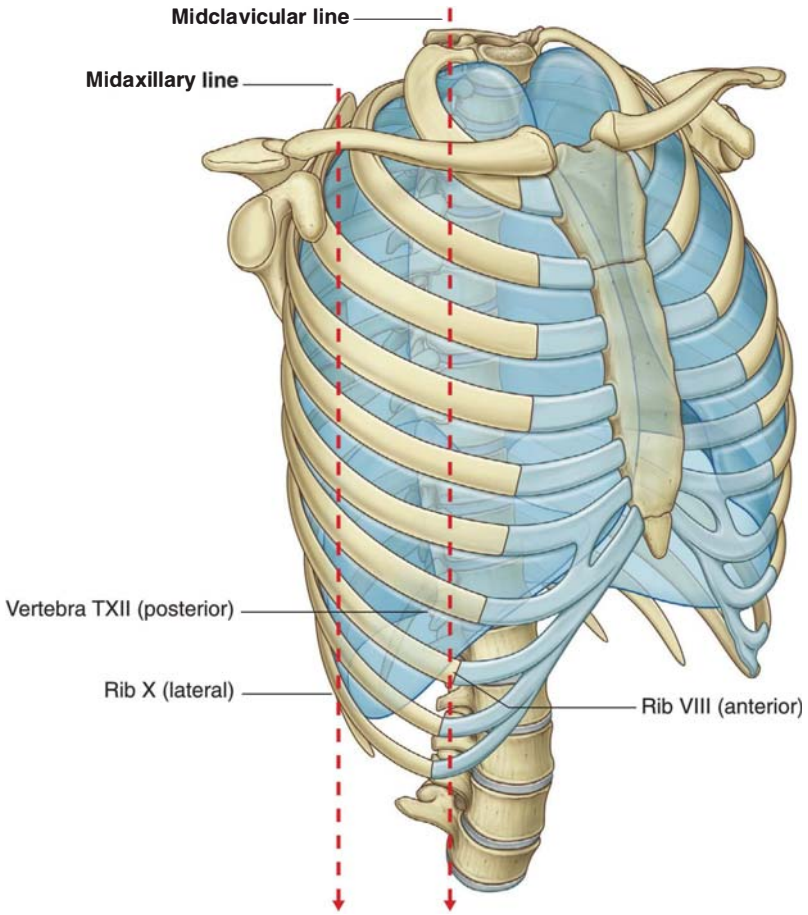


Fig. 3.37 Pleural reflections.

SUBASH KC/NMC-15TH/2014

boundary of the pleura can be approximated by a line that runs between rib VIII, rib X, and vertebra TXII.

Visceral pleura

The visceral pleura is continuous with the parietal pleura at the hilum of each lung, where structures enter and leave the organ. The visceral pleura is firmly attached to the surface of the lung, including both opposed surfaces of the fissures that divide the lungs into lobes.

Although the visceral pleura is innervated by visceral afferent nerves that accompany bronchial vessels, pain is generally not elicited from this tissue.

Pleural recesses

The lungs do not completely fill the anterior or posterior inferior regions of the pleural cavities (Fig. 3.38). This results in recesses in which two layers of parietal pleura become opposed. Expansion of the lungs into these spaces usually occurs only during forced inspiration; the recesses also provide potential spaces in which fluids can collect and from which fluids can be aspirated.

Costomediastinal recesses

Anteriorly, a **costomediastinal recess** occurs on each side where costal pleura is opposed to mediastinal pleura. The largest is on the left side in the region overlying the heart (Fig. 3.38).

Costodiaphragmatic recesses

The largest and clinically most important recesses are the **costodiaphragmatic recesses**, which occur in each pleural cavity between the costal pleura and diaphragmatic pleura (Fig. 3.38). The costodiaphragmatic recesses are the regions between the inferior margin of the lungs and inferior margin of the pleural cavities. They are deepest after forced expiration and shallowest after forced inspiration.

During quiet respiration, the inferior margin of the lung crosses rib VI in the midclavicular line and rib VIII in the

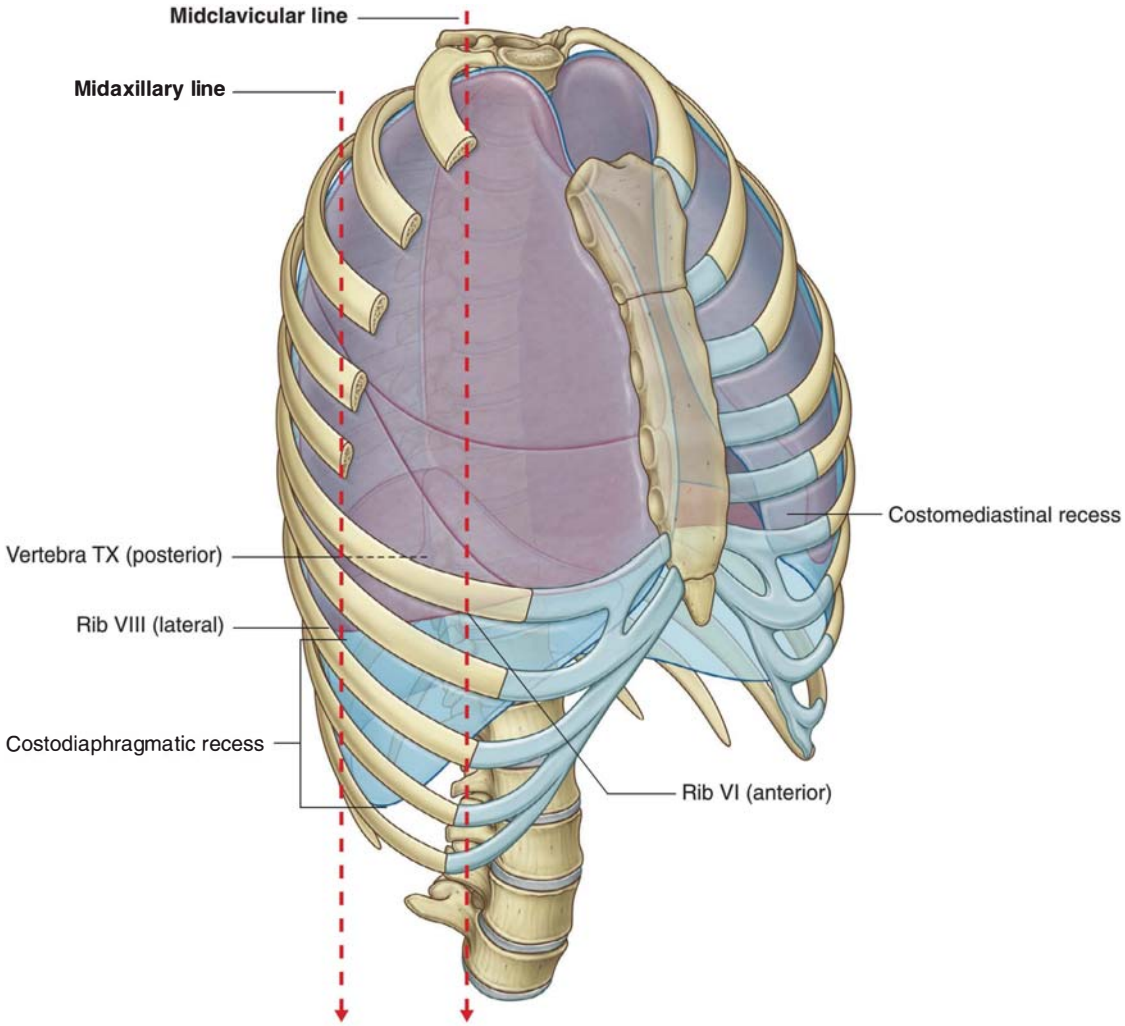


Fig. 3.38 Parietal pleural reflections and recesses.

SUBASH KC/NMC-15TH/2014

midaxillary line, and then courses somewhat horizontally to reach the vertebral column at vertebral level TX. Thus, from the midclavicular line and around the thoracic wall to the vertebral column, the inferior margin of the lung can be approximated by a line running between rib VI, rib VIII, and vertebra TX. The inferior margin of the pleural cavity at the same points is rib VIII, rib X, and vertebra TXII. The costodiaphragmatic recess is the region between the two margins.

During expiration, the inferior margin of the lung rises and the costodiaphragmatic recess becomes larger.

In the clinic

Pleural effusion

A pleural effusion occurs when excess fluid accumulates within the pleural space. As the fluid accumulates within the pleural space the underlying lung is compromised and may collapse as the volume of fluid increases. Once a pleural effusion has been diagnosed, fluid often will be aspirated to determine the cause, which can include infection, malignancy, cardiac failure, hepatic disease, and pulmonary embolism.

In the clinic

Pneumothorax

A pneumothorax is a collection of gas or air within the pleural cavity. When air enters the pleural cavity the tissue elasticity of the parenchyma causes the lung to collapse within the chest, impairing the lung function. Occasionally, the gas within the pleural cavity may accumulate to such an extent that the mediastinum is "pushed" to the opposite side, compromising the other lung. This is termed a tension pneumothorax and requires urgent treatment.

Most pneumothoraces are spontaneous (i.e., they occur in the absence of no known pathology and no known lung disease). In addition, pneumothoraces may occur as a result of trauma, inflammation, smoking, and other underlying pulmonary diseases.

The symptoms of pneumothorax are often determined by the degree of air leak and the rate at which the accumulation of gas occurs and the ensuing lung collapses. They include pain, shortness of breath, and cardiorespiratory collapse, if severe.

Lungs

The two lungs are organs of respiration and lie on either side of the mediastinum surrounded by the right and left pleural cavities. Air enters and leaves the lungs via main bronchi, which are branches of the trachea.

The pulmonary arteries deliver deoxygenated blood to the lungs from the right ventricle of the heart. Oxygenated blood returns to the left atrium via the pulmonary veins.

The right lung is normally a little larger than the left lung because the middle mediastinum, containing the heart, bulges more to the left than to the right.

Each lung has a half-cone shape, with a base, apex, two surfaces, and three borders (Fig. 3.39).

- The **base** sits on the diaphragm.
- The **apex** projects above rib I and into the root of the neck.
- The two surfaces—the **costal surface** lies immediately adjacent to the ribs and intercostal spaces of the thoracic wall. The **mediastinal surface** lies against the mediastinum anteriorly and the vertebral column posteriorly and contains the comma-shaped hilum of the lung, through which structures enter and leave.
- The three borders—the **inferior border** of the lung is sharp and separates the base from the costal surface. The **anterior** and **posterior borders** separate the costal surface from the medial surface. Unlike the anterior and inferior borders, which are sharp, the posterior border is smooth and rounded.

The lungs lie directly adjacent to, and are indented by, structures contained in the overlying area. The heart and major vessels form bulges in the mediastinum that indent the medial surfaces of the lung; the ribs indent the costal surfaces. Pathology, such as tumors, or abnormalities in one structure can affect the related structure.

Root and hilum

The **root** of each lung is a short tubular collection of structures that together attach the lung to structures in the mediastinum (Fig. 3.40). It is covered by a sleeve of mediastinal pleura that reflects onto the surface of the lung as visceral pleura. The region outlined by this pleural reflection on the medial surface of the lung is the **hilum**, where structures enter and leave.

A thin blade-like fold of pleura projects inferiorly from the root of the lung and extends from the hilum to the mediastinum. This structure is the **pulmonary ligament**. It may stabilize the position of the inferior lobe and may also accommodate the down-and-up translocation of structures in the root during breathing.

In the mediastinum, the vagus nerves pass immediately posterior to the roots of the lungs, while the phrenic nerves pass immediately anterior to them.

SUBASH KC/NMC-15TH/2014

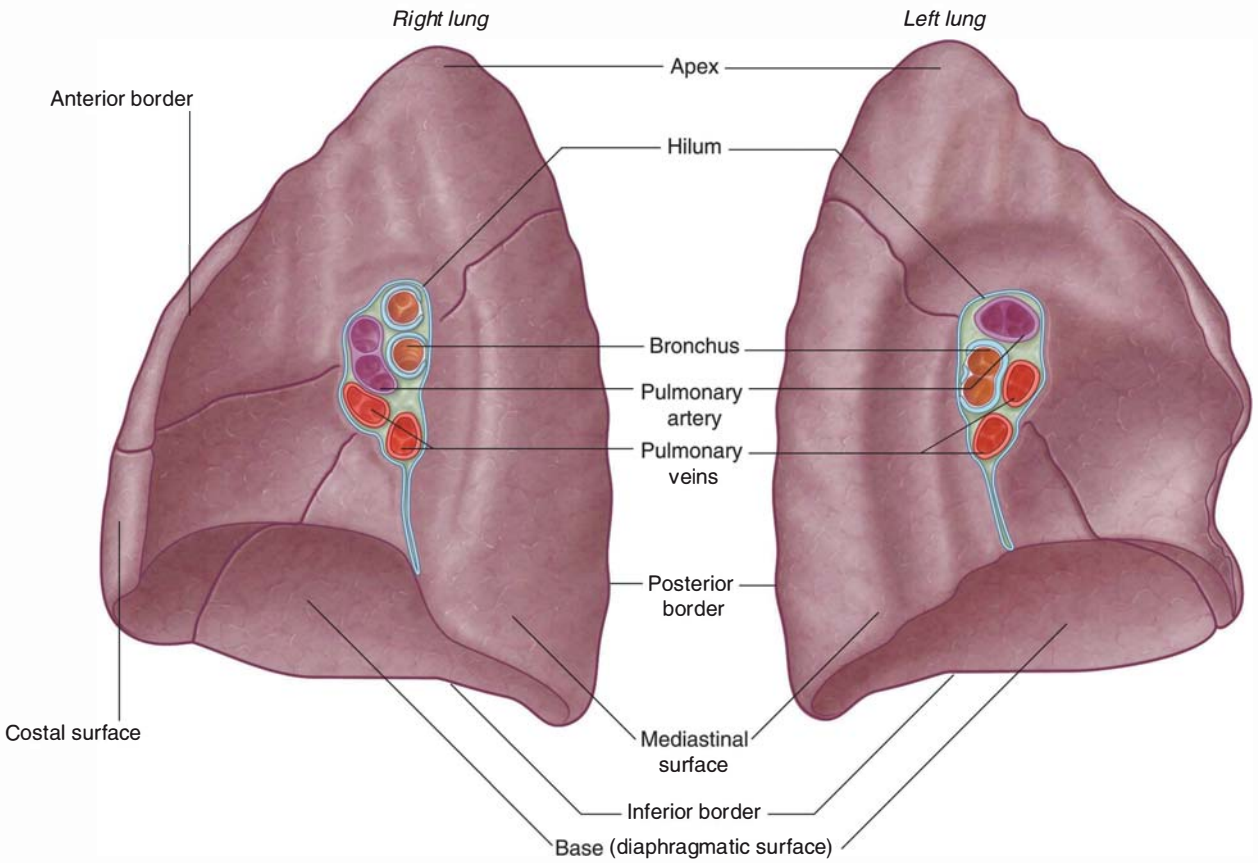


Fig. 3.39 Lungs.

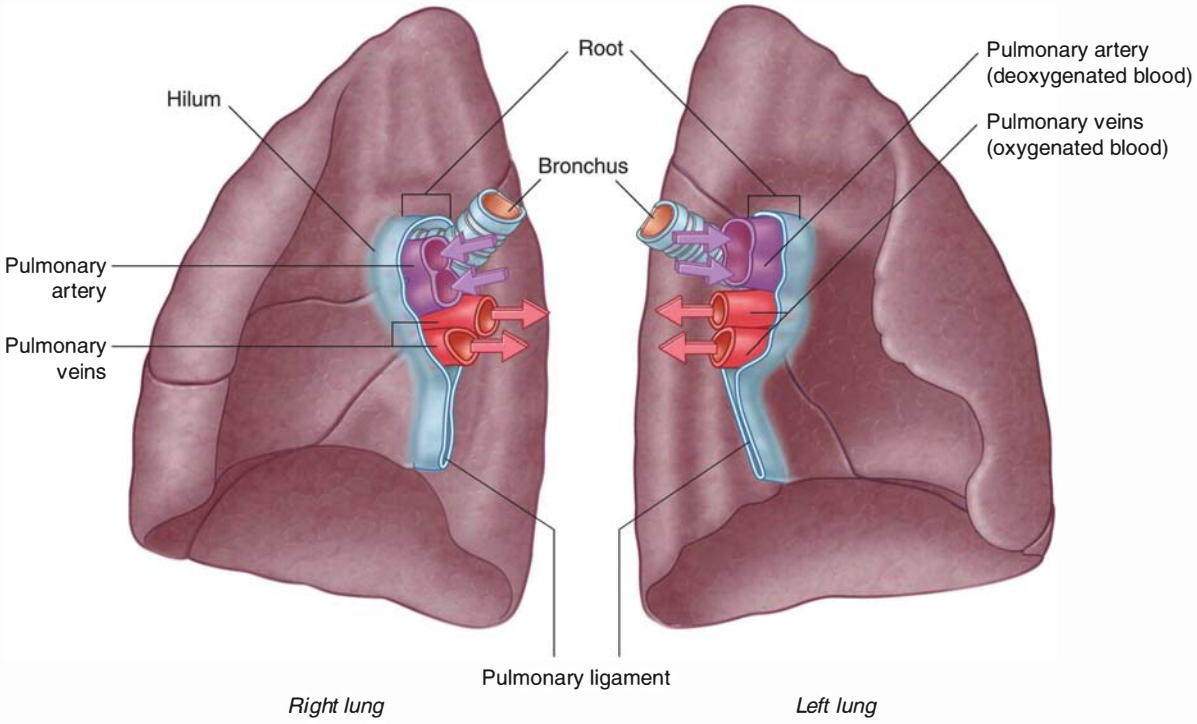


Fig. 3.40 Roots and hila of the lungs.

SUBASH KC/NMC-15TH/2014

Within each root and located in the hilum are:

- a pulmonary artery,
- two pulmonary veins,
- a main bronchus,
- bronchial vessels,
- nerves, and
- lymphatics.

Generally, the pulmonary artery is superior at the hilum, the pulmonary veins are inferior, and the bronchi are somewhat posterior in position.

On the right side, the lobar bronchus to the superior lobe branches from the main bronchus in the root, unlike on the left where it branches within the lung itself, and is superior to the pulmonary artery.

Right lung

The **right lung** has three lobes and two fissures (Fig. 3.41A). Normally, the lobes are freely movable against each other because they are separated, almost to the hilum, by invaginations of visceral pleura. These invaginations form the fissures:

- The **oblique fissure** separates the **inferior lobe** (**lower lobe**) from the **superior lobe** and the **middle lobe of the right lung**.
- The **horizontal fissure** separates the superior lobe (**upper lobe**) from the middle lobe.

The approximate position of the oblique fissure on a patient, in quiet respiration, can be marked by a curved line on the thoracic wall that begins roughly at the spinous process of the vertebra TIV level of the spine, crosses the fifth interspace laterally, and then follows the contour of rib VI anteriorly (see p. 239).

The horizontal fissure follows the fourth intercostal space from the sternum until it meets the oblique fissure as it crosses rib V.

The orientations of the oblique and horizontal fissures determine where clinicians should listen for lung sounds from each lobe.

The largest surface of the superior lobe is in contact with the upper part of the anterolateral wall and the apex of this lobe projects into the root of the neck. The surface of the middle lobe lies mainly adjacent to the lower anterior and lateral wall. The costal surface of the inferior lobe is in contact with the posterior and inferior walls.

When listening to lung sounds from each of the lobes, it is important to position the stethoscope on those areas of the thoracic wall related to the underlying positions of the lobes (see p. 240).

The medial surface of the right lung lies adjacent to a number of important structures in the mediastinum and the root of the neck (Fig. 3.41B). These include the:

- heart,
- inferior vena cava,
- superior vena cava,
- azygos vein, and
- esophagus.

The right subclavian artery and vein arch over and are related to the superior lobe of the right lung as they pass over the dome of the cervical pleura and into the axilla.

Left lung

The **left lung** is smaller than the right lung and has two lobes separated by an oblique fissure (Fig. 3.42A). The **oblique fissure** of the left lung is slightly more oblique than the corresponding fissure of the right lung.

During quiet respiration, the approximate position of the left oblique fissure can be marked by a curved line on the thoracic wall that begins between the spinous processes of vertebrae TIII and TIV, crosses the fifth interspace laterally, and follows the contour of rib VI anteriorly (see pp. 237–238).

As with the right lung, the orientation of the oblique fissure determines where to listen for lung sounds from each lobe.

The largest surface of the superior lobe is in contact with the upper part of the anterolateral wall, and the apex of this lobe projects into the root of the neck. The costal surface of the inferior lobe is in contact with the posterior and inferior walls.

When listening to lung sounds from each of the lobes, the stethoscope should be placed on those areas of the thoracic wall related to the underlying positions of the lobes (see p. 240).

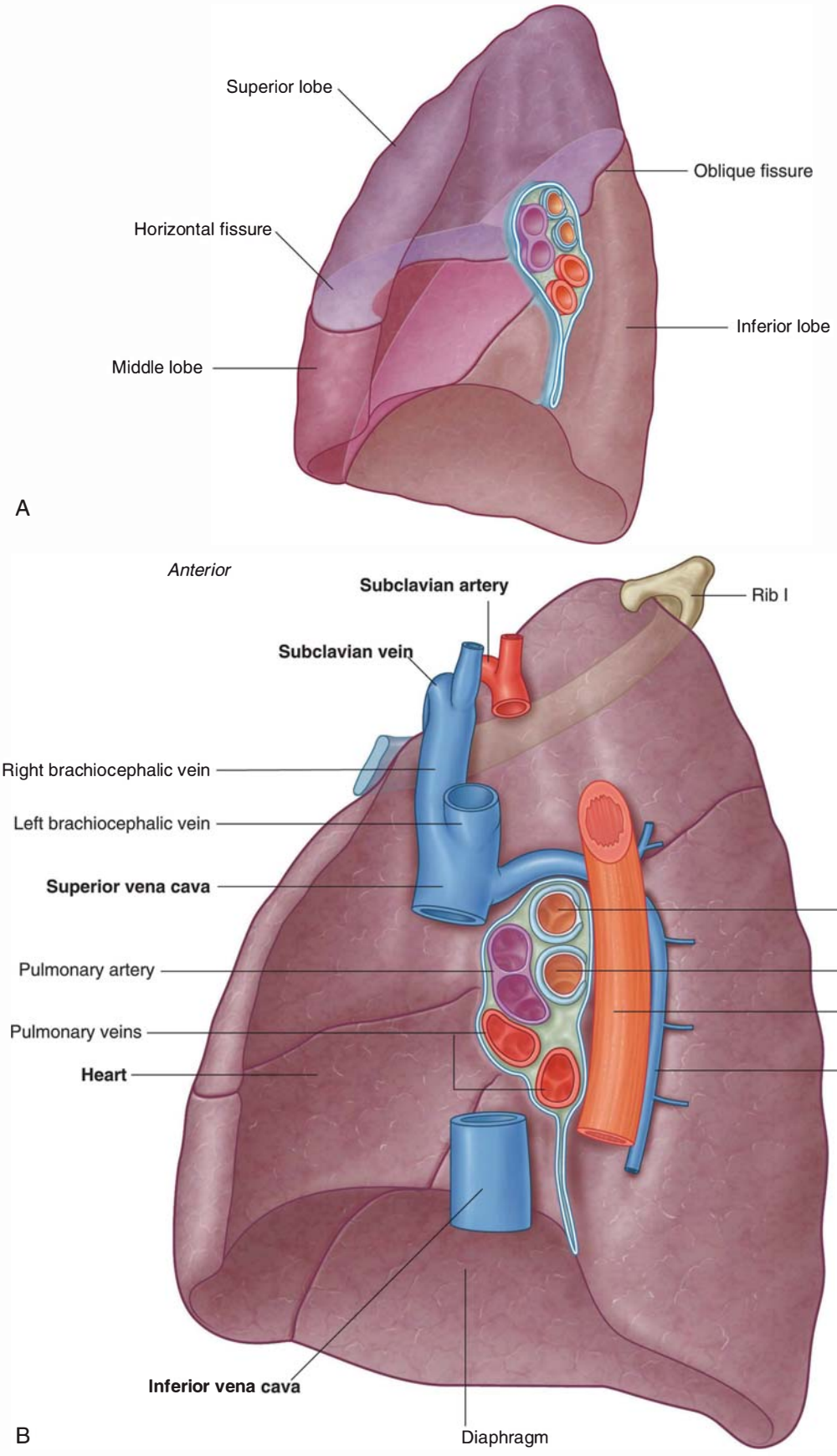
The inferior portion of the medial surface of the left lung, unlike the right lung, is notched because of the heart's projection into the left pleural cavity from the middle mediastinum.

From the anterior border of the lower part of the superior lobe a tongue-like extension (the **lingula of the left lung**) projects over the heart bulge.

The medial surface of the left lung lies adjacent to a number of important structures in the mediastinum and root of the neck (Fig. 3.42B). These include the:

- heart,
- aortic arch,
- thoracic aorta, and
- esophagus.

SUBASH KC/NMC-15TH/2014



SUBASH KC/NMC-15TH/2014

Fig. 3.41 A. Right lung. **B.** Major structures related to the right lung.

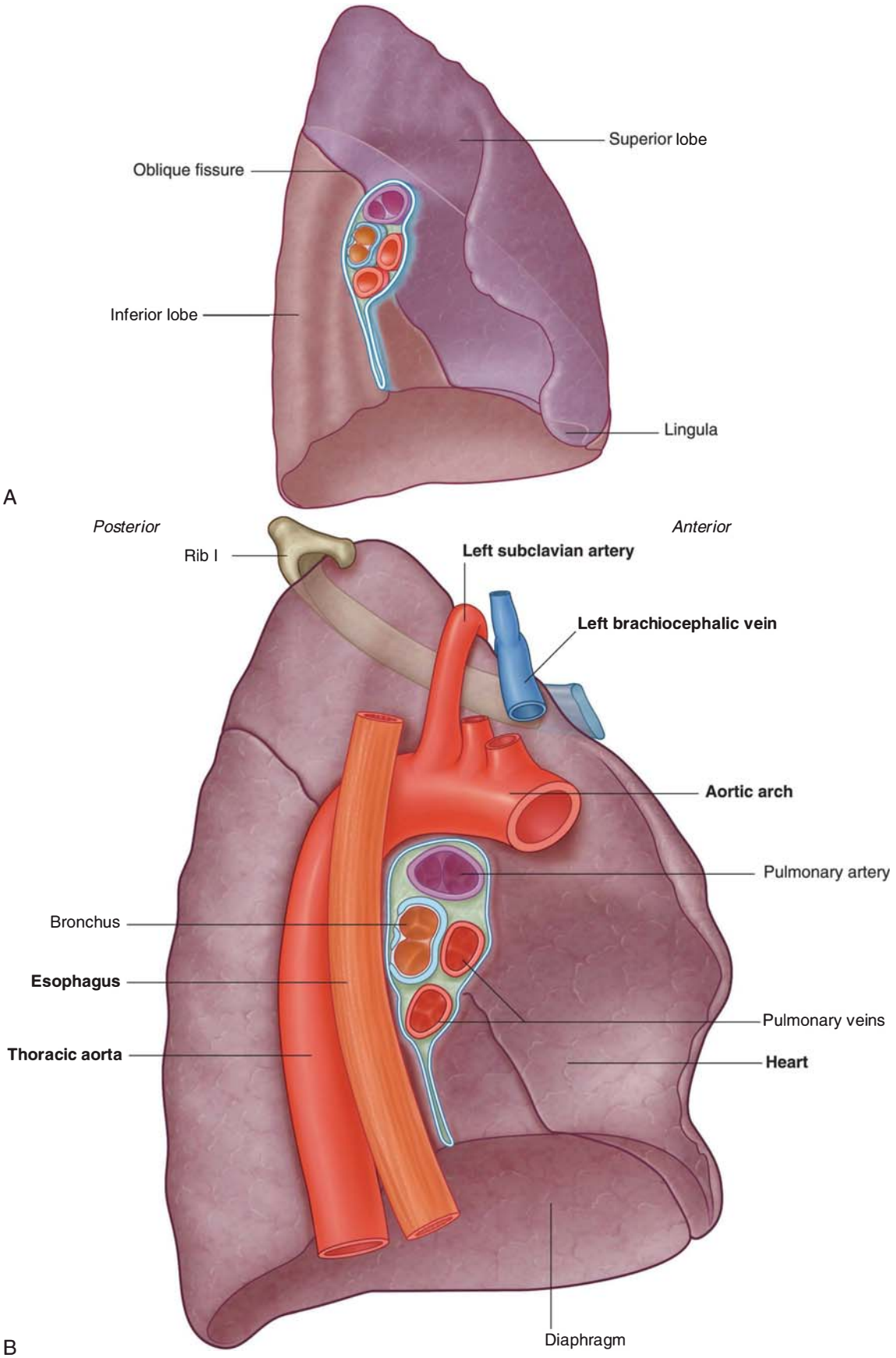


Fig. 3.42 **A.** Left lung. **B.** Major structures related to the left lung.

SUBASH KC/NMC-15TH/2014

The left subclavian artery and vein arch over and are related to the superior lobe of the left lung as they pass over the dome of the cervical pleura and into the axilla.

Bronchial tree

The **trachea** is a flexible tube that extends from vertebral level CVI in the lower neck to vertebral level TIV/V in the mediastinum where it bifurcates into a right and a left main bronchus (Fig. 3.43). The trachea is held open by C-shaped transverse cartilage rings embedded in its wall—the open part of the C facing posteriorly. The lowest tracheal ring has a hook-shaped structure, the carina, that projects backwards in the midline between the origins of the two main bronchi. The posterior wall of the trachea is composed mainly of smooth muscle.

Each main bronchus enters the root of a lung and passes through the hilum into the lung itself. The right main bronchus is wider and takes a more vertical course

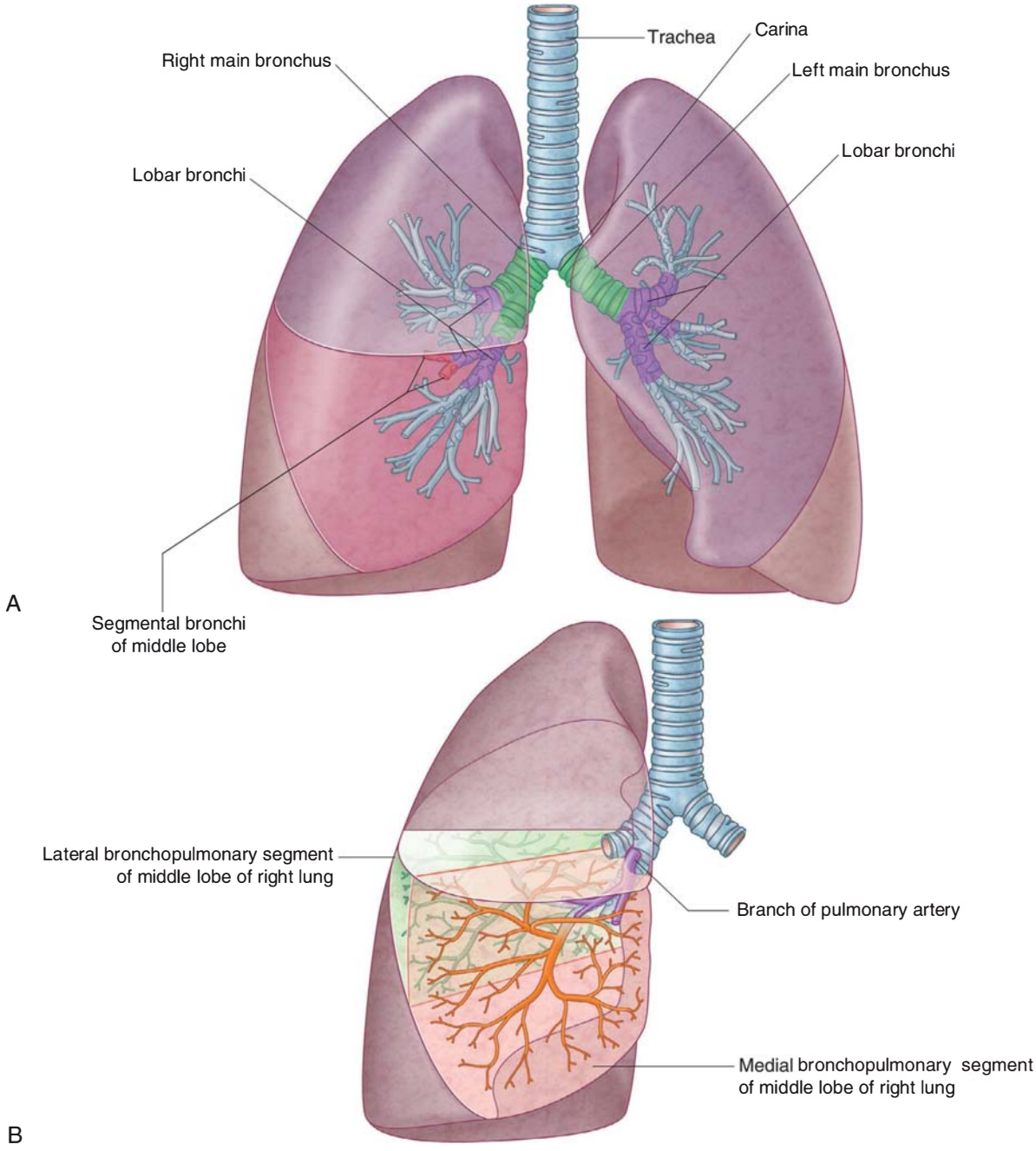


Fig. 3.43 A. Bronchial tree. **B.** Bronchopulmonary segments.

SUBASH KC/NMC-15TH/2014